Appendix 3C: Development of a Fleets-as-Areas (FAA) Stock Assessment Model for Alaska Sablefish

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Executive Summary

Given the complex ontogenetic spatial patterns exhibited by sablefish in Alaska as well as the change in NOAA Longline Survey design in 2025 (where only ~50% of the population is being sampled in a given year), a fleets-as-areas (FAA) model building exercise was undertaken to explore alternatives to current single region assessment models. FAA models disaggregate data by region and use model selectivity and catchability assumptions to alias differential age-based availability across regions to implicitly account for spatial heterogeneity. Thus, an FAA model was identified as a potential suitable structure for sablefish, as it could disaggregate survey data by region and avoid the need for 'carryover' extrapolation as is used in the single region assessment (Model *26.1_1Reg*; to ensure a consistent scaling of the index given inconsistent sampling across years, see Appendix 3B). Ultimately, the choice of model structure for sablefish is a tradeoff between model stability, assessment complexity, and assumed data representativeness (Table 3C.1). Although an FAA model would forego the need for data extrapolation, it may be more unstable and might overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high).

The final FAA model, Model *26.2_FAA*, appears to provide a robust and stable model for interpreting sablefish population dynamics. Although Model *26.2_FAA* leads to a moderate downward scaling of SSB and catch advice, general population dynamics and stock status remain consistent with Model *26.1_1Reg*. Similarly, fits to the available data retain similar issues, where survey indices are well fit at the expense of composition data, primarily due to the Francis reweighting approach of 'right-weighting' the data to ensure adequate fit to the indices. Although residual patterns in the composition data remain, the aggregated fits continue to appear adequate given associated model tension between fitting the NOAA Longline Survey indices versus age compositions. Fundamentally, both the single region and FAA models have trouble reconciling the much larger year classes and population growth suggested by the age composition data compared with the rapid, but more tempered, population growth observed in the survey RPNs. All model diagnostics, including retrospective patterns, indicated that Model *26.2_FAA* is relatively stable and showed no signs of significant misspecification. Although other FAA models were explored, data sparsity issues resulted when attempting to regionally disaggregate the fishery data, suggesting that Model *26.2_FAA* was a suitable model of intermediate complexity.

An exploratory FAA model run using the 'carryover' approach was enlightening as it showed the potential downside of using extrapolated data in the assessment, given that it led to a much more optimistic estimate of stock status and catch advice. Although the sablefish MSE work indicates that the carryover extrapolation is not likely to have a major detrimental impact on management advice from the single region model, it does appear that using extrapolated values in the index should be considered judiciously. Thus, Model *26.2_FAA* may be a good candidate for the

sablefish operational assessment and management, given that it does not rely on any data extrapolation and appears to provide a stable and robust assessment.

Sensitivity runs demonstrated application of the Dirichlet-multinomial likelihood, which effectively upweighted composition data relative to the Francis reweighting approach, as an attempt to provide a more statistically rigorous approach to data weighting (and to better fit the age compositions). However, using the Dirichlet-multinomial likelihood led to degraded fits to abundance indices, more variable recruitment estimates, and inflated population scale. Additionally, the combination of continuous time-varying selectivity and the Dirichlet-multinomial likelihood was explored but failed to reconcile the respective issues encountered when each approach was applied independently, particularly the positive retrospective patterns. Collectively, sensitivity run explorations suggest that none of the alternative configurations examined performed better than Model *26.2_FAA*. However, they provide a useful foundation for future model development and help clarify key areas of model tension and structural sensitivity.

The following document summarizes model performance and results of select FAA model runs, primarily in comparison to the final single region model (*26.1_1Reg*). Full model results and diagnostics are available at the Alaska sablefish assessment CIE review GitHub site, where individual folders are provided for every model run produced for the CIE Review. Folders contain SPoRC input and output R files along with PDFs of results (folders for FAA model runs are available here). Full model outputs for the final FAA model (*26.2_FAA*) and the final single region model (*26.1_1Reg*) are provided in the Supplementary Material (SM3C).

1. Fleets-as-Areas Model Development

Introduction

A change to the design of the NOAA Longline Survey starting in 2025 has led to a situation where approximately only 50% of the sablefish abundance/distribution in a given year is sampled. Since 1990, the Gulf of Alaska (GOA) has been sampled annually, while the Aleutian Islands (AI) were sampled in even years starting in 1996 and the Bering Sea (BS) was sampled in odd years starting in 1997 (Siwicke and Malecha, 2026). Beginning in 2025, only the GOA is being sampled in odd years, and both the BS and AI are being sampled in even years. However, the abundance in the BS and AI appears to be increasing more rapidly than other regions over the last decade (e.g., based on regional NOAA Longline Survey Relative Population Numbers, RPNs; Figure 3C.1), apparently due to extremely large recent year classes primarily recruiting in these regions before moving to other regions as they mature (see the results of the spatial model, Appendix 3D; Figure 3C.2).

To maintain a consistent scaling of the index given the inconsistent survey design, an extrapolation of index RPNs is required for years when a given region is not surveyed -- if the index is to be used in a single region assessment and assumed to be representative of the entire population. Historically, the year over year change in the GOA was used to extrapolate growth in the BS or AI for survey off years (Siwicke and Malecha, 2026). However, that approach is no longer feasible because the GOA is not sampled annually. In 2025, the 'carryover' approach was introduced wherein a constant hold-over value of the RPN from the previous sampled year was used for regions that were not sampled. Regardless of extrapolation approach, around 50% of the index relies on extrapolated values under the new survey design. Concomitantly, aggregated age compositions (using area RPN-weighted average proportions) are either dominated by the GOA historically (given larger summed RPNs across the GOA compared to either the BS or AI in a given year) or undergo strong interannual variability due to the survey switching between the GOA one year and BS and AI the other (where these regions demonstrate strong differences in compositions; see Figures SM3C.18, SM3C.24, and SM3C.31). Thus, resulting single region models are highly reliant on extrapolated data and likely prone to retrospective patterns in recruitment caused by interannual variability in age compositions due to sampling different segments of the population that aren't representative of the entire population on a yearly basis.

An alternative to developing a single index and extrapolating for a region that is not sampled is to develop multiple indices, disaggregated by region, such that the survey footprint is consistent within each region and off years are simply treated as missing data. Similarly, the associated composition data can be aggregated at the same regional scale, ensuring that each regional index has concordant composition data. Within a stock assessment, there are two approaches to fitting regional data: spatially implicit fleets-as-areas (FAA) models or spatially explicit multi-region models. The primary difference in assumptions is that FAA models assume a single population and use 'regional' (modeled as fleets) selectivity to alias spatial processes, while spatial models explicitly model spatial processes such as movement and regional recruitment dynamics (Goethel et al., 2023).

Similar to a single region assessment, an FAA model assumes that the 'regional' trends and compositions represent a single underlying population, where survey-specific catchability (q) and

selectivity (and/or fishing mortality, *F*, and fishery selectivity) estimates are used to implicitly account for varying age-specific availability (e.g., due to the survey footprint being smaller than the distribution of the population of inference or caused by ontogenetic or seasonal migration patterns; Berger et al., 2012; Waterhouse et al., 2014). FAA models tend to perform better than single region models when there are patterns in the composition data due to seasonal or ontogenetic availability that single region models are otherwise unable to explain (since they cannot account for spatial structure; Lee et al., 2017). Conversely, FAA may perform poorly compared to spatial assessments when underlying spatial population structure and complex movement patterns exist (Bosley et al., 2022). Moreover, FAA models may be prone to model instability due to issues estimating selectivity parameters as sample sizes decrease when regional fleets are modeled independently, particularly for lesser sampled regions (e.g., the BS and AI for sablefish, particularly in the fishery data). Interpreting the dynamics of FAA models can also be less intuitive, because each fleet is still operating on a single, homogeneous population, such that variability in indices and compositions can only be interpreted as changes in availability. Thus, when signals in the data (e.g., survey indices or compositions) differ for fundamental population processes (e.g., recruitment) across space, the model must attempt to rectify these differences primarily through selectivity differences, potentially leading to non-intuitive predictions (e.g., predicted survey abundance trends).

When there is complex population structure or spatial processes present, development of a spatially explicit model may be preferred. Spatial models fit the data at regional scales and, unlike FAA models, are better able to differentiate regionally varying data signals as changes in population composition (e.g., age structure or biological differences) or availability across space. In simulation studies, spatial models are often found to be the most robust to complex spatial dynamics (e.g., Lee et al., 2017; Goethel et al., 2021). However, real-world implementation is often difficult as data sparsity may cause severe model instability, parametrization is often complex, and tagging data is typically required to estimate movement yet comes with numerous additional complications (e.g., dealing tag mixing and data weighting; Goethel et al., 2023). A spatial model for sablefish has been developed (Cheng et al., 2025; see Appendix 3D for updated results), but it is primarily a research tool. Ongoing work with the sablefish spatial model has been exploring how best to loosen simplifying model assumptions (e.g., spatially invariant selectivity and time-invariant movement estimates), handle tag mixing, and deal with tag data weighting in relation to other data sources (e.g., indices and compositions). Despite relatively good model convergence and general diagnostics for the spatial model, instability still exists. Therefore, the spatial model requires further refinement and evaluation before being explored for management decision-making.

To better understand the potential tradeoffs among assessment model structures for sablefish, particularly for dealing with the recent survey design changes, a sablefish-conditioned closed loop simulation study was developed (see Appendix 3E). The desktop management strategy evaluation (MSE) utilized a spatially explicit, five region operating model with movement and spatial dynamics matching those estimated by the spatial assessment model (Appendix 3D). Moreover, the study emulated the real-world survey design both historically and moving forward, which enabled thorough evaluation of the expected impacts of survey changes in 2025 on each assessment structure. Three model structures were explored, including: a single region assessment (generally matching Model *26.1_1Reg*) utilizing an extrapolated, carryover longline survey index

(from 2025 onwards), a FAA model (generally matching Model *26.2a_FAA_Fsh+Srvy*) fitting regionally (BS, AI, GOA) disaggregated longline survey indices and comps as well as fixed gear fishery catch and compositions, and a three region spatial assessment (generally matching model *26.3a_Spat_3Reg*) that estimated regionally-varying parameters and movement. Overall, the FAA assessment configuration performed best, especially as survey coverage declined (emulating the real-world survey dynamics; Appendix 3E; Table 1). The single region model tended to demonstrate higher yet moderate SSB bias, associated with using the extrapolated longline survey index and utilizing aggregated survey composition data that did not always equitably reflect dynamics across all regions (e.g., it was over representative of the GOA). The spatial model performed well, on average, but was extremely unstable and prone to unrealistic estimates (e.g., no recruitment in a given region and year). From a management standpoint, all models tended to perform similarly, with low long-term risk to the population or foregone yield, despite potential assessment bias across structures. Thus, the FAA model was generally the 'best' assessment framework from a simulation standpoint, but differences were subtle. Conversely, the simulation study did not likely adequately represent potential regional data sparsity that could have detrimental impacts for the more complex assessment structures (i.e., FAA and spatial models).

Ultimately, the choice of model structure for sablefish is a tradeoff between model stability, assessment complexity, and assumed data representativeness (Table 3C.1). The relatively stable single region model performs adequately but requires extensive extrapolation assumptions and has difficulty simultaneously fitting the indices and composition data (indicating potential model tension and data inconsistencies; see Appendix 3B). An FAA model would forego the need for using any extrapolated data and would be able to fit regionally disaggregated data but may be more unstable and might overinterpret signals from sparse composition data (e.g., if the noise to signal ratio is high). Given the need for a sex-specific model for sablefish, data sparsity is of concern as compositions are partitioned by sex and region. A spatial model could better represent spatial dynamics but is more complex and likely much less stable (and is not considered further here).

Given the potential benefits of using a FAA model to address ontogenetic patterns and NOAA Longline Survey design changes, an extensive model development process was undertaken to identify a suitable FAA structure for sablefish. The results of the simulation and MSE work were used to first identify the FAA structures that were likely to perform the best. Then, a suite of FAA models was built and applied to real-world data. Here, the results of the two most robust FAA models are presented in comparison to the final single region assessment (Model *26.1_1Reg*). Although the FAA framework demonstrates promise and is especially appealing since it utilizes the regional survey data without the need for extrapolation, there remain performance issues and uncertainties on par with the single region assessment.

Methods

In addition to the final single region model (*26.1_1Reg*), two FAA models are described, which were winnowed from an extensive model exploration exercise. The results of the sablefish simulation and MSE study (Appendix 3E) were used to guide model exploration and target final model structure (Figure 3C.3). Key findings from the MSE work indicated that FAA models performed relatively better than other structures as survey coverage decreased or data quality declined but required adequate complexity in selectivity parametrization to address ontogenetic

changes in availability across regions (Appendix 3E; Table 3C.1). For instance, domed selectivity in the BS, particularly for the NOAA Longline Survey, often improved bias, while time-varying non-parametric selectivity helped improve simultaneous fits to the regional indices. However, the simulation work also demonstrated that a relatively simple FAA model (i.e., utilizing only FAA structure in the survey while maintaining aggregated fishery fleets) may perform as well as more complex models that disaggregated survey and fishery fleets and/or included domed selectivity (Figure 3C.3, column facet 2/row facet 2 model *1F_1T*).

The minimum criteria for the final choice of FAA model structure was to be able to fit the disaggregated NOAA Longline Survey indices and age compositions by region (BS, AI, GOA), such that no extrapolation for non-surveyed regions would be necessary (see Appendix 3B) and signals of recruitment from the BS and AI regions (where many of the recent large cohorts appear to be originating; Figure 3C.2) would be better represented. Model building explored three primary parametrization options: 1) whether to disaggregate the fixed gear fishery fleets by region (since this is the primary fleet representing a large majority of fishery removals); 2) whether to include domed selectivity in the BS or AI fishery or survey fleets; and 3) how many time blocks to include in the fixed gear fishery.

FAA model building began by adding incremental complexity to Model *26.1_1Reg* until adequate data fits (e.g., residual patterns from fits to indices and composition data), convergence criteria (e.g., maximum gradient, invertible Hessian, and no major parameter correlations), and general diagnostics (e.g., retrospective patterns and likelihood profiles) were obtained. Two FAA base models were identified that generally passed the desired convergence and diagnostic criteria: *26.2_FAA* and *26.2a_FAA_Fsh+Srvy*. Model *26.2_FAA*, which only disaggregated the NOAA Longline Survey index and age compositions, was deemed the minimally complex model necessary to address the primary goals driving the desire for exploring a FAA model. Model *26.2a_FAA_Fsh+Srvy*, which disaggregated the NOAA Longline Survey and fixed gear fishery fleets by region, was then deemed the least complex and most stable model that allowed disaggregating both the survey and fixed gear fleet (but required assuming a single fixed gear fishery time block due to instability in selectivity estimates when assuming two time blocks). Alternatives to these two base models were explored for the three primary parametrization options noted above, including the number of fishery fleets to model (e.g., 1, 2, or 3 fleets), whether to allow for domed selectivity in the BS fleets (fishery or survey), and whether to assume 1 or 2 fishery time blocks. Additionally, time-varying survey selectivity approaches were explored (see Section 2: Sensitivity Runs). However, most alternatives had extensive stability issues (e.g., due to trying to estimate domed selectivity or tradeoffs with sample sizes when fishery time blocks were added), while others portrayed poor diagnostics or data fits, and all were excluded from further consideration (only pertinent and select exploratory results are presented here).

We provide a comparison between the two primary FAA models (Model *26.2_FAA* and Model *26.2a_FAA_Fsh+Srvy*) in terms of data fits and diagnostics, model stability, and biological dynamics (see SM3C for full model outputs from each model). However, the summarized results focus primarily on comparisons of Model *26.2_FAA* to Model *26.1_1Reg*. See Table 3C.2 for a summary of models and results. Further model explorations that were deemed mostly exploratory for 2026 are then provided as sensitivity runs (see Section 2: Sensitivity Runs). A full mathematical description of the underlying SPoRC assessment framework is provided in Appendix 3A.

All models presented here utilize the same parametrization as the final single region model (*26.1_1Reg*; see Appendix 3B), with added complexity solely via fleet structure. For all model runs in this document, the final data through the end of 2025 were utilized as would have been available for the 2025 operational assessment. All models underwent Francis reweighting of compositional data to account for correlation among composition bins (Francis, 2011, 2017). All models are built in the R-TMB based generalized Spatial Populations over Regional Components (SPoRC) stock assessment R package (Cheng et al., 2026; https://chengmatt.github.io/SPoRC/index.html).

Primary model updates included:

- *26.1_1Reg*:
    - The final single region model for 2026 (see Appendix 3B for a full model description).

- *26.2_FAA*:
    - Matches the structure, settings, and data inputs as Model *26.1_1Reg* with adjustments to the NOAA Longline Survey fleet structure, where the fleets are disaggregated by 'region' including the BS, AI, and GOA.
    - Both the NOAA Longline Survey indices and age compositions are disaggregated by region and fit at the same scale.
    - The GOA data streams are aggregated (within the GOA) in the same way as the single region models, where area RPNs are summed across the GOA for the index and age compositions are aggregated using the same RPN-weighted aggregation approach as the single region model.
    - For each NOAA Longline Survey fleet, an independent catchability (q) parameter and two logistic selectivity parameters per sex (a_{50} and a_{95}) for each survey time block are estimated (i.e., compared to Model *26.1_1Reg* two additional q parameters and 16 additional selectivity parameters are estimated).
    - All other fleets (i.e., fixed gear fishery, trawl fishery, and Japanese Longline Survey) are aggregated to a single region matching the structure of Model *26.1_1Reg*.

- *26.2a_FAA_Fsh+Srvy*:
    - Matches the structure, settings, and data inputs as Model *26.2_FAA* with adjustments to the fixed gear fishery fleet structure, where the fleets are disaggregated by 'region' including the BS, AI, and GOA.
    - Both the fixed gear fishery catch and age compositions are disaggregated by region and fit at the same scale.
    - The GOA data streams are aggregated (within the GOA) in the same way as the single region models, where catch is summed across the GOA and age compositions are aggregated using the same catch-weighted (in numbers) aggregation approach as the single region model.

- For the fixed gear fishery, the associated time blocks for selectivity are removed due to stability issues because of reduced sample sizes (especially in the BS and AI) when composition data are disaggregated.
- For each fixed gear fishery fleet, an independent fishing mortality (F) parameter, yearly F deviations, and two logistic selectivity parameters per sex (a_{50} and k) are estimated (i.e., compared to Model *26.1_1Reg* two additional F parameters and four additional selectivity parameters – because a selectivity time block is removed -- along with two additional time series of F deviations are estimated).
- All other fleets (i.e., trawl fishery and Japanese Longline Survey) are aggregated to a single region matching the structure of Model *26.1_1Reg*.

Results

Disaggregating the NOAA Longline Survey indices and age compositions and using an FAA structure (Model *26.2_FAA*), led to moderate downward rescaling in SSB (Figure 3C.4, Table 3C.3) and average recruitment (Figure 3C.5), primarily due to a decline in M estimates (Figure 3C.6) as well as other scaling parameters (Table 3C.4). The result is a strong decline in $B_{40\%}$ and catch advice (Table 3C.3). Generally, the SSB trends remain consistent, though stock rebuilding increases more rapidly since 2018 in Model *26.2_FAA* compared to Model *26.1_1Reg*. The primary difference between models is the interpretation of recent recruitment events, where Model *26.2_FAA* now predicts that the 2017 year class is much larger than the 2016 year class and all subsequent recruitment has been considerably lower than in Model *26.1_1Reg* (Figure 3C.5).

Disaggregating the fixed gear fishery data in Model *26.2a_FAA_Fsh+Srvy* led to minor increases in SSB scaling as well as other scaling parameters (R_0, M; Tables 3C.3 – 3C.4), but results were generally similar to Model *26.2_FAA*. The major difference was that Model *26.2a_FAA_Fsh+Srvy* was very unstable, due to the need to estimate fishing mortality and selectivity parameters for two additional fleets with sparse data (e.g., the BS; Figure SM3C.39) as well as difficult to interpret trends (e.g., the enormous plus group in the AI in the 2010s; Figure SM3C.42). Moreover, Model *26.2a_FAA_Fsh+Srvy* had difficulty with initial scaling, as demonstrated in retrospective (Figure SM3C.91) and jitter (Figures SM3C.102 and SM3C.105) diagnostics. Given model instability, Model *26.2a_FAA_Fsh+Srvy* is not recommended, and the remaining results focus on comparisons between Model *26.2_FAA* and *26.1_1Reg*.

Despite disaggregating the NOAA Longline Survey indices and age comps in Model *26.2_FAA*, the relative data weights after Francis reweighting remain similar to Model *26.1_1Reg* (Table 3C.5). However, the fixed gear fishery age compositions are moderately downweighted, given the comparatively higher weights and increase in number of NOAA Longline Survey age composition data sets (Table 3C.5). Moreover, the AI age compositions receive moderate upweighting relative to other age composition data. It is likely that the AI compositions better align with the trends in the indices, such that the Francis reweighting upweights the AI because it allows simultaneously good fit to the indices compared to other composition data sets. Similarly, the AI appears to provide a mixture of recruitment signals (strong 2014, 2016, and 2017 year classes; Figure SM3C.24) that balances those in the BS (strong 2014 and 2016 and very strong 2017 year classes; Figures SM3C.18) and GOA (strong 2014 and very strong 2016 year classes; Figures SM3C.31), which may serve as a mediary when trying to fit the various composition data. However, there may also

be a data-year interaction related to the timing of the AI survey in even years (potentially matching years when some of the large recent recruitment events have occurred; e.g., 2014 and 2016). The resulting fit to the mean composition bins after reweighting do not drastically change compared to the single region model (Figures SM3C.2 – SM3C.3). However, disaggregation of the NOAA Longline Survey age compositions better elucidates the rapid reduction in mean age in recent years, particularly in the BS and AI, as well as the comparatively higher variation in mean age in these regions compared to the GOA.

The disaggregated NOAA Longline Survey indices demonstrate general agreement in long-term population trends across regions, with all indicating rapid population growth since 2015 (Figure 3C.7). Model *26.2_FAA* provides adequate fits to each index, given the slight discrepancies in signals, tension in fitting indices and age compositions from all three regions simultaneously, and the need to alias availability with catchability and selectivity estimates when assuming a single underlying population in a FAA model. Although fishery selectivity does not alter much (aside from trawl selectivity becoming a bit steeper; Figure 3C.8), survey selectivity for the GOA becomes much less steep (Figure 3C.9). The GOA selectivity shape reflects the model assuming that the younger age classes (< age-20 and aside from age-2) were less available to the GOA survey. Ultimately, the tension between fitting the age compositions and indices, an underlying issue in Model *26.1_1Reg*, remains in Model *26.2_FAA*, because better fit to the age compositions would result in a strong overestimation of GOA survey RPNs in recent years (see Model *26.2_FAA_DM* which effectively upweights composition data leading to overestimation of the GOA index; Figure 3C.27).

It is also worth noting that the strong differences in the shape of the survey selectivity curves across regions (i.e., much steeper in BS and AI compared to GOA) leads to an interesting divergent trend in the last few years in the predicted index. The BS and AI predicted indices begin to decline despite observed RPNs continuing to increase, while the GOA continues to increase and better fits the increasing observed index (Figure 3C.7). The reason for the divergence across regions is due to the tradeoff between survey selectivity, cohort mortality, and reductions in estimated year class sizes since the 2017 year class. Mainly, for the BS and AI survey indices, maximum selectivity occurs around age-4, so a cohort's maximum contribution to RPNs will decline after this age due to ongoing mortality (i.e., the size of the cohort can only decline). Conversely, in the GOA, survey selectivity is maximum around age-20, so there is a tradeoff between slowly increasing selectivity and increasing cohort removals due to mortality across the lifespan, such that maximum contributions to RPNs likely occur at intermediate ages (but much later than in the BS and AI). Because the 2017 year class has reached full selection in the BS and AI surveys, it is the largest cohort in the time series, and subsequent cohorts have been much smaller, the model predicts a decline in BS and AI RPNs starting around 2022 (i.e., when the cohort reaches age-5). For the GOA, the predicted survey index continues to increase because the tipping point between mortality and selectivity has not yet been reached. To verify that this is true, a sensitivity run was undertaken where the GOA NOAA Longline Survey selectivity was fixed at the AI values. As expected, when the GOA survey selectivity was much steeper, the resulting predicted index increased much more rapidly, greatly overestimated the recent observed index, and began to decline after 2023 (Figure 3C.10)

For composition data in Model *26.2_FAA*, fits tend to show similar patterns as Model *26.1_1Reg*, particularly for the spatially aggregated fleets (Figures SM3C.45 – SM3C.46). A notable change is that the fixed gear fishery male age compositions demonstrate a noticeably worse fit for Model *26.2_FAA*, where expected values are much higher than observed for ages 4 – 6 (Figure SM3C.49). The pattern was present in Model *26.1_1Reg* for certain years (e.g., 2012 -- 2013) but has become much more exacerbated in Model *26.2_FAA*. Fixed gear fishery male selectivity in the first time block is estimated to be much steeper in Model *26.2_FAA* (Figure SM3C.6), which is likely driving the overestimation of young male fish. The relative downweighting of the fishery age compositions relative to the multiple NOAA Longline Survey age composition data sets is probably a primary factor in the poorer fit to the fishery age compositions in Model *26.2_FAA* (Table 3C.5).

Given the disaggregation by region of the NOAA Longline Survey age compositions in Model *26.2_FAA*, it is not possible to make a direct comparison to Model *26.1_1Reg* survey composition fits. However, because the GOA compositions are more highly sampled and receive higher relative weight in the aggregation method for the single region model, the GOA compositions in the FAA model are quite similar. Generally, the observed and predicted survey compositions are quite similar (Figures SM3C.30 – SM3C.31), with generally similar residual patterns (Figures SM3C.33 – SM3C.34). A noticeable difference is that Model *26.2_FAA* strongly overestimates age-2 compositions across sexes but has a slightly improved estimate for intermediate ages (e.g., ages 10 to 20). Fits to the BS and AI NOAA Longline Survey age compositions tend to be adequate in Model *26.2_FAA*, especially considering the relatively lower sample sizes and increased sparsity across ages (Figures SM3C.18 and SM3C.24). The AI age data tends to be fit slightly better, given the relatively higher final ESS. Residual patterns tend to reflect the pervasive trend seen in Model *26.1_1Reg* of generally underestimating peak abundance at ages 4 – 6 and overestimating intermediate ages 10 – 20 as well as the plus group (Figures SM3C.20 – SM3C.26). No major changes in the fits to the Japanese Longline Survey or trawl fishery compositions were detected for Model *26.2_FAA* compared to Model *26.1_1Reg* (see the SM3C figures for all plots of compositional fits as well as One Step Ahead, OSA, and Pearson residuals). Given the pervasive tension between simultaneously fitting the indices versus the compositions, and the resulting lower weights given to the composition data, the fits across data sources for Model *26.2_FAA* were deemed adequate and on par with those from Model *26.1_1Reg*.

Similarly, no major changes were noted in the other key model diagnostics. Model *26.2_FAA* demonstrates little retrospective patterning in SSB, with a slight tendency to underestimate SSB as data is removed similar to Model *26.1_1Reg* (Figures SM3C.89- SM3C.90). Similarly, large recruitment year classes are often initially underestimated, but estimates tend to stabilize more quickly for Model *26.2_FAA* (Figures SM3C.92 – SM3C.93). However, the internal tension and difference in recruitment signals across age composition data sources (e.g., the different NOAA Longline Survey regional compositions) for Model *26.2_FAA* is apparent, given the switch between 2016 and 2017 being the dominant year class as more data is integrated over time, as well as the downsizing of the 2014 year class (Figure SM3C.90). As noted, the GOA compositions tend to favor the 2016 year class, whereas the BS favors the 2017 year class and the AI demonstrates strong signals for both. Aside from conflicting signals within the age compositions themselves, there is also a year-data interaction, given the historical odd-even year survey design for the BS and AI regions, respectively. Therefore, the retrospective pattern partially reflects a data

availability issue, where initial recruitment signals are probably dominated by patterns in the GOA, then shift slightly towards the BS and AI signals as those data become available.

Likelihood profiles on key scaling parameters show similar trends across models, although those from Model *26.2_FAA* tend to be more stable and demonstrate a stronger minima for the age composition data (i.e., the primary source of information on recruitment and M; Figures SM3C.95 – SM3C.98). Not surprisingly, given the final estimates (Table 3C.4), the age composition data indicate lower values for M and R_0 in Model *26.2_FAA*. Jitter diagnostics remain essentially unchanged across models, demonstrating that both have likely found a robust solution and demonstrate strong stability and convergence (Figures SM3C.100 – SM3C.104).

As noted, model building involved hundreds of exploratory runs to understand model performance, identify robust model parametrizations, and better understand key assumptions. Given the results of the simulation study demonstrating that domed selectivity for BS fleets may be warranted to address availability (Figure 3C.1), an exploratory run that assumed domed BS NOAA Longline Survey selectivity for the recent survey time block was developed. Results demonstrated improved fits to BS mean age, compositions, and the associated BS index (Figure 3C.11). However, estimating domed selectivity led to moderate model instability and worse fits to the survey indices for the other regions (Figure 3C.11).

To better understand the impact of using extrapolated ('carryover') NOAA Longline Survey index RPN values for years when a region was not surveyed, an exploratory model was run that used similar assumptions to regional index construction as Model *26.1_1Reg*. Mainly, for a survey off-year in a given region, the previous (carryover) RPN was used for the off-year value (instead of assuming a missing data point, as in Model *26.2_FAA*), such that there were no data gaps in the regional indices (aside from 2024 when there was no survey). Interestingly, the use of extrapolated carryover values led to a strong upward scaling of SSB, particularly since the start of the NOAA Longline Survey in the mid-1990s (Figure 3C.12). The increase in recent scaling is driven primarily by major increases in the estimates of the large year classes (e.g., 2014, 2016, and 2017). Because carryover RPNs are not accompanied by associated age compositions, there is likely a non-intuitive interaction between fitting these static index values for the BS or AI and fitting the survey age compositions for the regions that were surveyed while assuming all these data simultaneously provide signals from a single population. Using carryover values may also implicitly act to upweight each survey data point, by effectively including it twice in a given index (since a given value is repeated in the following year when a region is not sampled). Ultimately, this exploratory run demonstrates the potential downsides of using extrapolated data in the assessment, indicating that Model *26.1_1Reg* (and any single region model for sablefish) could be artificially inflating the resulting recruitment, SSB, and catch advice due to the need for using extrapolated survey values.

Notable changes in model dynamics included (Table 3C.2):

- *26.1_1Reg* (full results here):
    - A stable model that provides good fits to the NOAA Longline Survey index.
    - Mediocre fits to age compositions.

- Requires extrapolating (using carryover method) ~50% of the NOAA Longline Survey index in a given year (to maintain a consistent scale for the single region assessment) for non-surveyed regions.
- GOA composition data are implicitly over-represented during data aggregation due to higher sampling and catch which likely under-represents recruitment signals from the BS and AI, resulting in extremely large 2016 year class.

- *26.2_FAA* (full results here):
  - Model remains relatively stable and demonstrates adequate fits to the disaggregated NOAA Longline Survey indices.
  - Shows similar tension as single region model when simultaneously trying to fit indices and age compositions and demonstrates poor fit to the fixed gear fishery male age compositions.
  - GOA NOAA Longline Survey selectivity becomes less steep, reflecting need for model to alias regional availability with selectivity.
  - Recruitment trends better reflect signals from BS and AI as evident by increasing estimates of 2014/2017 year classes and decreases in the 2016 year class.
  - Moderate downward scaling of SSB due to decreasing estimates of M and R_0, likely reflecting the removal of extrapolated index data.
  - Retrospective patterns are minimal for SSB, while continued underestimation of large recruitment events occurs as well as switching between large year classes (2016 vs. 2017) as new data become available.

- *26.2a_FAA_Fsh+Srvy* (full results here):
  - Due to stability issues with selectivity estimation, the fishery time blocks needed to be removed to ensure convergence.
  - Increased stability issues, especially for initial scale and selectivity estimation.
  - Data sparsity issues are present for BS and AI fishery age compositions.
  - There are no major changes in fits to the data, with GOA fixed gear fishery age compositions showing similar overestimation of age 4 to 6 compositions as the fixed gear fishery compositions in Model *26.2_FAA*.
  - Demonstrates a minor upward scaling of SSB, but stock status and catch advice remain similar to Model *26.2_FAA*.
  - Minor retrospective patterns emerge for SSB and recruitment, primarily indicating model instability, particularly with initial scale.
  - Jitter diagnostics indicate a consistent solution was not found, again associated with uncertainties in the initial population scale, further demonstrating model stability issues.

Conclusion

Model *26.2_FAA* appears to provide a robust and stable model for interpreting sablefish population dynamics. Although Model *26.2_FAA* assumes a moderate downward scaling of SSB and catch advice, general population dynamics and stock status remain consistent with Model *26.1_1Reg*. Similarly, fits to the available data retain similar issues, where survey indices are well fit at the

expense of composition data, as a result of the Francis reweighting approach. Although residual patterns in the composition data remain, the aggregated fits continue to appear adequate given associated model tension between fitting the NOAA Longline Survey indices versus age compositions. Fundamentally, both models have trouble reconciling the much larger year classes and population growth suggested by the age composition data compared with the rapid, but more tempered, population growth observed in the survey RPNs. All model diagnostics, including retrospectives, indicated that Model *26.2_FAA* is relatively stable and showed no signs of significant misspecification. Although other FAA models were explored, data sparsity issues resulted when attempting to regionally disaggregate the fishery data, suggesting that Model *26.2_FAA* was a suitable model of intermediate complexity.

Ultimately, the choice of model structure for the sablefish assessment necessitates a tradeoff between data quality and quantity. For instance, utilizing a single region assessment and aggregating data provides improved sample sizes, but requires extrapolating NOAA Longline Survey index values to ensure consistent scaling and may over-represent GOA age compositions. Conversely, disaggregating the NOAA Longline Survey data into regional 'fleets' in an FAA model avoids the need to extrapolate indices, but greatly reduces sample sizes and may increase the noise to signal ratio in the age compositions. There is not a 'silver bullet' solution to this conundrum. However, the exploratory FAA model run using the 'carryover' approach was enlightening as it showed the potential downside of using extrapolated data in the assessment, given that it led to a much more optimistic estimate of stock status and catch advice. Although the sablefish MSE work indicates that the carryover extrapolation is not likely to have a major detrimental impact on management advice from the single region model, it does appear that using extrapolated values in the index should be considered judiciously. Thus, Model *26.2_FAA* should be considered as a good candidate for the sablefish operational assessment and management, given that it does not rely on any data extrapolation and appears to provide a stable and robust assessment.

Tables

Table 3C.1. Summary of tradeoffs between using a single region, Fleets-as-Areas (FAA), and muti-region (spatial) models for sablefish.

Model	*Benefits*	*Issues*	*MSE Conclusions*	*Tradeoffs*
Single Region Model	High sample sizes. Low model complexity. Intuitive and transparent results. High inertia towards moving away from status quo assessment methodology.	Requires fitting extrapolated data for the Longline Survey Index. Implicit historical over-weighting of GOA composition data during aggregation or high interannual variability recently (GOA then BS/AI). Little signal from BS/AI compositions, because GOA given higher weight (RPNs or catch) in data aggregation. Ignores spatial structure.	Moderate SSB bias and moderate instability due to the survey extrapolation and aggregated composition data but impacts on resulting management advice may be negligible.	Higher sample sizes and improved stability but requires fitting the extrapolated longline survey index and primarily represents the recruitment signals from the GOA due to data aggregation assumptions.
Fleets-as-Areas (FAA)	Fitting non-extrapolated Longline Survey index. Fitting disaggregated survey compositions. Better accounts for signals from BS/AI compositions on recent year class strength. Implicitly accounts for ontogenetic movement dynamics	Low sample sizes and sparse data for BS/AI compositions. Increased instability from added complexity (estimating extra q and selectivity parameters for 'regional' survey fleets and/or extra F and selectivity parameters for fishery fleets). Non-intuitive estimates that are difficult to interpret (e.g., due to using multiple fleets that are interacting with a single underlying population – not multiple populations/regions). Selectivity needs to be flexible to address ontogenetic dynamics.	Low SSB bias and instability with improved performance over other structures when survey coverage decreases, but adequate flexibility in selectivity parametrizations is likely needed to ensure good performance.	Able to fit the regional longline survey indices without aggregation and likely provides more robust outputs when survey coverage decreases, but sample size limitations may prevent adequate selectivity flexibility to address ontogenetic availability issues due to model instability.
Spatial (3 or 5 Region)	Fit data at the regional scale it was collected. Better account for signals from each region. More intuitive than FAA. Explicitly account for ontogenetic movement dynamics.	Increased sample size limitations. Increased parametrization decisions related to spatial processes. Lack of best practices for fitting tagging data (e.g., dealing with tag mixing issues and data weighting). Increased parameter correlation, particularly between regional recruitment parameters and movement. Diagnostics can be difficult to interpret. Increased instability.	High instability due to overparameterization and low sample sizes but best represents localized dynamics.	Best represents spatial dynamics but suffers from high model instability due to data sparsity that requires extremely careful parametrization, while appropriate handling of tagging data is difficult and potentially subjective.

Table 3C.2. Primary FAA model runs, including major differences, rationale, and summary of results.

Model Group	*Model Name*	*Major Changes*	*Rationale*	*Conclusion*
Final 1 Region Model	***26.1_1Reg***	(*26.1i*) None.	The 2026 final proposed single region model with all updates implemented.	• A stable model that provides good fits to the NOAA Longline Survey index. • Mediocre fits to age compositions. • Requires extrapolating (using carryover method) ~50% of the NOAA Longline Survey index in a given year (to maintain a consistent scale for the single region assessment) for non-surveyed regions. • GOA composition data are implicitly upweighted during data aggregation due to higher sampling and catch which likely under-represents recruitment signals from the BS and AI, resulting in extremely large 2016 year class.
Final FAA Model	*26.2_FAA_Srvy*	(*26.1+*) Split NOAA Longline Survey into 3 independently modeled fleets, with unique parameters (*q*, selectivity).	Balance ability to fit regional indices and comps against rapidly increasing complexity from adding fleets.	• Model remains relatively stable and demonstrates adequate fits to the disaggregated NOAA Longline Survey indices. • Continues to show similar tension with simultaneously fitting age compositions, with poor fit to the fixed gear fishery male age compositions. • GOA NOAA Longline Survey selectivity becomes less steep, reflecting need for model to alias regional availability. • Recruitment trends better reflect signals from BS and AI as evident by increasing estimates of 2014/2017 year classes and decreases in the 2016 year class. • Moderate downward scaling of SSB due to decreasing estimates of M and R0, likely reflecting the removal of extrapolated index data.
Alternate FAA	*26.2a_FAA_Fsh+Srvy*	(*26.2+*) Split the fixed gear fishery into 3 independently modeled fleets, with unique parameters (*F*, selectivity) and remove fishery time blocks.	Further parse out recruitment signals from BS and AI by disaggregating fishery fleets.	• Increased stability issues, especially for initial scale. • Data sparsity issues for BS and AI fishery age compositions. • Minor upward scaling of SSB.

Table 3C.3. Key management quantity outputs. SSB is in kt, B_Ref_Pt is the associated $B_{40\%}$ reference point in kt, and F_Ref_Pt is the associated $F_{40\%}$. Catch advice is the 2026 Acceptable Biological Catch (ABC) resulting from fishing at $F_{40\%}$.

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.1_1Reg	1	216.4521	369.6689	0.04642913	40.85173	139.0172	0.07903	1.55702	0.58553	0.58749
26.2_FAA	1	193.1500	358.1881	0.05874255	31.69570	122.4267	0.08054	1.57768	0.53924	0.72940
26.2a_FAA_Fsh+Srvy	1	196.3059	353.5244	0.05864900	33.14317	120.4123	0.08484	1.63028	0.55528	0.69129

Table 3C.4. Estimates of primary scaling parameters (e.g., catchability, q, unfished recruitment, R_0, and natural mortality, M) along with associated standard error (SE) estimates from each model. All parameters and associated SE are in natural logarithmic (ln) space. The maximum gradient value across all parameters is also shown as a metric of model convergence.

Parameter	**Model *26.1_1Reg***		**Model *26.2_FAA***		**Model *26.2_FAA_Fsh+Srvy***	
	Estimate	*SE*	*Estimate*	*SE*	*Estimate*	*SE*
ln_R0	2.97	0.13	2.76	0.11	2.88	0.12
ln_M_female	-2.39	0.07	-2.43	0.07	-2.39	0.07
ln_M_male	-2.41	0.07	-2.44	0.08	-2.36	0.08
ln_q_LLS	1.91	0.10	--	--	--	--
ln_q_LLS_BS	--	--	0.55	0.10	0.50	0.10
ln_q_LLS_AI	--	--	0.34	0.10	0.30	0.10
ln_q_LLS_GOA	--	--	2.20	0.08	2.13	0.08
Max. Gradient	2.91 e -10		7.64 e -10		5.86 e -12	

Table 3C.5. Assumed mean (across years) statistical distribution variance terms for each data source defined in terms of standard error (SE) for indices or input sample size (ISS) for compositional data for Model *26.1_1Reg* (left), Model 26.2_FAA (middle), and Model *26.2a_FAA_Fsh+Srvy* (right). The compositional data weights (λ, Data_Wt) are calculated using the Francis reweighting method (Francis, 2011). The effective values (Eff_SE or effective sample size, ESS) represent the input values multiplied by the data weights and are the final variance terms used to fit the data in the assessment likelihoods. All compositional data are fit using the 'joint' approach (Cheng et al., 2025b), so only a single ISS/ESS or weighting term is provided across both sexes.

Source	SE	Eff_SE	Data_Wt	ISS	ESS
LL Survey Index	0.07	0.07	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.12	145.15	17.69
LL Survey Ages	NA	NA	0.28	171.43	47.31
JPN LL Survey Ages	NA	NA	0.17	83.33	14.45
Trawl Fish Lengths	NA	NA	0.25	45.39	11.51
JPN LL Survey Lengths	NA	NA	0.34	54.62	18.84

Source	SE	Eff_SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	0.20	1.00	NA	NA
AI LL Survey Index	0.24	0.24	1.00	NA	NA
GOA LL Survey Index	0.08	0.08	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.04	145.15	6.08
BS LL Survey Ages	NA	NA	0.51	37.43	19.15
AI LL Survey Ages	NA	NA	1.91	39.14	74.82
GOA LL Survey Ages	NA	NA	0.29	77.14	22.66
JPN LL Survey Ages	NA	NA	0.18	83.33	15.33
Trawl Fish Lengths	NA	NA	0.14	45.39	6.53
JPN LL Survey Lengths	NA	NA	0.33	54.62	17.98

Source	SE	Eff_SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	0.20	1.00	NA	NA
AI LL Survey Index	0.24	0.24	1.00	NA	NA
GOA LL Survey Index	0.08	0.08	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
BS Fixed Gear Fish Ages	NA	NA	0.33	26.92	8.81
AI Fixed Gear Fish Ages	NA	NA	0.04	29.65	1.14
GOA Fixed Gear Fish Ages	NA	NA	0.11	64.15	6.96
BS LL Survey Ages	NA	NA	0.56	37.43	20.87
AI LL Survey Ages	NA	NA	1.81	39.14	70.68
GOA LL Survey Ages	NA	NA	0.33	77.14	25.84
JPN LL Survey Ages	NA	NA	0.17	83.33	14.56
Trawl Fish Lengths	NA	NA	0.15	45.39	6.73
JPN LL Survey Lengths	NA	NA	0.34	54.62	18.61

Figures

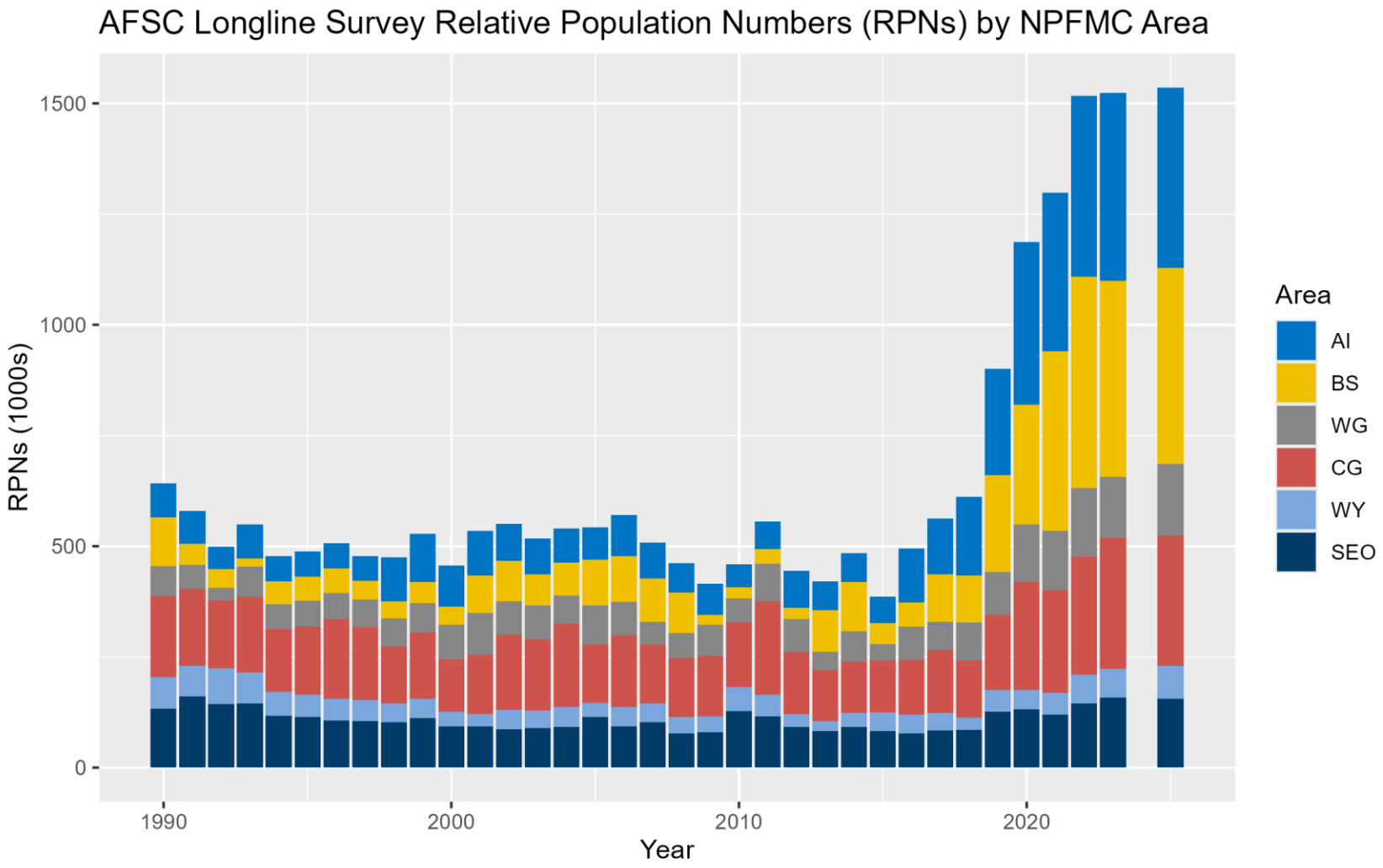

Figure 3C.1. NOAA Longline Survey relative population numbers (RPNs, 1000s of fish) by region. Note that the Western Gulf (WG), Central Gulf (CG), Western Yakutat (WY), and Southeast Outside (SEO) are all part of the Gulf of Alaska (GOA) region.

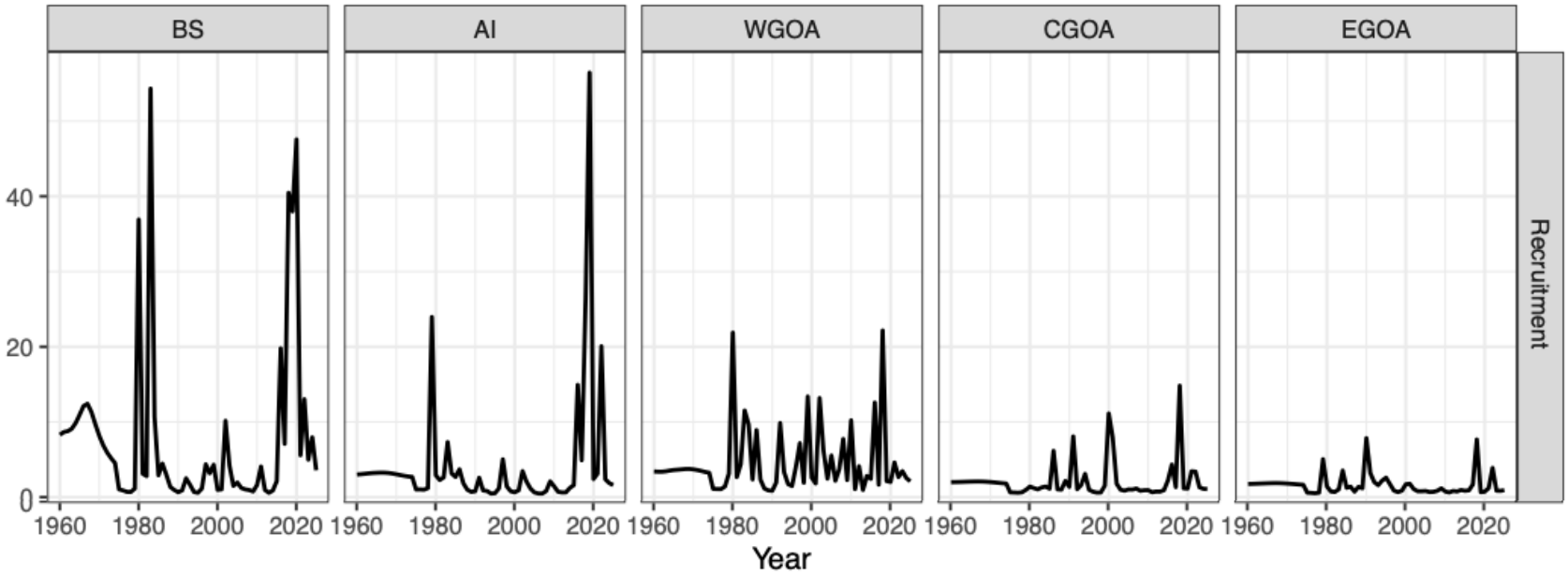

Figure 3C.2. Time-series of recruitment (millions of age-2 fish) by region from the five-region spatial model (See Appendix 3D).

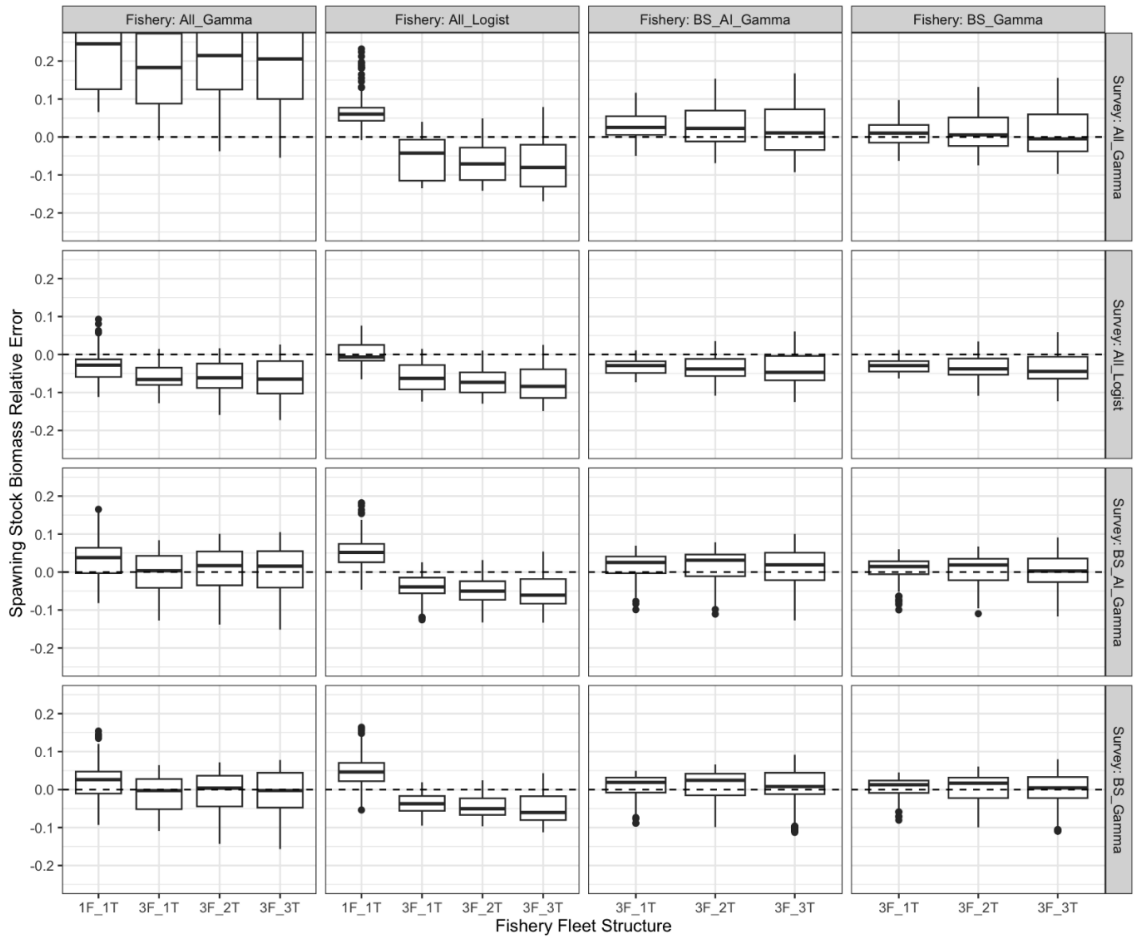

Figure 3C.3. Relative error in estimates of SSB (median across all years) of the simulation experiment that explored assessment bias for different Fleets-as-Areas structures in the fishery (x-axis) as well as selectivity functional forms (fixed gear fishery – column facets; NOAA Longline Survey – row facets). Note that all models assumed three NOAA Longline Survey fleets and a single (aggregated) Japanese Longline Survey fleet. For the x-axis, F stands for fixed gear (longline) fishery, T stands for the trawl fishery and the preceding number is the number of fleets for that fishery (3 means 3 fleets split by region -- BS, AI, and GOA; 2 means 2 fleets split by BS and AI + GOA, and 1 means a single aggregated fleet across all regions). The intersection of the logistic fishery and logistic survey facets provides models matching the FAA models described in this document, where 1F_1T generally matches Model *26.2_FAA* and 3F_1T generally matches model *26.2a_FAA_Fsh+Srvy*. See Appendix 3E for full results.

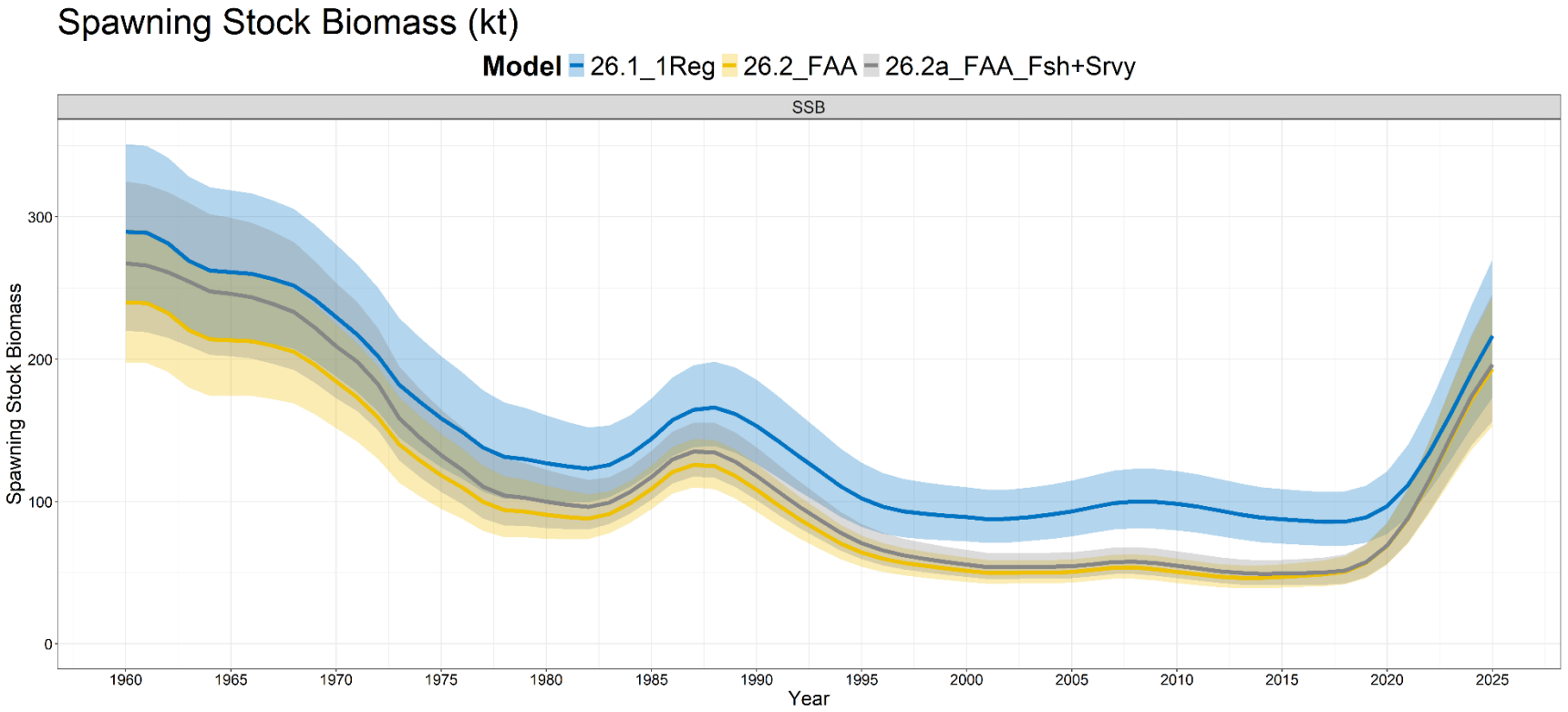

Figure 3C.4. Comparison of spawning stock biomass (SSB, in kilotons) across models, where shaded regions represent the 95% confidence intervals.

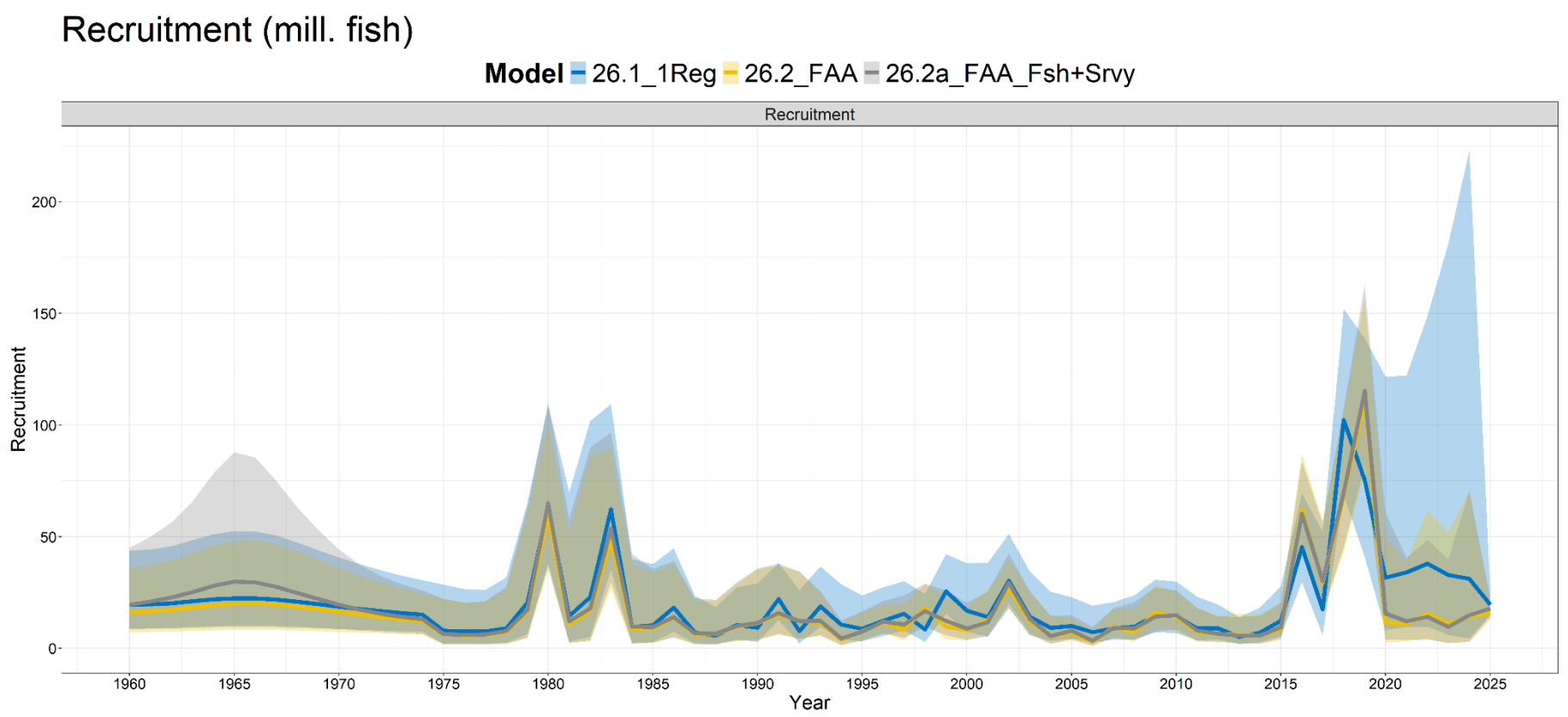

Figure 3C.5. Comparison of recruitment (in millions of fish) across models, where shaded regions represent the 95% confidence intervals.

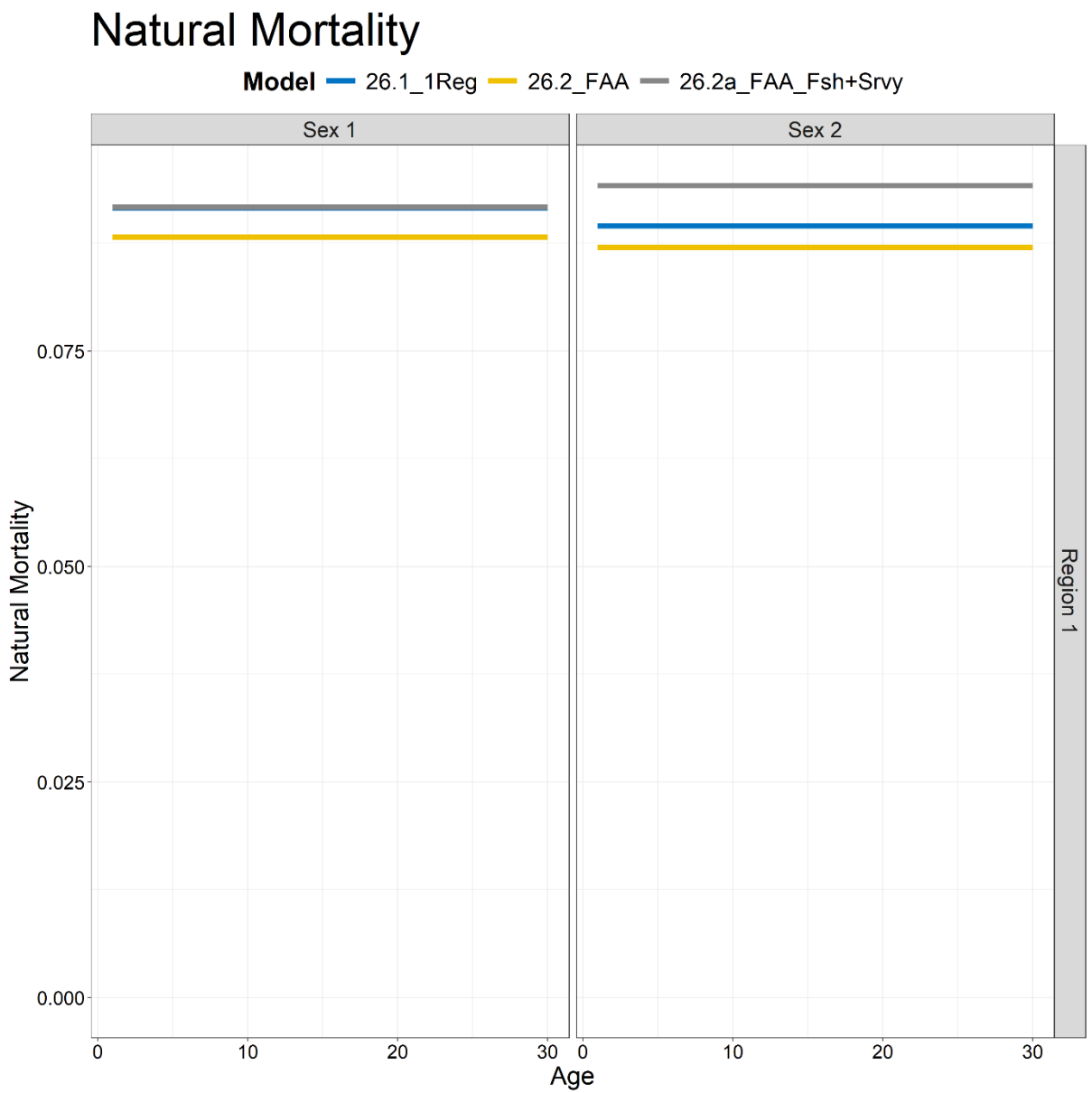

Figure 3C.6. Comparison of natural mortality estimates across models. Sex 1 is females and sex 2 is males.

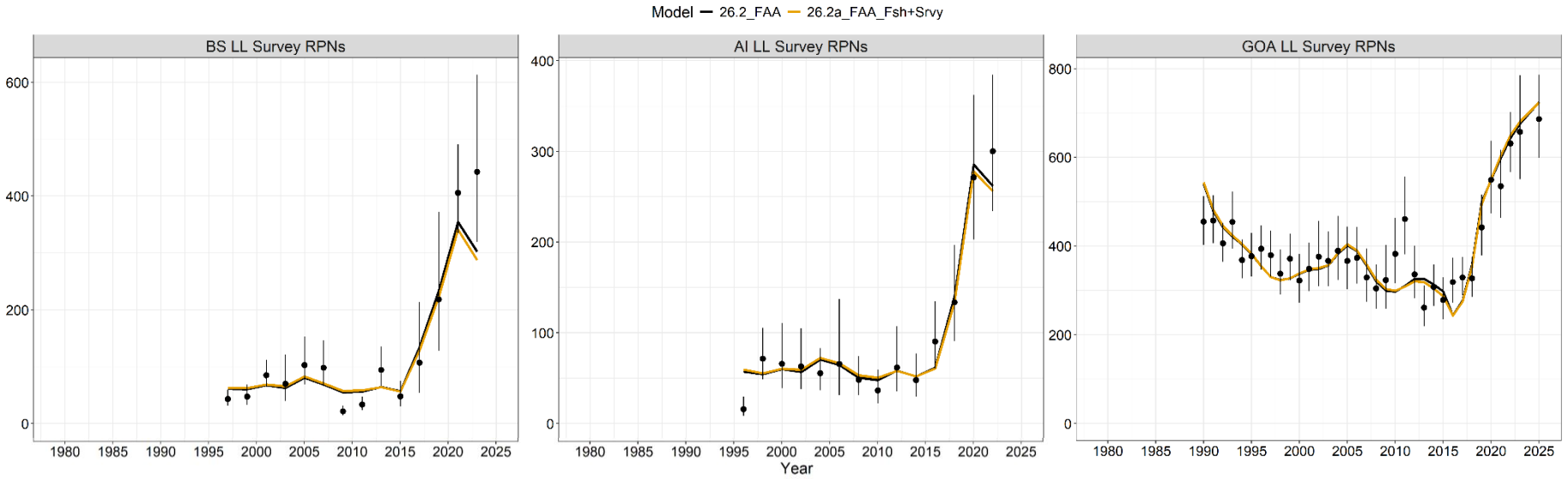

Figure 3C.7. Fits (lines) to the NOAA Longline Survey abundance indices in relative population numbers (RPNs – 1000s of fish, y-axis; black dots with vertical lines representing 95% confidence intervals) for the FAA models.

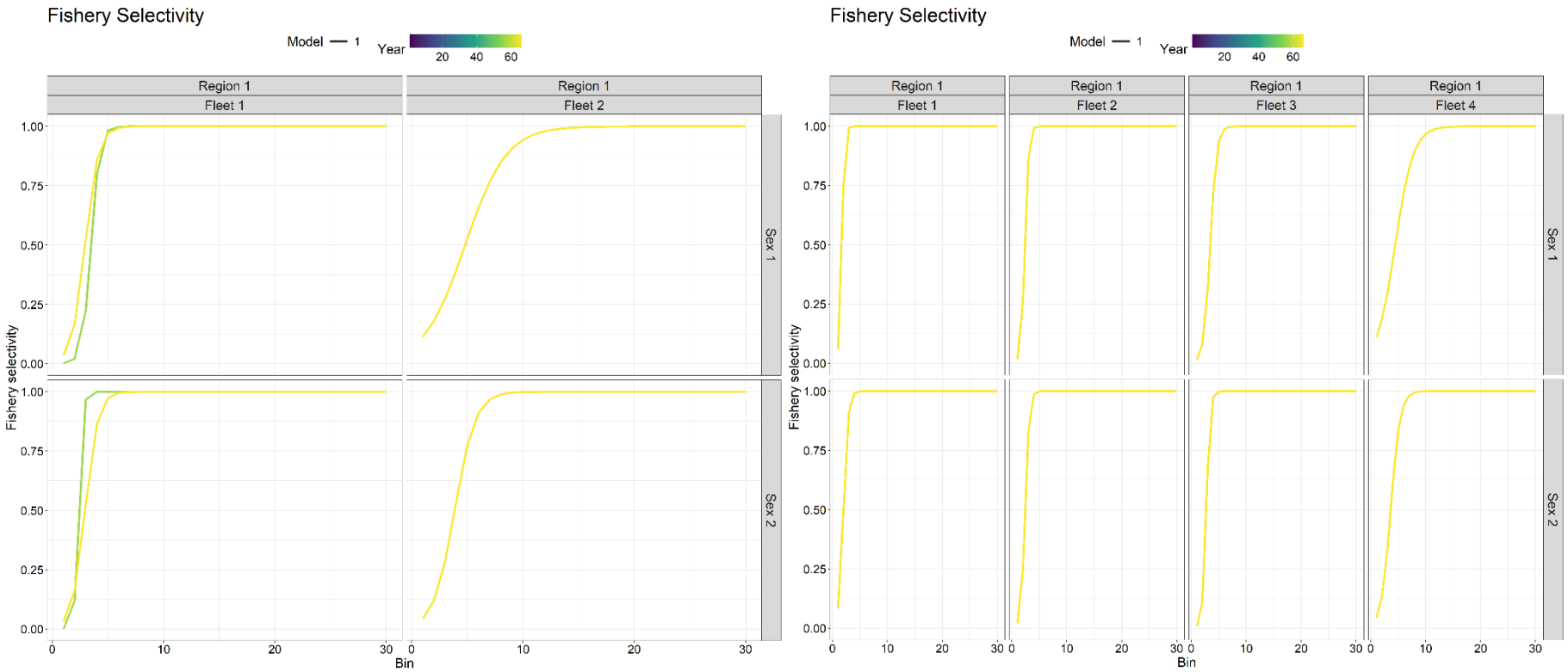

Figure 3C.8. Comparison of fishery selectivity for Model *26.2_FAA* (left panels) and Model *26.2a_FAA_Fsh+Srvy* (right panel). Sex 1 is females and sex 2 is males. For Model *26.2_FAA* Fleet 1 is the fixed gear fishery and Fleet 2 is the trawl fishery. For Model *26.2a_FAA_Fsh+Srvy* Fleet 1 is the Bering Sea fixed gear fishery, Fleet 2 is the Aleutian Islands fixed gear fishery, Fleet 3 is the Gulf of Alaska fixed gear fishery, and Fleet 4 is the trawl fishery. All models assume logistic selectivity for fishery fleets.

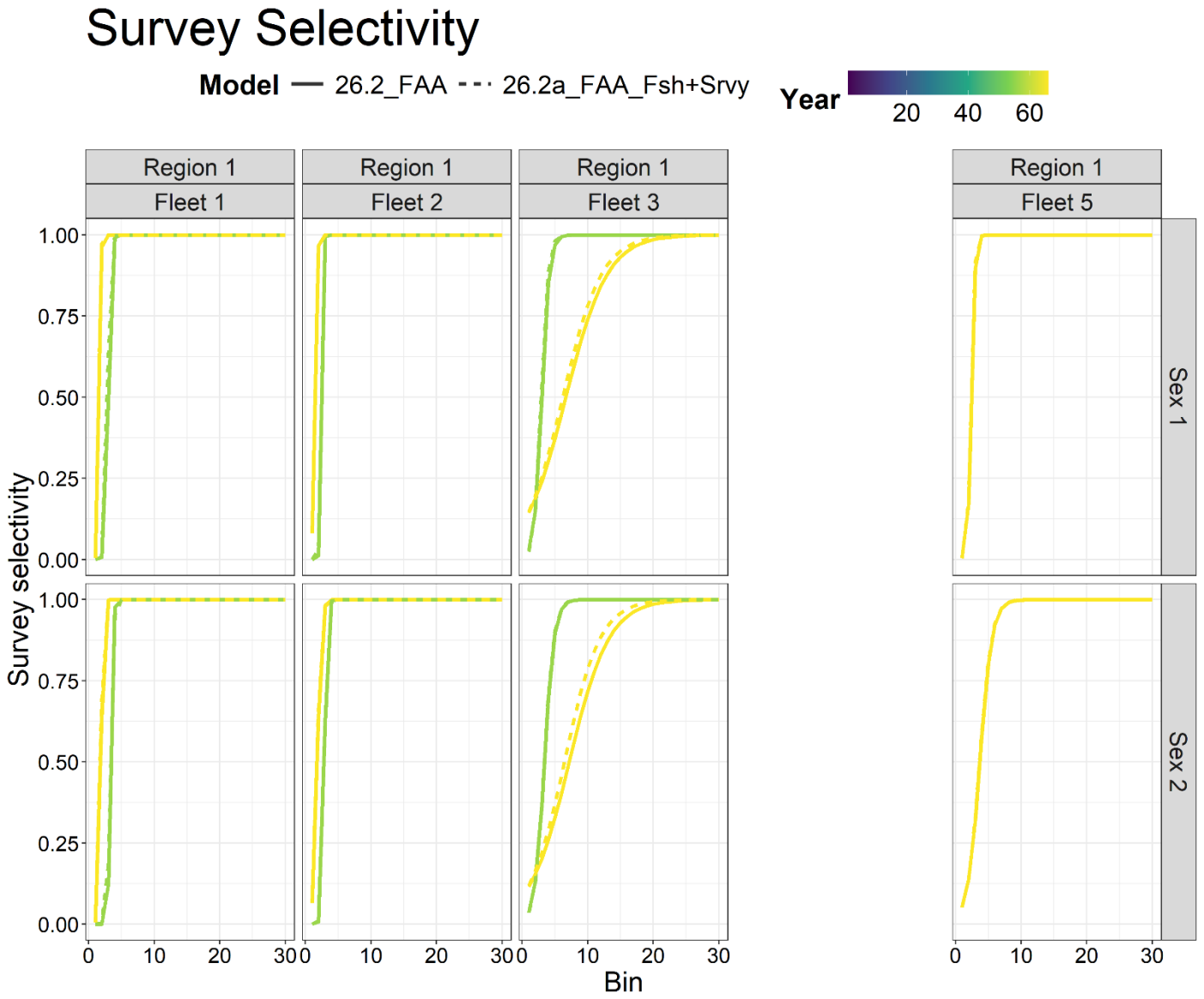

Figure 3C.9. Comparison of survey selectivity for fleets-as-areas (FAA) model runs. Sex 1 is females and sex 2 is males. Fleet 1 is the Bering Sea NOAA Longline Survey, Fleet 2 is the Aleutian Islands NOAA Longline Survey, Fleet 3 is the Gulf of Alaska NOAA Longline Survey, and Fleet 5 is the Japanese Longline Survey.

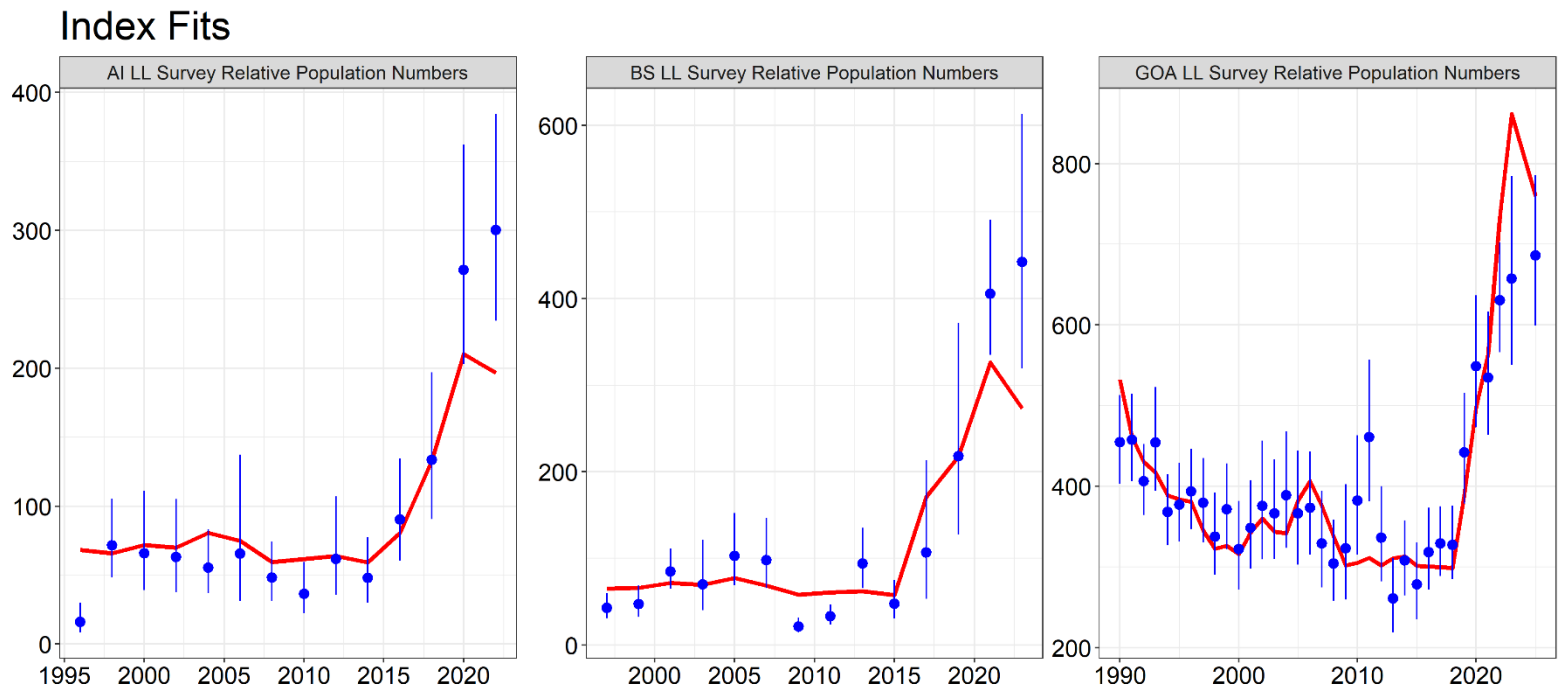

Figure 3C.10. Fits (red lines) to the NOAA Longline Survey abundance indices in relative population numbers (RPNs – 1000s of fish, y-axis; blue dots with vertical lines representing 95% confidence intervals) for an exploratory run matching Model *26.2_FAA*, but fixing the GOA NOAA Longline Survey selectivity at similar values as the BS and AI survey selectivity.

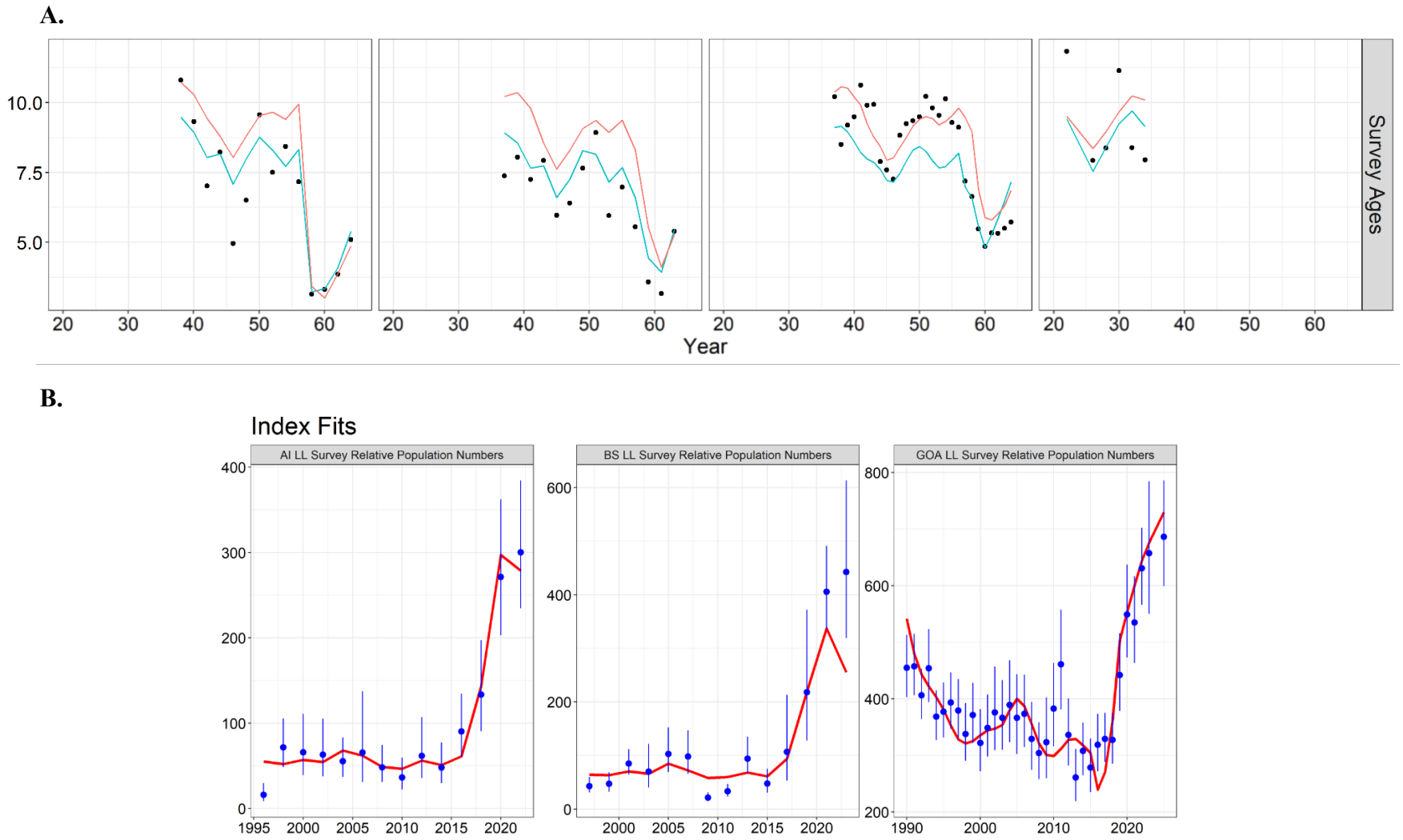

Figure 3C.11. Results from an exploratory model with the same parametrization as Model *26.2_FAA*, but assuming domed BS survey selectivity for the recent time block, including fit to mean survey age (panel A) and NOAA Longline Survey indices (panel B). In Panel A (y-axis is mean age), red lines represent the model fit to the mean age under the assumption of equal weighting (data weights equal to one for all data sources) and the blue lines represent model fit to the mean age after Francis reweighting is complete (and resulting data weights have been applied). In Panel B, red lines are model predicted fits to the NOAA Longline Survey abundance indices in relative population numbers (RPNs – 1000s of fish, y-axis) and blue dots are observed RPNs with vertical lines representing 95% confidence intervals.

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2_FAA	1	193.1500	358.1881	0.05874255	31.6957	122.4267	0.08054	1.57768	0.53924	0.72940
26.2_FAA_Carry	1	301.1013	462.3924	0.03737749	53.9853	145.3347	0.08519	2.07178	0.65118	0.43876

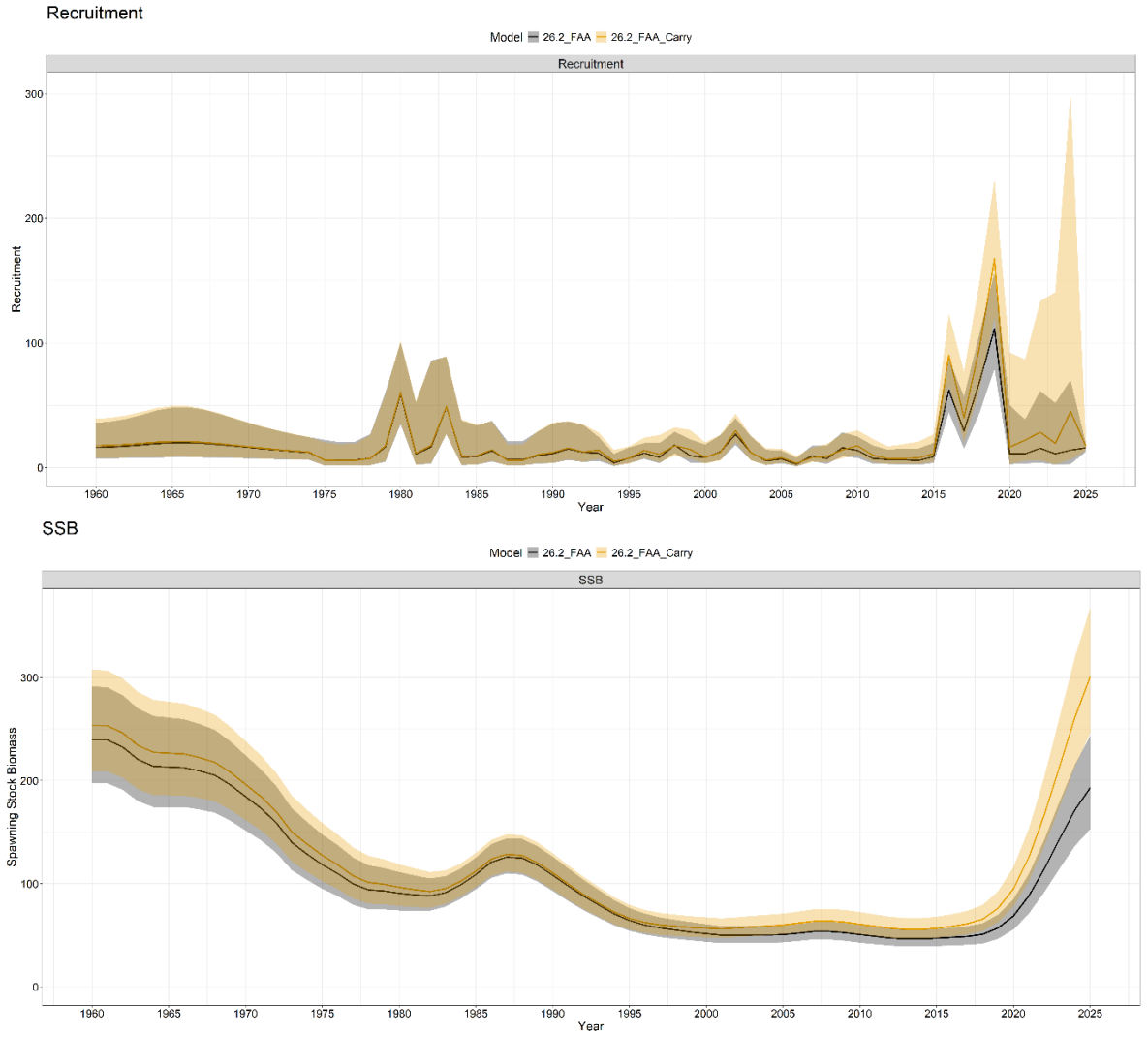

Figure 3C.12. For an exploratory run matching Model *26.2_FAA* but assuming using 'carryover' RPN values for regional indices when a given region is not surveyed, estimates of stock status (top), recruitment (millions of fish, middle), and SSB (kt, bottom).

2. Sensitivity Runs

Introduction

Following similar rationale for single-region models, several analogous sensitivity runs were developed for fleets-as-areas (FAA) models. In particular, the use of empirical weight-at-age (EWAA) was explored to account for potential density-dependent growth associated with recent large year classes (*26.2_FAA_TVWAA*; Cheng et al., 2024). Moreover, while the use of an FAA model theoretically accounts for average differences in availability across regions, there may also be a need to address time-varying availability given the spatial dynamics of sablefish, where large recruitment events often appear in western regions and shift eastward as cohorts mature. Thus, a second sensitivity model implemented continuous semi-parametric time-varying survey selectivity for the NOAA longline survey in the BS, AI, and GOA fleets to evaluate whether a more flexible selectivity formulation could better capture temporal changes in survey availability (*26.2_FAA_ContTVSel*). Likewise, a model exploring the use of the Dirichlet-multinomial likelihood for composition data was developed to evaluate whether model tension arising from unaccounted-for process error, particularly associated with time-varying survey selectivity, could be better accommodated relative to the Francis re-weighting approach (*26.2_FAA_DM*). Lastly, a sensitivity run combining continuous semi-parametric time-varying survey selectivity and the Dirichlet-multinomial likelihood for composition data (*26.2_FAA_ContTVSel_DM*) was explored to evaluate the extent to which these approaches interacted.

Methods

The sensitivity model runs follow the parameterization of *26.2_FAA* with incremental updates as noted below. All results presented herein are compared to *26.2_FAA*. Francis reweighting was only employed for model *26.2_FAA_TVWAA*.

The configuration of sensitivity model runs was as follows:

- *26.2_FAA_TVWAA*:
    - Time-varying EWAA was incorporated into model *26.2_FAA* following the methods from Cheng et al. (2024), where annual weight-at-age values were estimated external to the assessment model. An initial set of EWAA estimates was first derived using bootstrapping, after which a model-based smoothing framework was applied to account for correlations among age, year, and cohort effects in bootstrapped estimates of weight-at-age.

- *26.2_FAA_ContTVSel*:
    - The NOAA domestic survey selectivity for the BS, AI, and GOA fleets was specified with continuous semi-parametric time-varying selectivity. For all fleets, the parametric form assumed a logistic shape, while semi-parametric lognormal deviations were specified for each age to be independent and identically distributed (iid) random variables following a normal distribution. These deviations were estimated within a penalized likelihood framework, with the standard deviation fixed at 0.5 to balance fit to composition data as well as indices (based on initial

explorations). Note that deviations were estimated for all age, sex, and fleet combinations.

- *26.2_FAA_DM*:
    - Model *26.2_FAA* assumes that all composition datasets arise from a Multinomial distribution. To incrementally add to this model, composition likelihoods were specified assuming the linear parameterization of a Dirichlet-multinomial distribution Thorson *et al.* (2017). The Dirichlet-multinomial likelihood was applied to all composition datasets except those associated with the Japanese longline survey length compositions, given the sparse nature of those data. A multinomial likelihood was instead assumed for these observations.

- *26.2_FAA_ ContTVSel_DM*:
    - Continuous time-varying selectivity and self-weighting likelihoods are known to interact, and has been found to provide improved assessment results when jointly considered (Xu *et al.* 2019). Accordingly, model *26.2_FAA_ContTVSel_DM* combines the attributes of models *26.2_FAA_ContTVSel* and *26.2_FAA_DM*, such that time-varying survey selectivity for the BS, AI, and GOA fleets was estimated concurrently with the implementation of Dirichlet-multinomial likelihoods.

Results

- *26.2_TVWAA* (full results here):
    - Recent declines in weight-at-age were notable for young to intermediate-aged individuals, likely reflecting density-dependent effects associated with recent large recruitment events (Appendix 3B, Fig. 3B.11). Potentially unrealistic increases in weight-at-age were also observed for older females, likely attributed to parameter correlation between the intrinsic growth rate and asymptotic length parameters, compounded by limited data availability at older ages (Appendix 3B, Fig. 3B.11).
    - Declines in *SSB* coincided with declines in EWAA, with a reduction in population scale as compared to Model *26.2_FAA* (Fig. 3C.13). However, data fits remained largely unchanged and estimates of stock status also maintained similar levels (although fishing mortality appears to be larger than $F_{40\%}$ in this model configuration) (Fig. 3C.14 and Table 3C.7).

- *26.2_FAA_ContTVSelex* (full results here):
    - Specifying continuous semi-parametric time-varying survey selectivity for the NOAA longline survey did not lead to substantive improvements in model fits to composition data or abundance indices relative to model *26.2_FAA* (and were somewhat degraded despite the added complexity) (Fig. 3C.15). Moreover, aggregate fits to young individuals appeared degraded for *26.2_FAA_ContTVSel*, with young fish consistently overpredicted across all domestic longline survey fleets relative to model *26.2_FAA* (Fig. 3C.16 – 3C.22).
    - Moderate variability in survey selectivity estimates was detected, with recent years exhibiting some degree of doming, likely reflecting the influence of large recruitment events. Dome-shaped selectivity patterns were particularly apparent for

the BS survey fleets, likely driven in part by large recruitment events in the BS, while the AI and GOA survey fleets generally exhibited selectivity patterns more closely resembling a logistic function (Fig. 3C.23). Moreover, GOA survey selectivity appeared to shift towards older ages in recent years, likely attributed to the movement into the GOA region of maturing fish from recent year classes in recent years. Similar to the single-region sensitivity runs, attempts to estimate time-varying selectivity within a state-space framework were also explored, but convergence issues were encountered.

- Implementing continuous time-varying survey selectivity led to degraded retrospective performance, resulting in moderate and consistent overestimation of *SSB* relative to the terminal assessment (likely due to consistent overestimation of large recruitment events in recent years; Mohn's rho of 0.189; Fig. 3C.24).
- Although survey selectivity appeared to vary over time, likely attributed to time-varying availability, model fits for both abundance indices and composition data appeared to worsen. Additionally, comparison of stock status and population dynamics did not appear to substantially differ between model *26.2_FAA* and model *26.2_FAA_ContTVSel* (Fig. 3C.25, 3C.26). Moderate positive retrospective patterns were detected, likely due to year-to-year changes in recruitment and interactions with continuous time-varying selectivity as cohorts aged into the population.

- *26.2_FAA_DM* (full results here):
  - The use of a Dirichlet-multinomial likelihood resulted in substantially poorer fits to abundance indices (Fig. 3C.27). Presumably, this occurred because composition data were effectively upweighted relative to the Francis re-weighting approach (Fig. 3C.28, 3C.34), thereby reducing the model's ability to adequately fit abundance indices simultaneously.
  - The introduction of additional overdispersion parameters resulted in degraded retrospective performance relative to model *26.2_FAA*, where estimates of large recruitment events were frequently overestimated relative to the terminal year, manifesting as a positive SSB retrospective pattern (Mohn's rho of approximately 0.147; Fig. 3C.35). These retrospective patterns may in part reflect conflicting recruitment signals within the composition data arising from spatial heterogeneity in recruitment as new composition data entered the model.
  - Recruitment estimates were substantially more variable relative to model *26.2_FAA*, with an associated upward revision in overall population scale. Presumably, this increase in population scale resulted from the effective upweighting of composition data relative to the Francis re-weighting approach, where the model compensated by increasing recruitment estimates to improve fits to composition data. The increased recruitment estimates then propagated into a larger overall population scale, which was necessary to maintain adequate fits to catch data (Fig. 3C.36, 3C.37).
  - While the Dirichlet-multinomial likelihood is theoretically more statistically rigorous than Francis re-weighting, the model was generally unable to adequately balance simultaneous fits to both composition and abundance index data.

- *26.2_FAA_ContTVSel_DM* (full results here):
    - Combining continuous semi-parametric time-varying survey selectivity with a Dirichlet-multinomial likelihood for composition data continued to produce degraded fits to both abundance indices and composition data (Fig. 3C.38 – 3C.45). Moreover, the combined approach was unable to adequately reconcile the respective model misfits observed when each method was applied independently.
    - Positive retrospective patterns in SSB were detected, likely reflecting the continual revision of recruitment estimates and differing recruitment signals across space as new composition data entered the model and cohorts aged into the population (Fig. 3C.46). These patterns were likely exacerbated by the more flexible survey selectivity parameterizations. Overall interpretation of population dynamics remained generally similar to model *26.2_FAA*, although recruitment estimates were more variable and overall population scale was larger (Fig. 3C.47, 3C.48). Thus, this model configuration does not appear to provide substantive improvement over model *26.2_FAA*.

Conclusions

Sensitivity analyses indicated that Model *26.2_FAA* with Francis reweighting for the composition data likely provided the most balanced and robust performance among the configurations examined. While incorporation of time-varying weight-at-age captured biologically plausible density-dependent growth reductions among younger age classes associated with recent large recruitment events, this approach also resulted in somewhat unrealistic estimates for older ages where data were sparse. Future work could evaluate methods to improve the smoothing of those older ages, as well as refine the bootstrapping procedure to better derive uncertainty of EWAA estimates. In evaluating the use of continuous semi-parametric time-varying survey selectivity, temporally variable selectivity patterns consistent with time-varying availability were detected. However, this approach resulted in degraded fits to both abundance indices and composition data, alongside elevated positive retrospective patterns. Moreover, application of the Dirichlet-multinomial likelihood effectively upweighted composition data relative to the Francis reweighting approach, leading to substantially more variable recruitment estimates, inflated population scale, and degraded fits to abundance indices. Finally, the combination of continuous time-varying selectivity and the Dirichlet-multinomial likelihood failed to reconcile the respective misfits observed under each approach when applied independently, with positive retrospective patterns persisting. Collectively, these results suggest that none of the alternative configurations examined performed better than Model *26.2_FAA*. However, they provide a useful foundation for future model development and help clarify key areas of model tension and structural sensitivity.

Tables

Table 3C.6. Summary of sensitivity model runs and associated results.

Model Group	*Model Name*	*Major Changes*	*Rationale*	*Conclusions*
Sensitivity Runs	*26.2_FAA_TVWAA*	(*26.2_FAA+*) Incorporate time-varying empirical weight-at-age.	Incorporating time-varying weight-at-age is expected to improve the biological realism of the model.	• Declines in weight-at-age were evident for young to intermediate-aged individuals in recent years, consistent with density-dependent effects. • Potentially unrealistic increases in weight-at-age for older females. • Declines in SSB coincided with declines in empirical weight-at-age. • Fishing mortality exceeded $F_{40\%}$ under this configuration.
	26.2_FAA_ContTVSel	(*26.2_FAA +*) Specify continuous semi-parametric time-varying survey selectivity for the domestic NOAA longline survey.	Better capture temporal changes in survey selectivity and associated changes in availability due to recruitment and ontogenetic movement.	• Fits to abundance indices and composition data were degraded despite added complexity. • Moderate variability in selectivity was detected, with dome-shaped patterns apparent for BS survey fleet and a shift towards older ages in the GOA survey fleet. • Retrospective performance was degraded, with moderate and consistent overestimation of *SSB*.
	26.2_FAA_DM	(*26.2_FAA +*) Specify Dirichlet-multinomial likelihoods for composition data.	Account for overdispersion in composition data and provide a more flexible statistical treatment than Francis reweighting.	• Composition data were effectively upweighted relative to the Francis reweighting approach. • Fits to abundance indices were substantially degraded. • Recruitment estimates were more variable with an associated upward revision in population scale. • Retrospective patterns of overestimating large recruitment events.
	26.2_FAA_ContTVSel_DM	(*26.2_FAA +*) Combines *26.2_FAA_ContTVSel* and *26.2_FAA_DM*.	This model is intended to address the interaction effects of both *26.2_FAA_ContTVSel* and *26.2_FAA_DM*.	• Degraded fits to abundance indices and composition data. • Positive retrospective patterns in SSB were detected, likely reflecting continual revision of recruitment estimates and differing spatial recruitment signals. • Recruitment estimates were more variable, and population scale was larger.

Table 3C.7. Key management quantity outputs for sensitivity runs. SSB is in kt, B_Ref_Pt is the associated $B_{40\%}$ reference point in kt, and F_Ref_Pt is the associated $F_{40\%}$. Catch advice is the 2026 Acceptable Biological Catch (ABC) resulting from fishing at $F_{40\%}$.

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2b_FAA_3Flt_LLS_Only	1	192.6658	357.5388	0.05878813	31.77084	122.3957	0.08070	1.57412	0.53887	0.72846
26.2b_FAA_3Flt_LLS_Only_DM	1	152.0593	317.7291	0.07041890	28.25891	126.1990	0.08206	1.20492	0.47858	0.85810
26.2b_FAA_3Flt_LLS_Only_ContTVSel	1	188.2890	362.7107	0.05456974	32.82219	141.1775	0.07484	1.33370	0.51912	0.72912
26.2b_FAA_3Flt_LLS_Only_ContTVSel_DM	1	175.5888	353.4455	0.05788491	33.65480	142.1871	0.07873	1.23491	0.49679	0.73524
26.2b_FAA_3Flt_LLS_Only_TVWAA	1	141.2156	296.2604	0.07934650	21.63043	102.7233	0.07571	1.37472	0.47666	1.04796

Figures

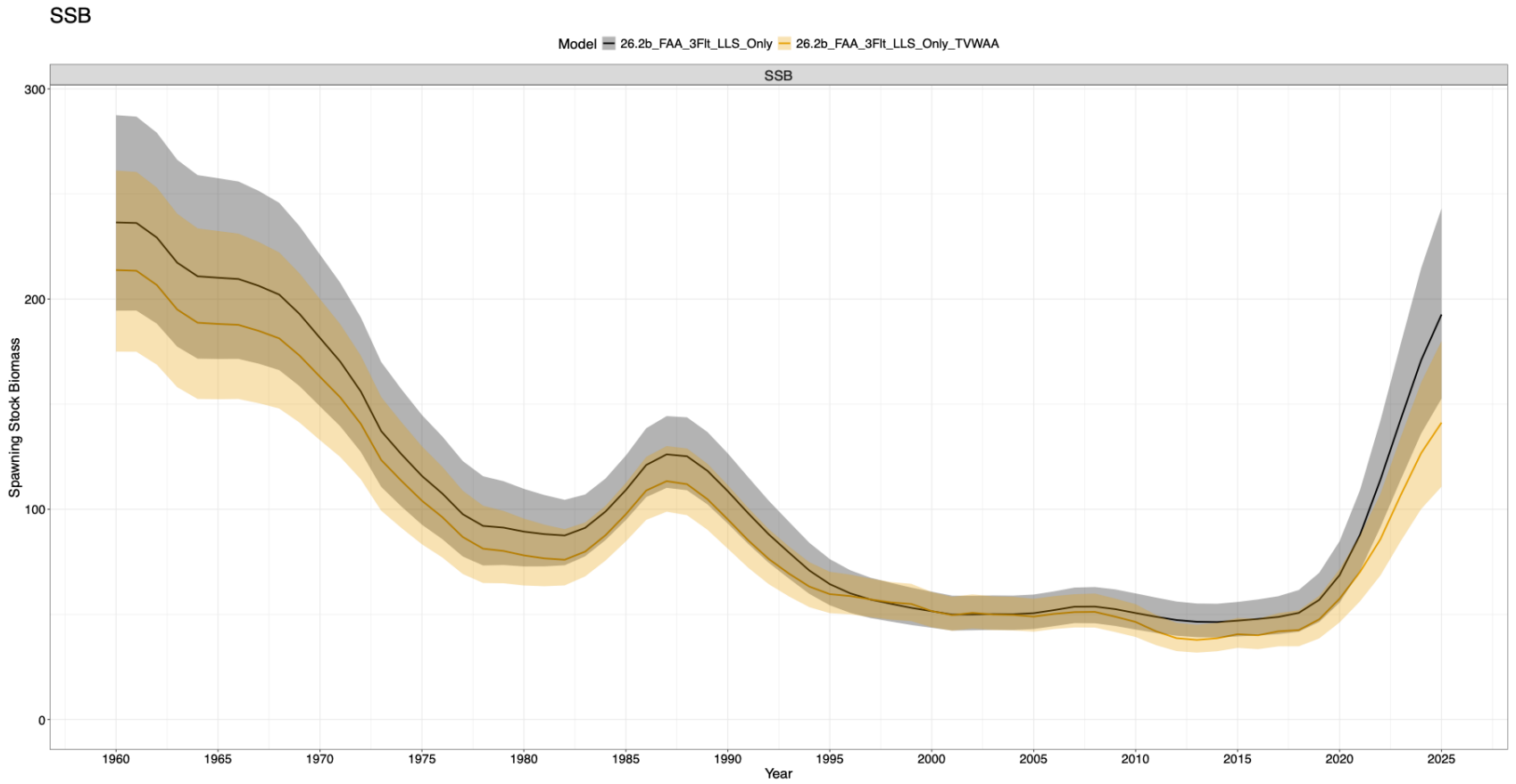

Figure 3C.13. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates time-varying empirical weight-at-age (*26.2b_FAA_3Flt_LLS_Only_TVWAA*). Shaded regions represent the 95% confidence intervals.

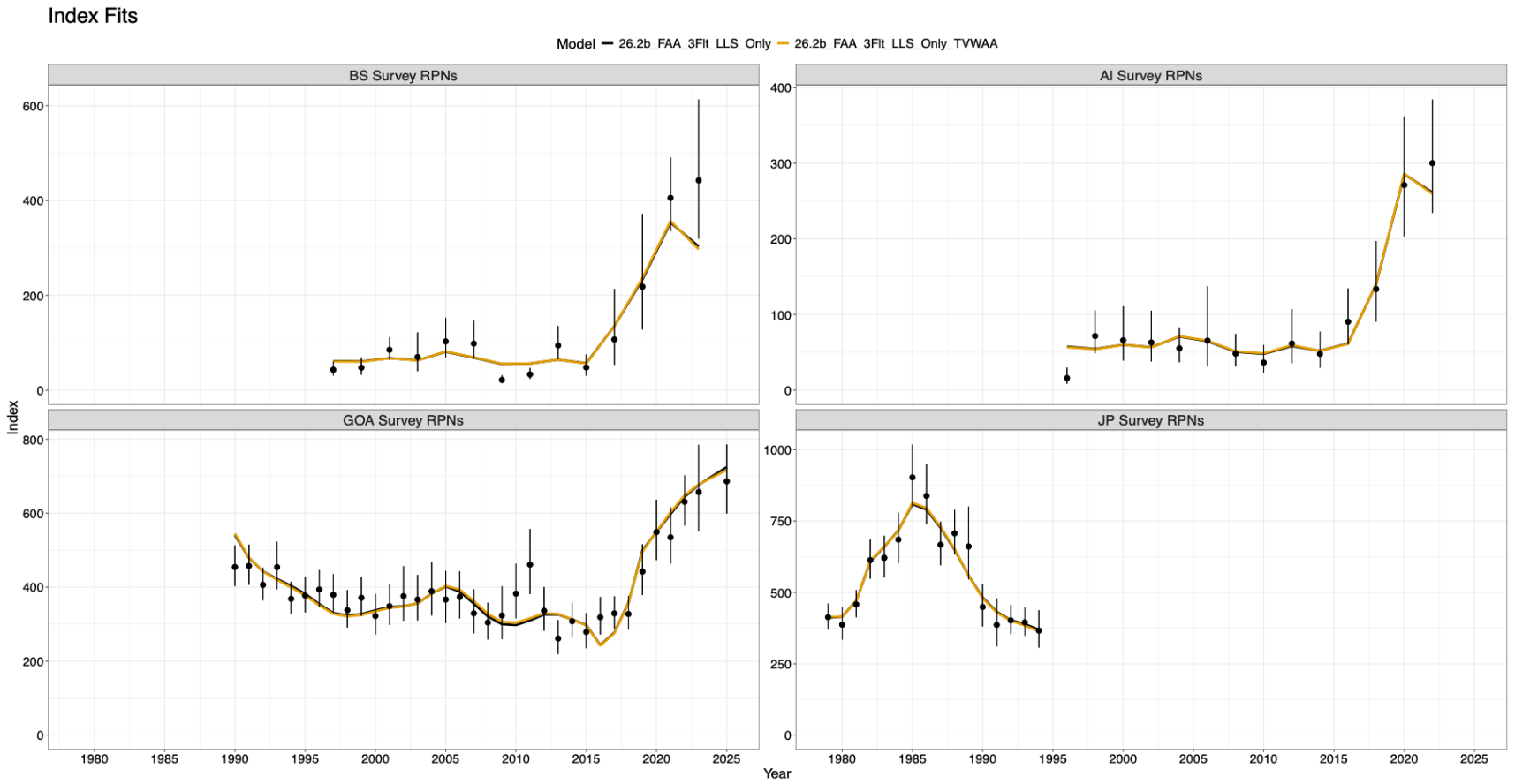

Figure 3C.14. Model fits to survey abundance indices between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates time-varying empirical weight-at-age (*26.2b_FAA_3Flt_LLS_Only_TVWAA*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

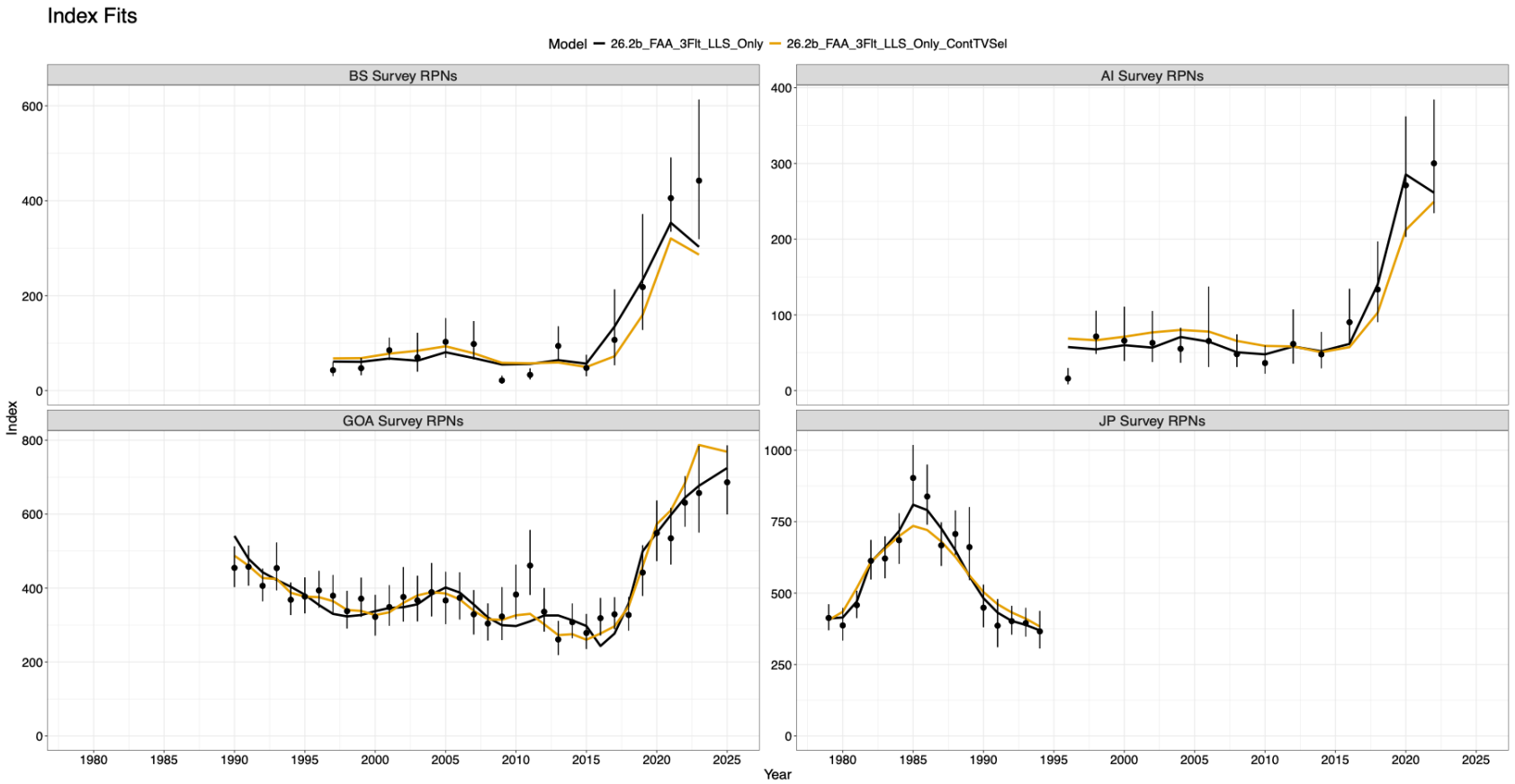

Figure 3C.15. Model fits to survey abundance indices between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous semi-parametric time-varying survey selectivity (*26.2b_FAA_3Flt_LLS_Only_ContTVSel*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

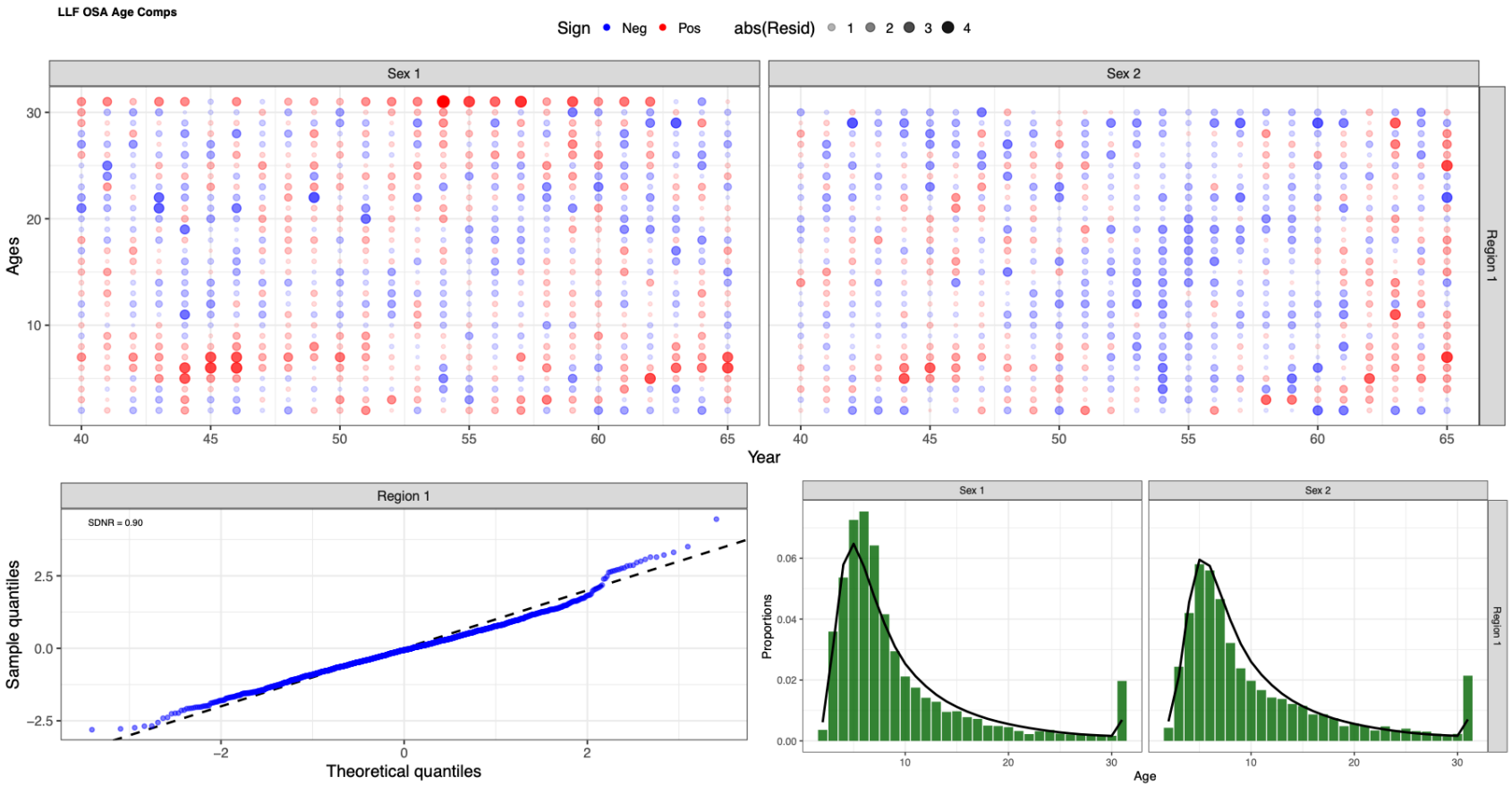

Figure 3C.16. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

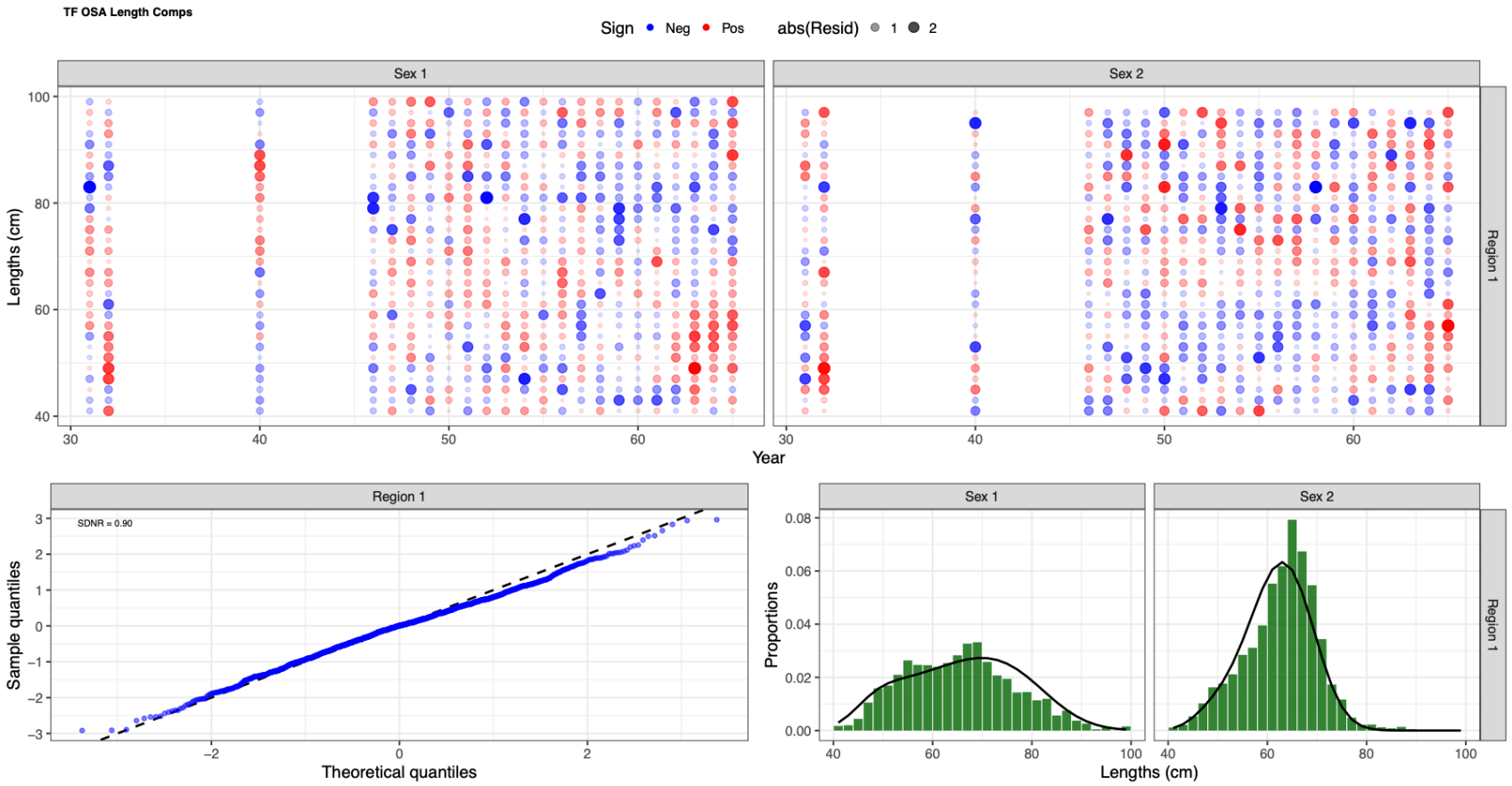

Figure 3C.17. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

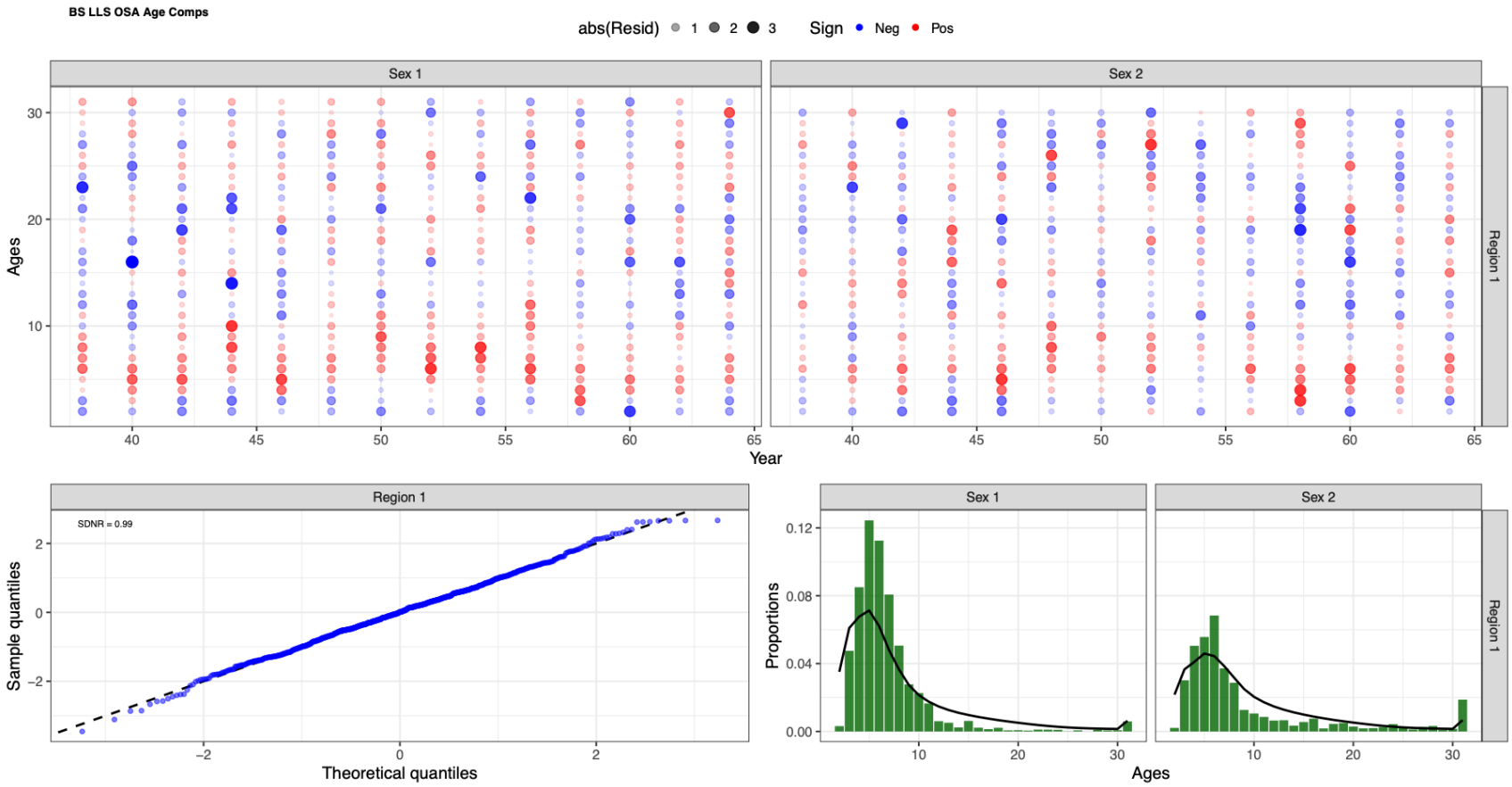

Figure 3C.18. Fits to NOAA longline Bering Sea (BS) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

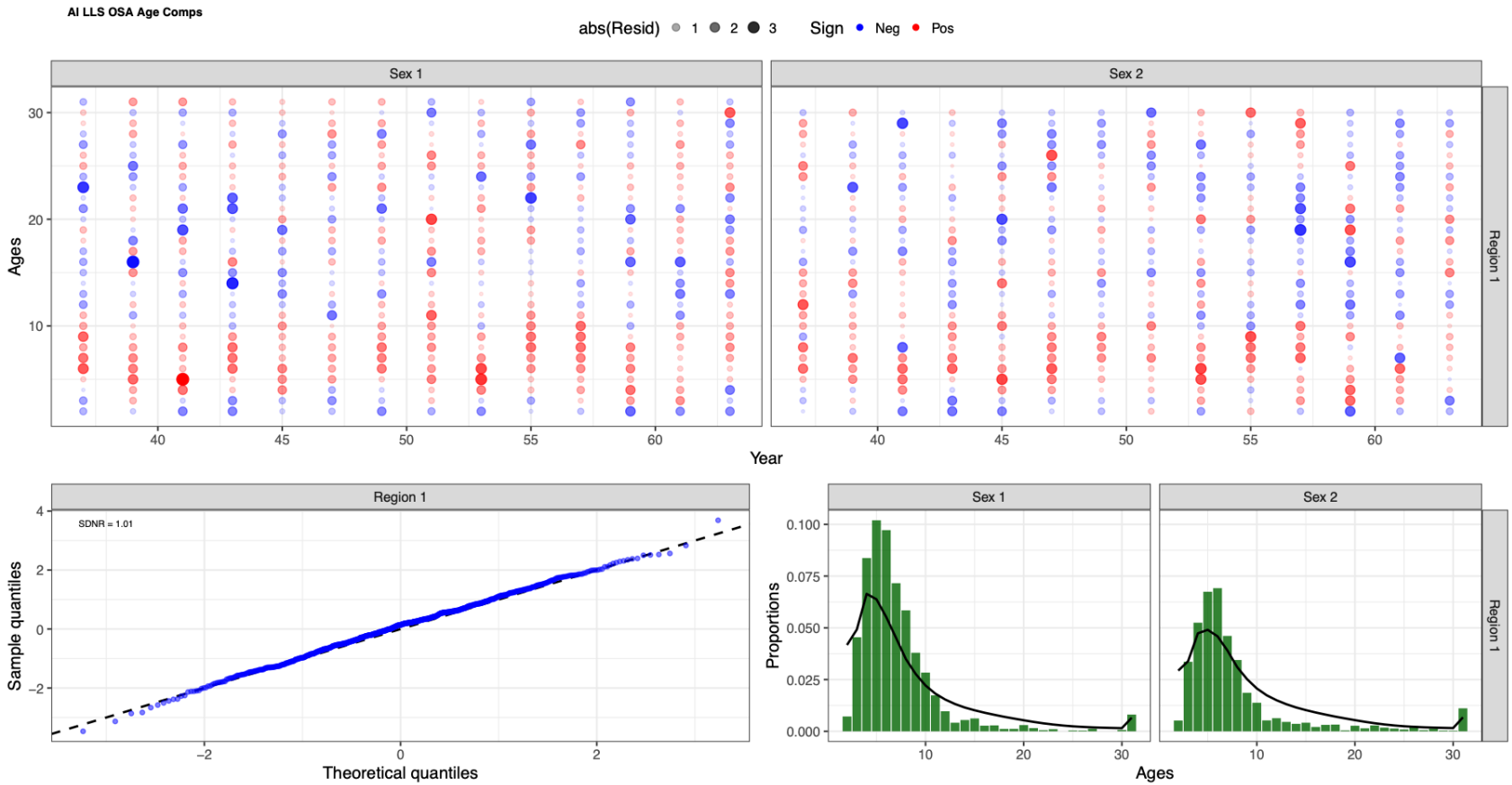

Figure 3C.19. Fits to NOAA longline Aleutian Islands (AI) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

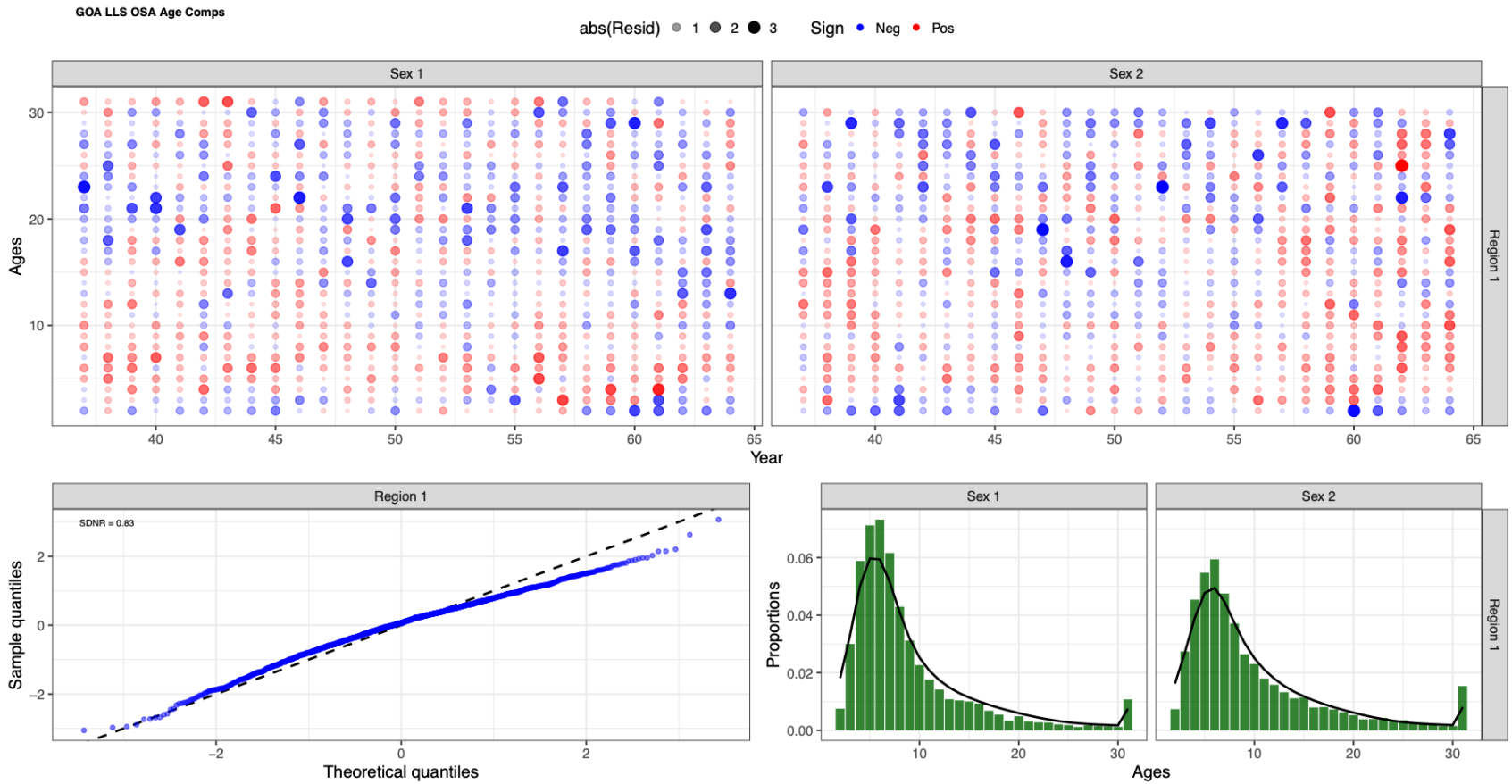

Figure 3C.20. Fits to NOAA longline Gulf of Alaska (GOA) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

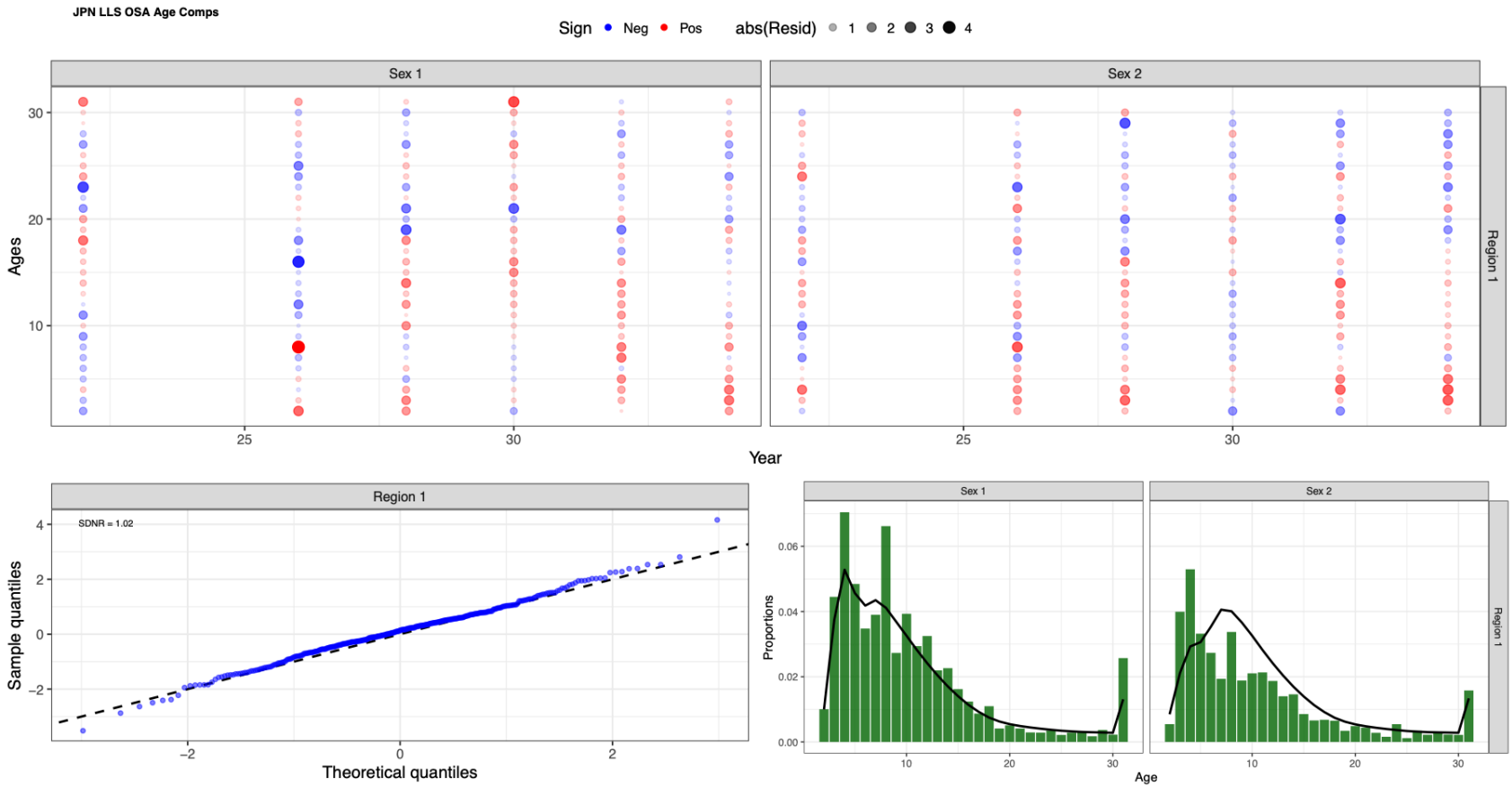

Figure 3C.21. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

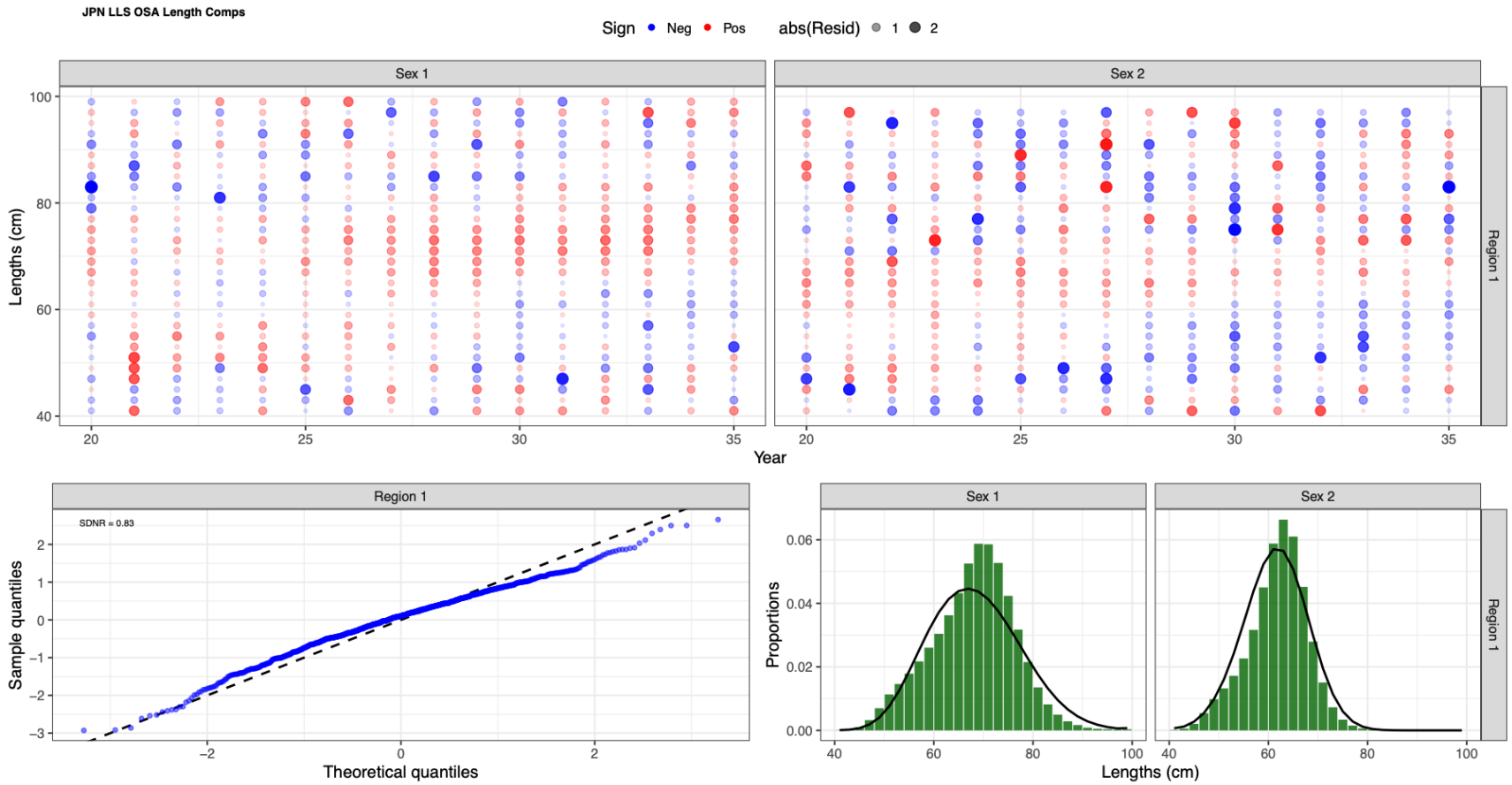

Figure 3C.22. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel*.

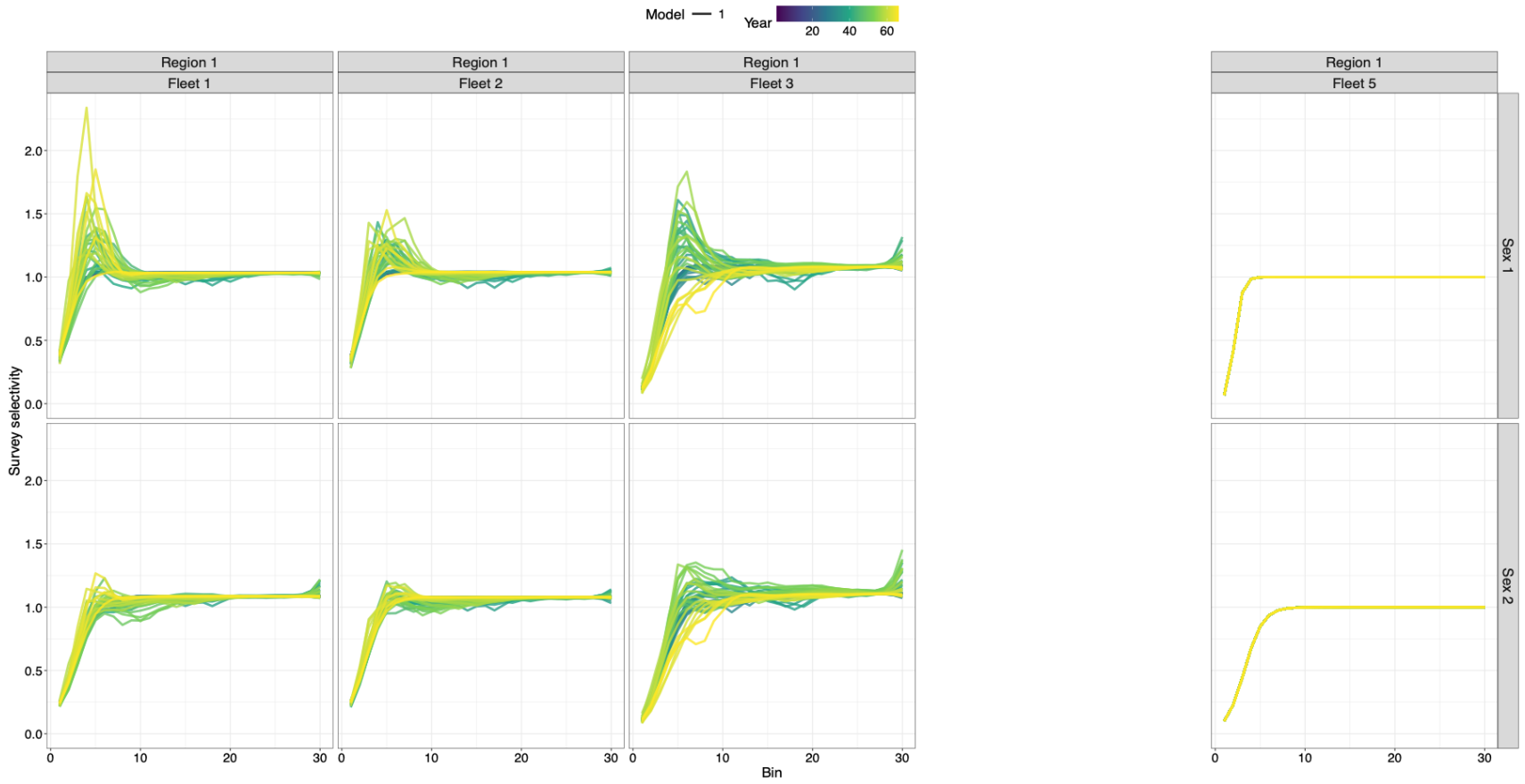

Figure 3C.23. Survey selectivity estimates from Model *26.2_FAA_ContTVSel*. Fleets 1 – 3 represent the Bering Sea, Aleutian Islands, and Gulf of Alaska domestic longline survey fleets. Fleet 5 represents the Japanese longline survey fleet. Sex 1 is females and sex 2 is males.

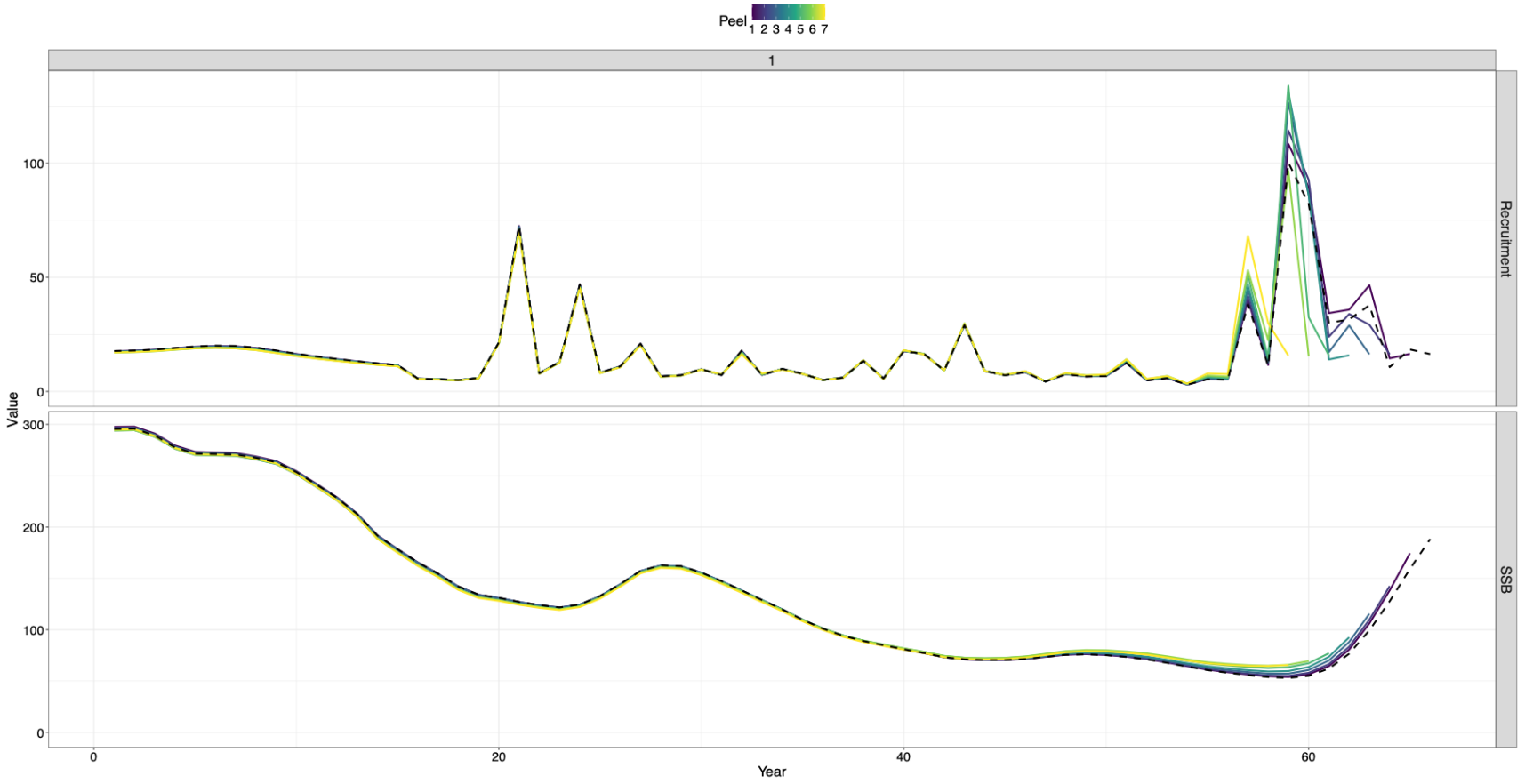

Figure 3C.24. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA_ContTVSel*. Mohn's rho for SSB was 0.189.

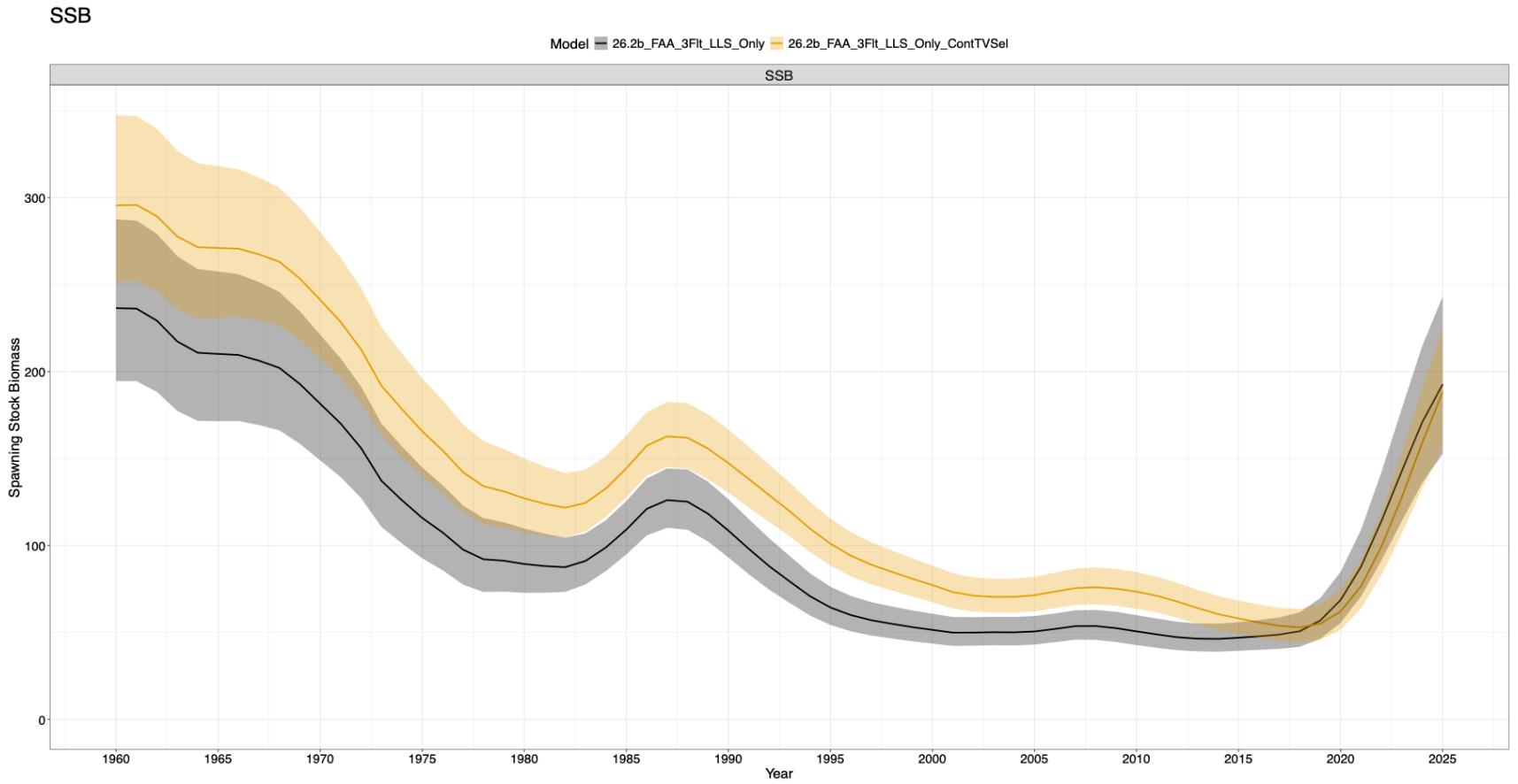

Figure 3C.25. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity (*26.2b_FAA_3Flt_LLS_Only_ContTVSel*). Shaded regions represent the 95% confidence intervals.

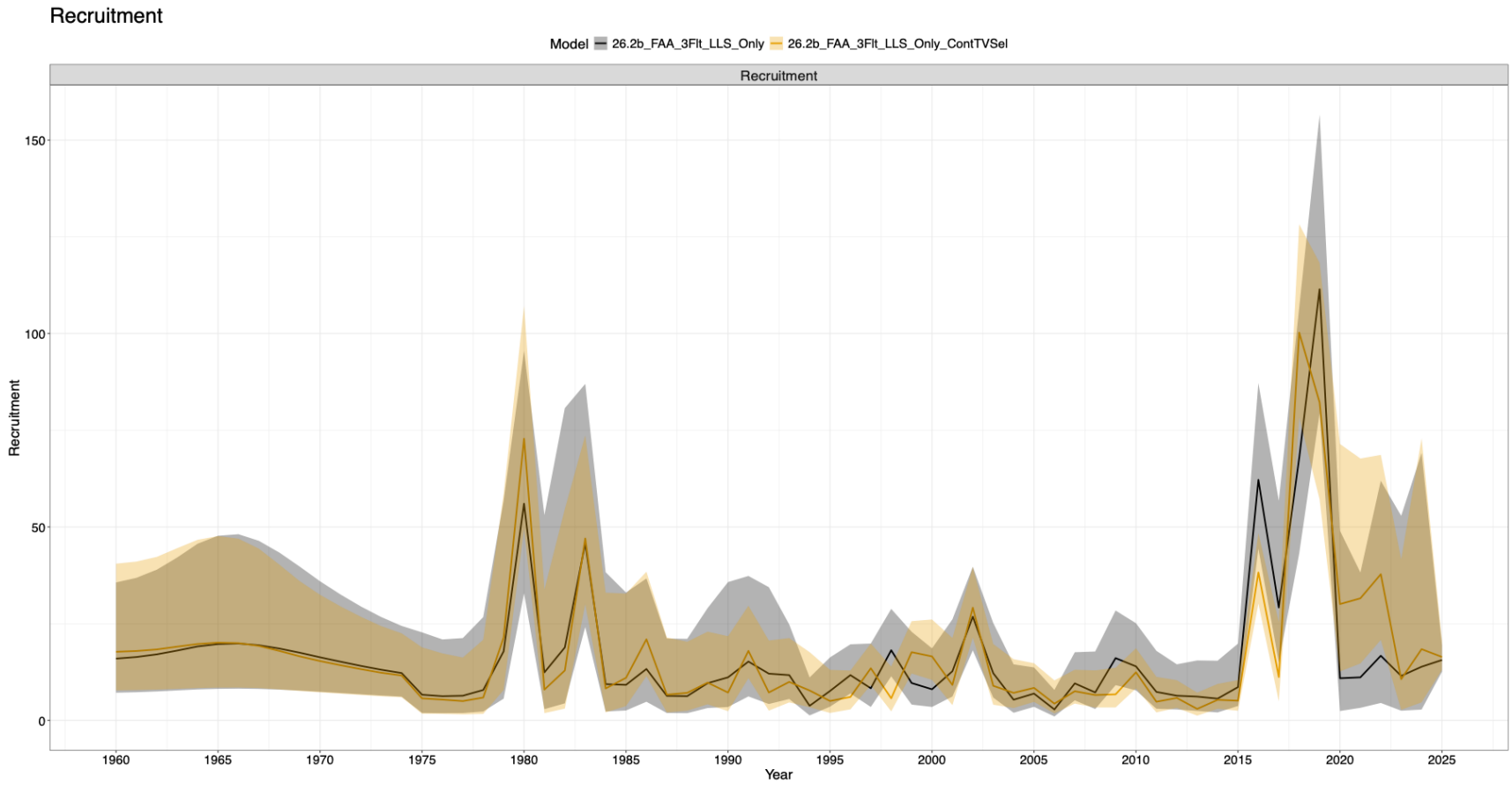

Figure 3C.26. Comparison of age-2 recruitment (in millions of individuals) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity (*26.2b_FAA_3Flt_LLS_Only_ContTVSel*). Shaded regions represent the 95% confidence intervals.

Figure 3C.27. Model fits to survey abundance indices between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.2b_FAA_3Flt_LLS_Only_DM*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

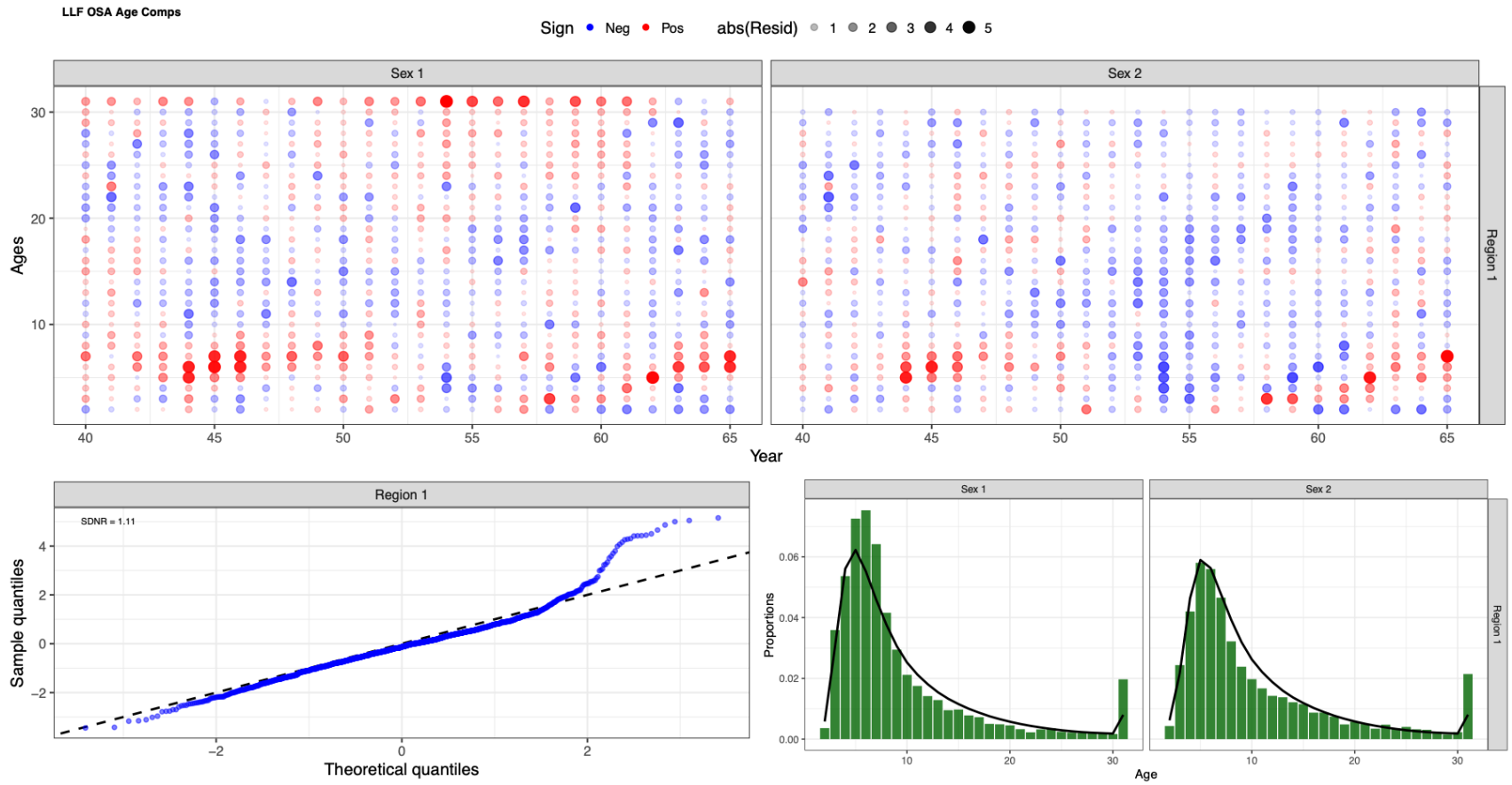

Figure 3C.28. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

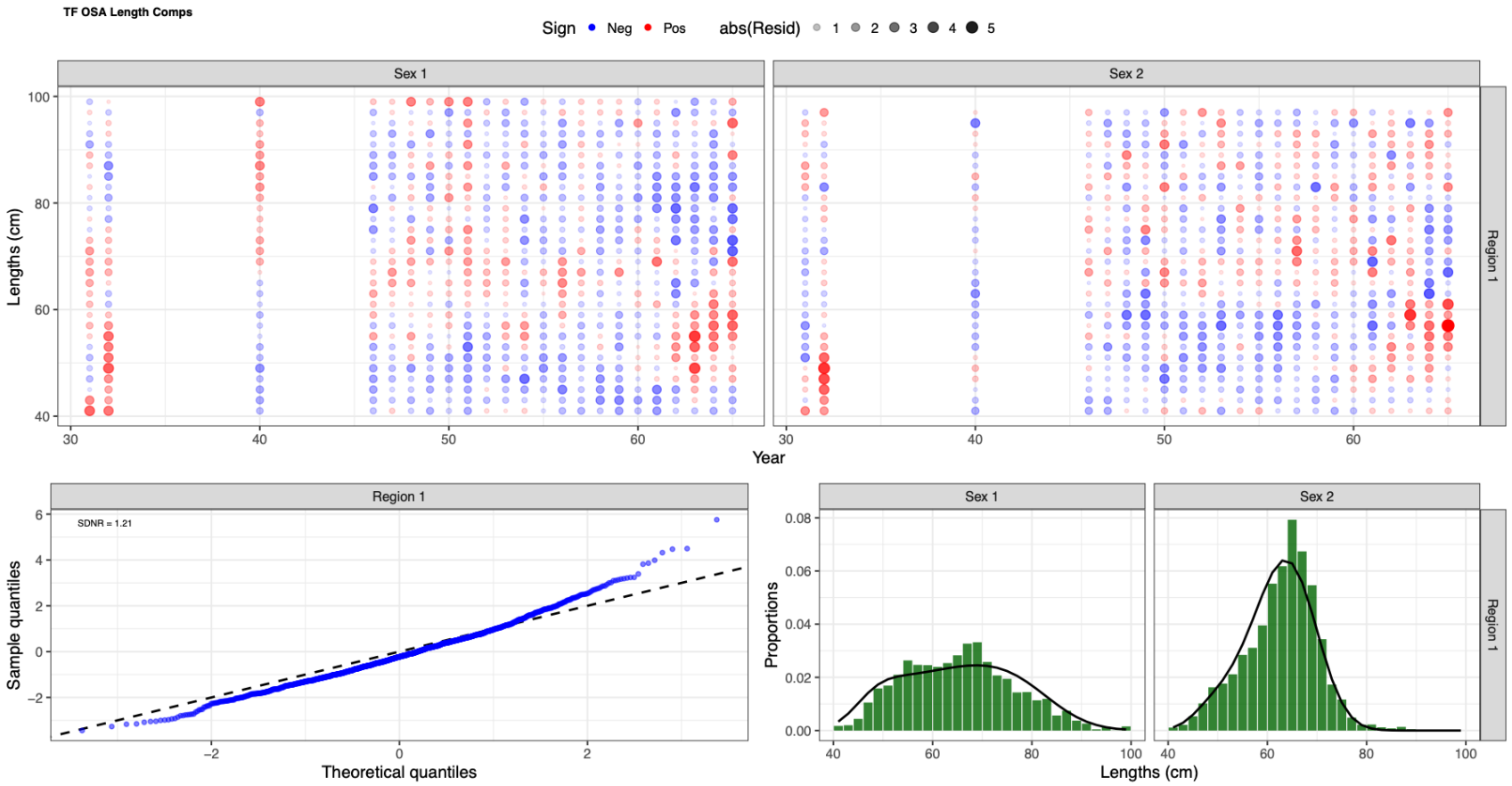

Figure 3C.29. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

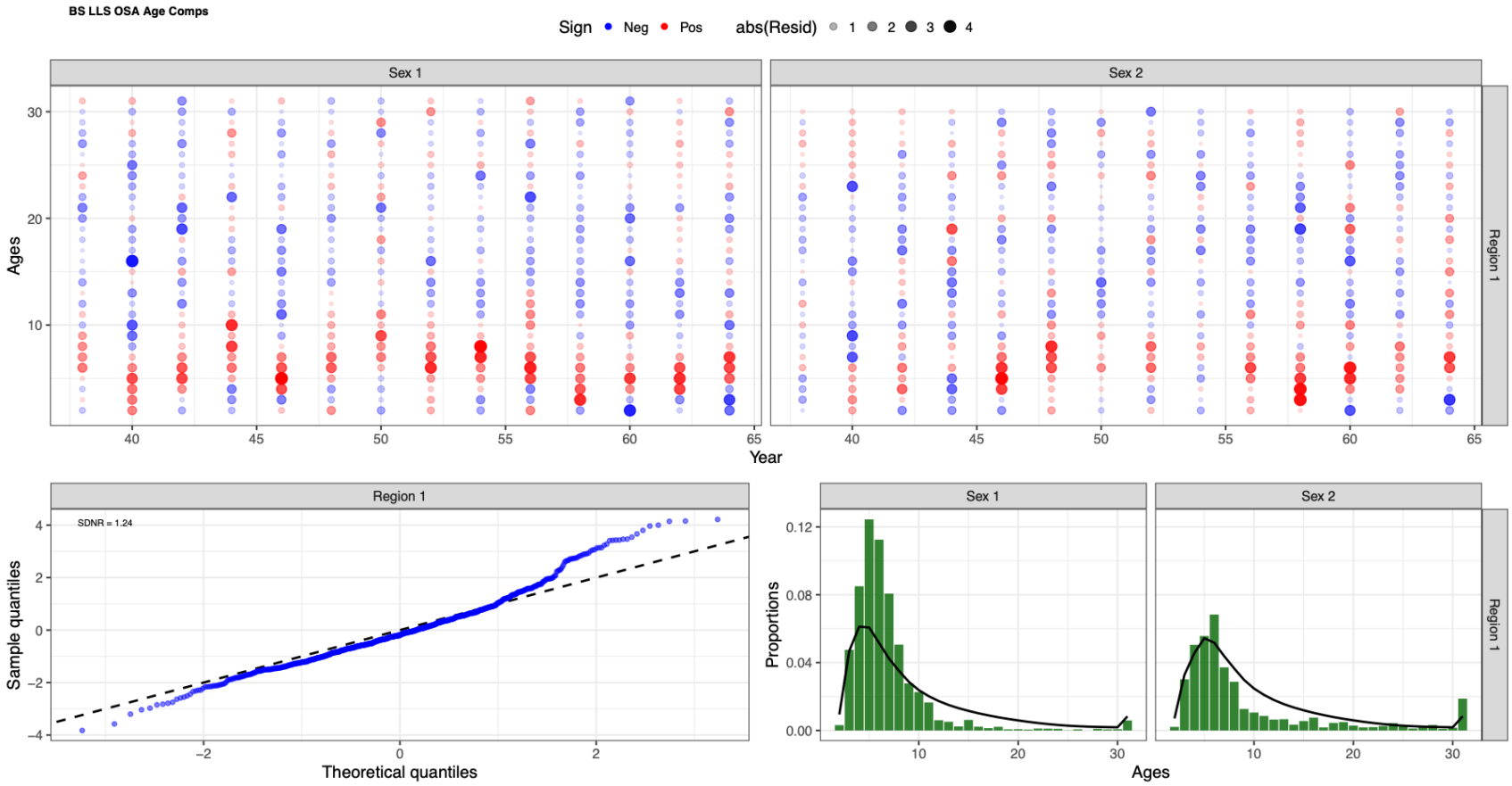

Figure 3C.30. Fits to NOAA longline Bering Sea (BS) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

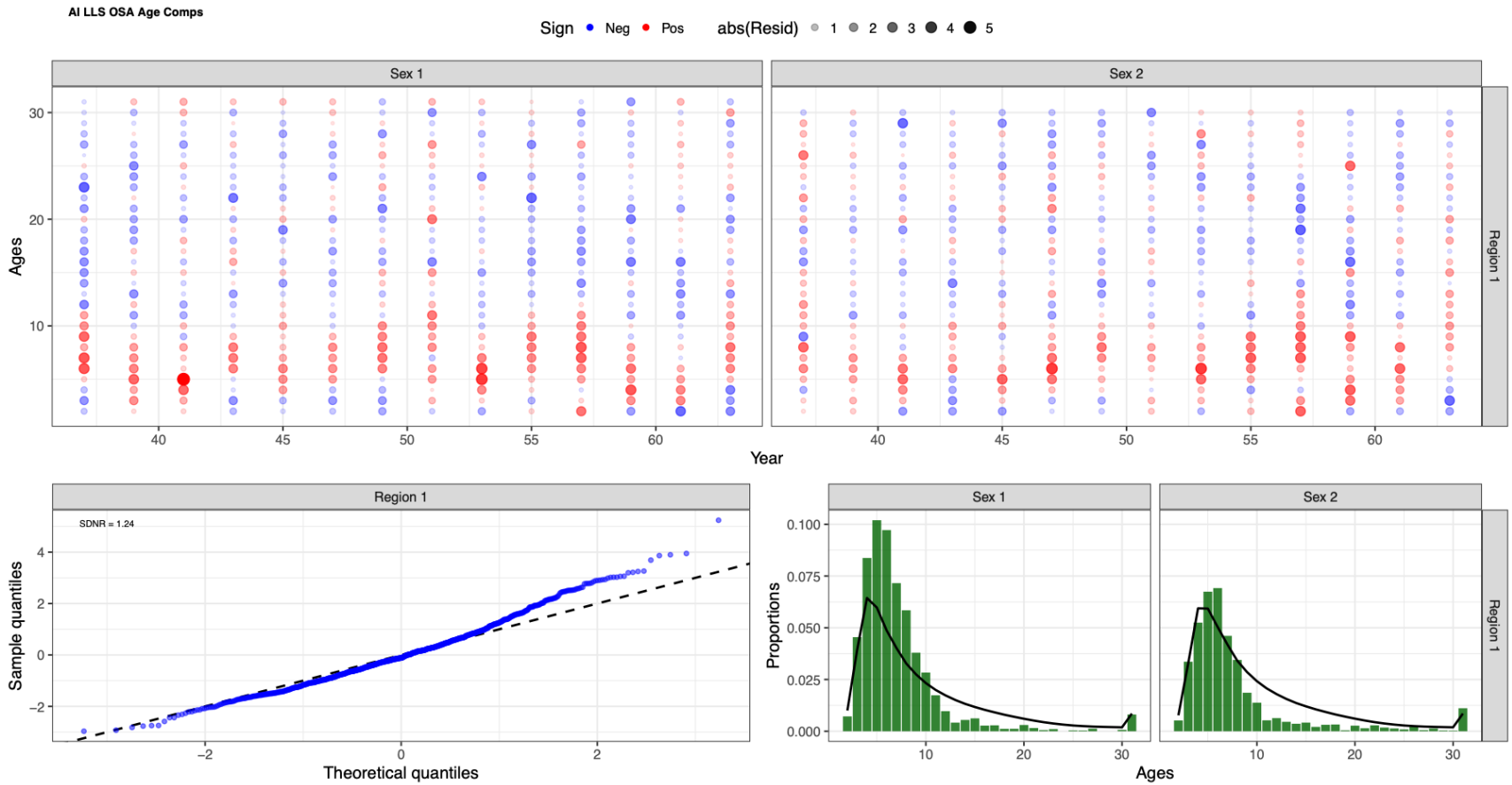

Figure 3C.31. Fits to NOAA longline Aleutian Islands (AI) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

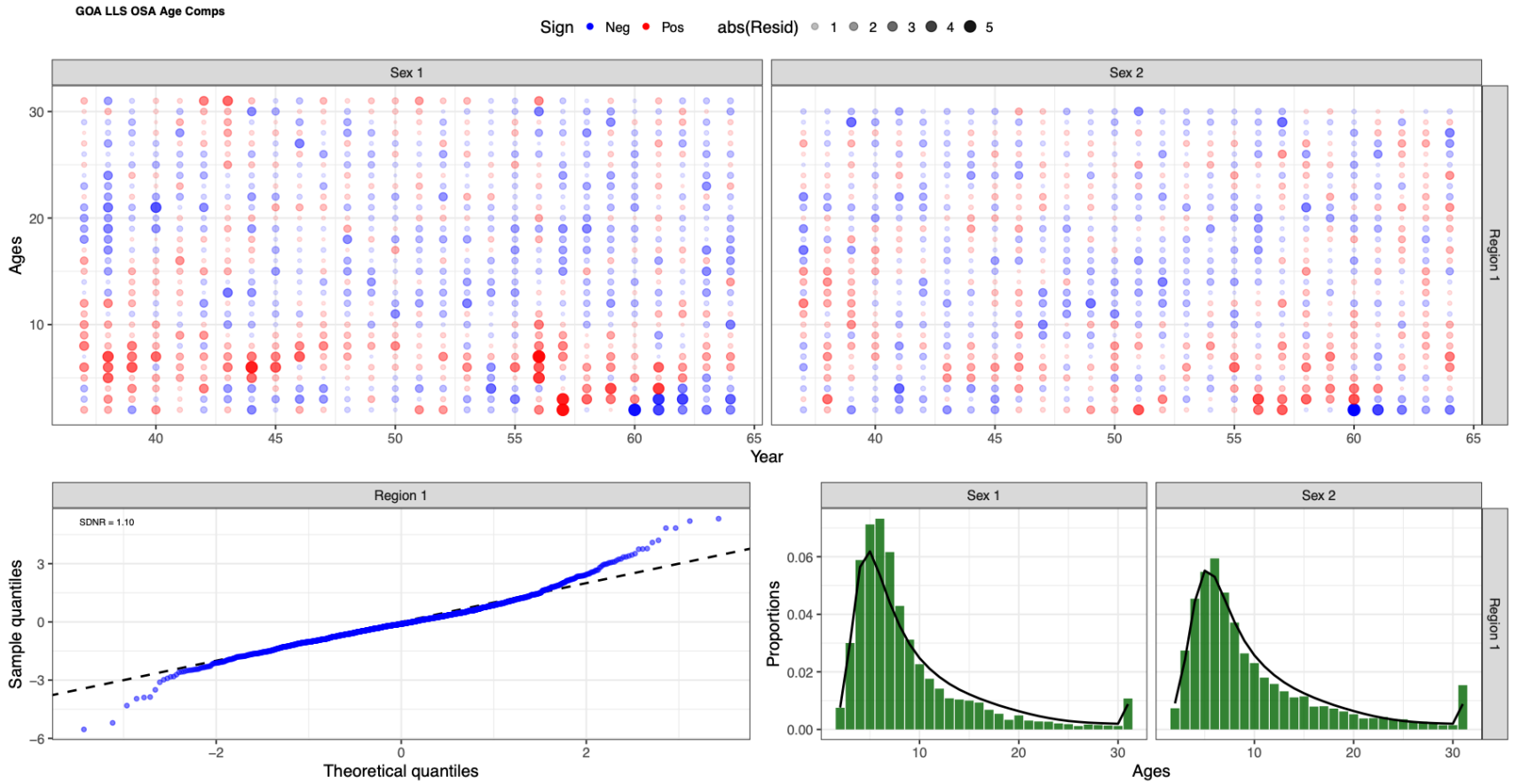

Figure 3C.32. Fits to NOAA longline Gulf of Alaska (GOA) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

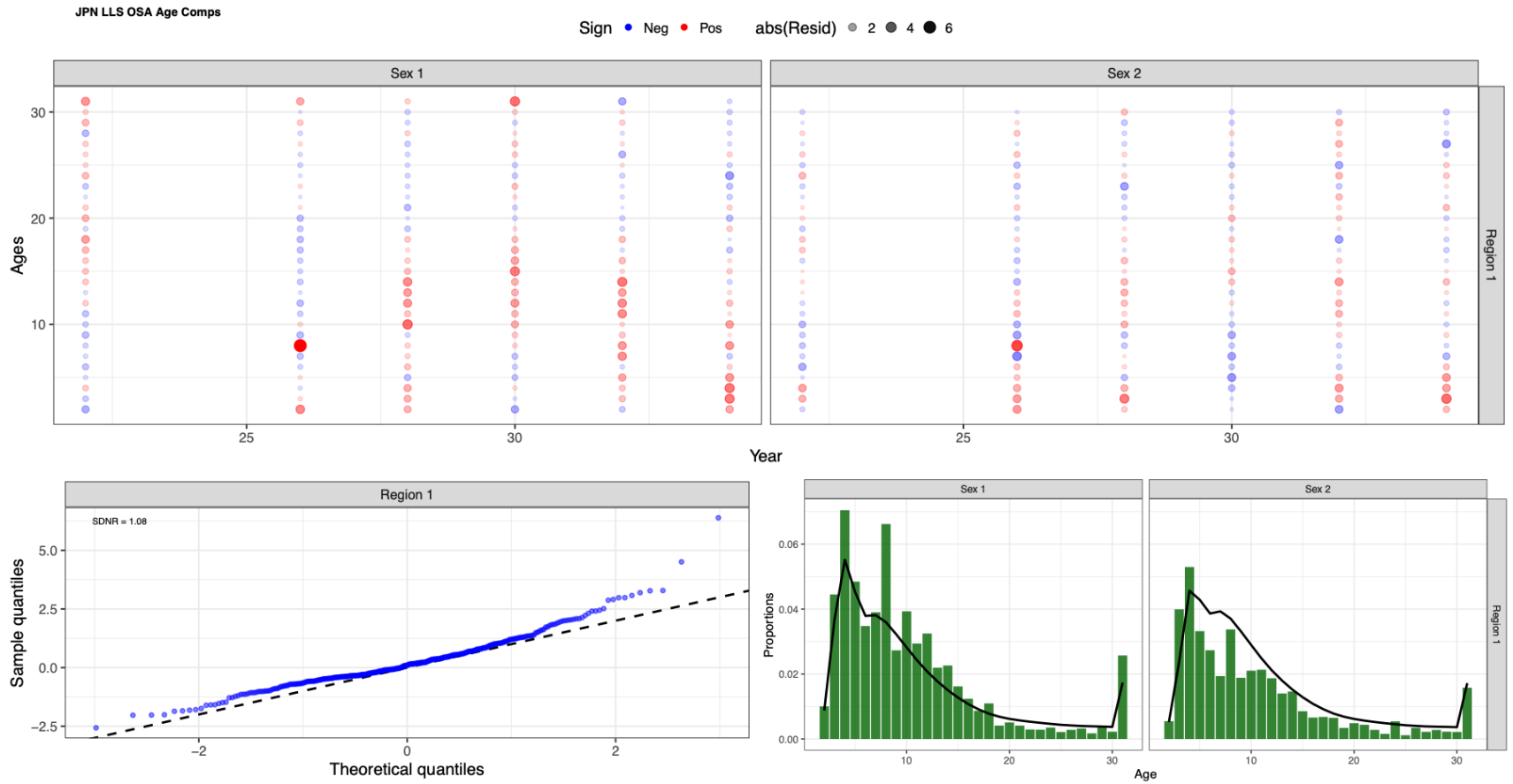

Figure 3C.33. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

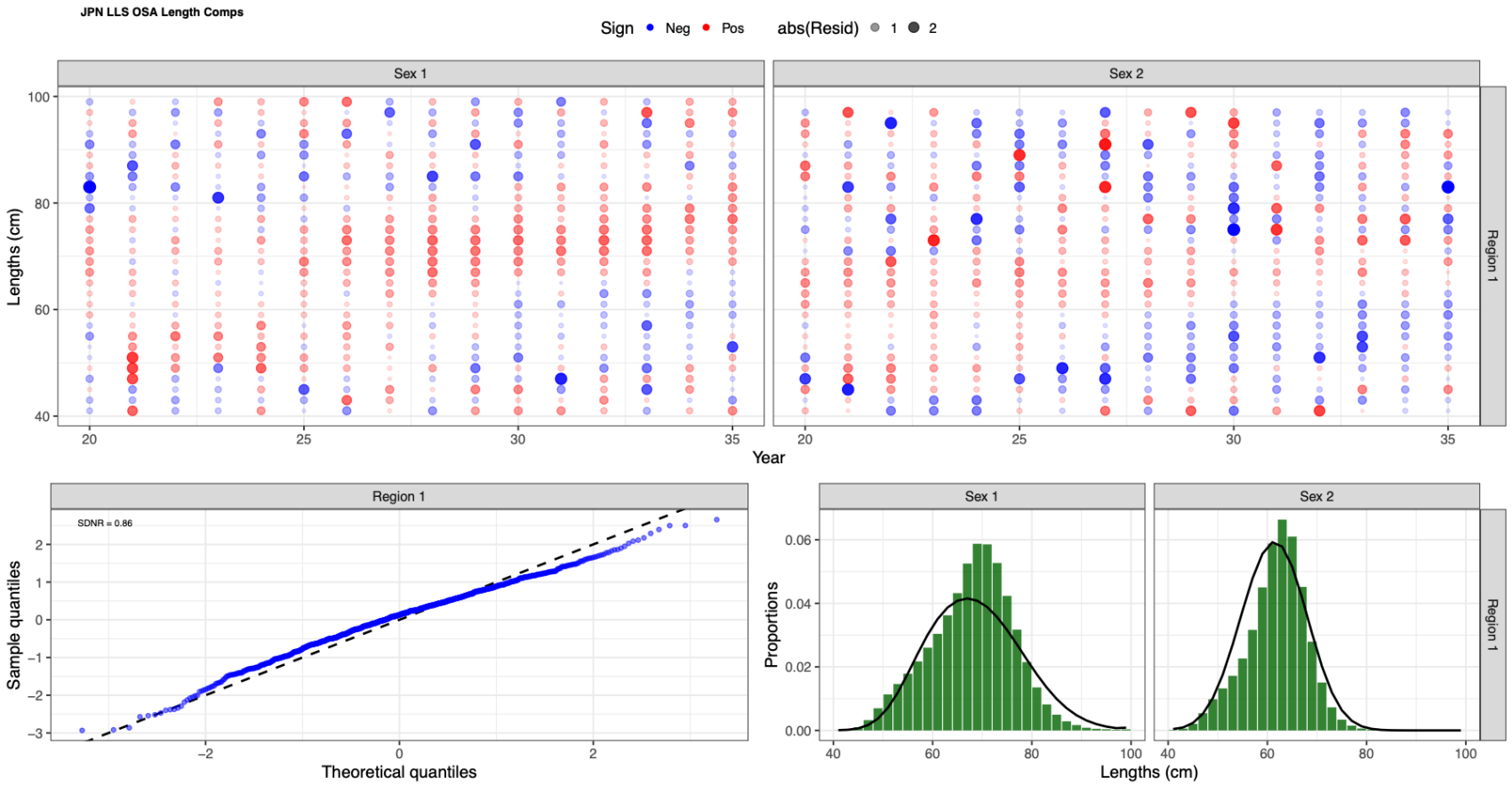

Figure 3C.34. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

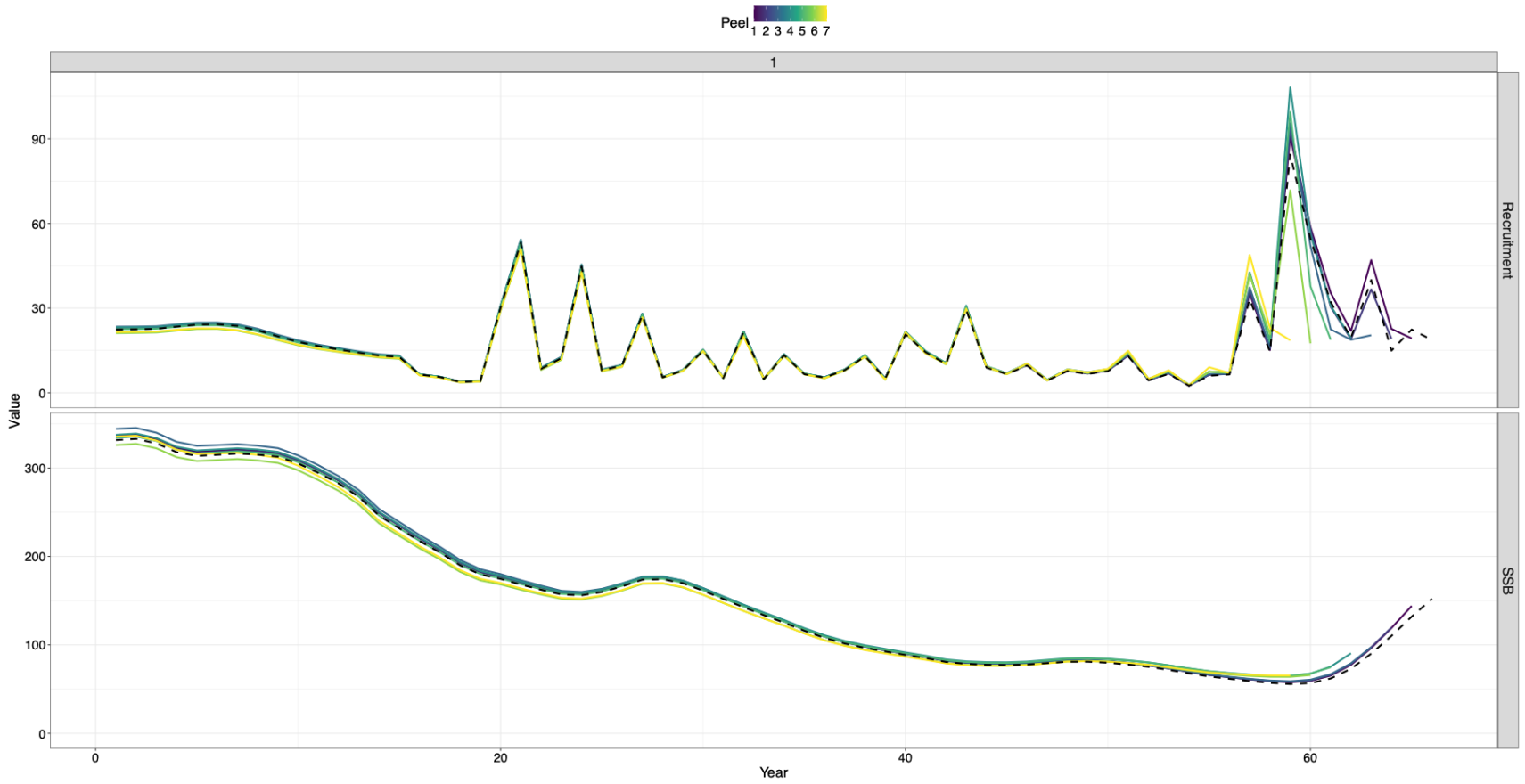

Figure 3C.35. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA_DM*. Mohn's rho for SSB was 0.147.

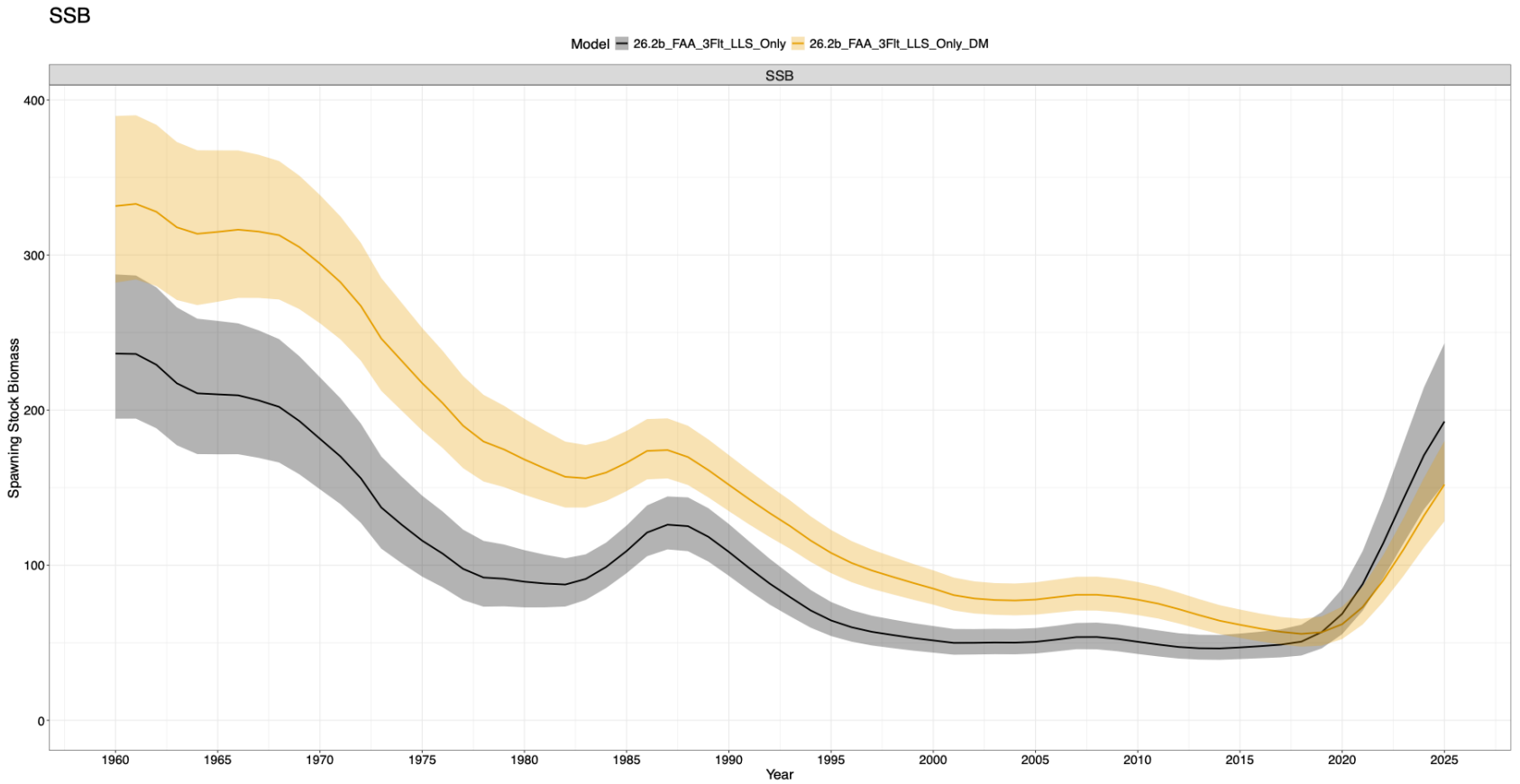

Figure 3C.36. Comparison of spawning stock biomass (SSB, in kilotons) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.2b_FAA_3Flt_LLS_Only_DM*). Shaded regions represent the 95% confidence intervals.

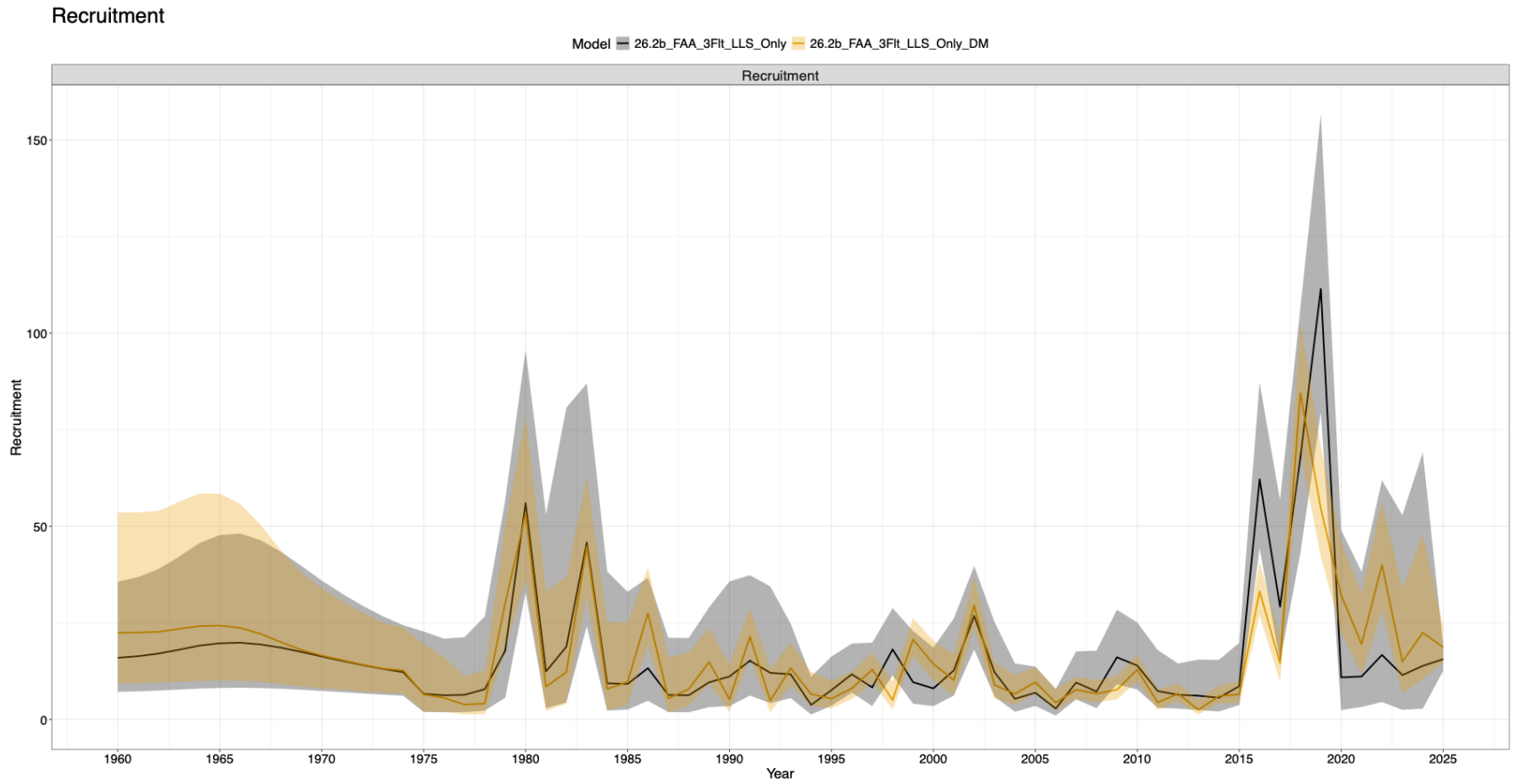

Figure 3C.37. Comparison of age-2 recruitment (millions of individuals) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates the Dirichlet-multinomial likelihood for composition data (*26.2b_FAA_3Flt_LLS_Only_DM*). Shaded regions represent the 95% confidence intervals.

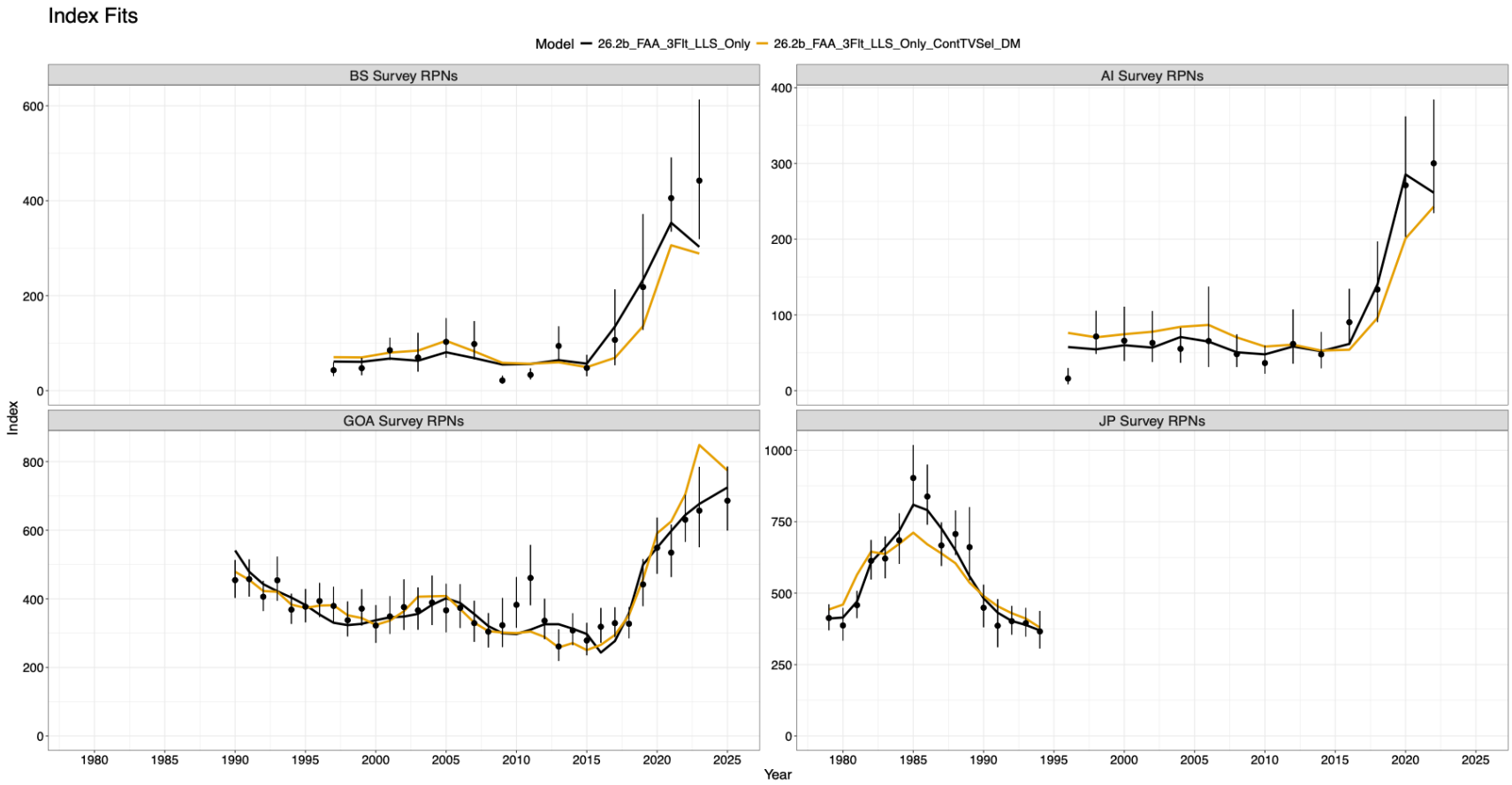

Figure 3C.38. Model fits to survey abundance indices between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous semi-parametric time-varying survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.2b_FAA_3Flt_LLS_Only_ContTVSel_DM*). Point ranges denote observed data and associated 95% confidence intervals. Solid colored lines represent the model expectation.

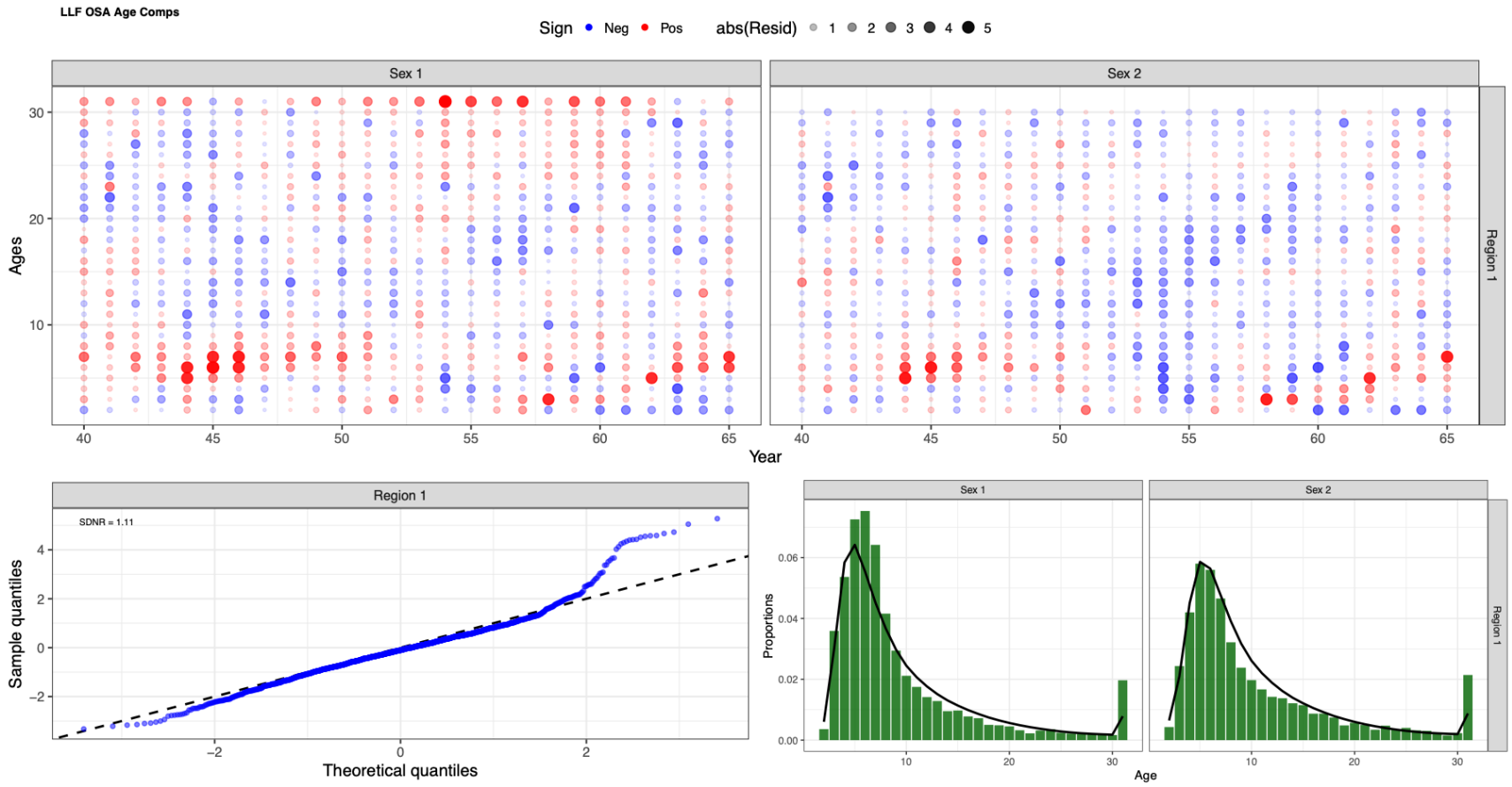

Figure 3C.39. Fits to longline fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

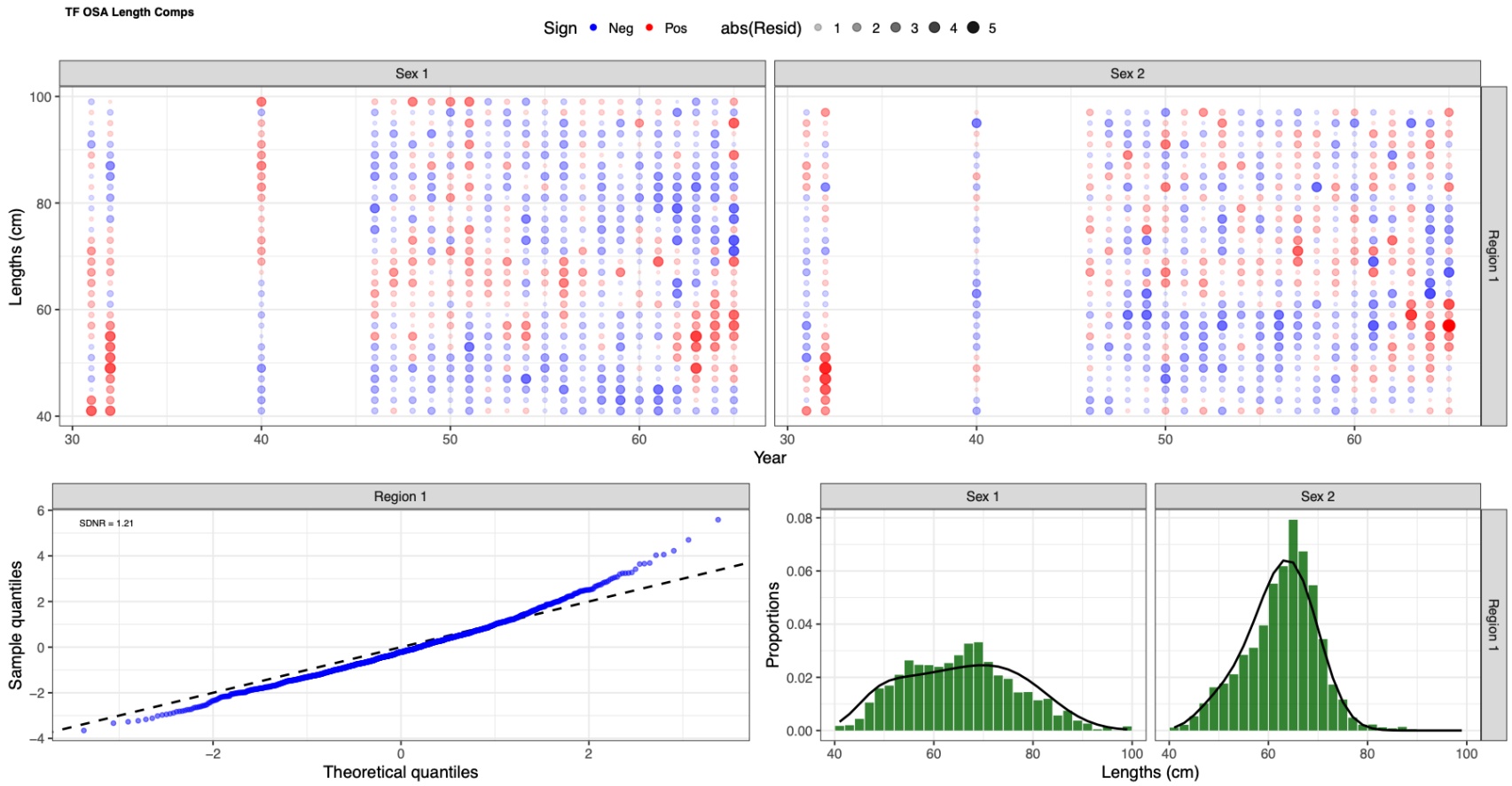

Figure 3C.40. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_DM*.

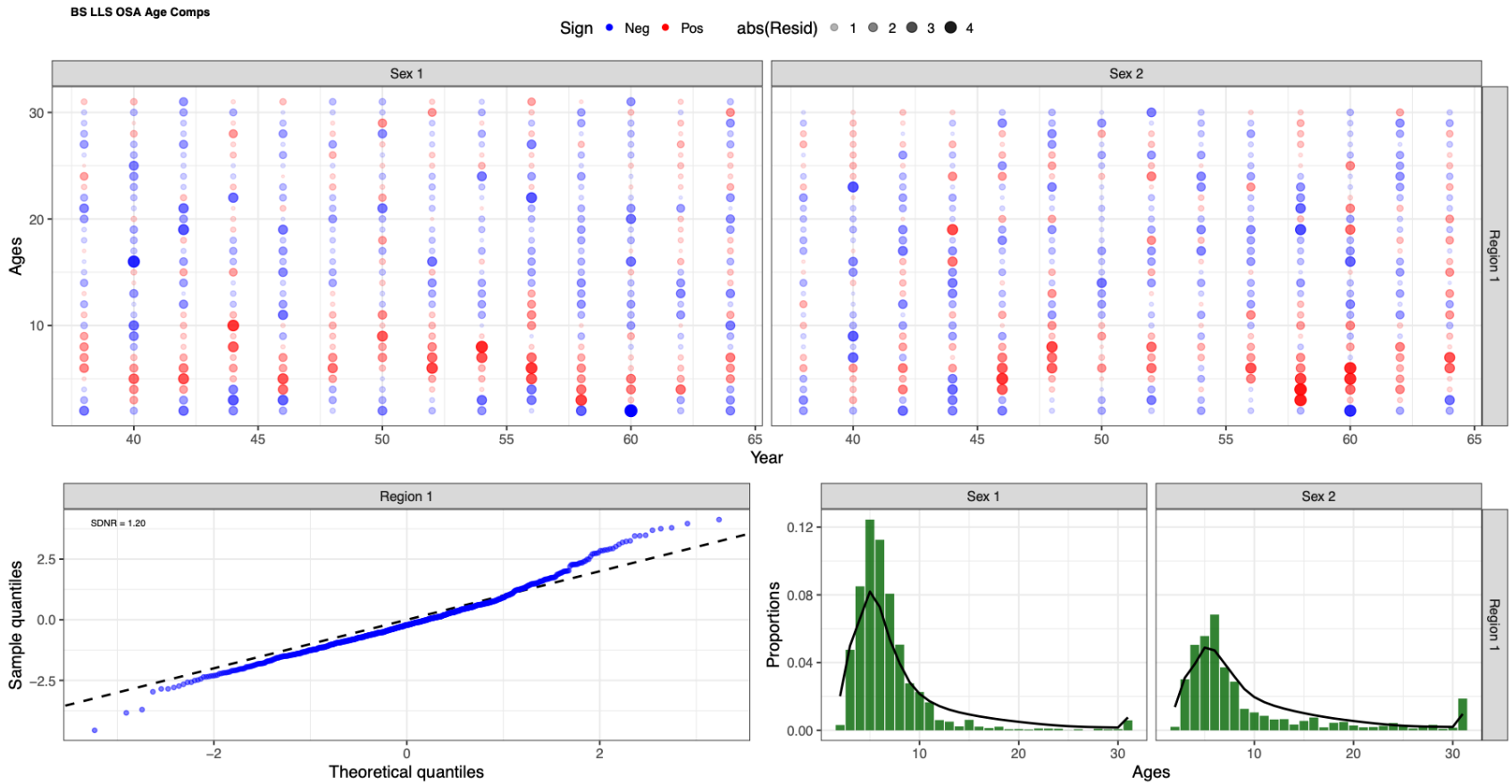

Figure 3C.41. Fits to NOAA longline Bering Sea (BS) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

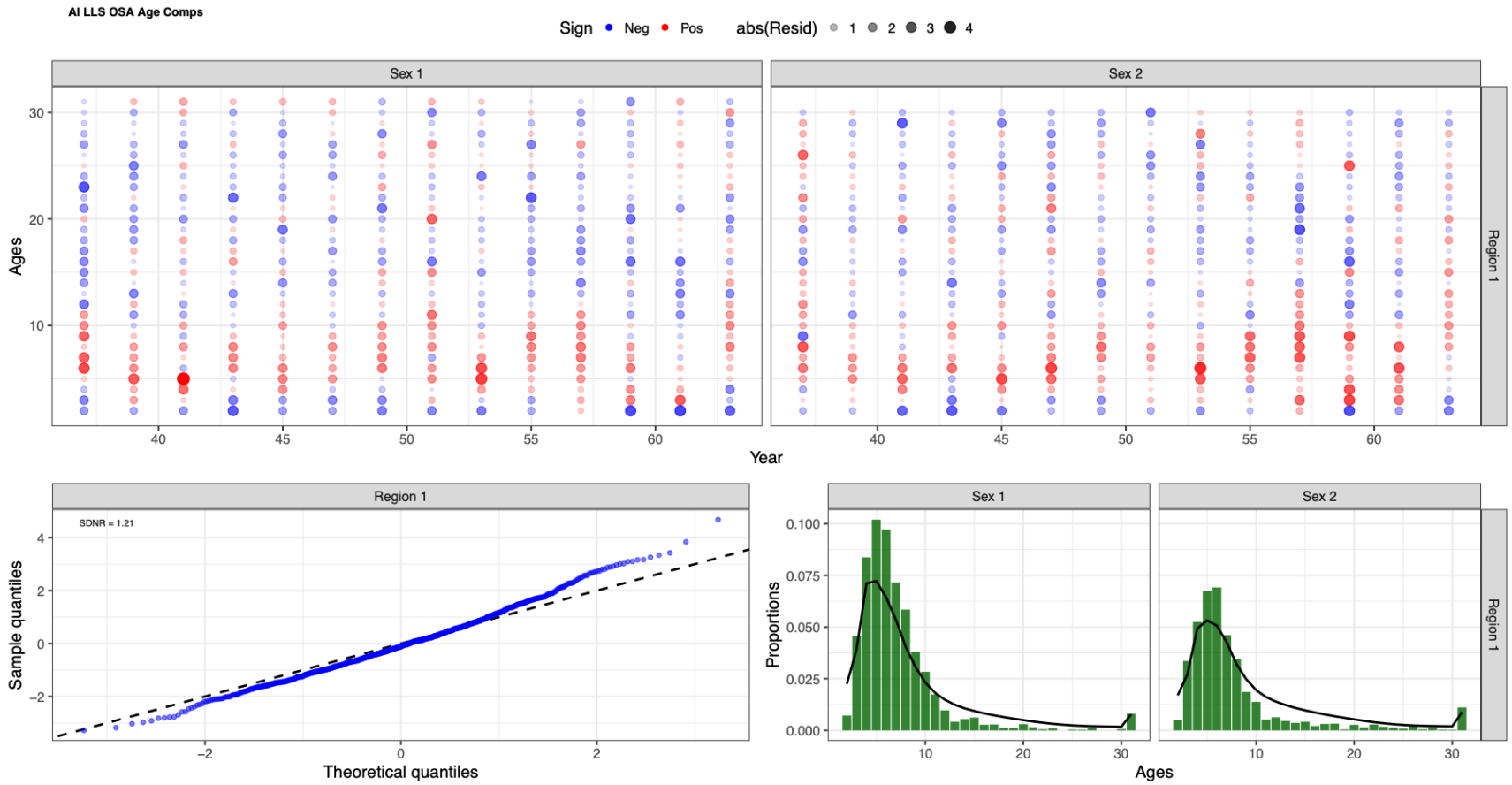

Figure 3C.42. Fits to NOAA longline Aleutian Islands (AI) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

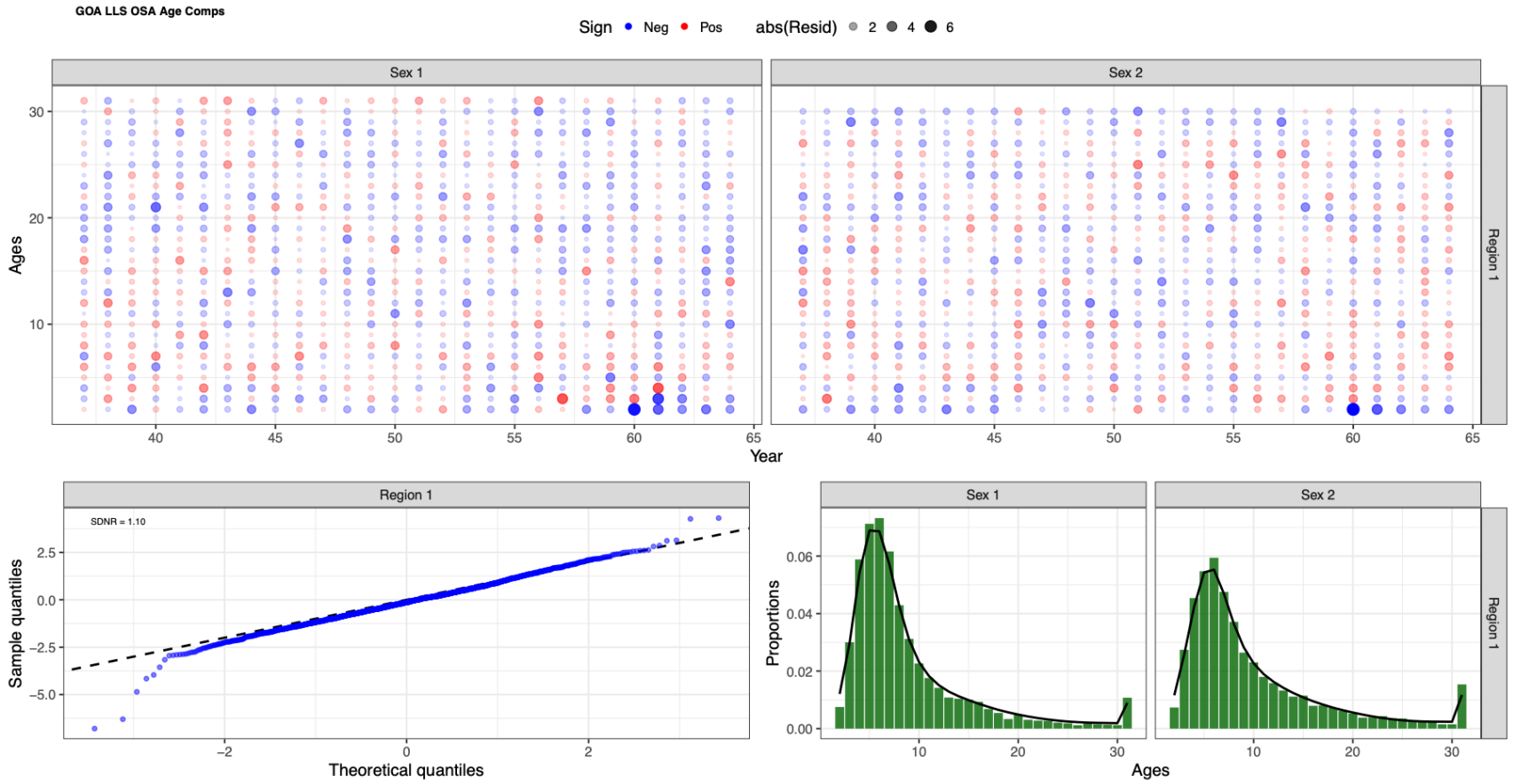

Figure 3C.43. Fits to NOAA longline Gulf of Alaska (GOA) survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

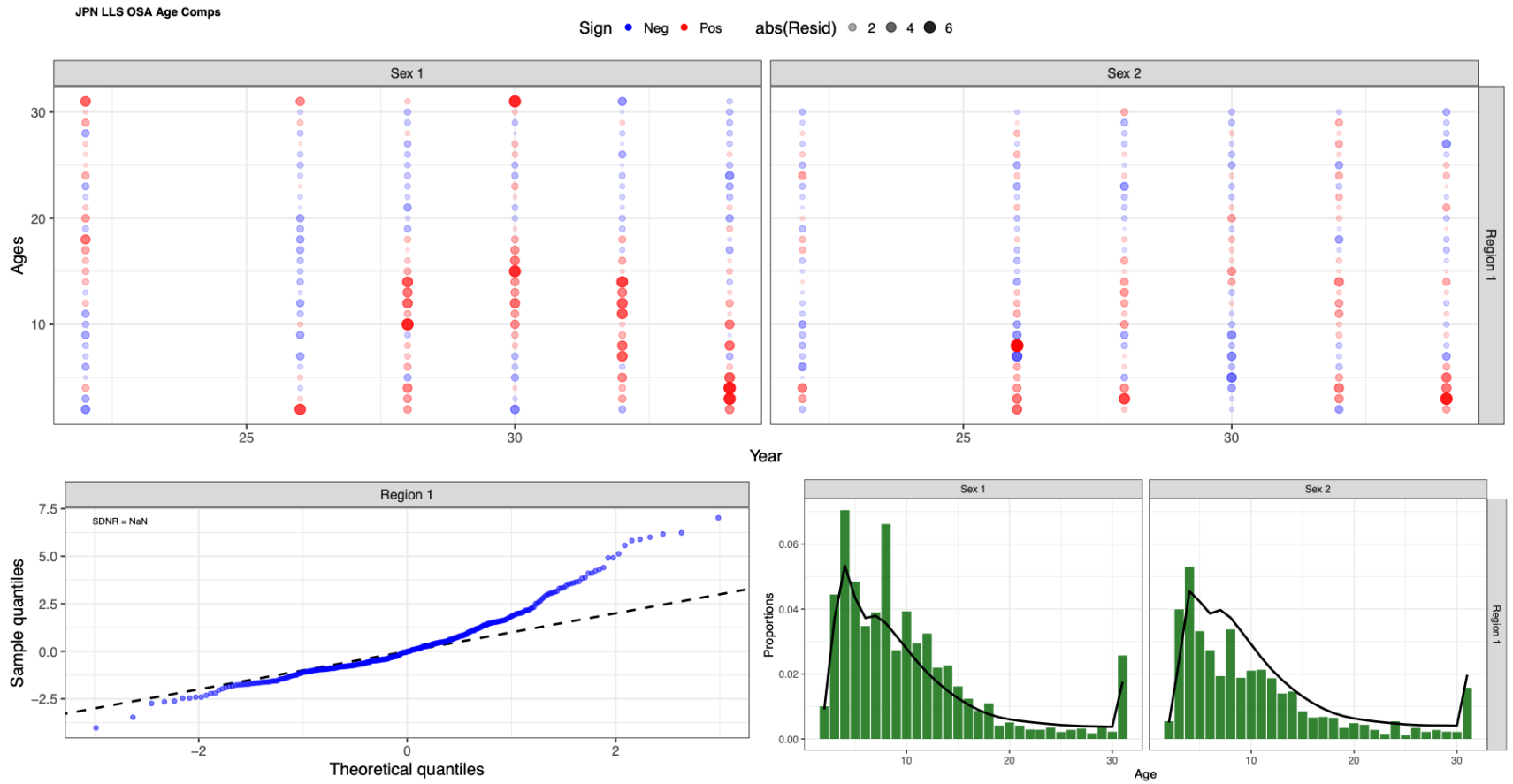

Figure 3C.44. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

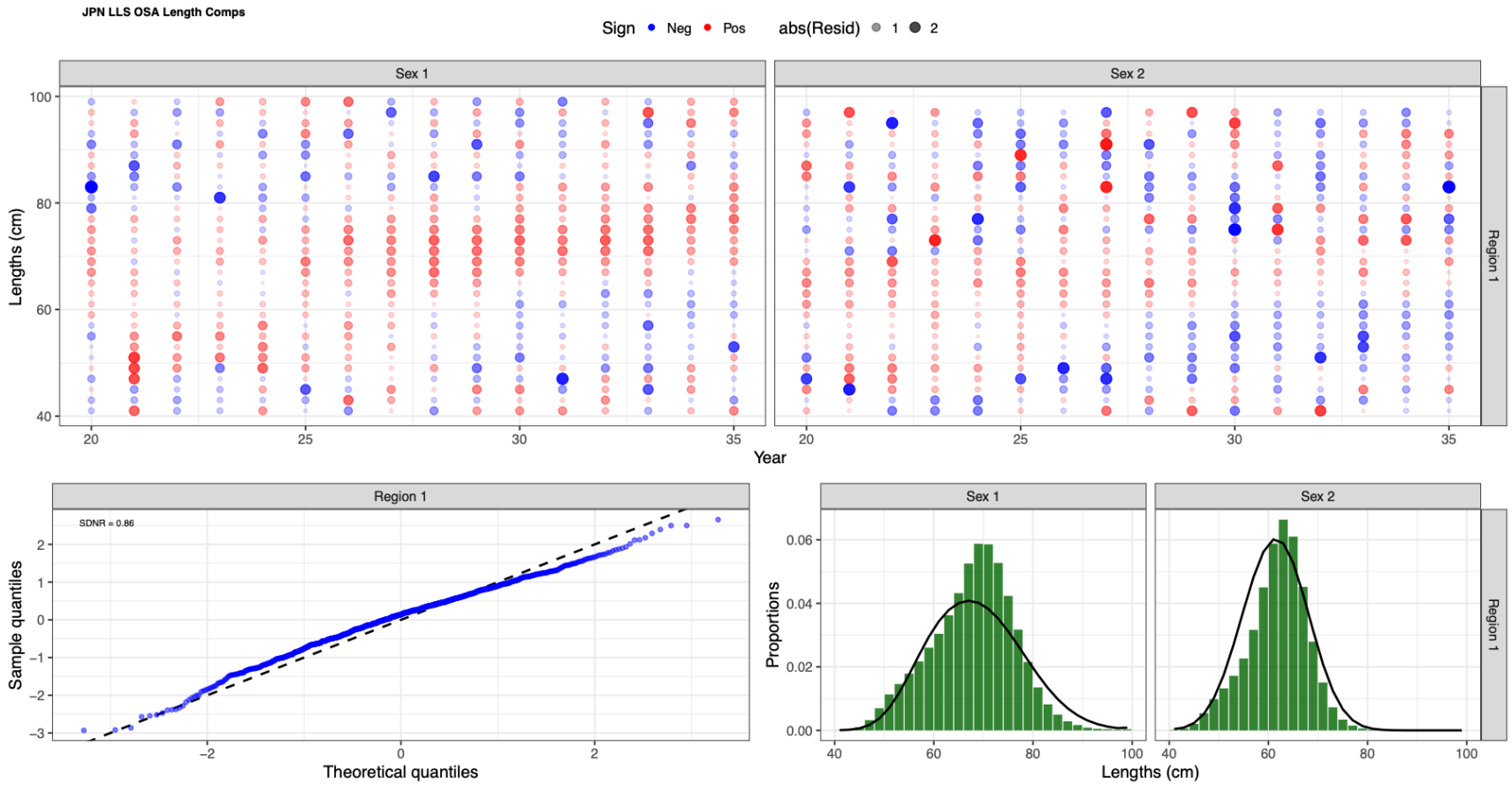

Figure 3C.45. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA_ContTVSel_DM*.

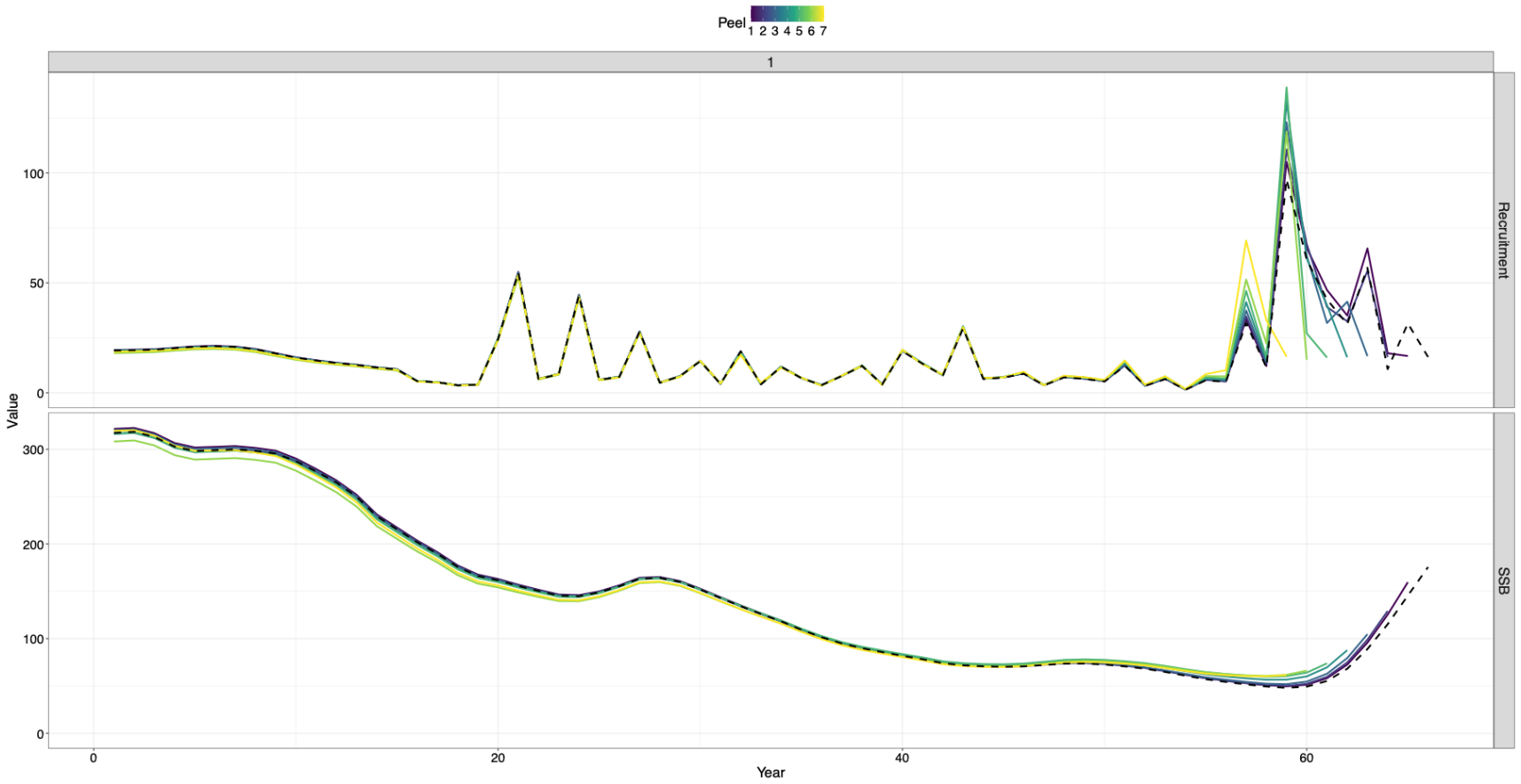

Figure 3C.46. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA_ContTVSel_DM*. Mohn's rho for SSB was 0.234.

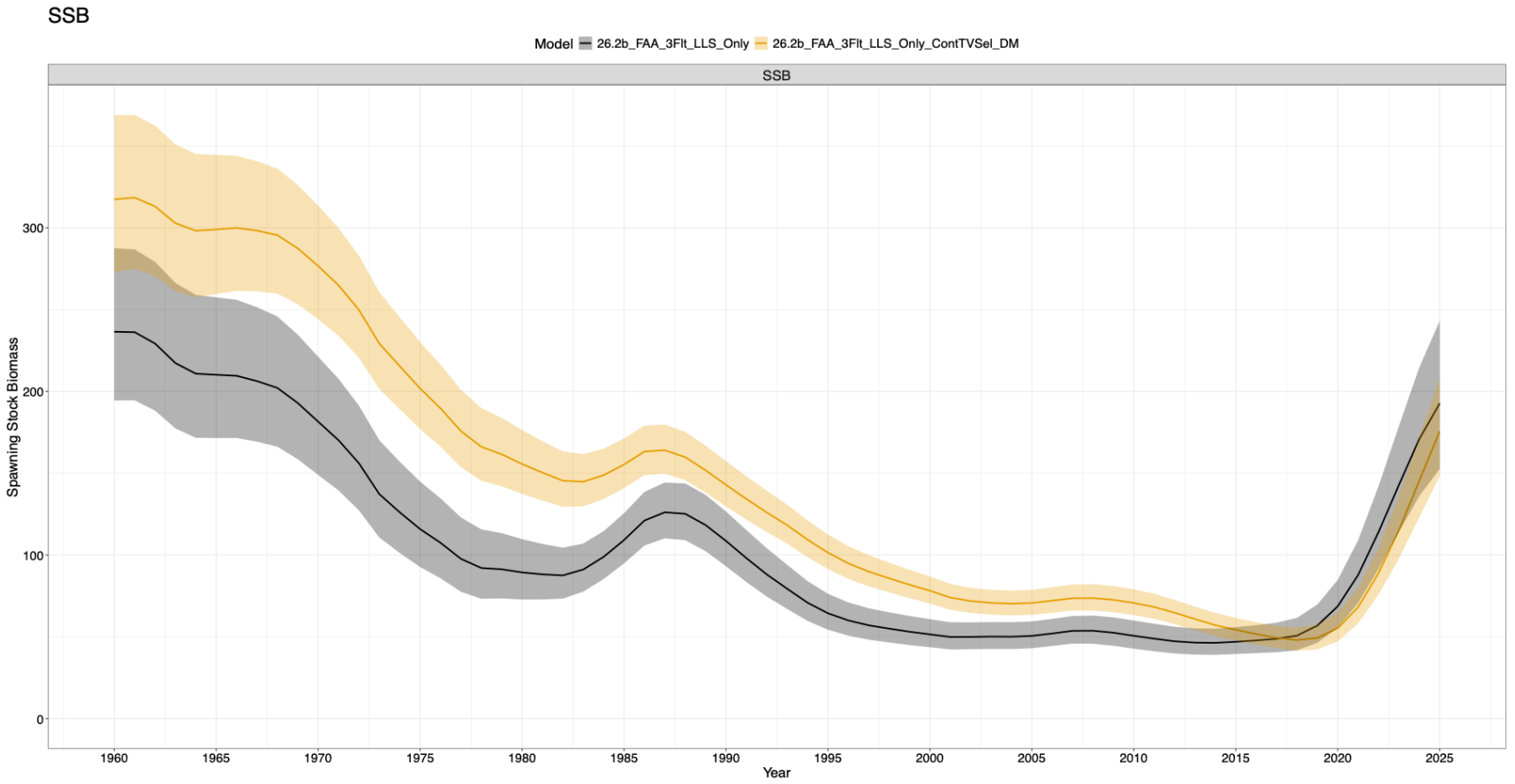

Figure 3C.47. Comparison of spawning stock biomass (SSB, in kilotons) Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.2b_FAA_3Flt_LLS_Only_ContTVSel_DM*). Shaded regions represent the 95% confidence intervals.

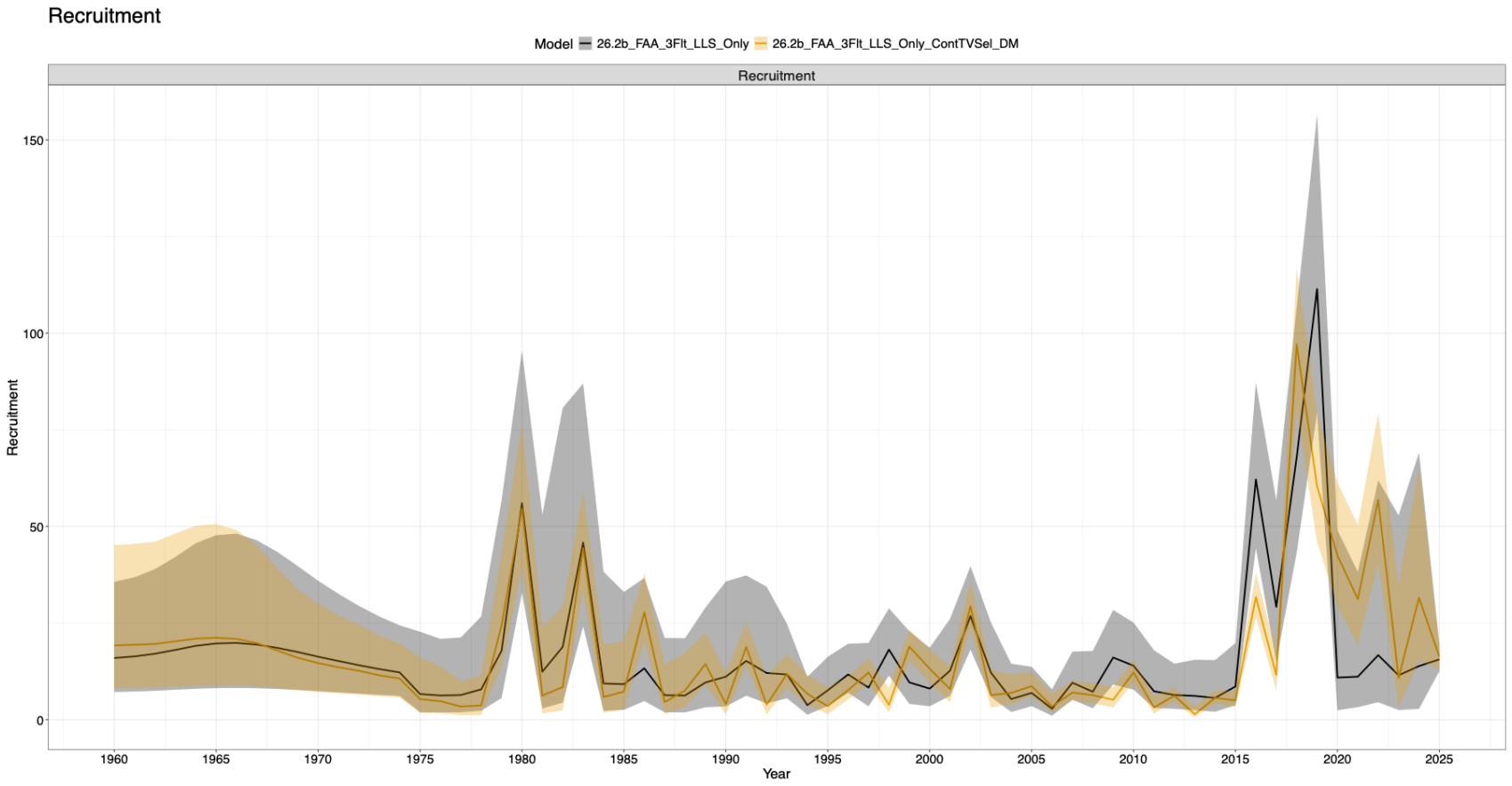

Figure 3C.48. Comparison of age-2 recruitment (millions of fish) between Model *26.2_FAA* (also termed *26.2b_FAA_3Flt_LLS_Only*), along with a configuration that incorporates continuous time-varying semi-parametric survey selectivity and Dirichlet-multinomial likelihoods for composition data (*26.2b_FAA_3Flt_LLS_Only_ContTVSel_DM*). Shaded regions represent the 95% confidence intervals.

3. References

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4. Supplemental Material 3C: Additional Figures

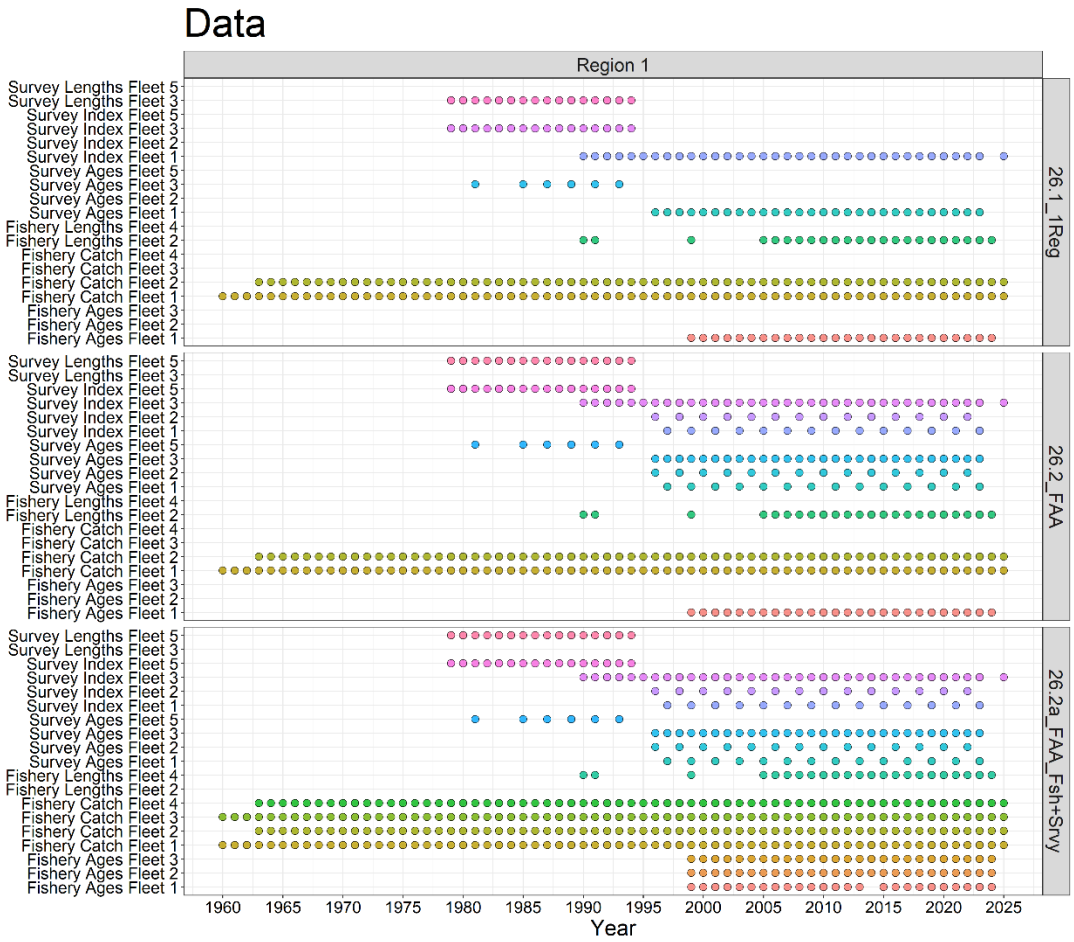

Figure SM3C.1. Data inputs used by each model. Note that fleet numbers vary across models. When fleets are not disaggregated by region then Fishery Fleet 1 is the fixed gear fleet and Fishery Fleet 2 is the trawl fleet, while Survey Fleet 1 is the NOAA Longline Survey and survey Fleet 3 is the Japanese Longline Survey. When fleets are disaggregated by region then Fishery Fleet 1 is the BS fixed gear fleet, Fishery Fleet 2 is the AI fixed gear fleet, Fishery Fleet 3 is the GOA fixed gear fleet, and Fishery Fleet 4 is the trawl fleet, while Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey.

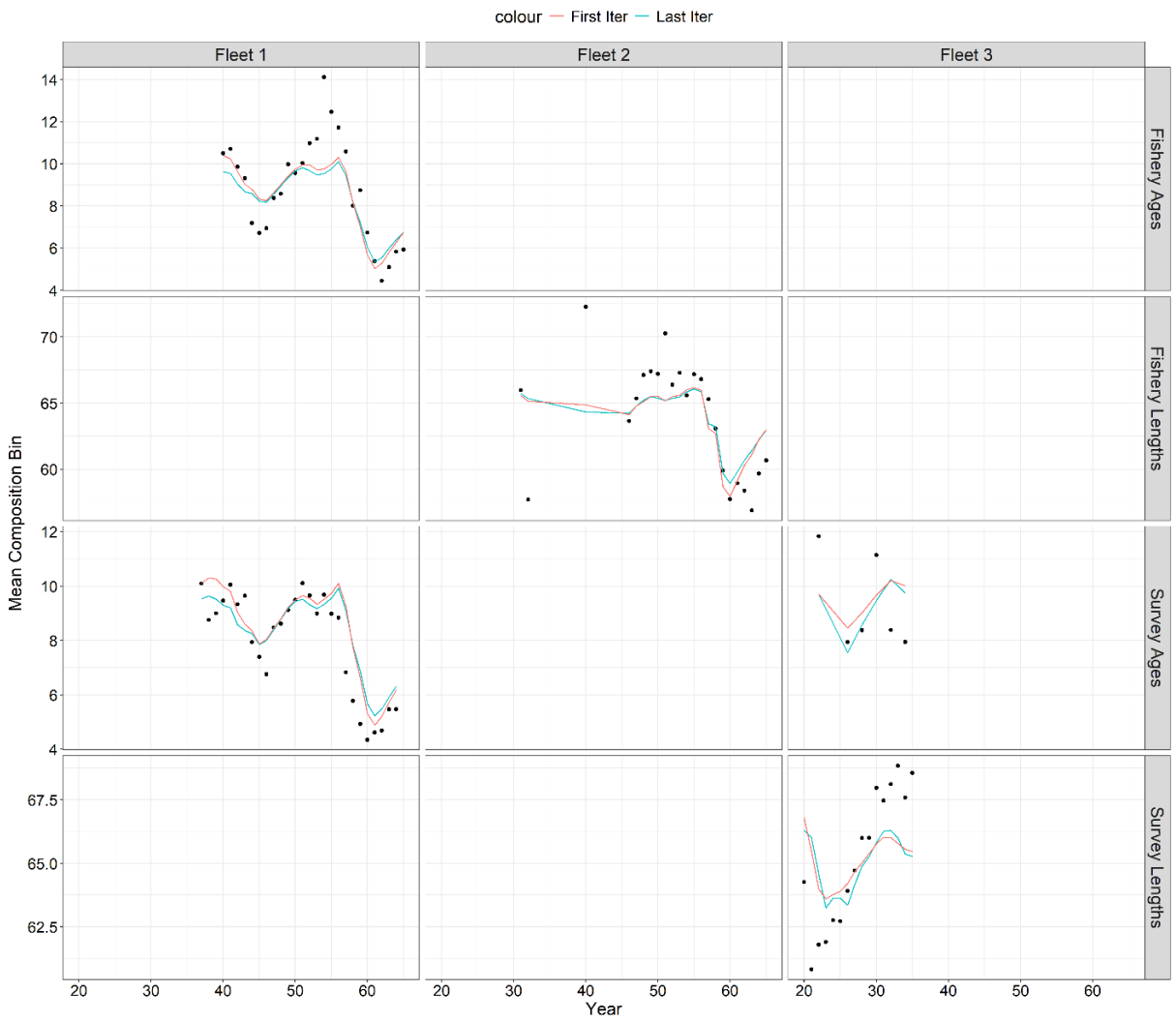

Figure SM3C.2. Fits to the mean age or mean length by year for each compositional data source at the start (using equal data weights, red line) and end (using final data weights, blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.1_1Reg*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey fleet 1 is the NOAA longline survey and fleet 3 is the Japanese longline survey. Year 1 is 1960.

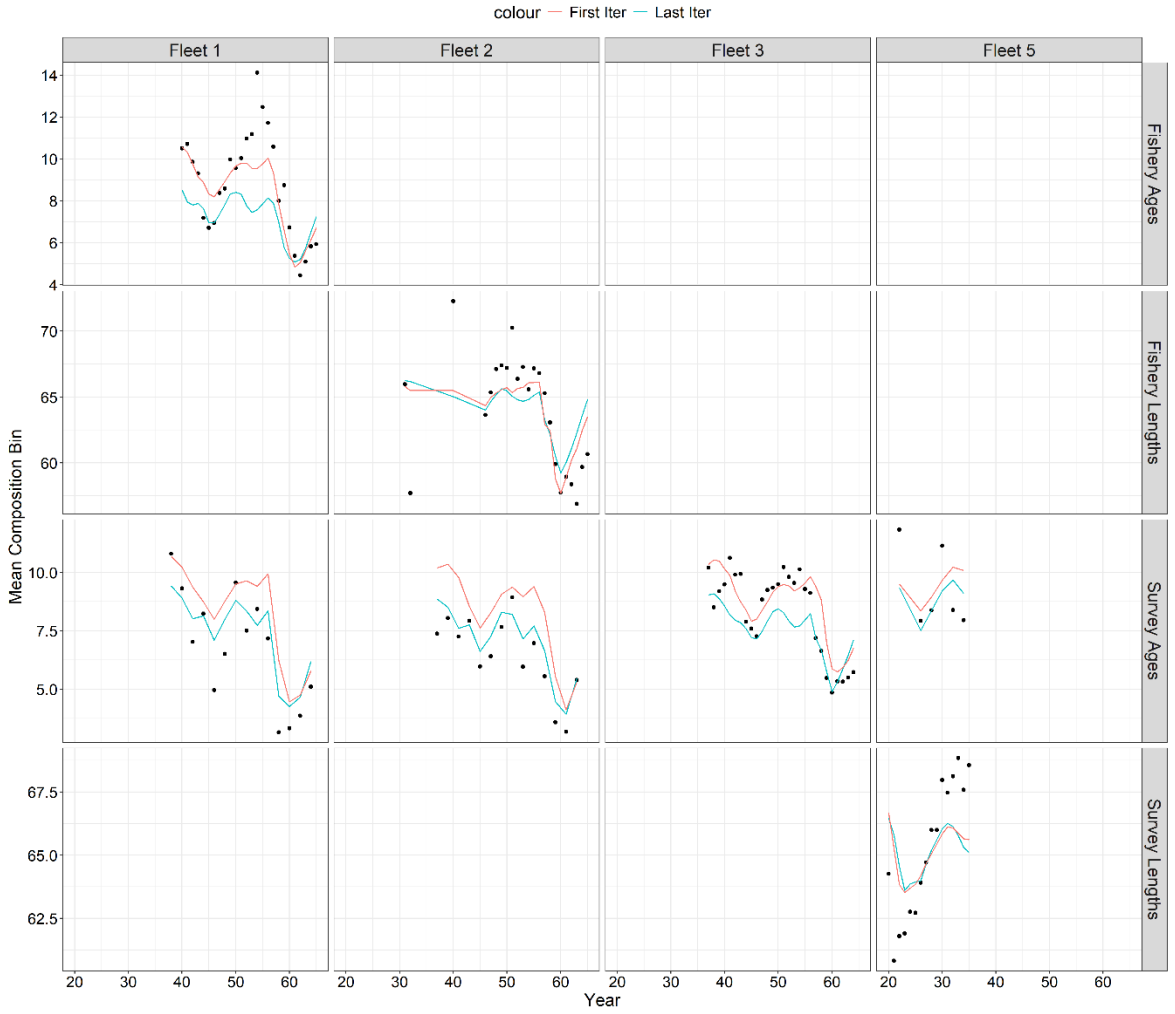

Figure SM3C.3. Fits to the mean age or mean length by year for each compositional data source at the start (using equal data weights, red line) and end (using final data weights, blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.2_FAA*. Fishery fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery. Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey. Year 1 is 1960.

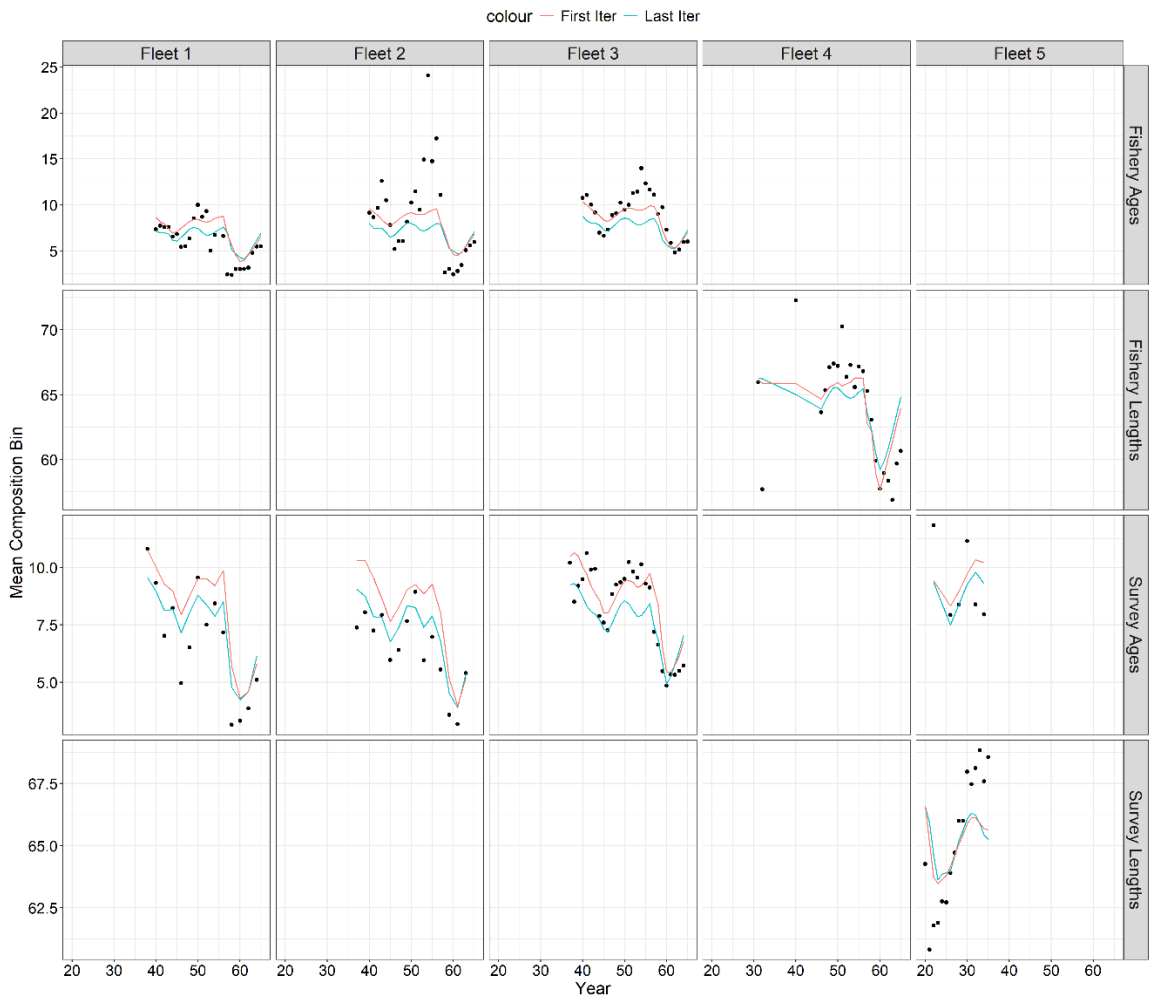

Figure SM3C.4. Fits to the mean age or mean length by year for each compositional data source at the start (using equal data weights, red line) and end (using final data weights, blue lines) of the Francis reweighting method (Francis, 2011) for Model *26.2a_FAA_Fsh+Srvy*. Fishery Fleet 1 is the BS fixed gear fleet, Fishery Fleet 2 is the AI fixed gear fleet, Fishery Fleet 3 is the GOA fixed gear fleet, and Fishery Fleet 4 is the trawl fleet. Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey. Year 1 is 1960.

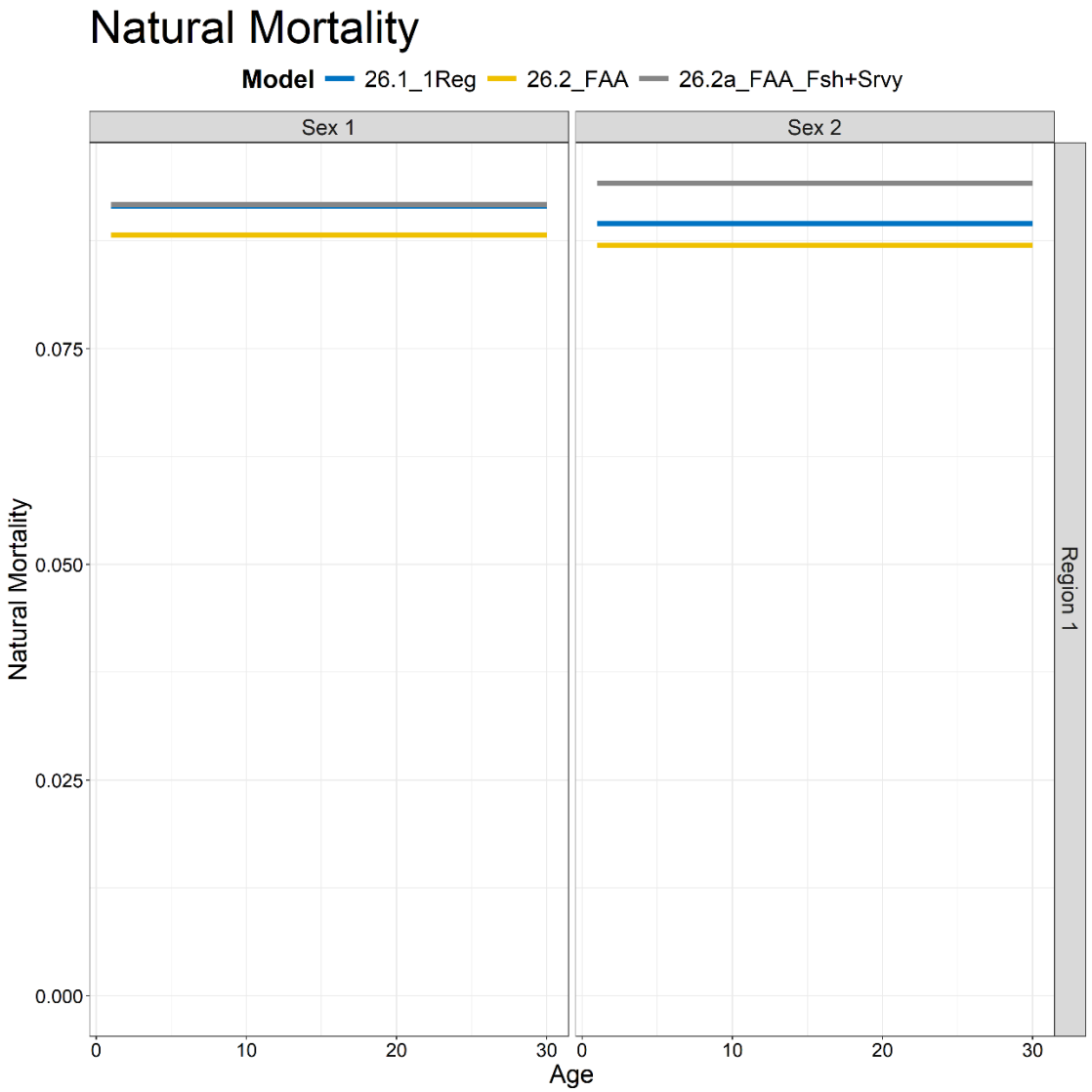

Figure SM3C.5. Natural mortality estimates from Model *25.12_1Reg_Cont* (black line) and Model *26.1_1Reg* (orange line). Sex 1 is females and sex 2 is males.

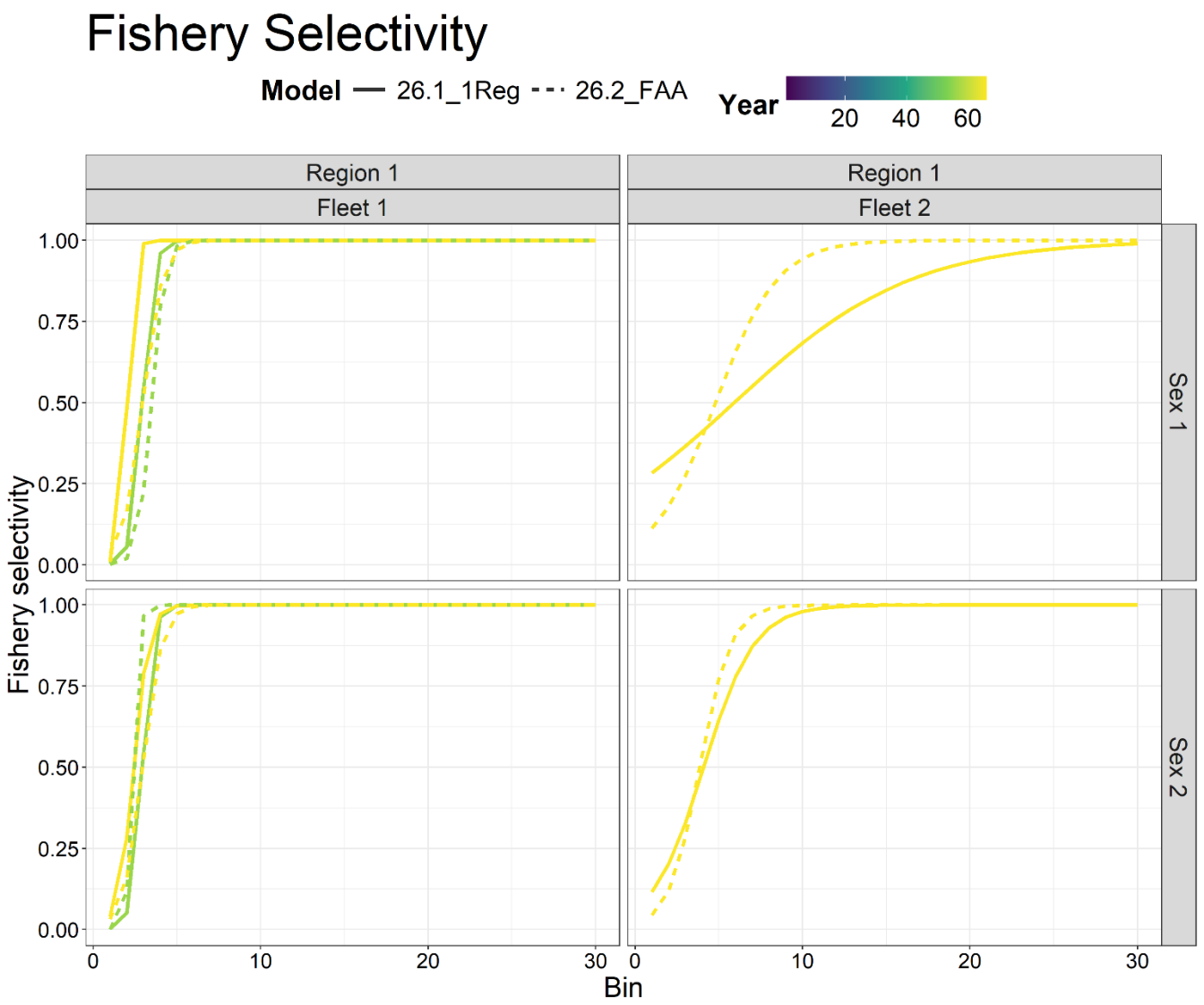

Figure SM3C.6. Fishery selectivity estimates from Model *26.1_1Reg* (solid line) and Model *26.2_FAA* (dashed line) where different colors represent the different time blocks. Fleet 1 is the fixed gear fleet and fleet 2 is the trawl fleet. Sex 1 is females and sex 2 is males.

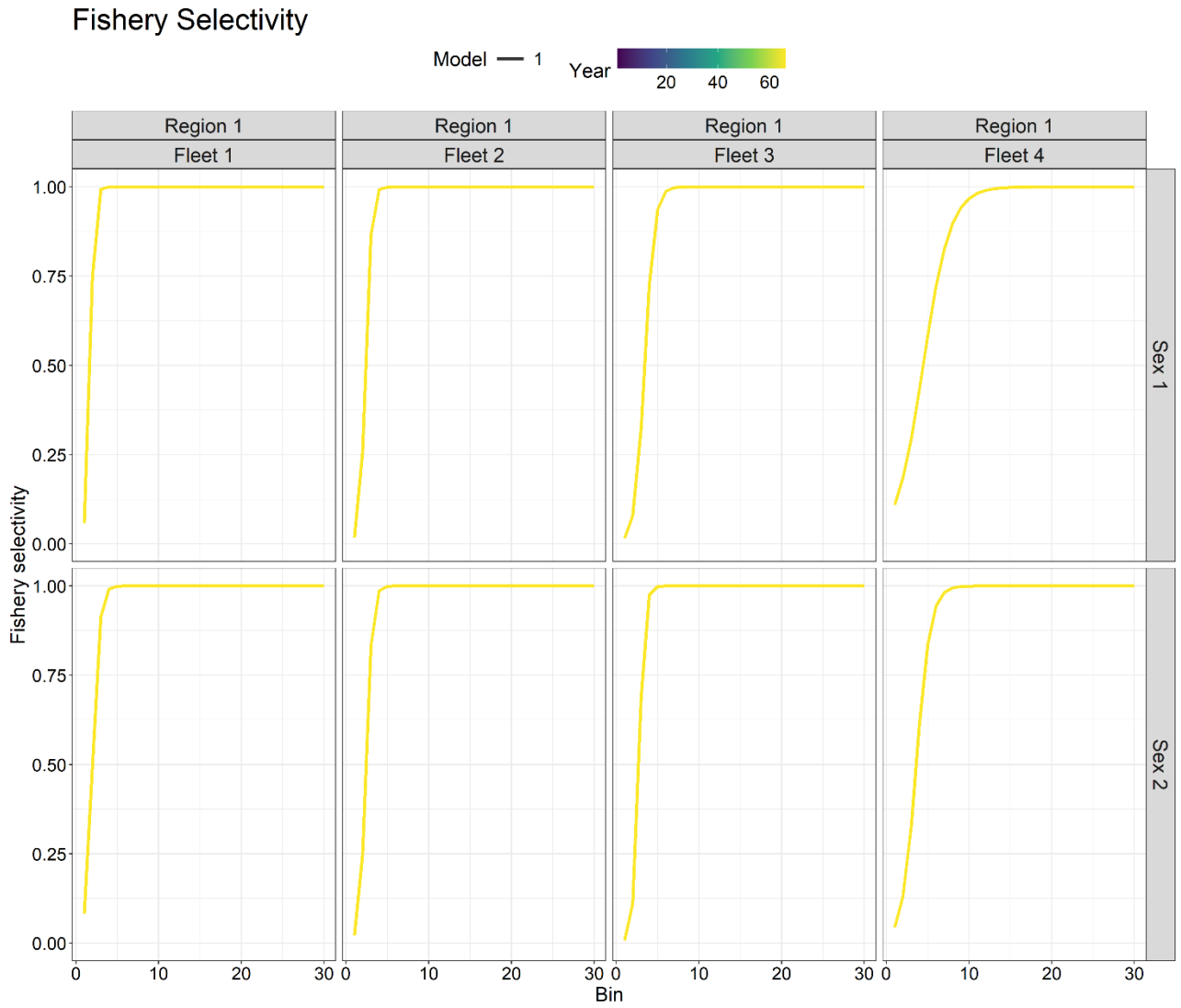

Figure SM3C.7. Fishery selectivity estimates from Model *26.2a_FAA_Fsh+Srvy* where different colors represent the different time blocks. Fishery Fleet 1 is the BS fixed gear fleet, Fishery Fleet 2 is the AI fixed gear fleet, Fishery Fleet 3 is the GOA fixed gear fleet, and Fishery Fleet 4 is the trawl fleet. Sex 1 is females and sex 2 is males.

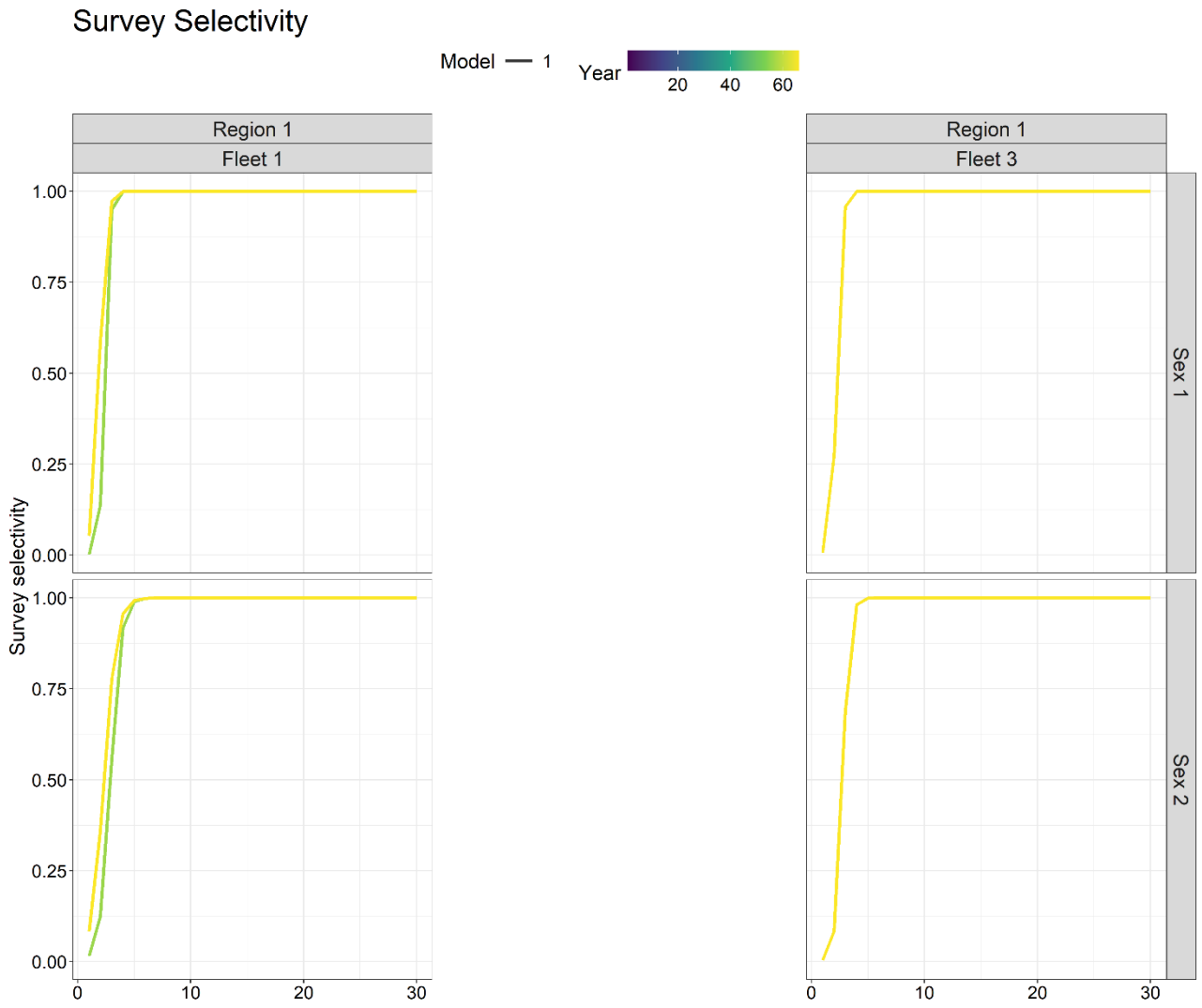

Figure SM3C.8. Survey selectivity estimates from Model *26.1_1Reg* where different colors represent the different time blocks. Fleet 1 is the NOAA Longline Survey and fleet 3 is the Japanese Longline Survey. Sex 1 is females and sex 2 is males.

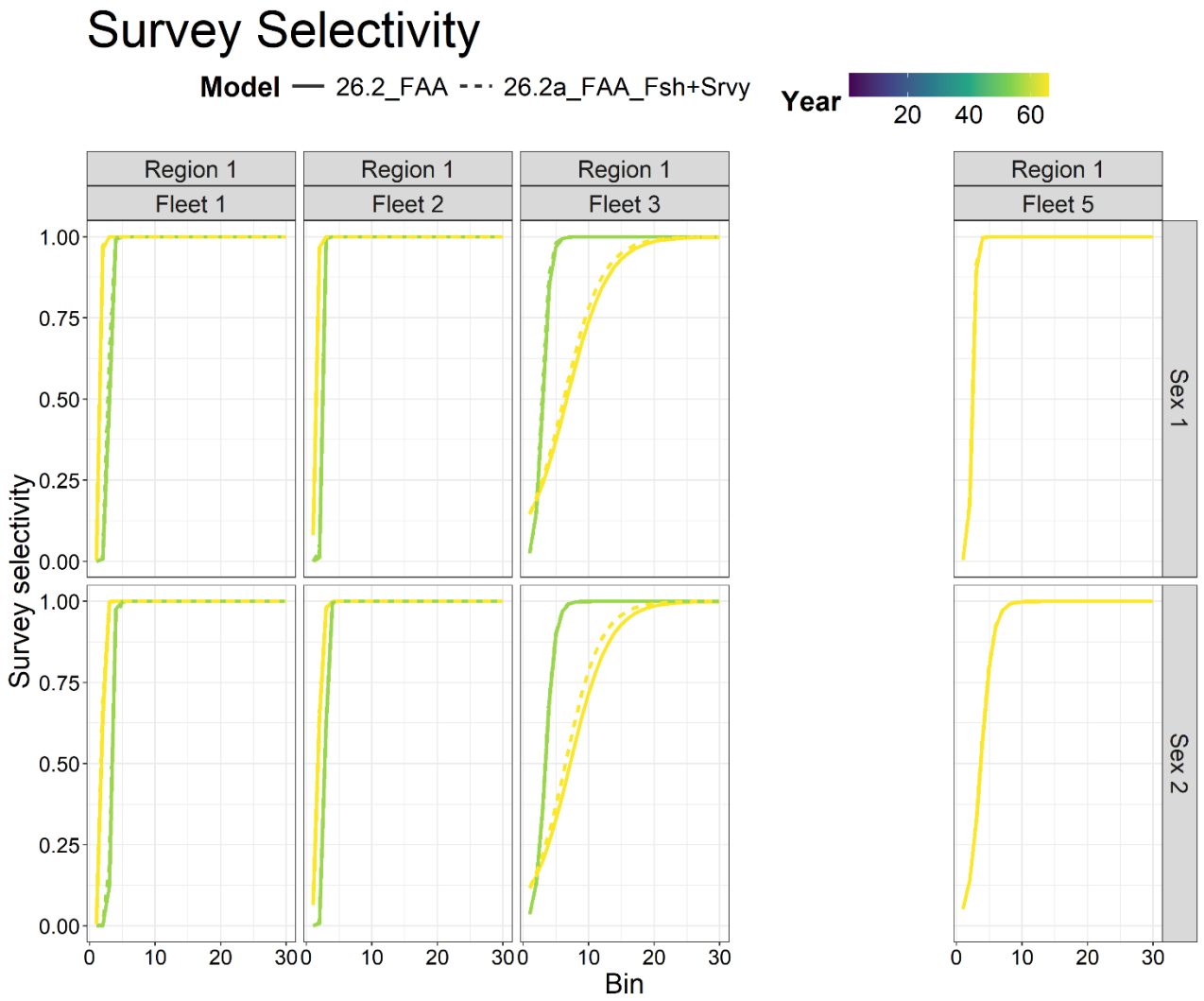

Figure SM3C.9. Survey selectivity estimates from Model *26.2_FAA* (solid lines) and Model *26.2a_FAA_Fsh+Srvy* (dashed lines) where different colors represent the different time blocks. Survey Fleet 1 is the BS NOAA Longline Survey, Survey Fleet 2 is the AI NOAA Longline Survey, Survey Fleet 3 is the GOA NOAA Longline Survey, and Survey Fleet 5 is the Japanese Longline Survey. Sex 1 is females and sex 2 is males.

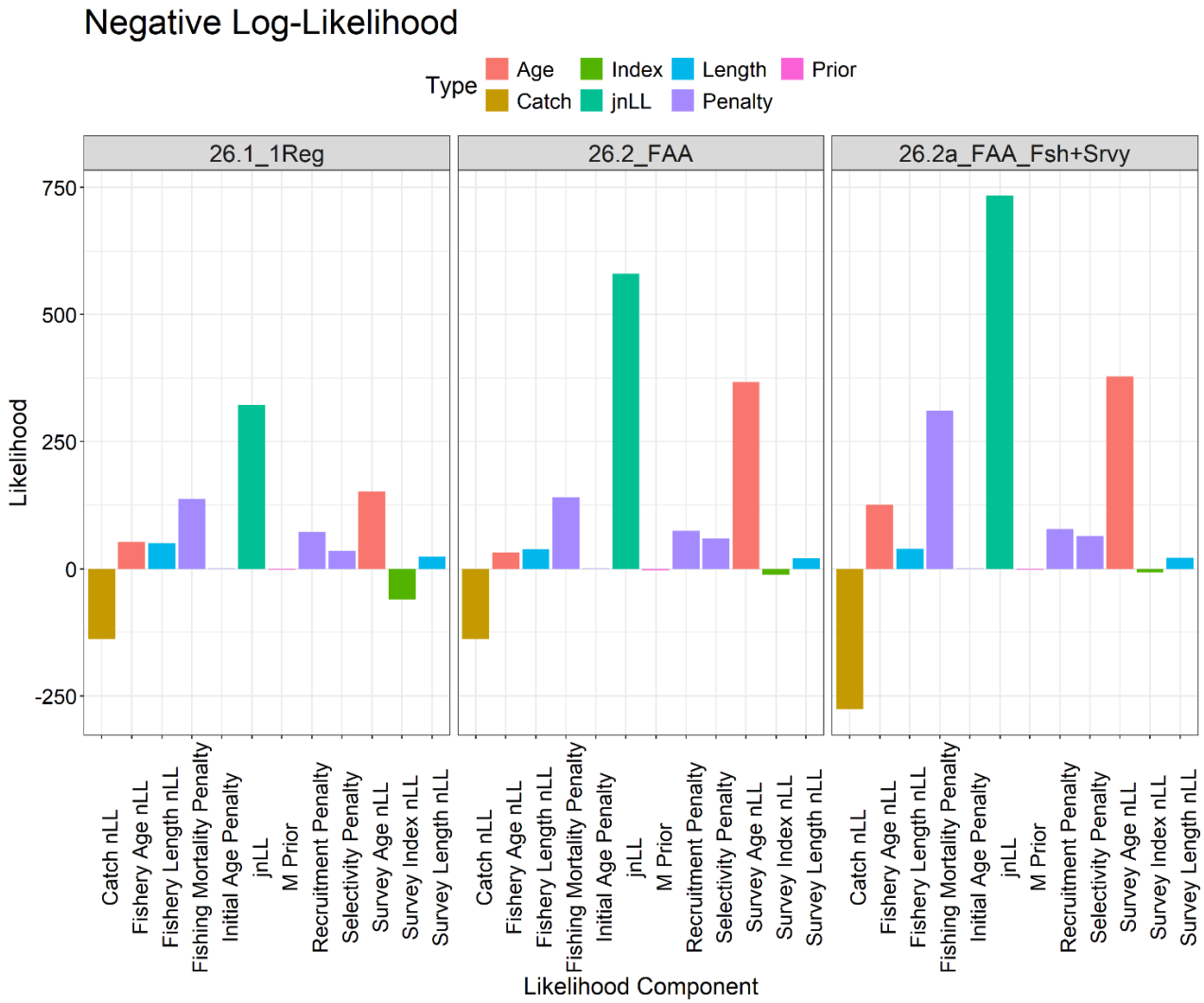

Figure SM3C.10. Joint negative log-likelihood (jnLL) components from Model *26.1_1Reg*, Model *26.2_FAA*, and Model 26.2a_FAA_Fsh+Srvy where different colors represent the various data sets and priors or penalty terms. Note that due to differences in likelihood structures and data inputs (regional disaggregation of fleets) used in each model, these figures are not directly comparable across models.

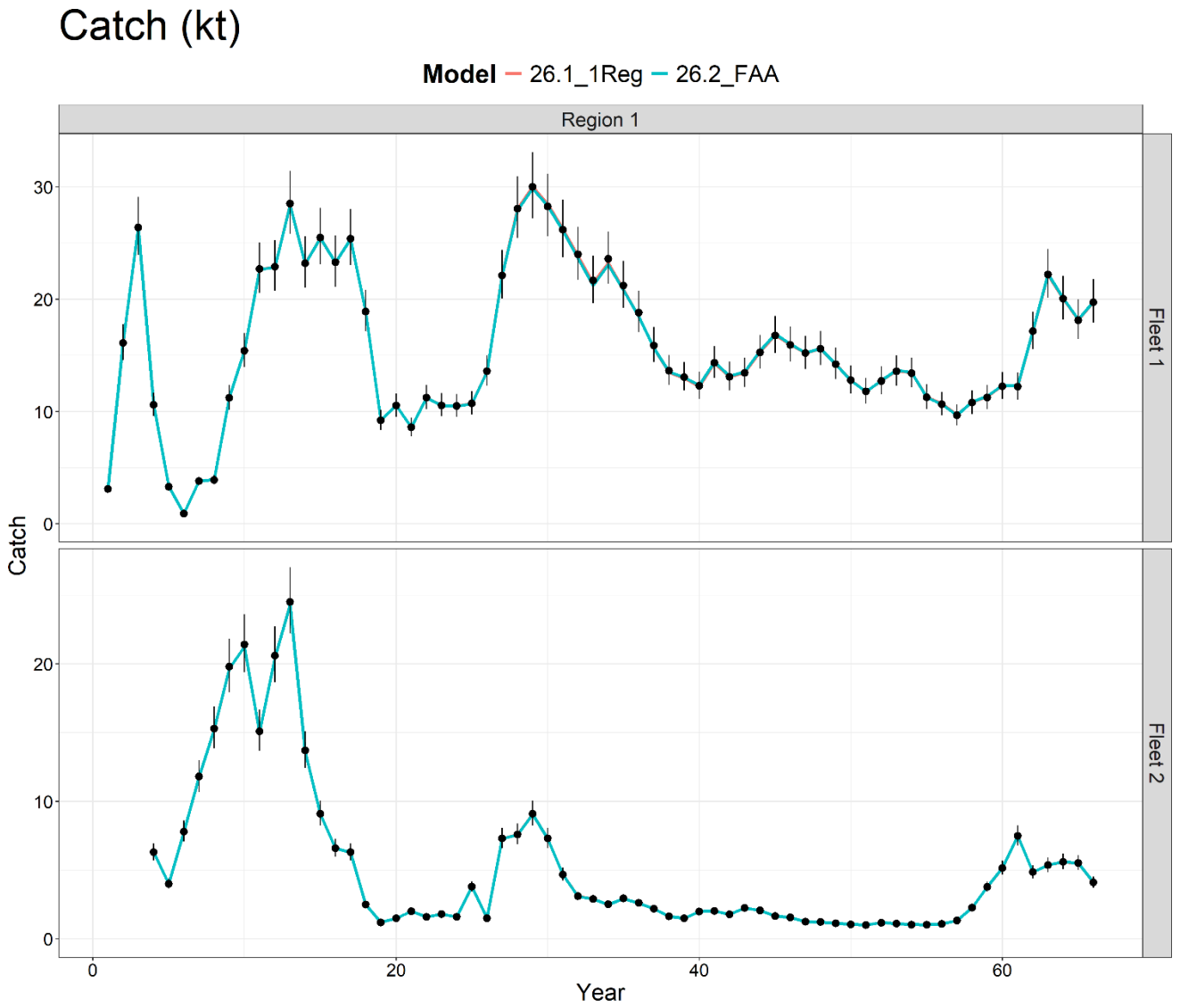

Figure SM3C.11. Fit to the catch data (black dots with 95% confidence intervals) by fleet (fleet 1 is the fixed gear fishery and fleet 2 is the trawl fleet) for Model *26.1_1Reg* (red line) and Model *26.2_FAA* (blue line). Note the varying y-axis scales.

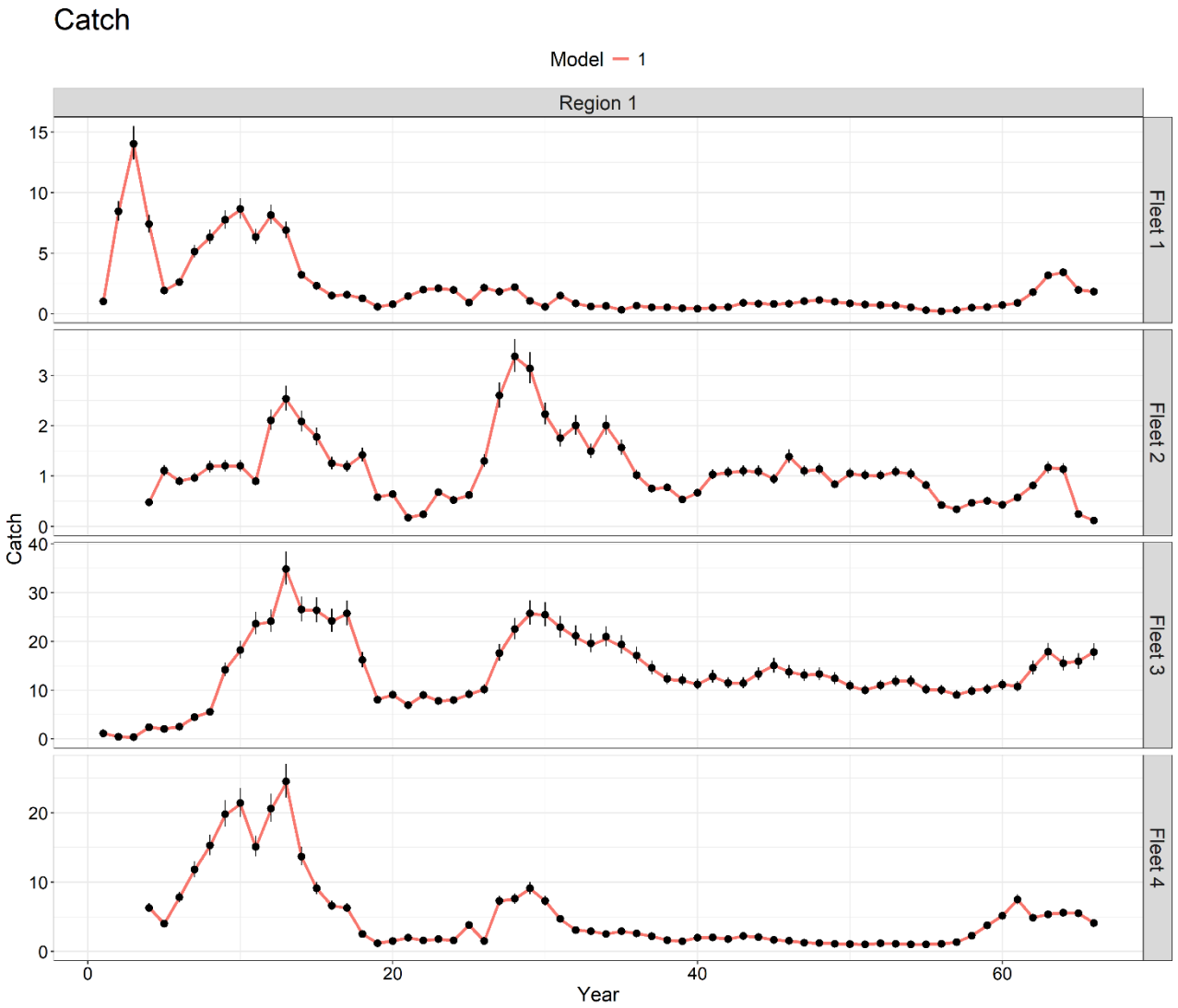

Figure SM3C.12. Fit to the catch data (black dots with 95% confidence intervals) by fleet (Fishery Fleet 1 is the BS fixed gear fleet, Fishery Fleet 2 is the AI fixed gear fleet, Fishery Fleet 3 is the GOA fixed gear fleet, and Fishery Fleet 4 is the trawl fleet) for Model *25.2_FAA_Fsh+Srvy* (red line). Note the varying y-axis scales.

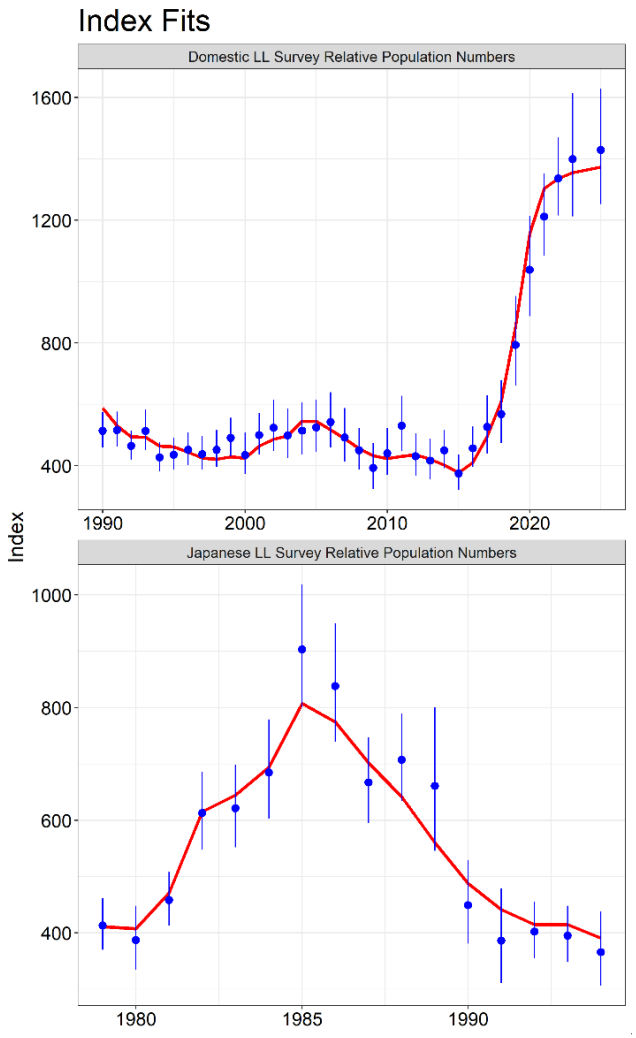

Figure SM3C.13. Model *26.1_1Reg* fit to indices, including the NOAA Longline Survey RPNs (top) and Japanese Longline Survey RPNs (bottom). Red lines are model predicted values, and blue dots are observed values (with 95% confidence intervals based on the assumed standard errors).

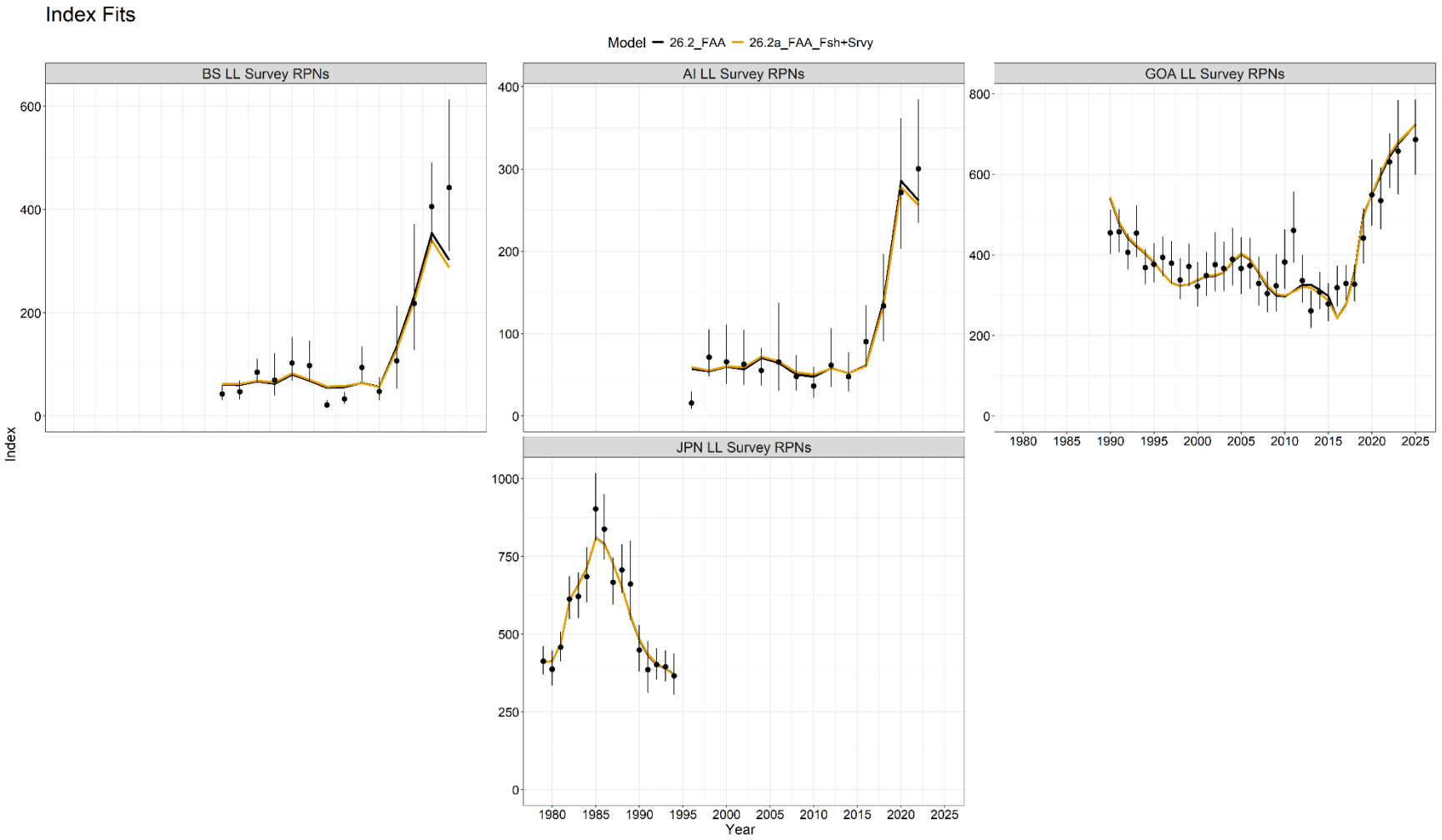

Figure SM3C.14. Model *26.2_FAA* (black line) and Model *26.2a_FAA_Fsh+Srvy* (orange line) fit to indices, including the NOAA Longline Survey RPNs (top panels) and Japanese Longline Survey RPNs (bottom panel). Lines are model predicted values, and black dots are observed values (with 95% confidence intervals based on the assumed standard errors).

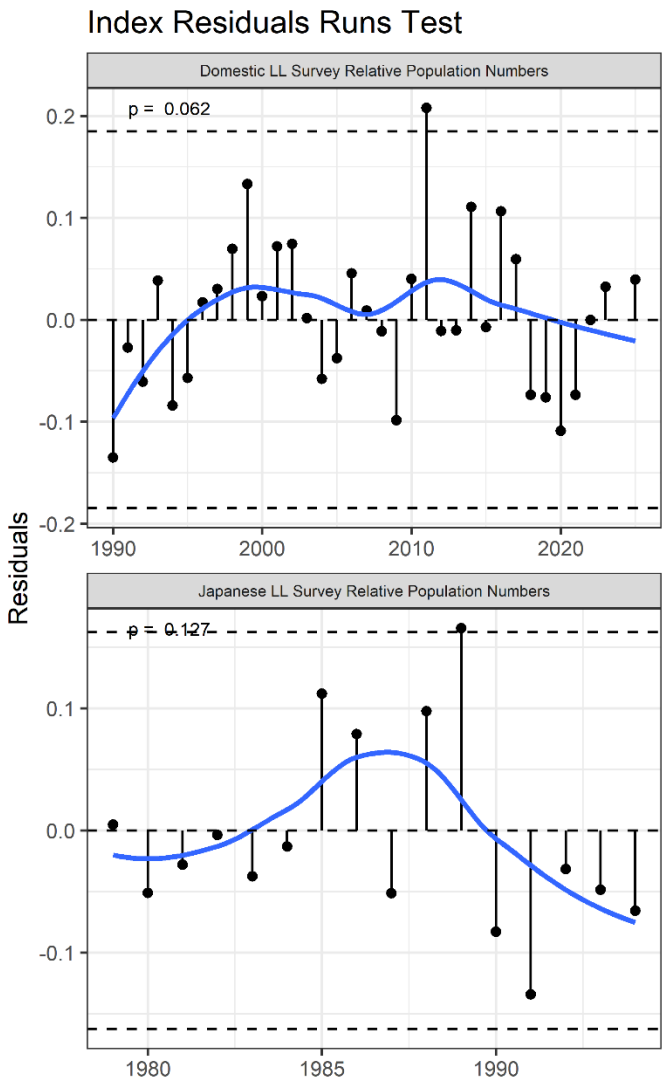

Figure SM3C.15. Results of an index residuals runs test for Model *26.1_1Reg*.

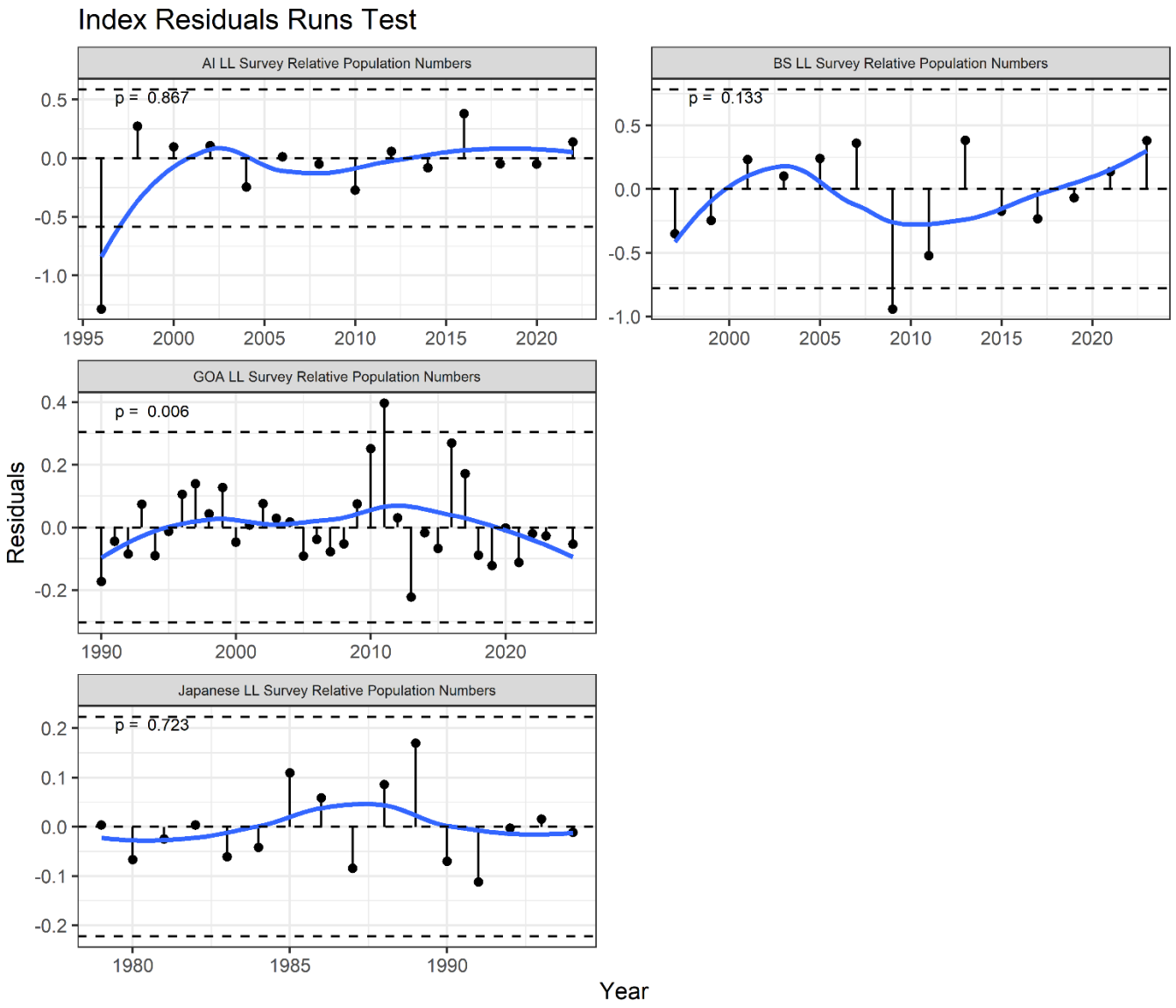

Figure SM3C.16. Results of an index residuals runs test for Model *26.2_FAA*.

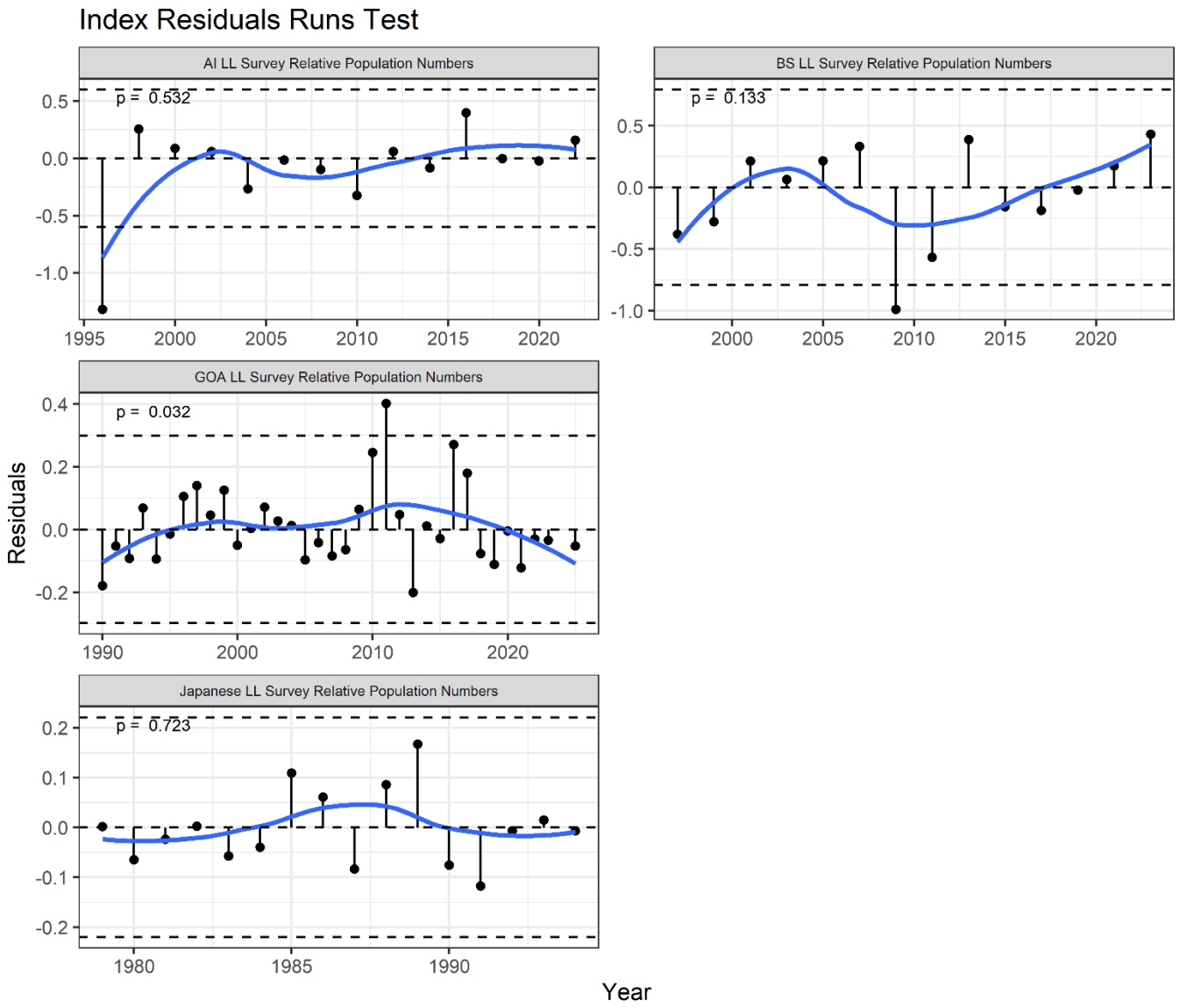

Figure SM3C.17. Results of an index residuals runs test for Model *26.2a_FAA_Fsh+Srvy*.

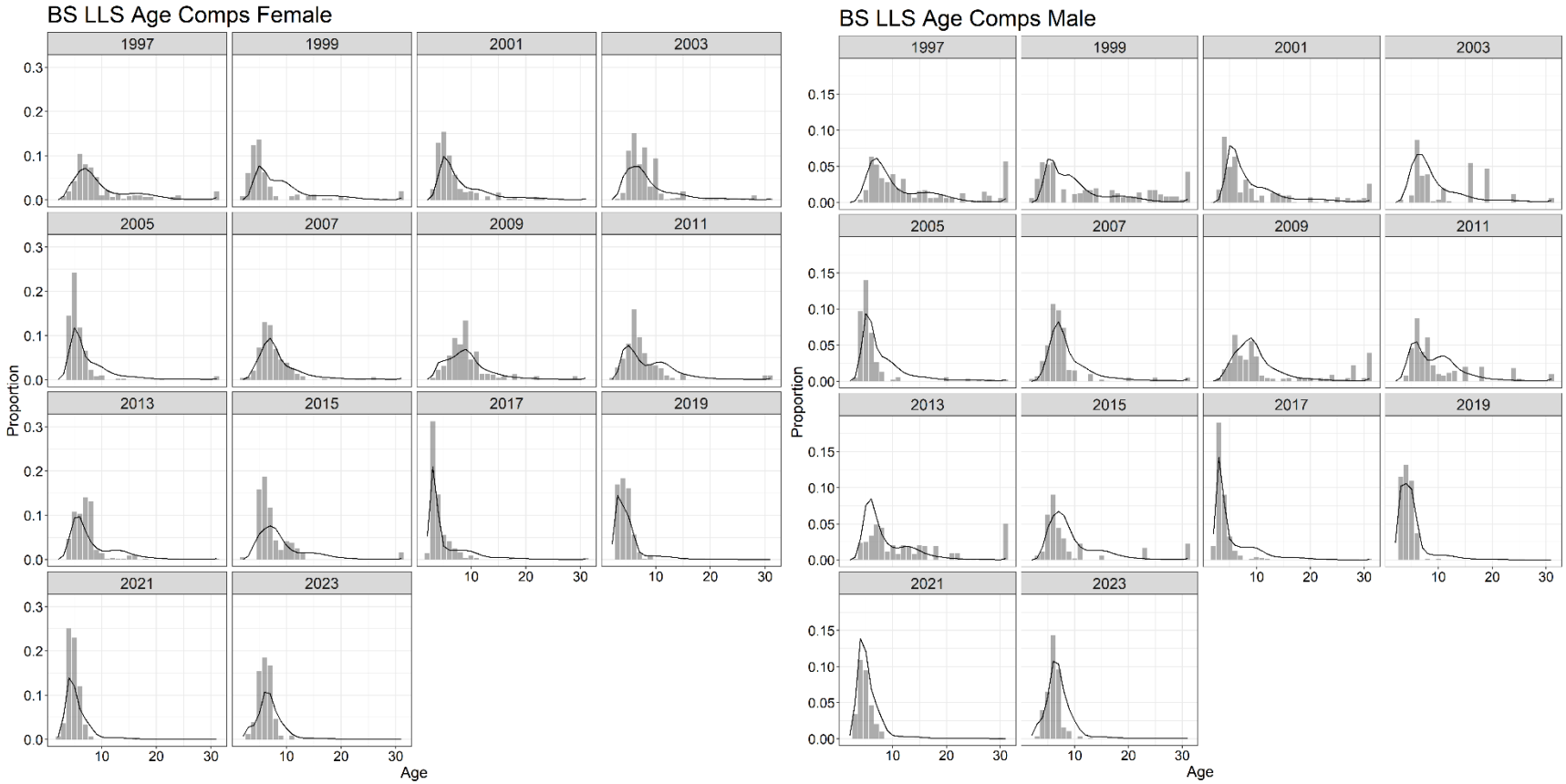

Figure SM3C.18. Fits to BS NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2_FAA*.

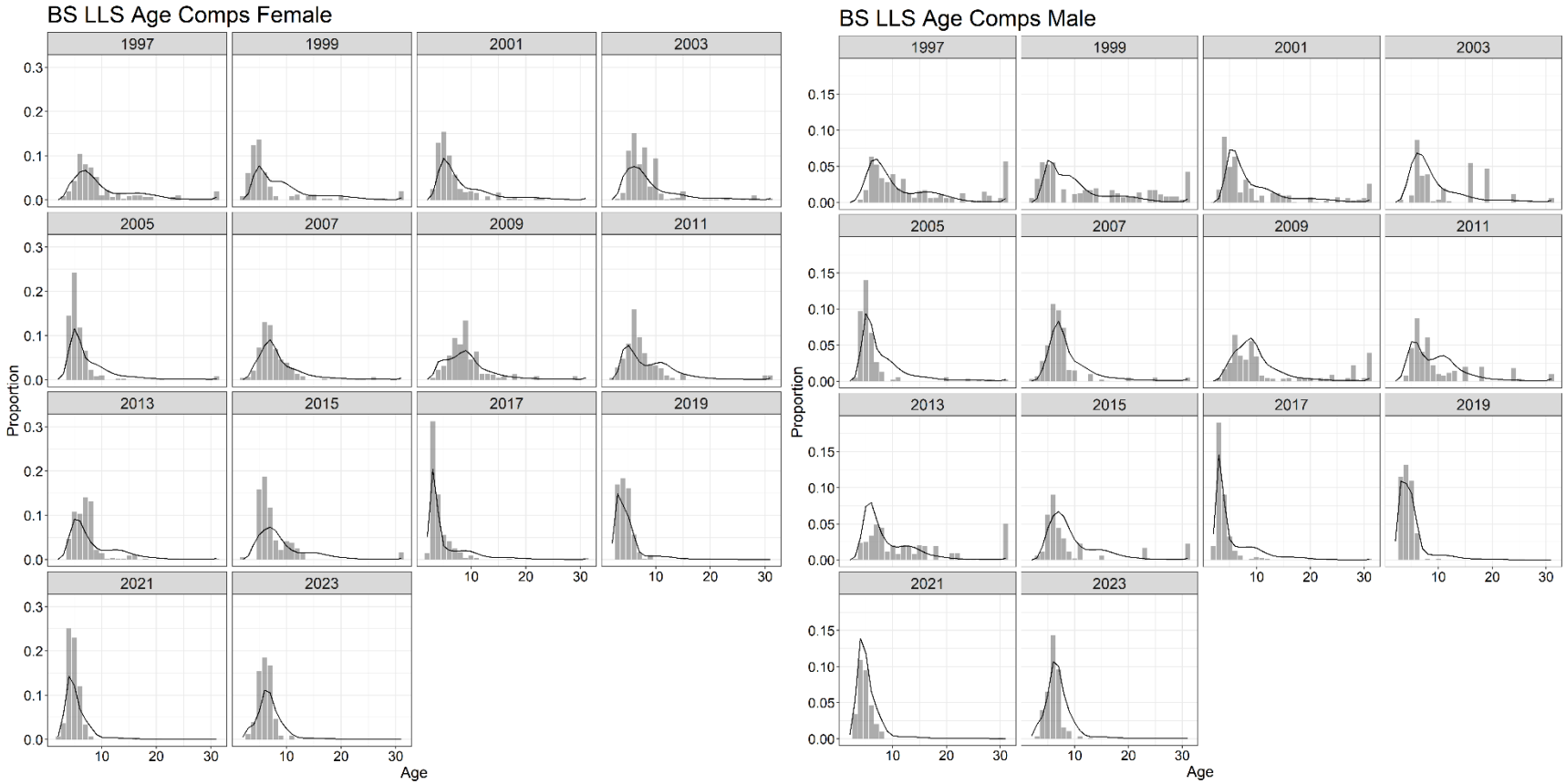

Figure SM3C.19. Fits to BS NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

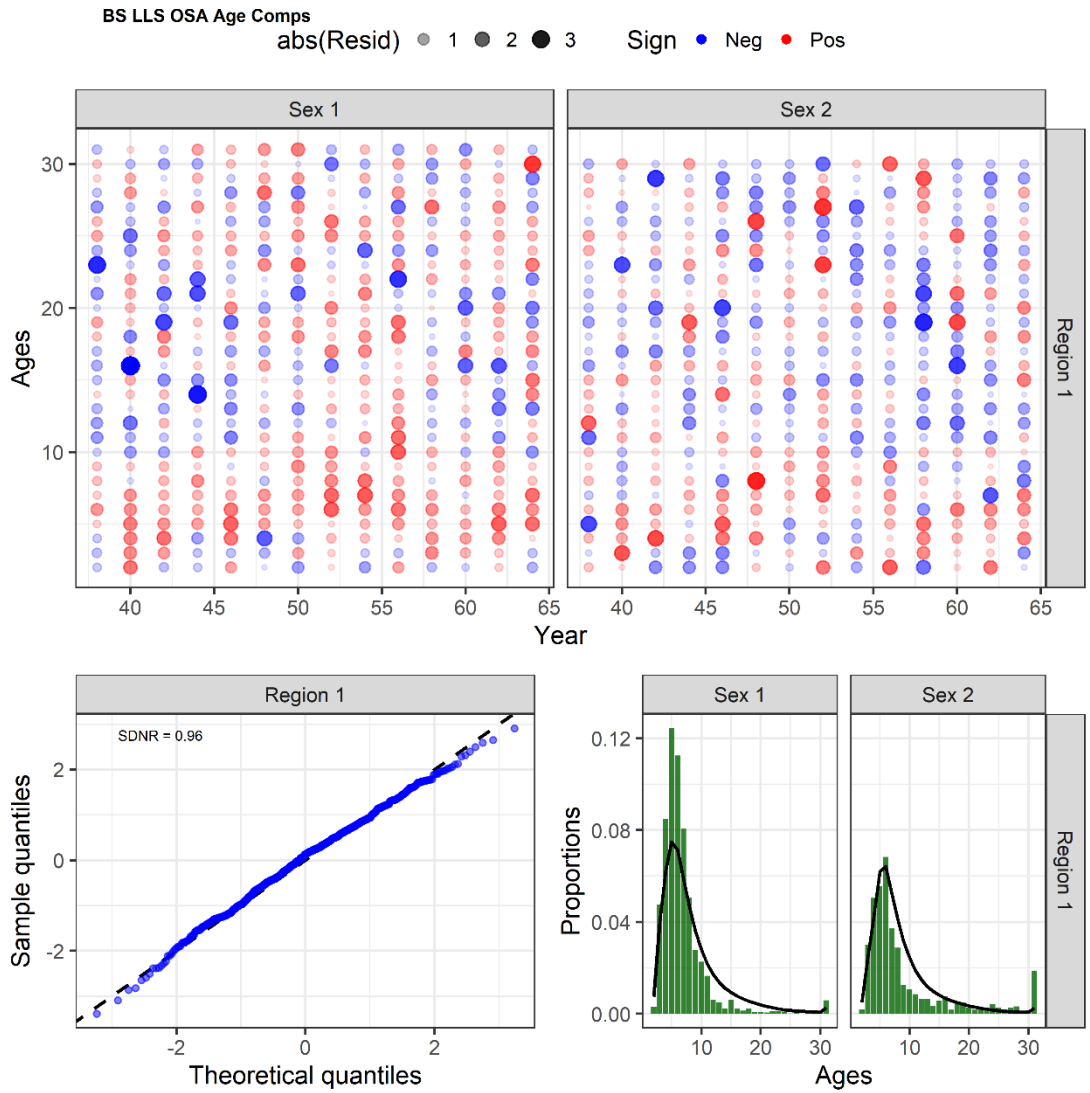

Figure SM3C.20. Fits to BS NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

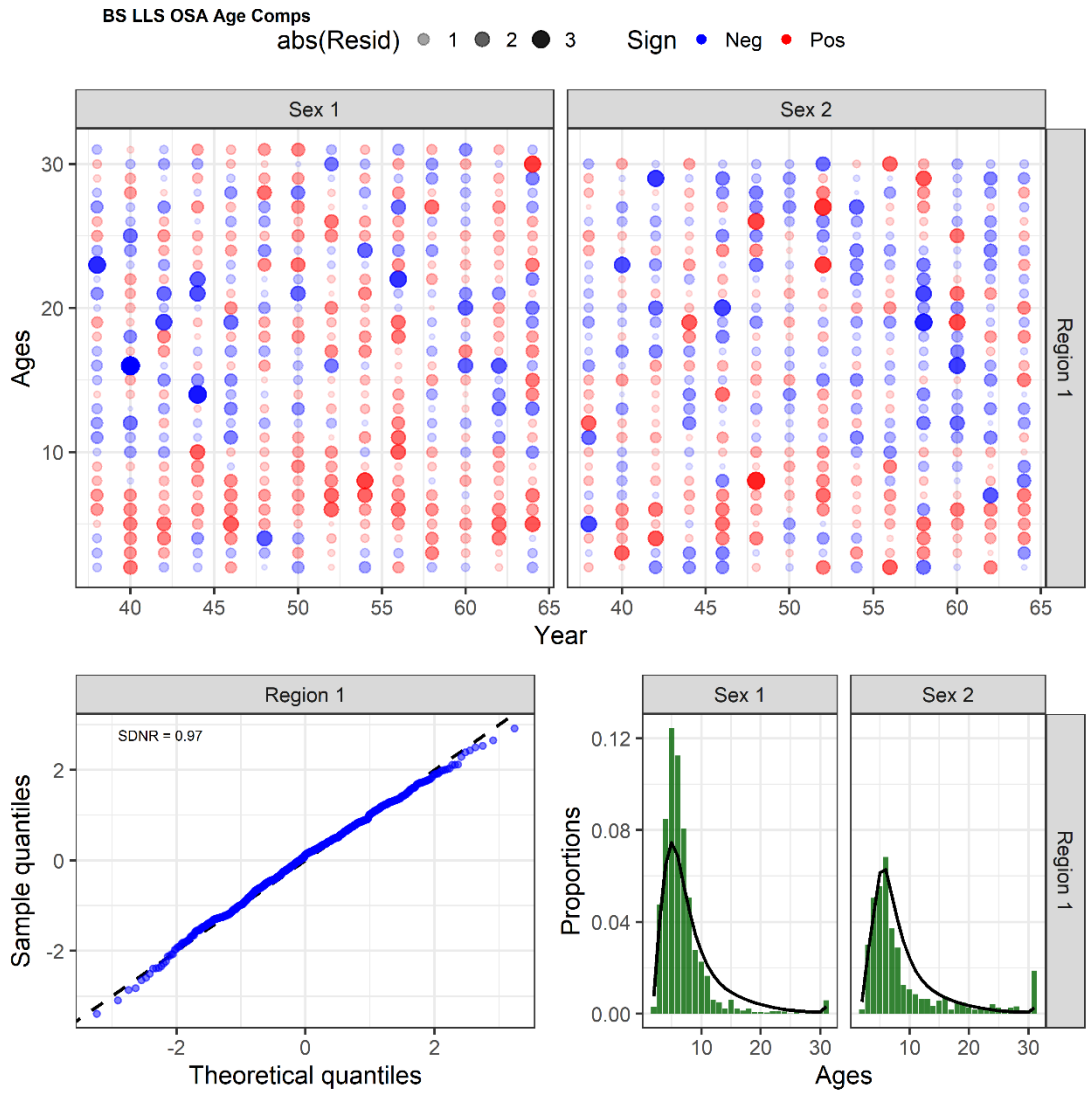

Figure SM3C.21. Fits to BS NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

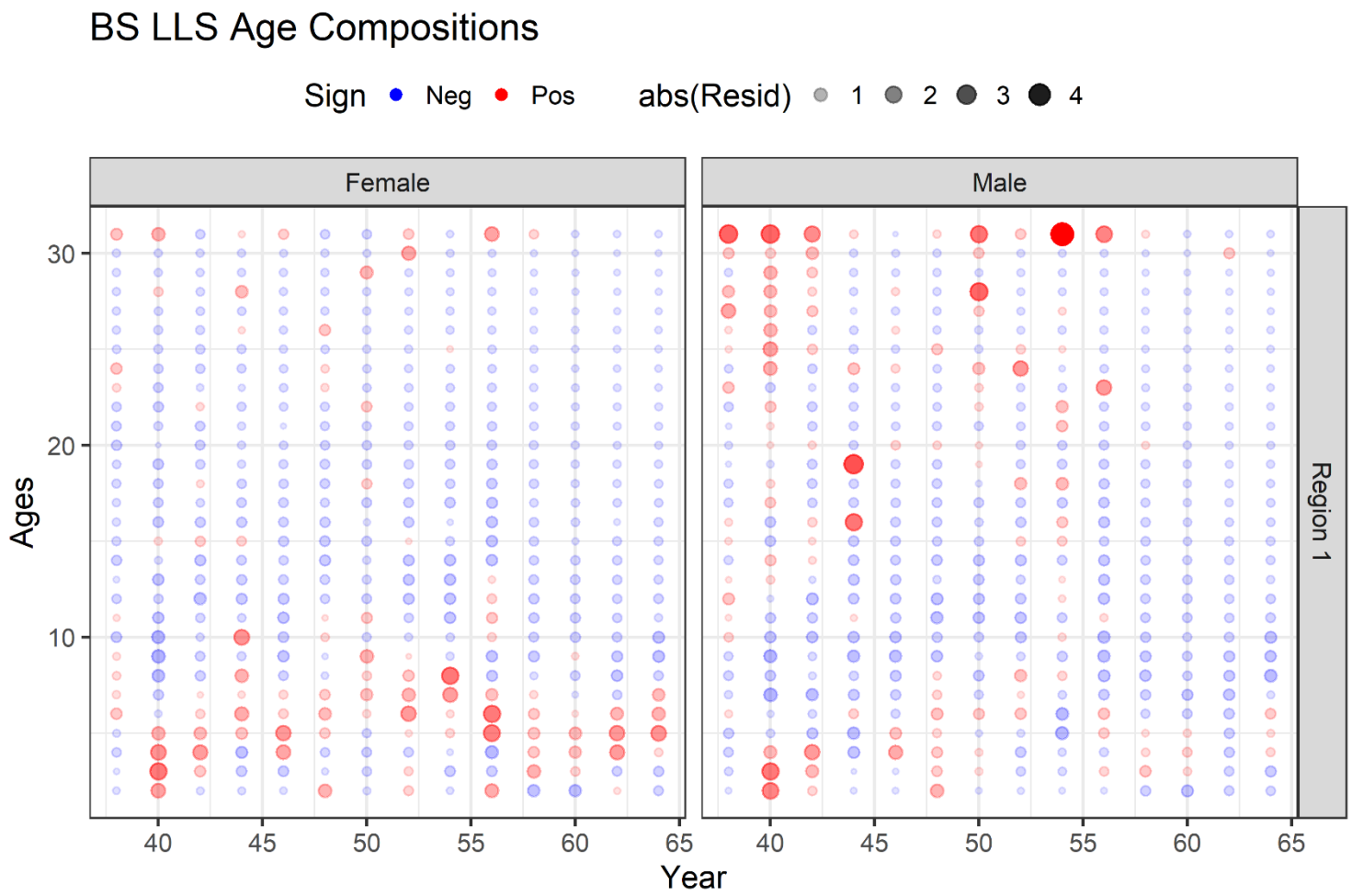

Figure SM3C.22. Fits to BS NOAA longline survey age composition data based on Pearson residuals for model *26.2_FAA*.

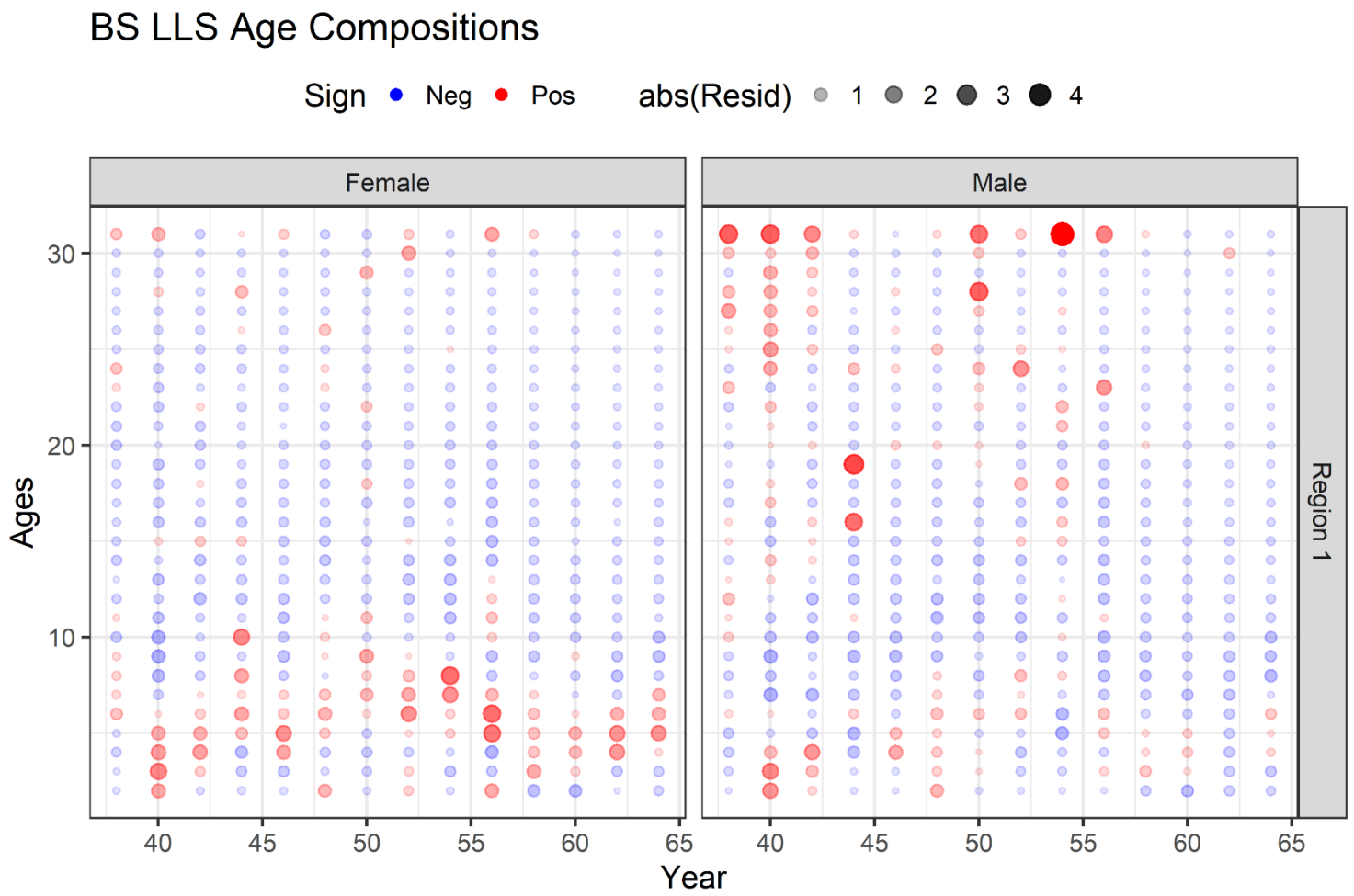

Figure SM3C.23. Fits to BS NOAA longline survey age composition data based on Pearson residuals for model *26.2a_FAA_Fsh+Srvy*.

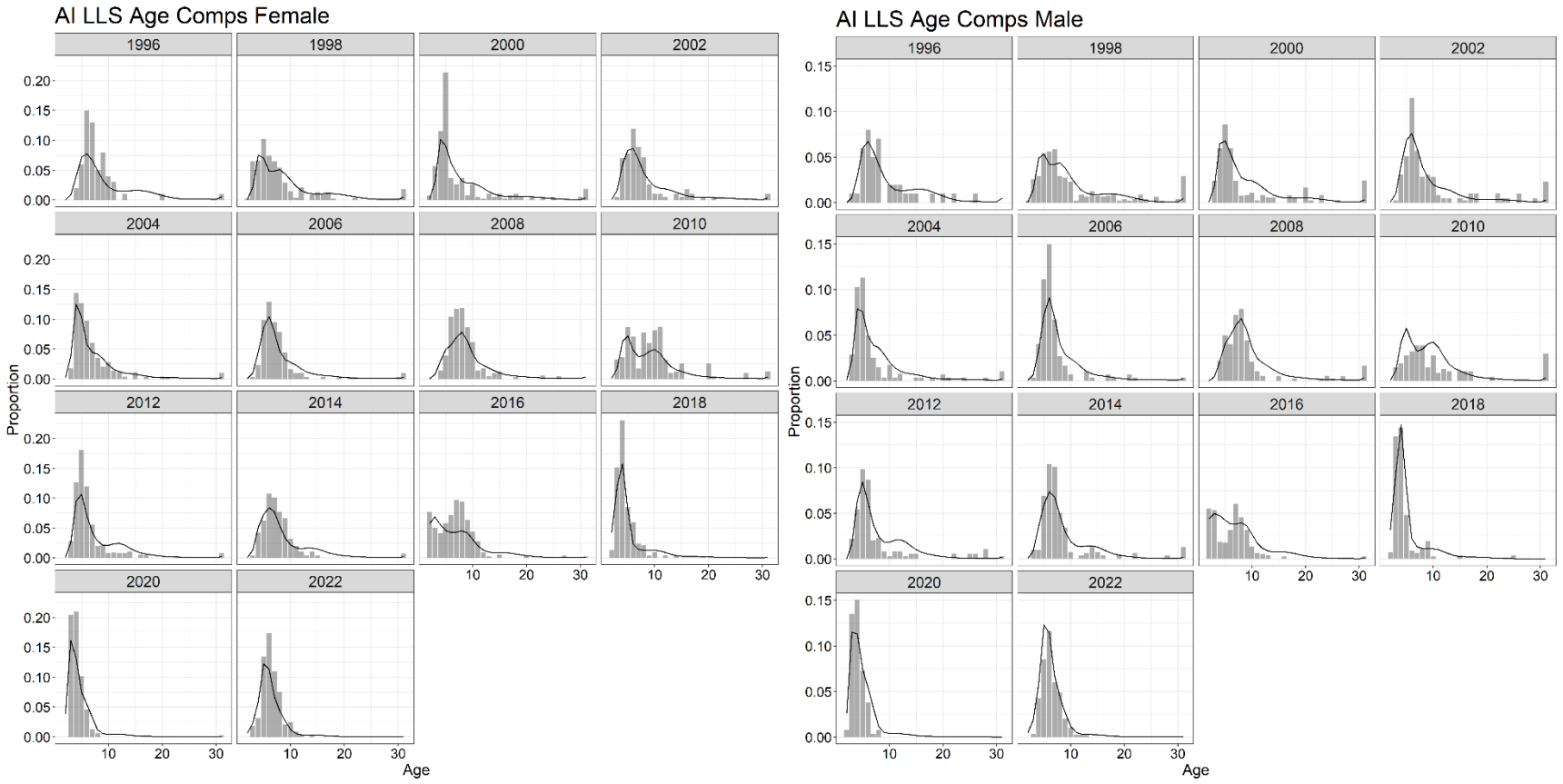

Figure SM3C.24. Fits to AI NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2_FAA*.

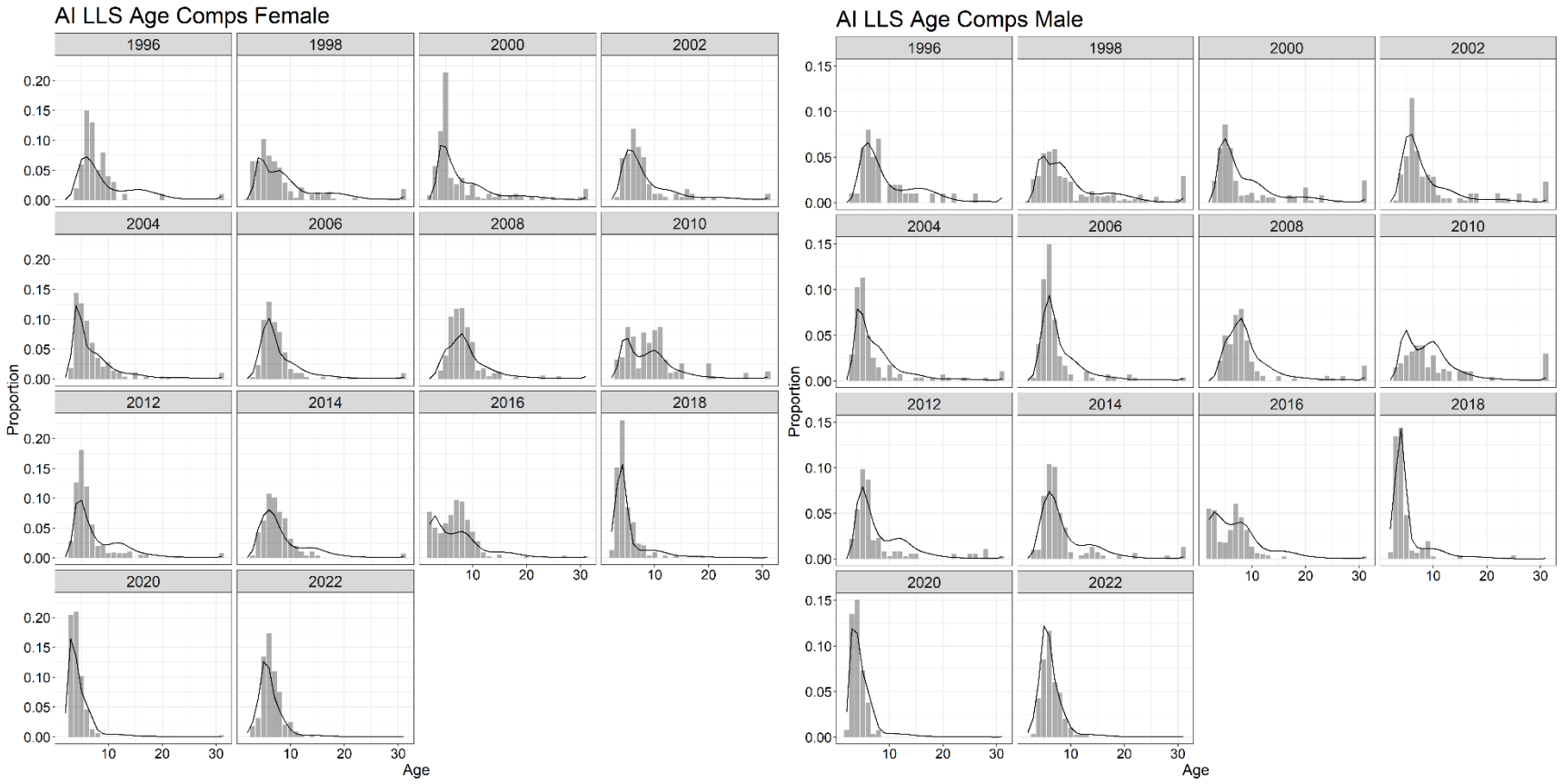

Figure SM3C.25. Fits to AI NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

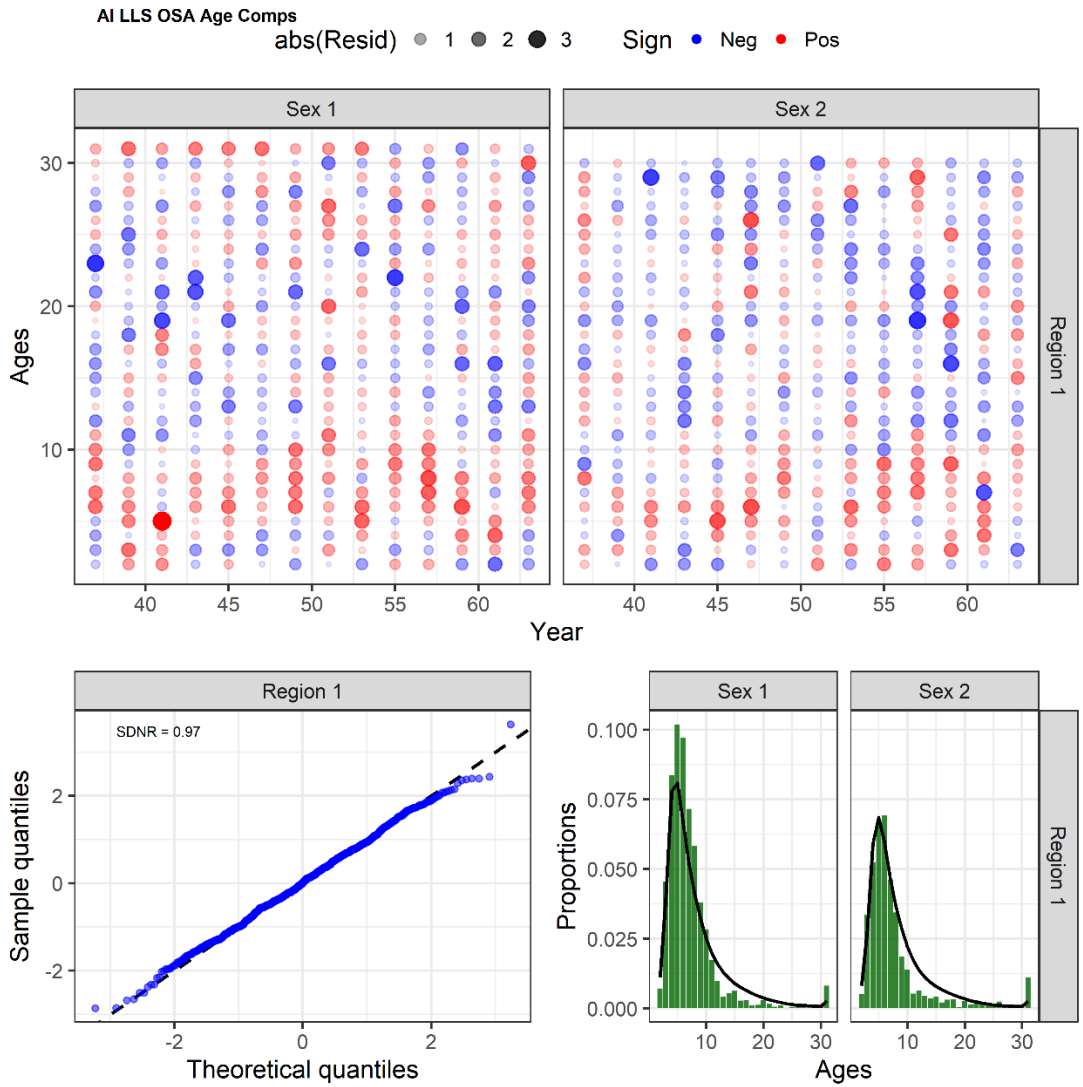

Figure SM3C.26. Fits to AI NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

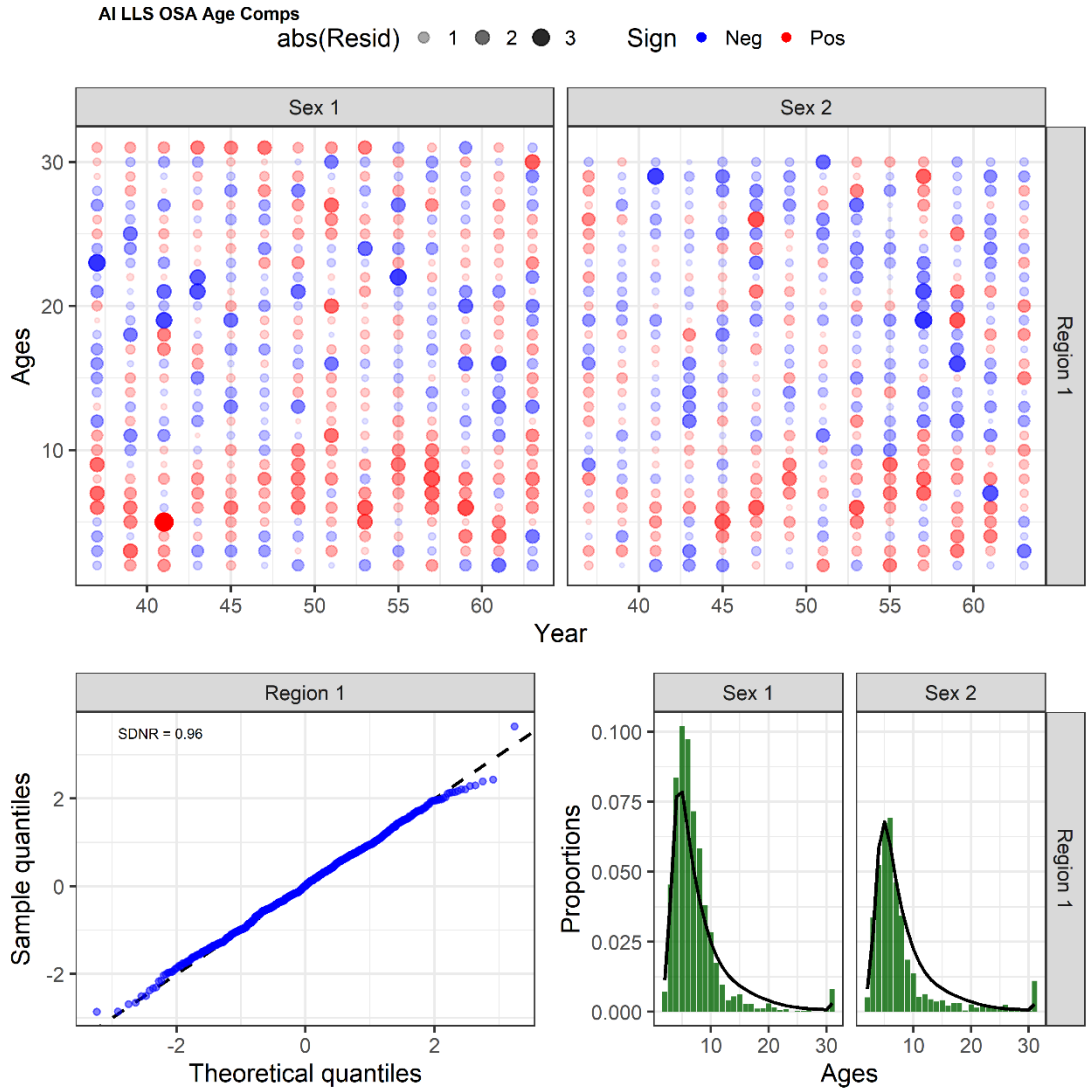

Figure SM3C.27. Fits to AI NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

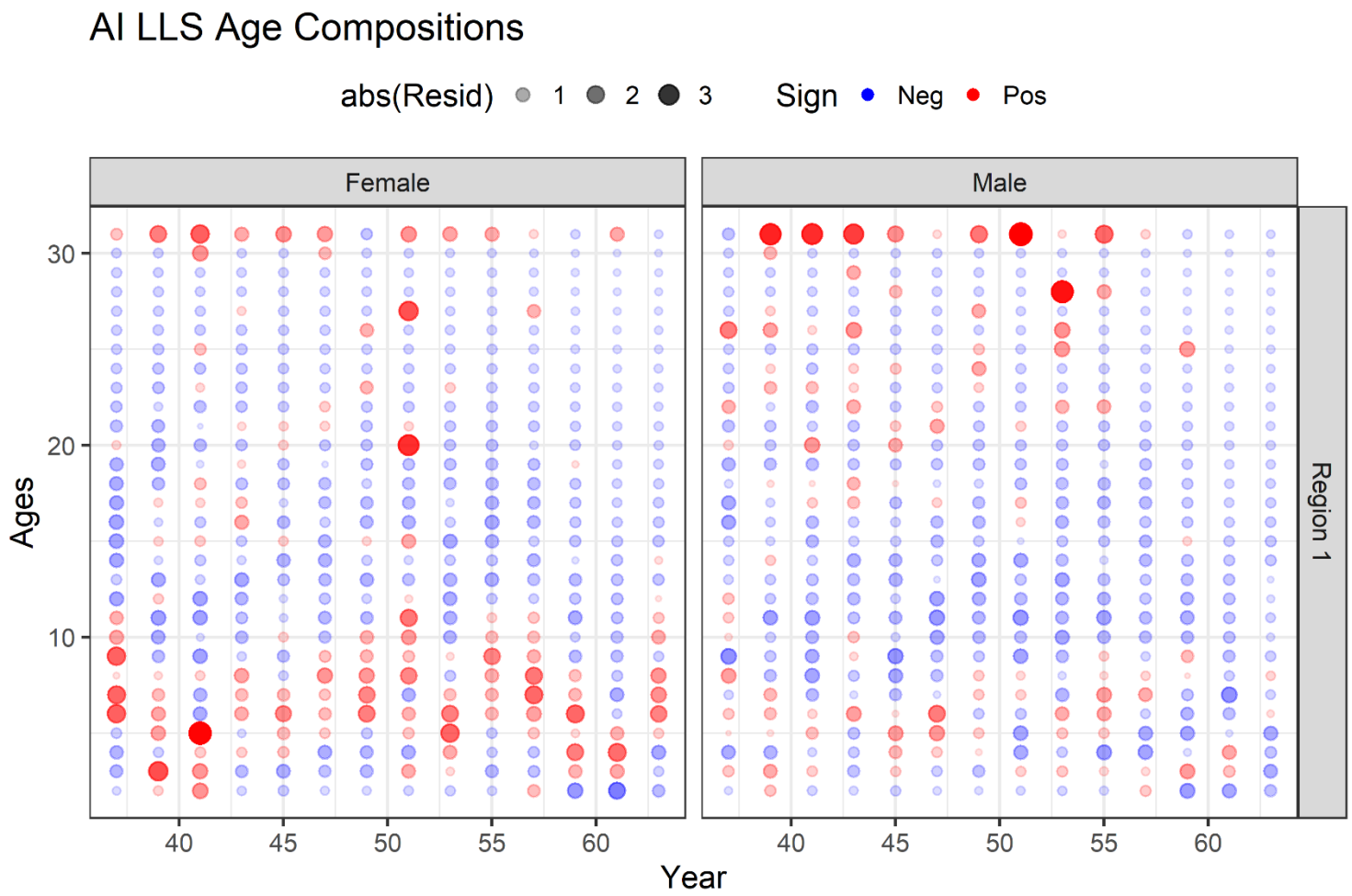

Figure SM3C.28. Fits to AI NOAA longline survey age composition data based on Pearson residuals for model *26.2_FAA*.

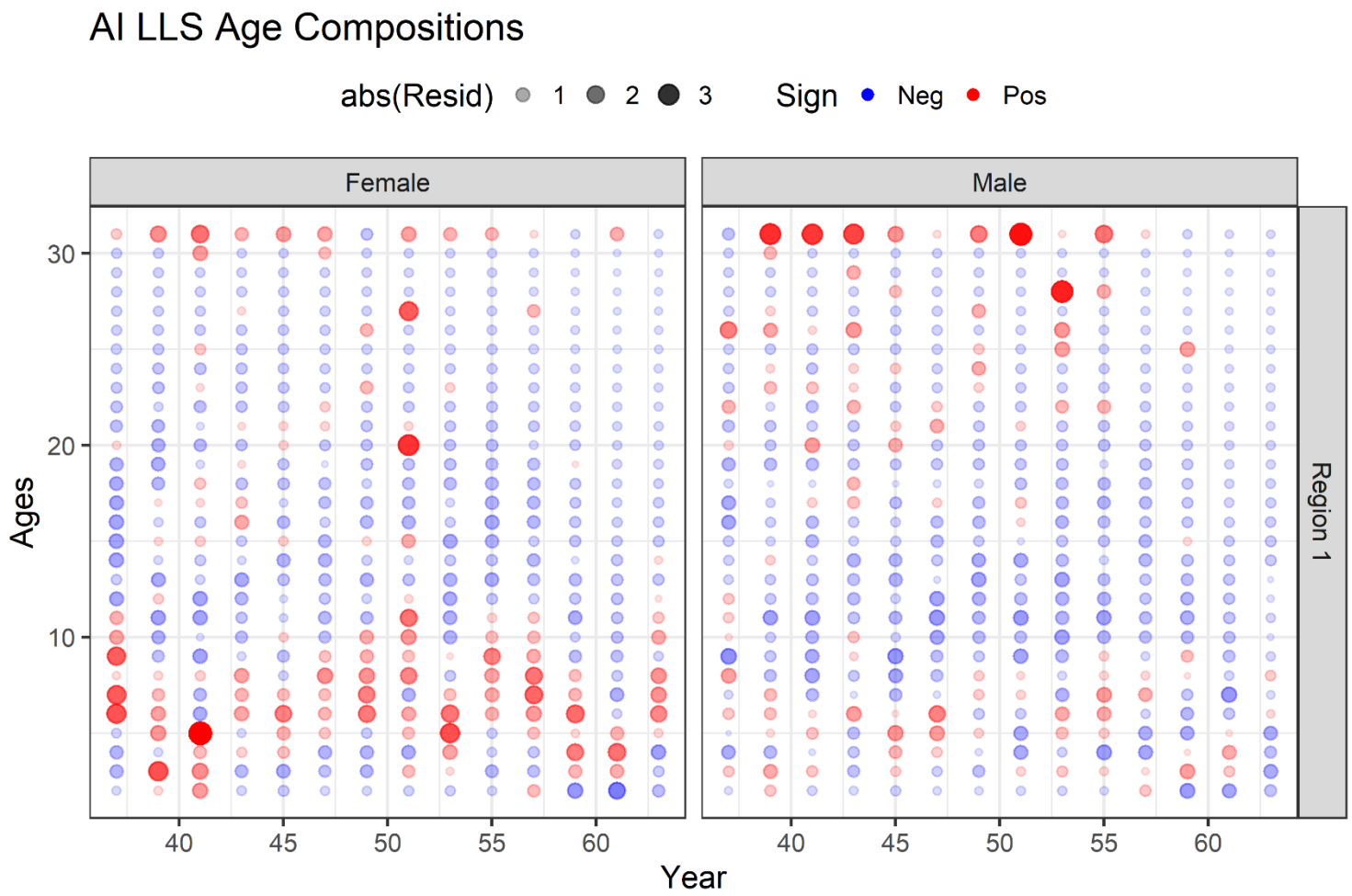

Figure SM3C.29. Fits to AI NOAA longline survey age composition data based on Pearson residuals for model *26.2a_FAA_Fsh+Srvy*.

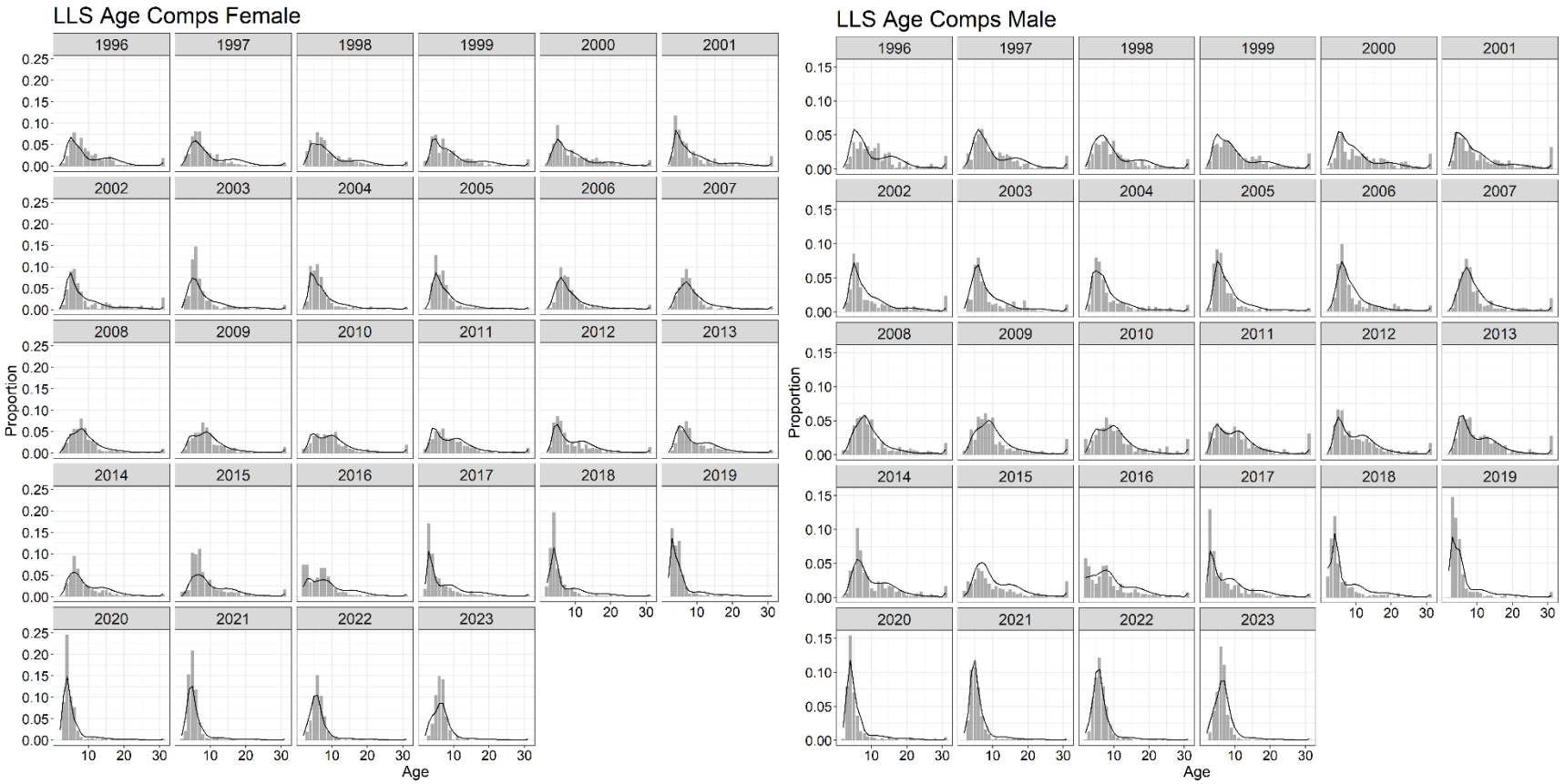

Figure SM3C.30. Fits to NOAA longline survey age composition data by year and sex (females left and males right) for model *26.1_1Reg*.

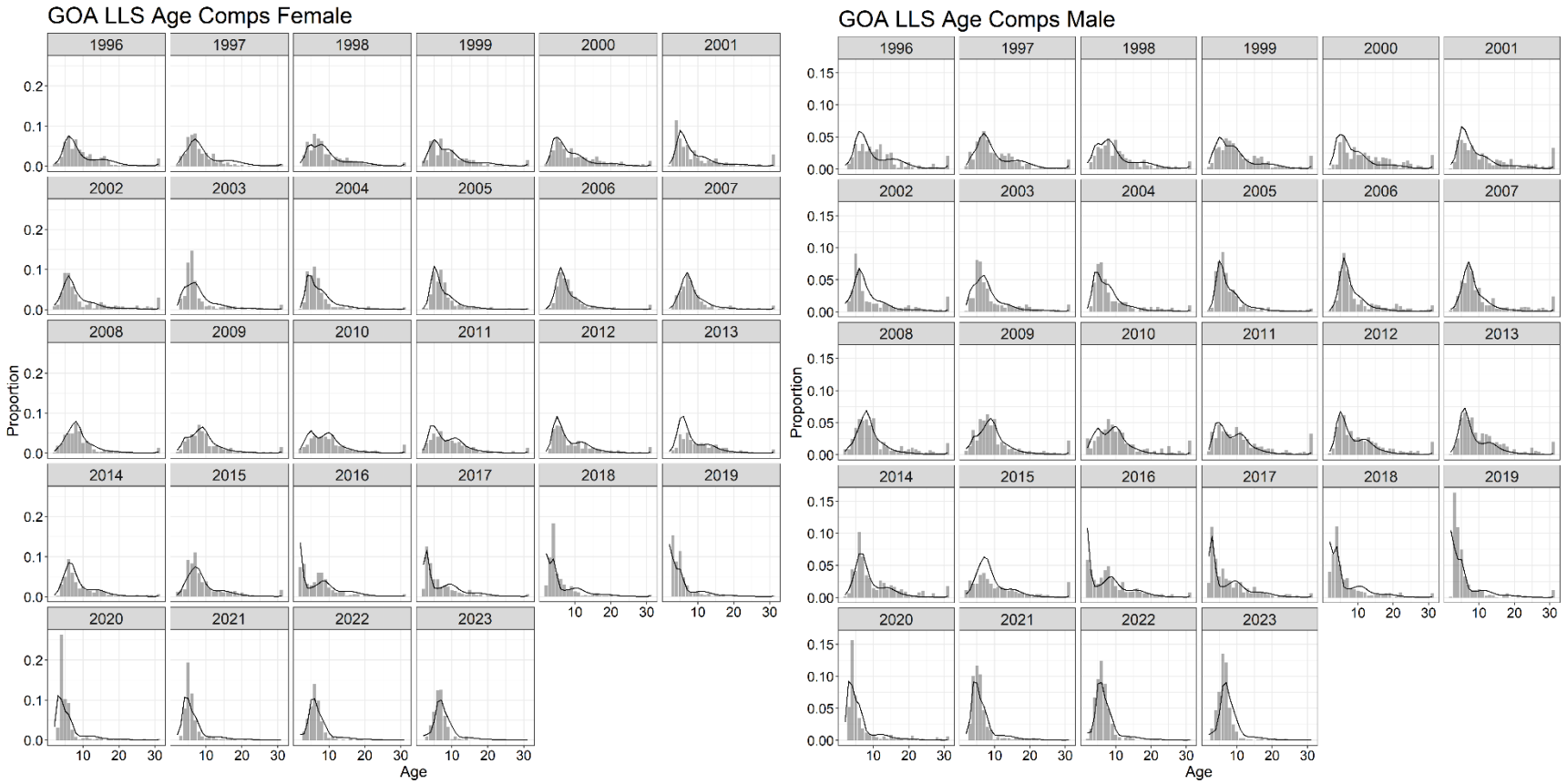

Figure SM3C.31. Fits to GOA NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2_FAA*.

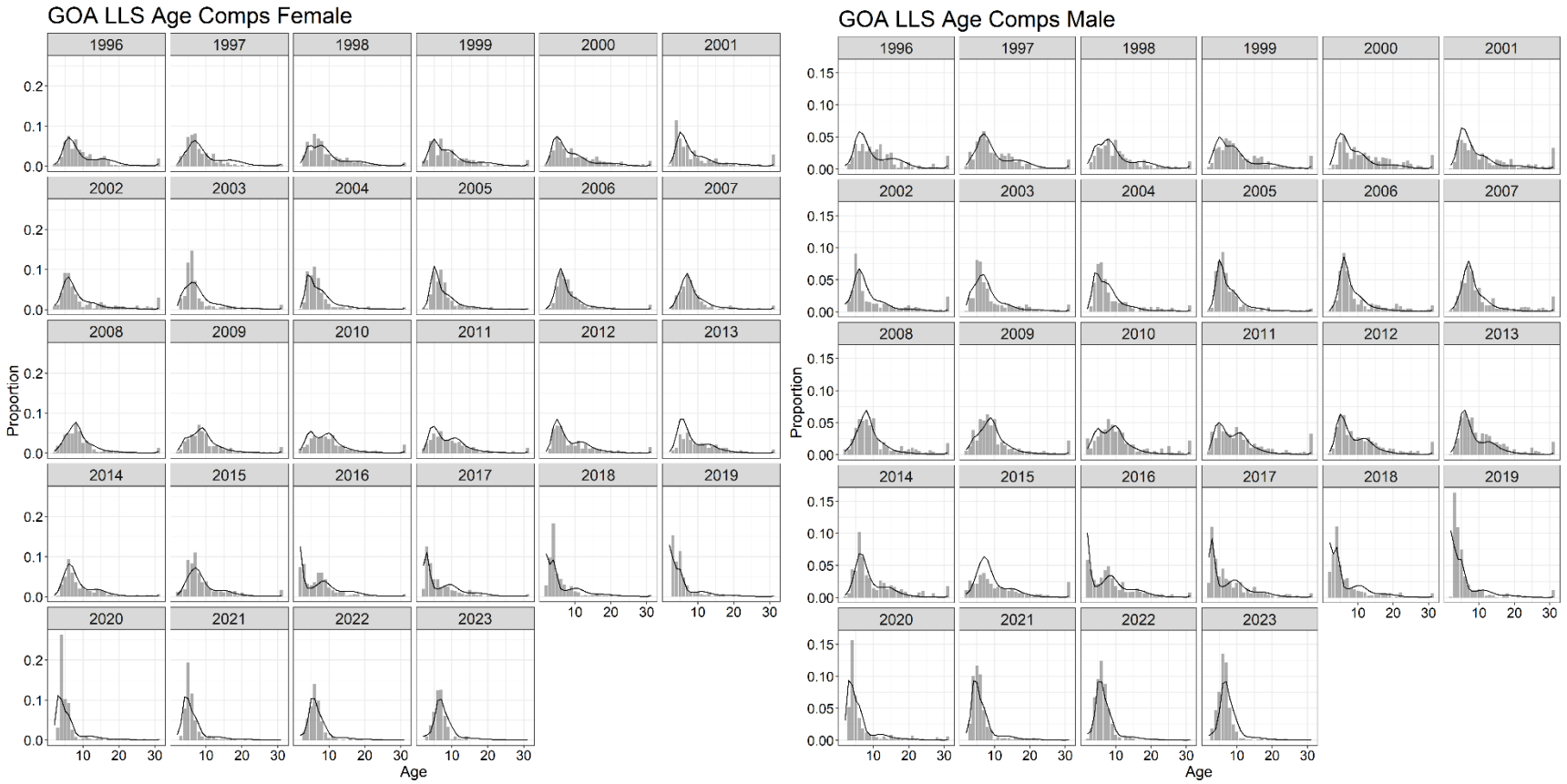

Figure SM3C.32. Fits to GOA NOAA longline survey age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

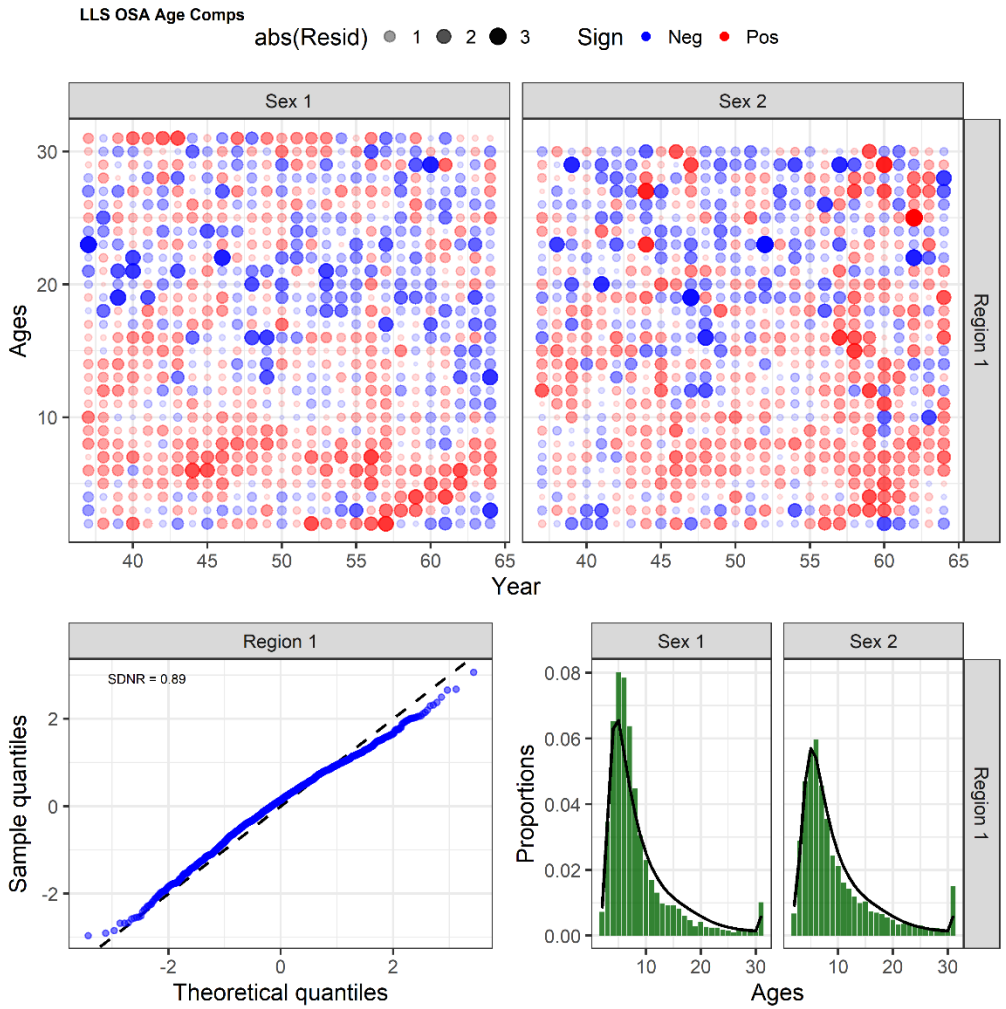

Figure SM3C.33. Fits to NOAA longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg.*

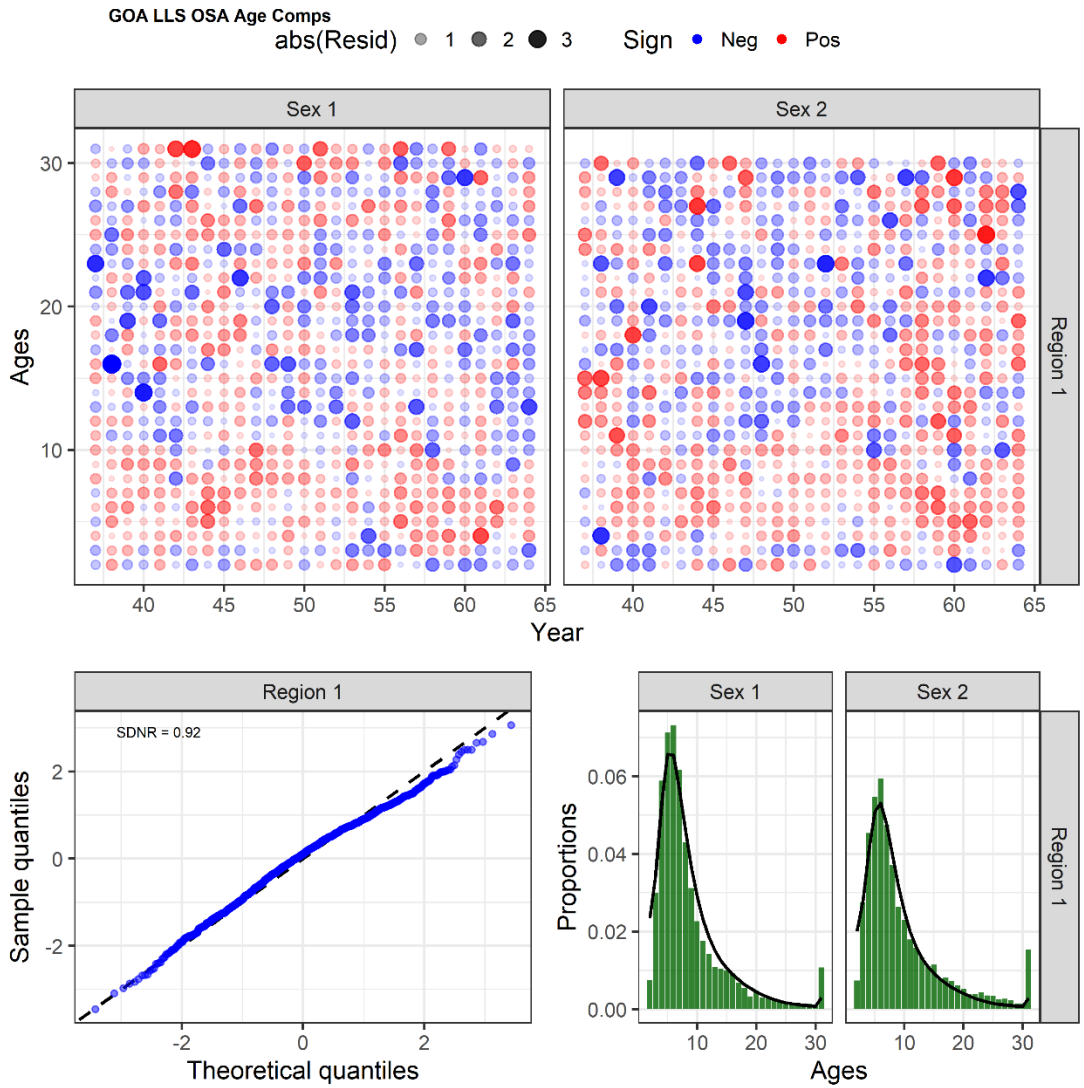

Figure SM3C.34. Fits to GOA NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

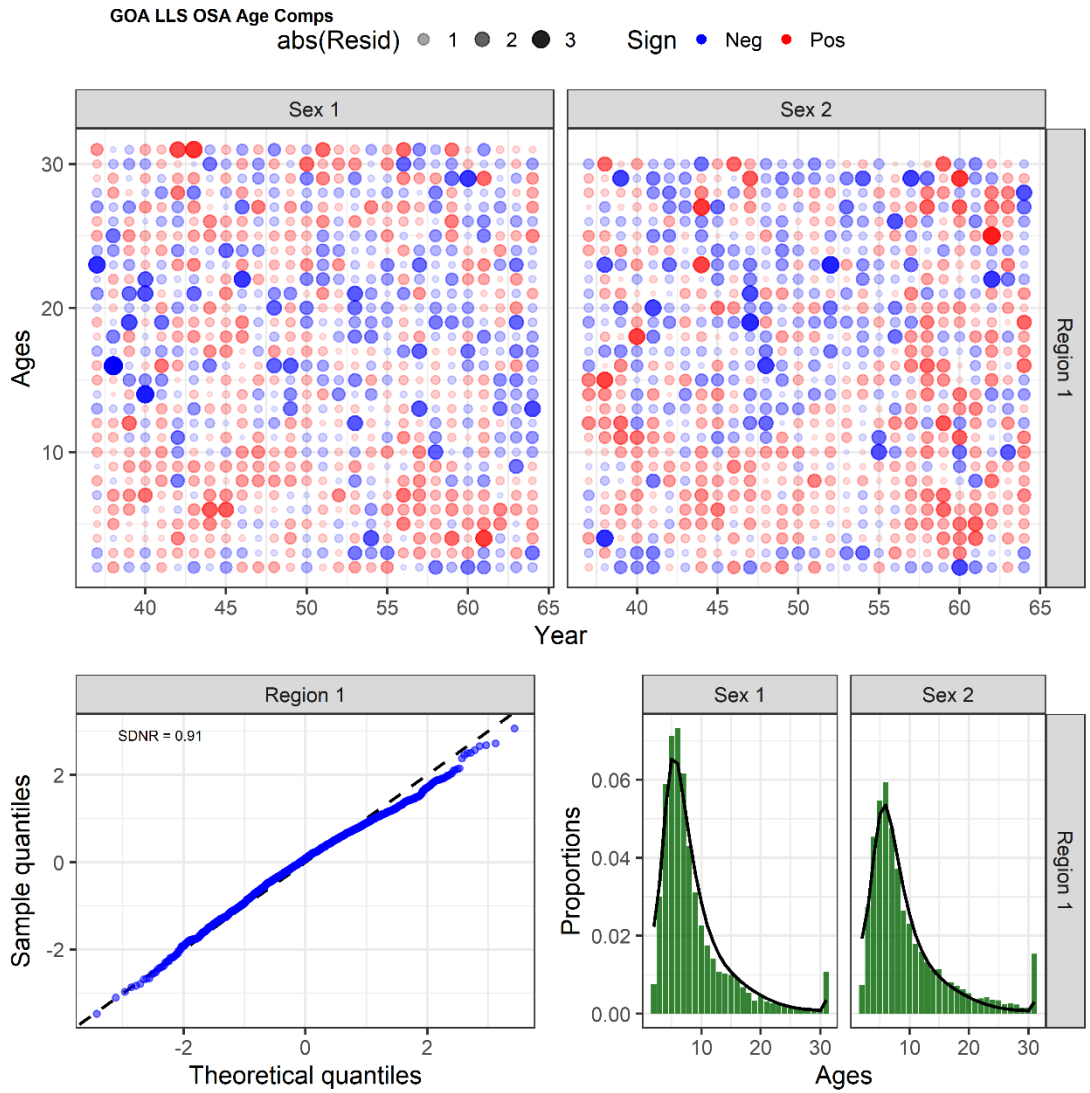

Figure SM3C.35. Fits to GOA NOAA Longline Survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

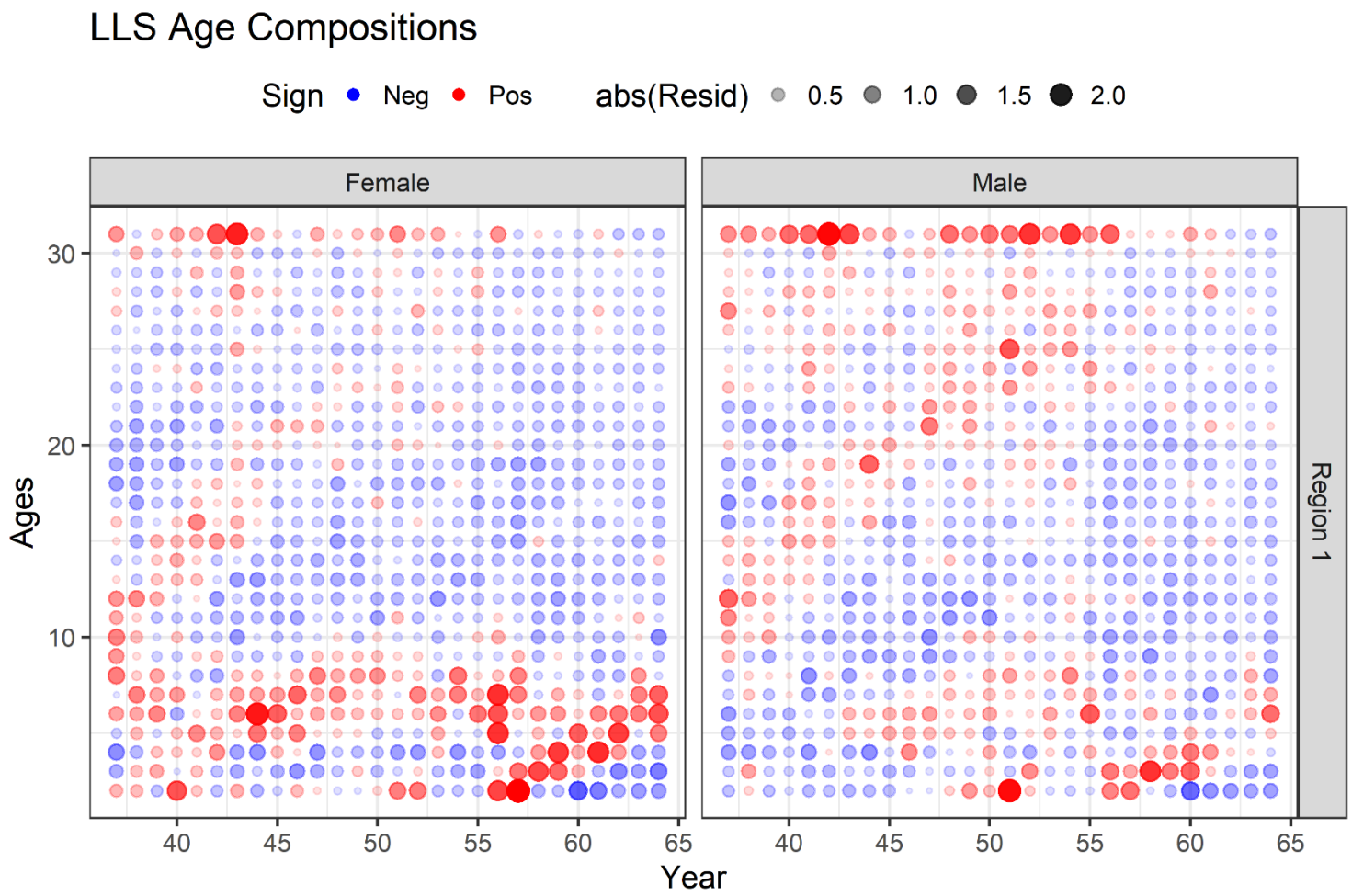

Figure SM3B.36. Fits to NOAA longline survey age composition data based on Pearson residuals for model *26.1_1Reg*.

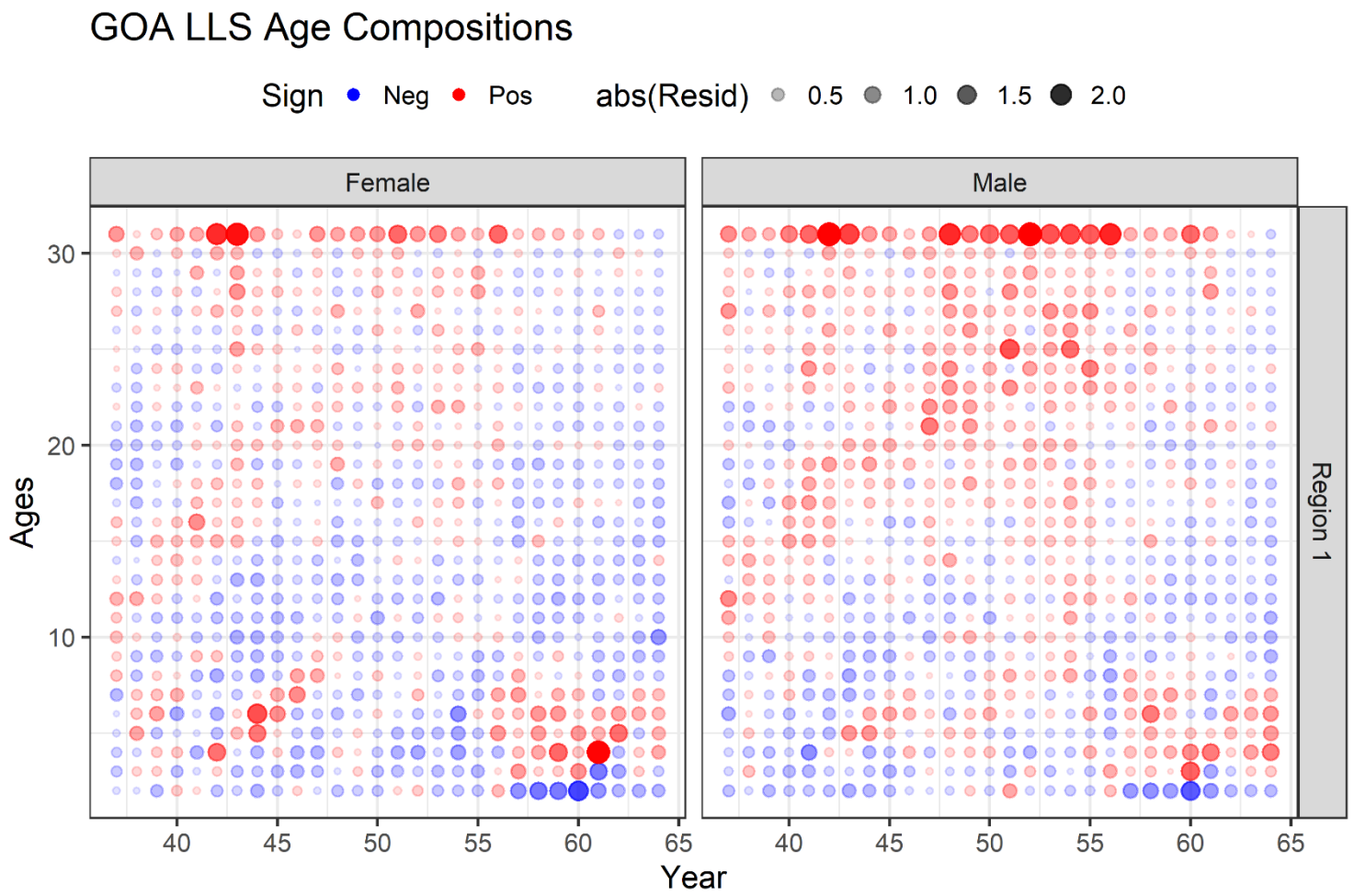

Figure SM3C.37. Fits to GOA NOAA longline survey age composition data based on Pearson residuals for model *26.2_FAA*.

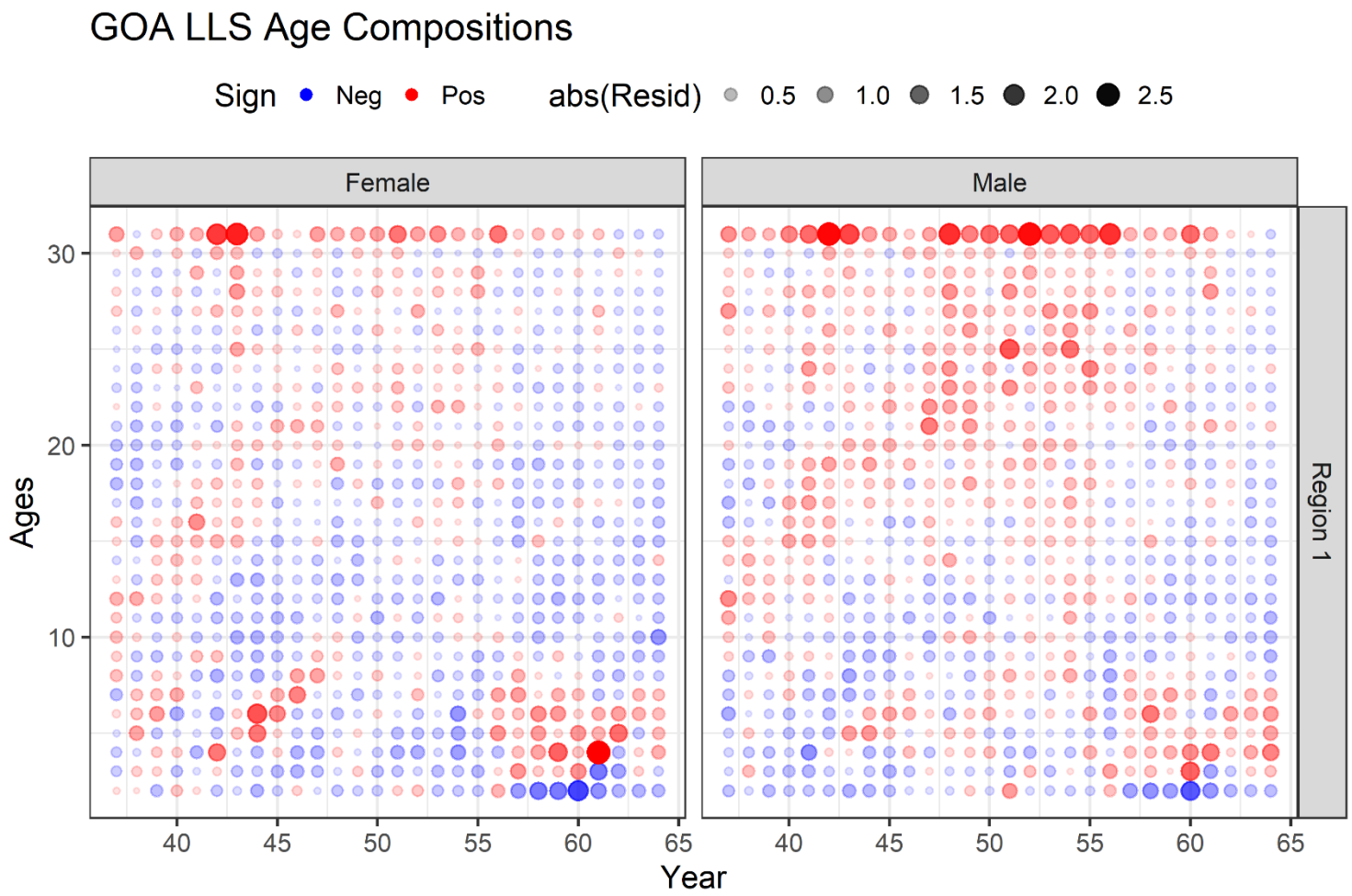

Figure SM3C.38. Fits to GOA NOAA longline survey age composition data based on Pearson residuals for model *26.2a_FAA_Fsh+Srvy*.

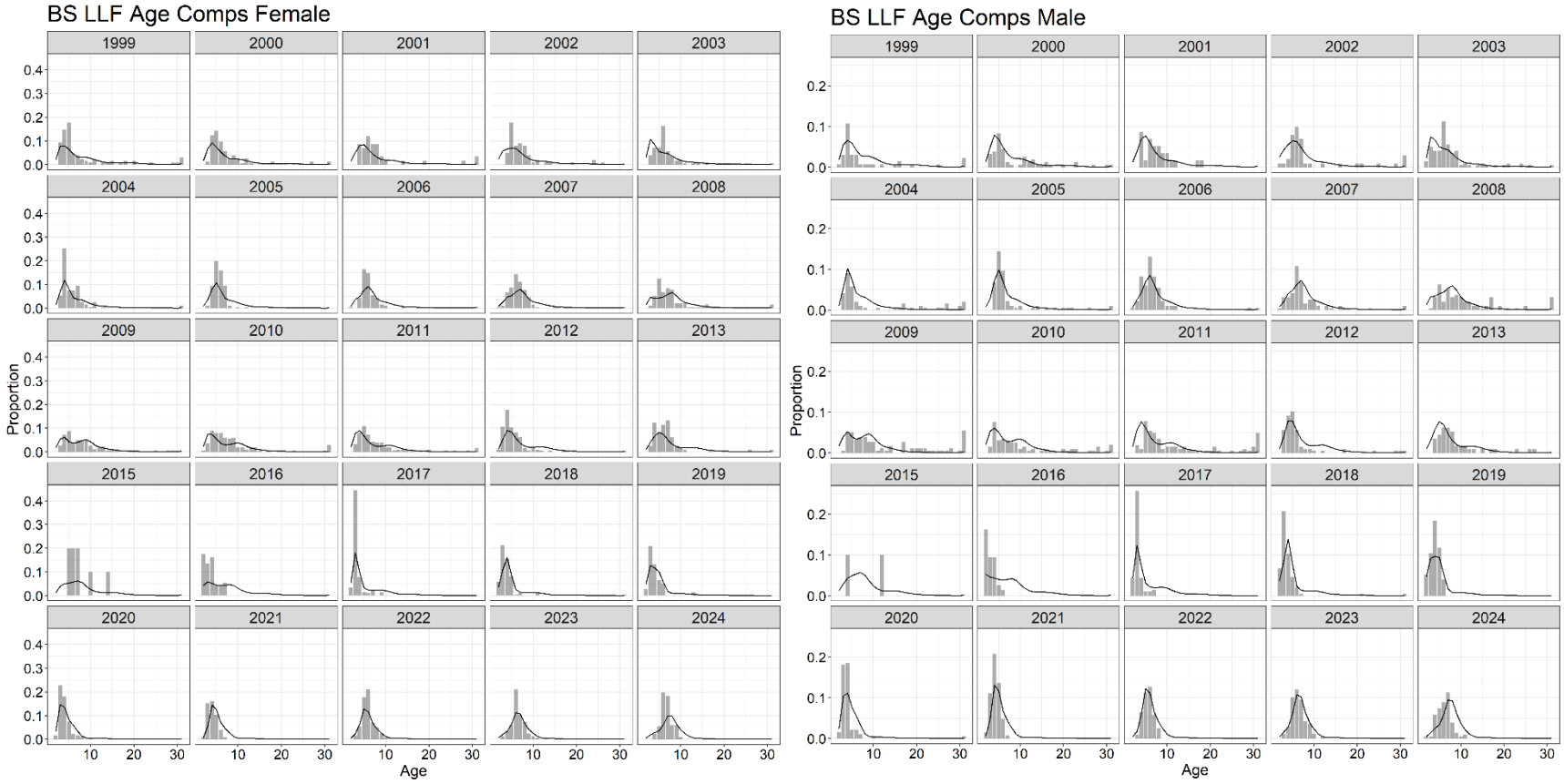

Figure SM3C.39. Fits to BS fixed gear fishery age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

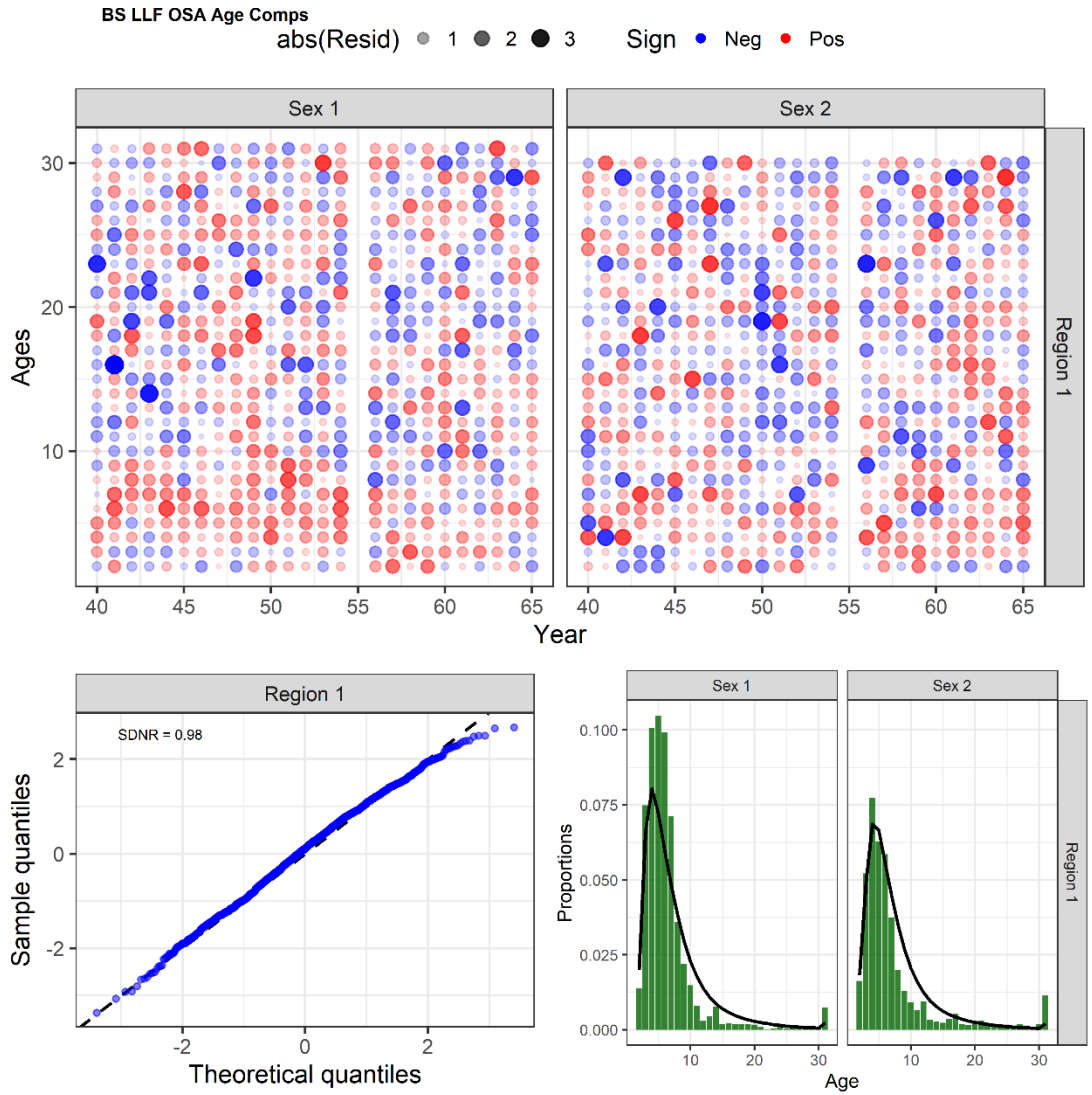

Figure SM3C.40. Fits to BS fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

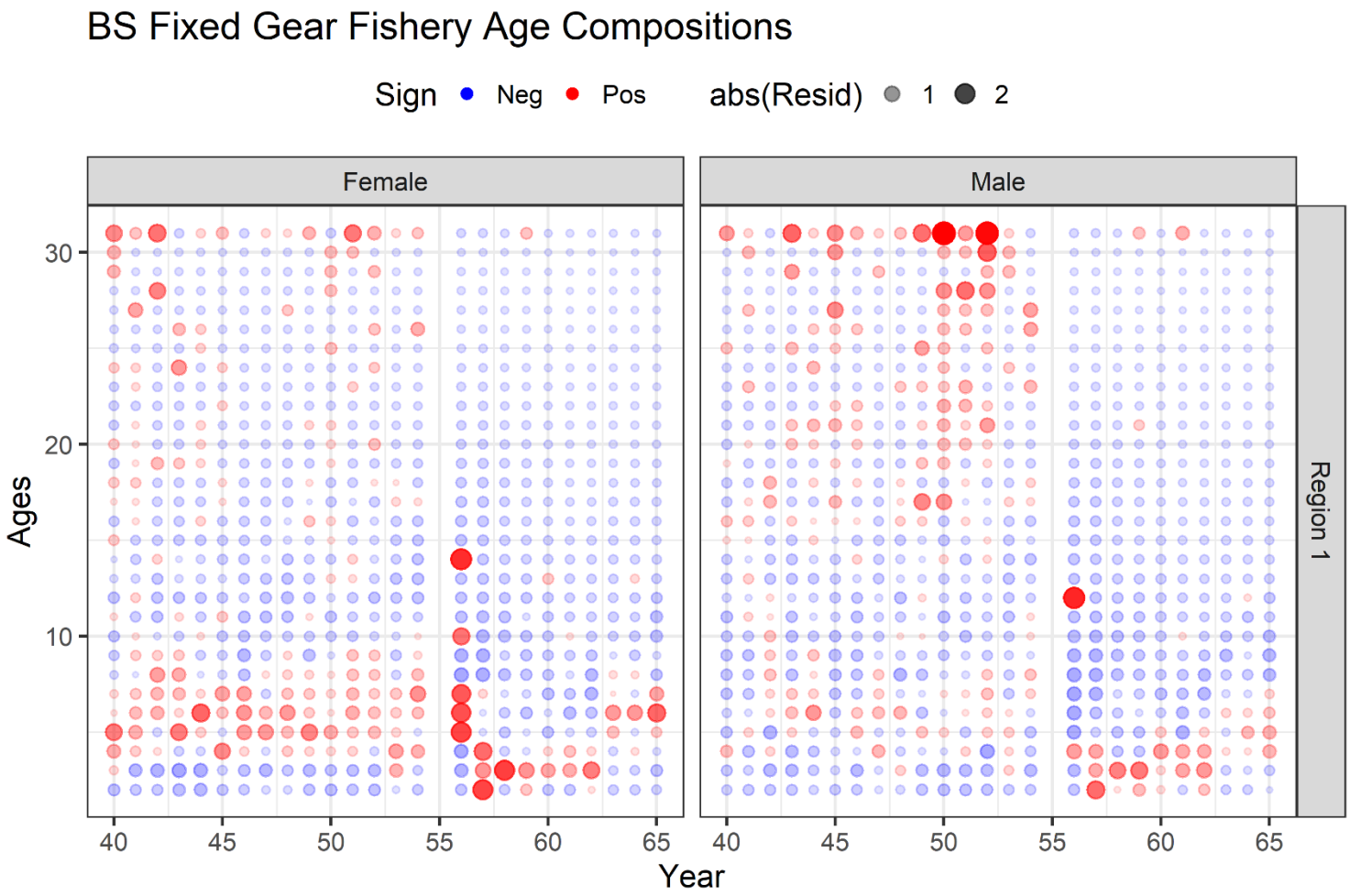

Figure SM3C.41. Fits to the BS fixed gear fishery age composition data based on Pearson residuals for model *25. 2_FAA_Fsh+Srvy*.

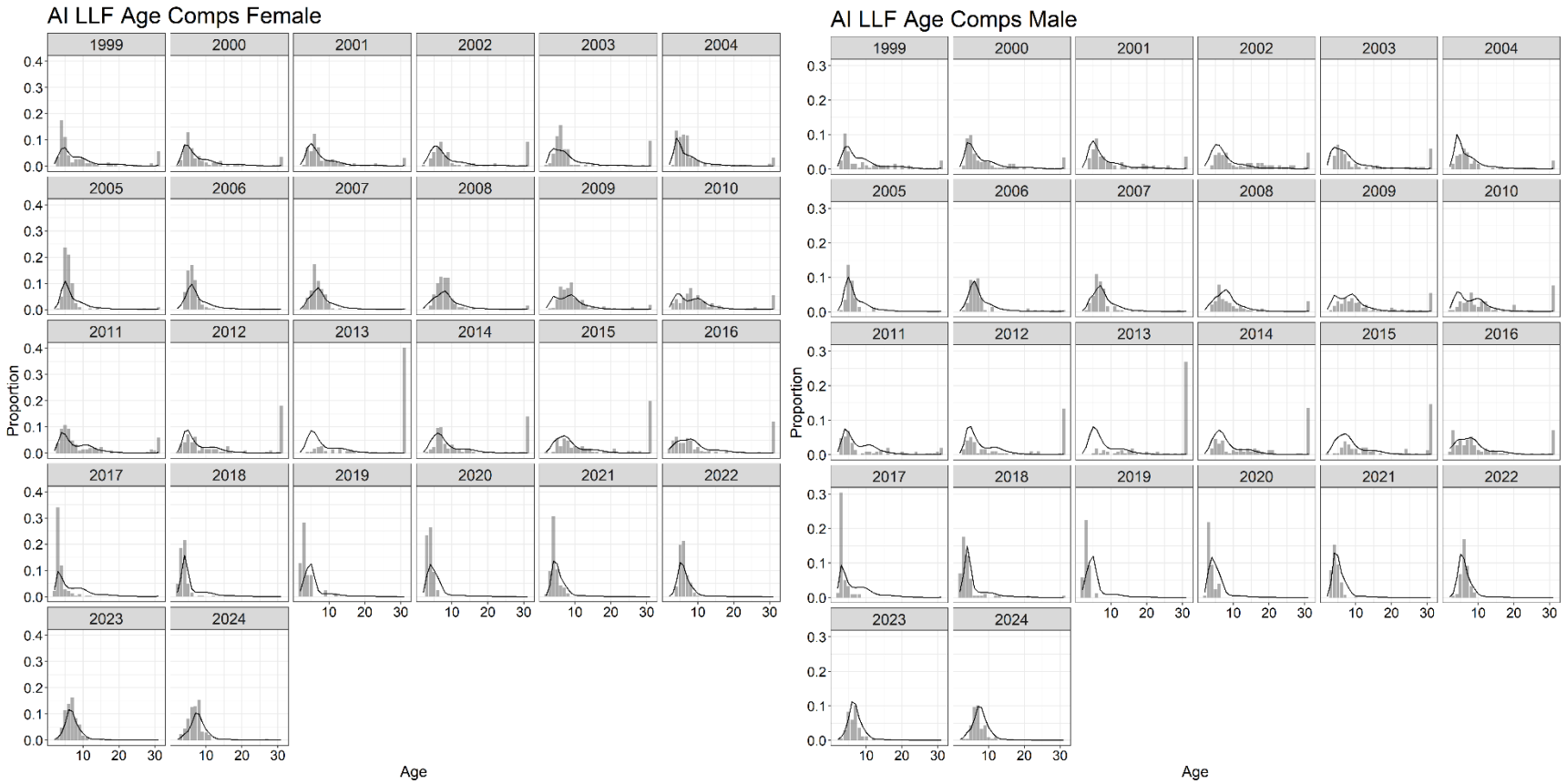

Figure SM3C.42. Fits to AI fixed gear fishery age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

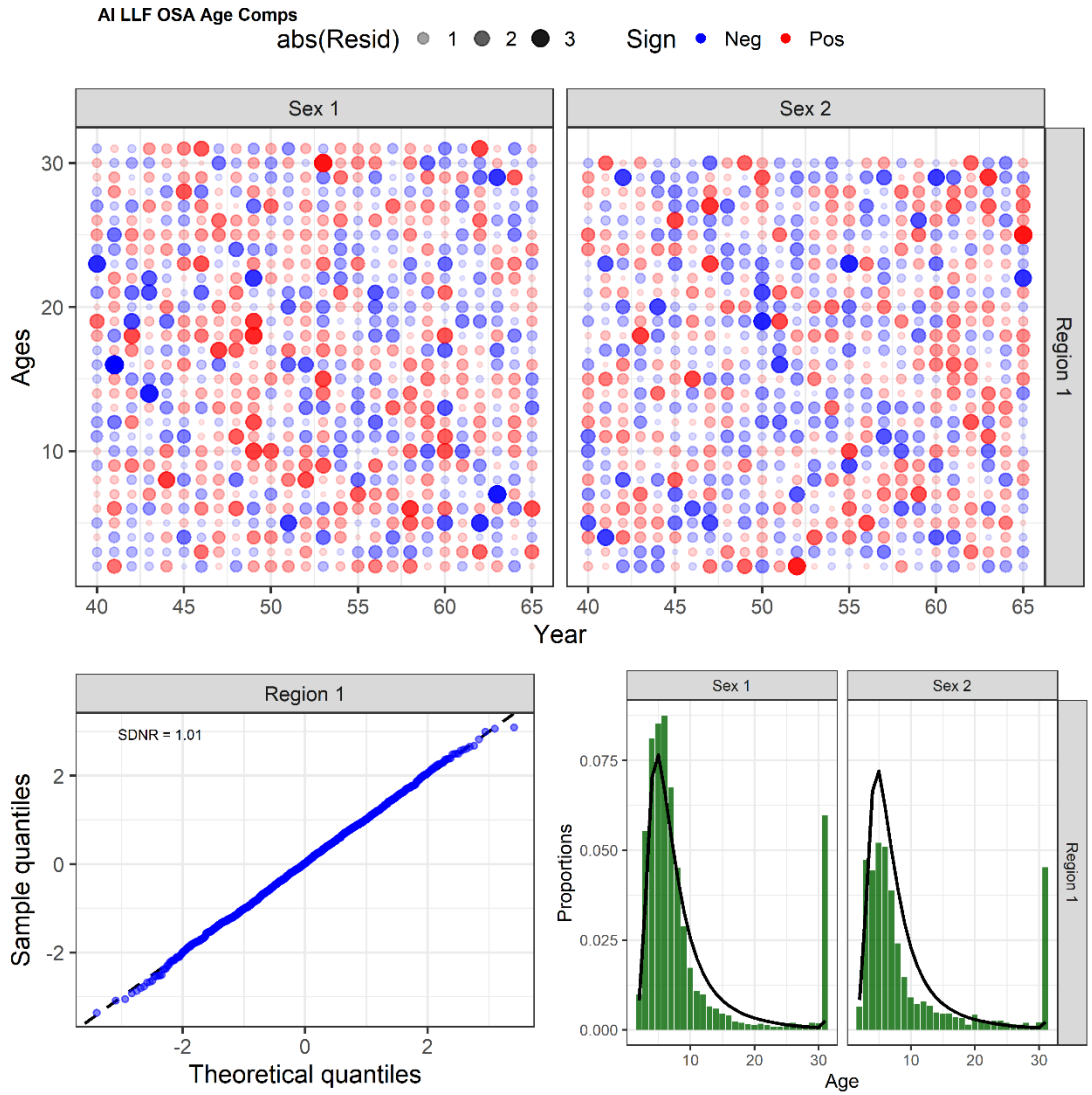

Figure SM3C.43. Fits to AI fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

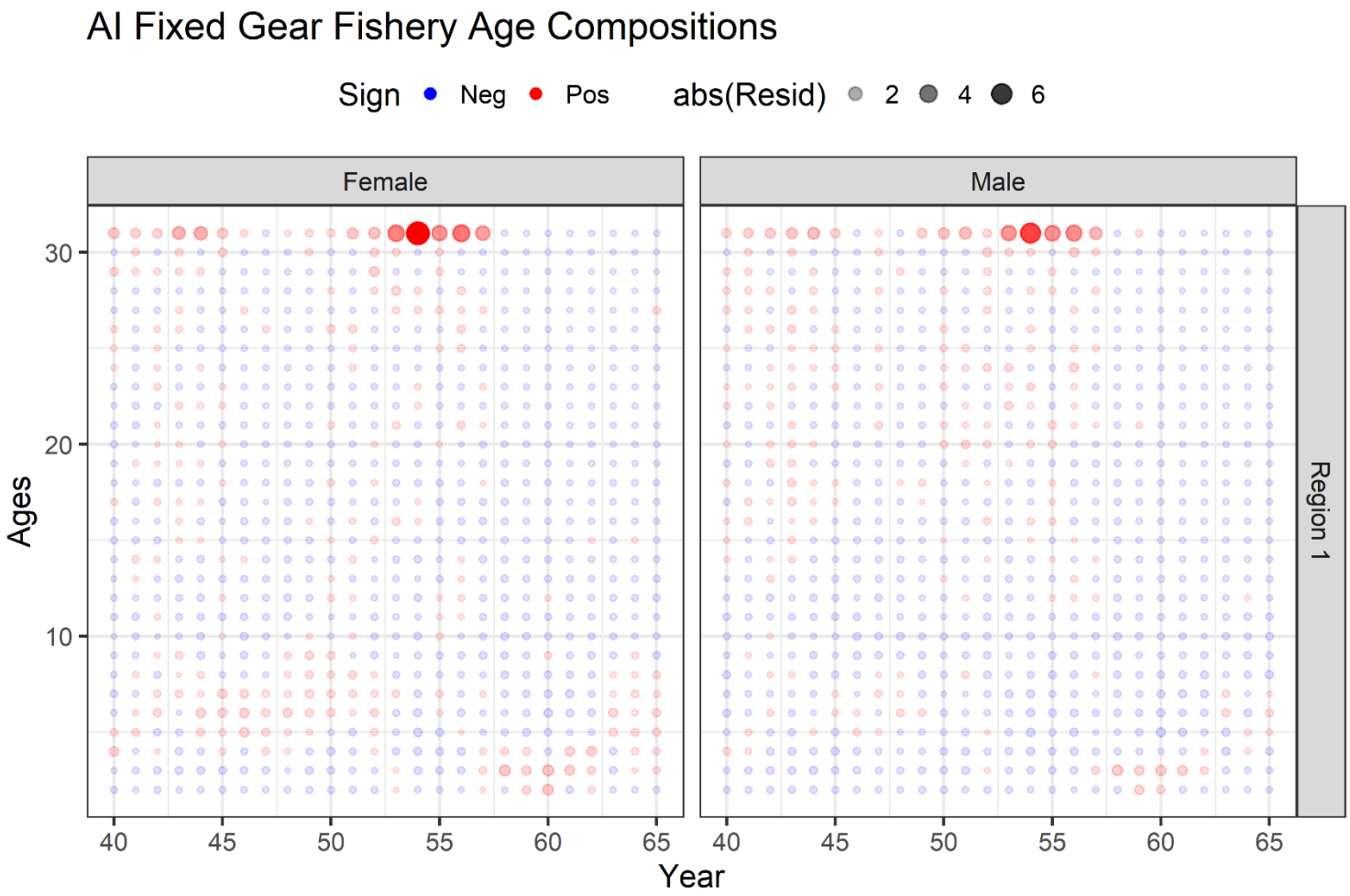

Figure SM3C.44. Fits to the AI fixed gear fishery age composition data based on Pearson residuals for model *25. 2_FAA_Fsh+Srvy*.

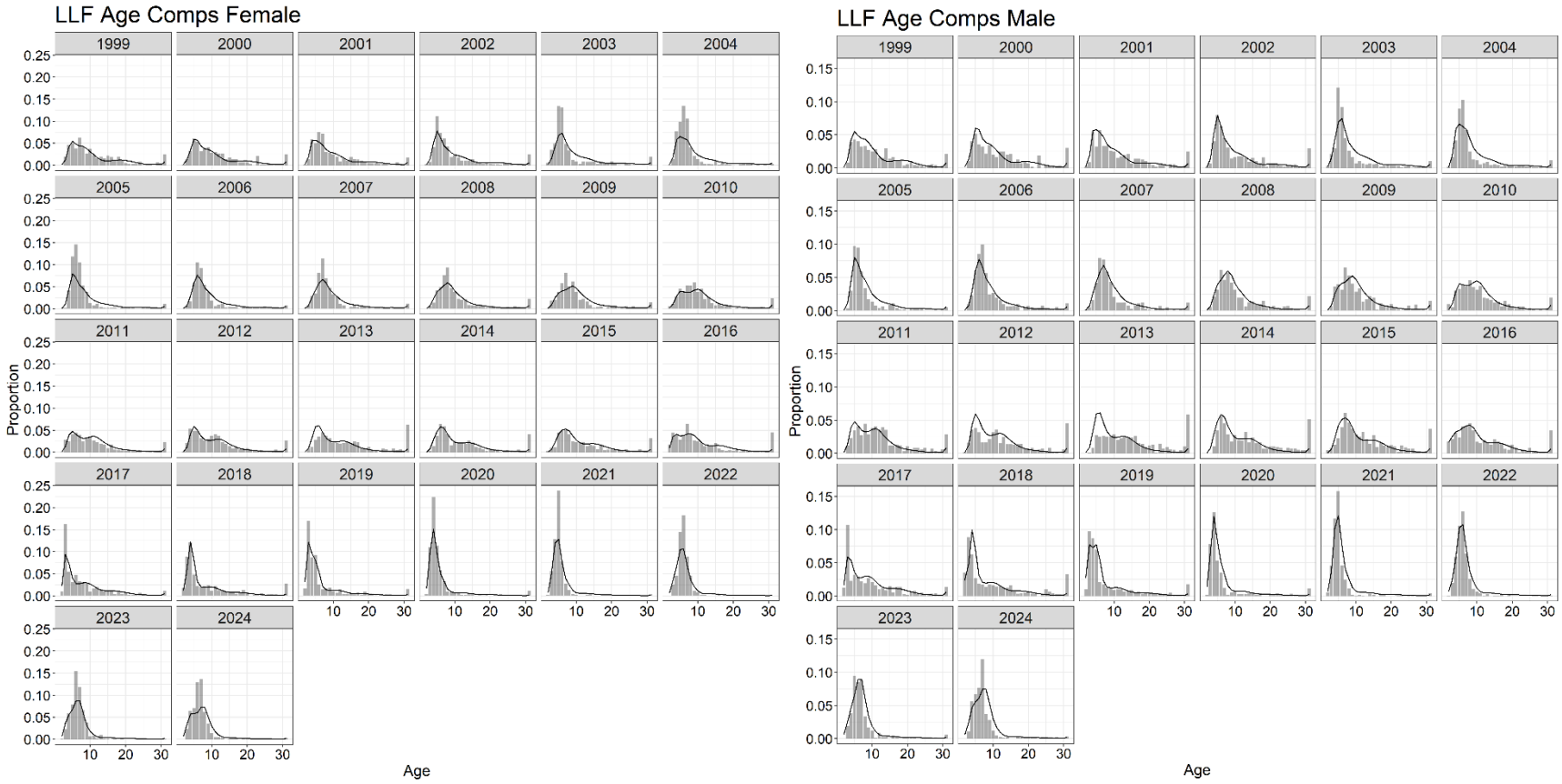

Figure SM3C.45. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

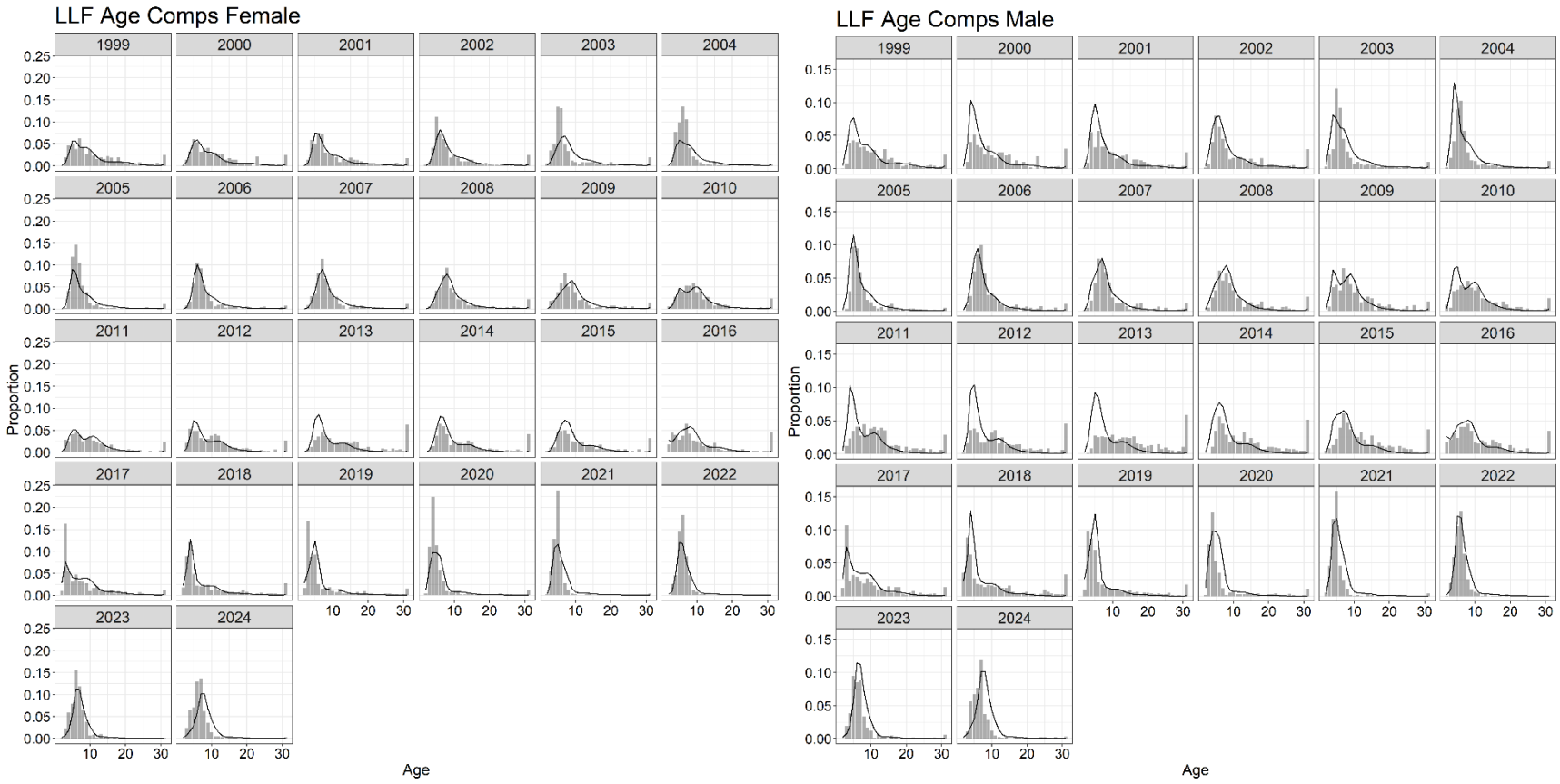

Figure SM3C.46. Fits to fixed gear fishery age composition data by year and sex (females left and males right) for model *26.2_FAA*.

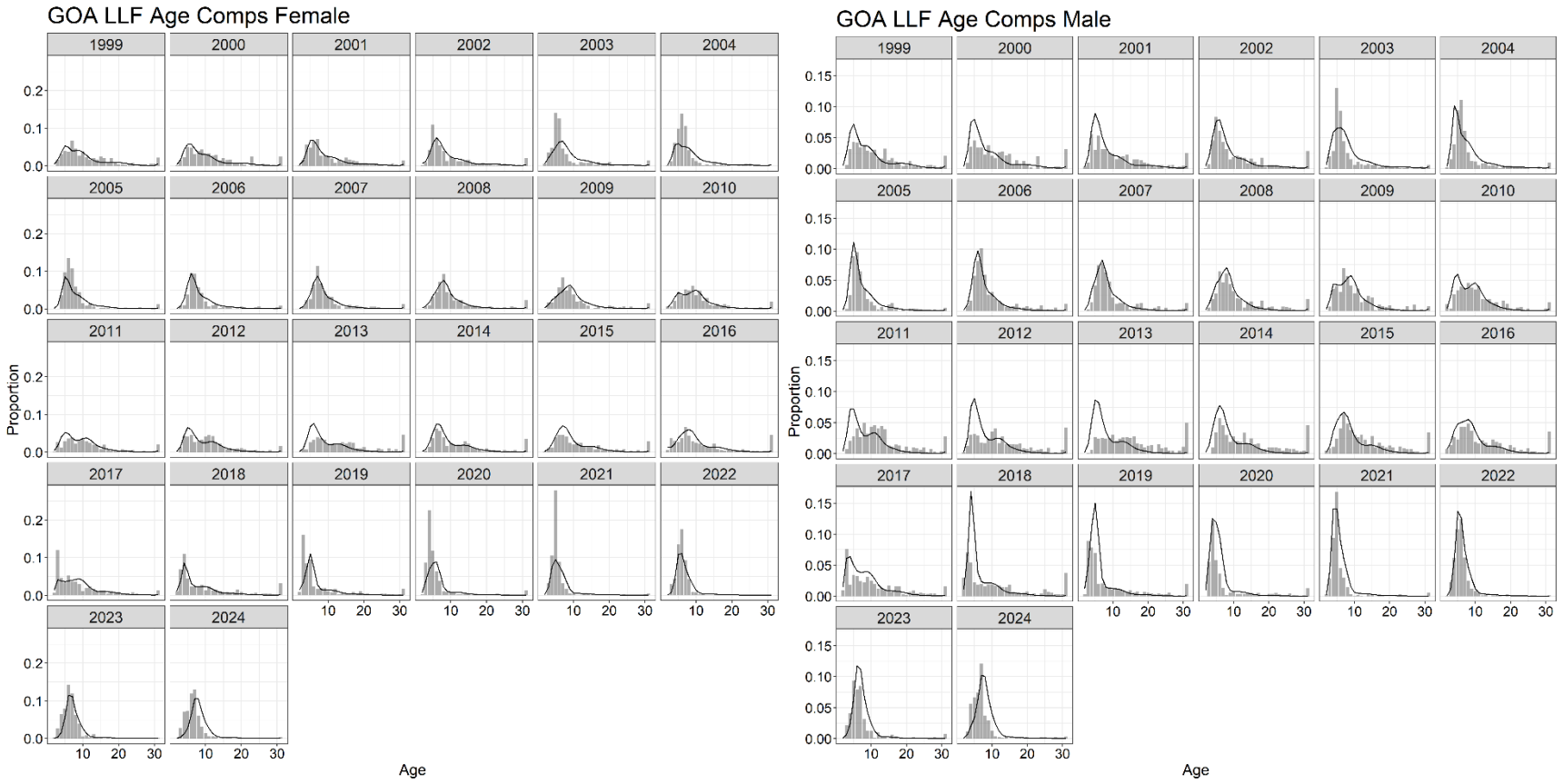

Figure SM3C.47. Fits to GOA fixed gear fishery age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

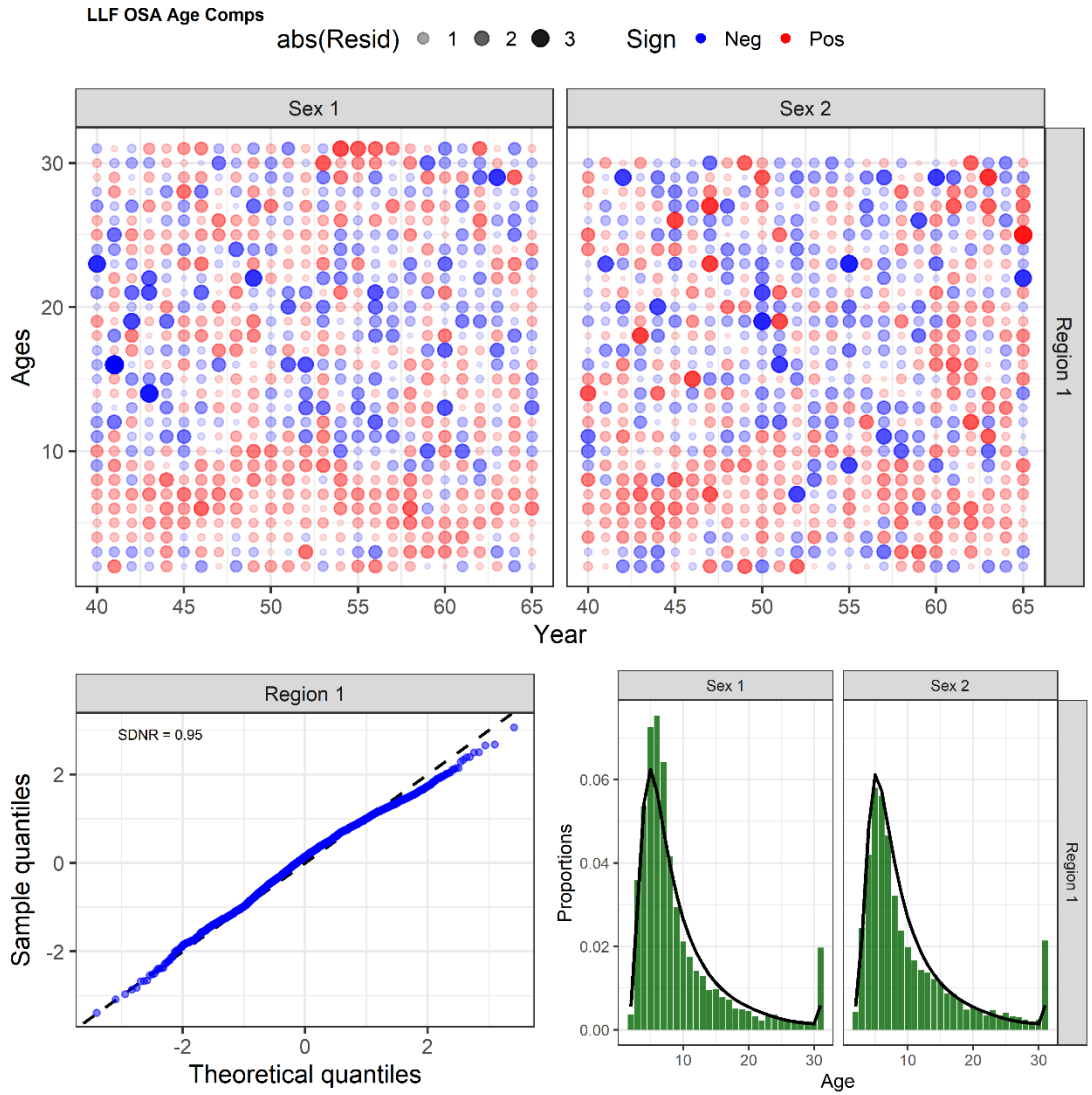

Figure SM3C.48. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

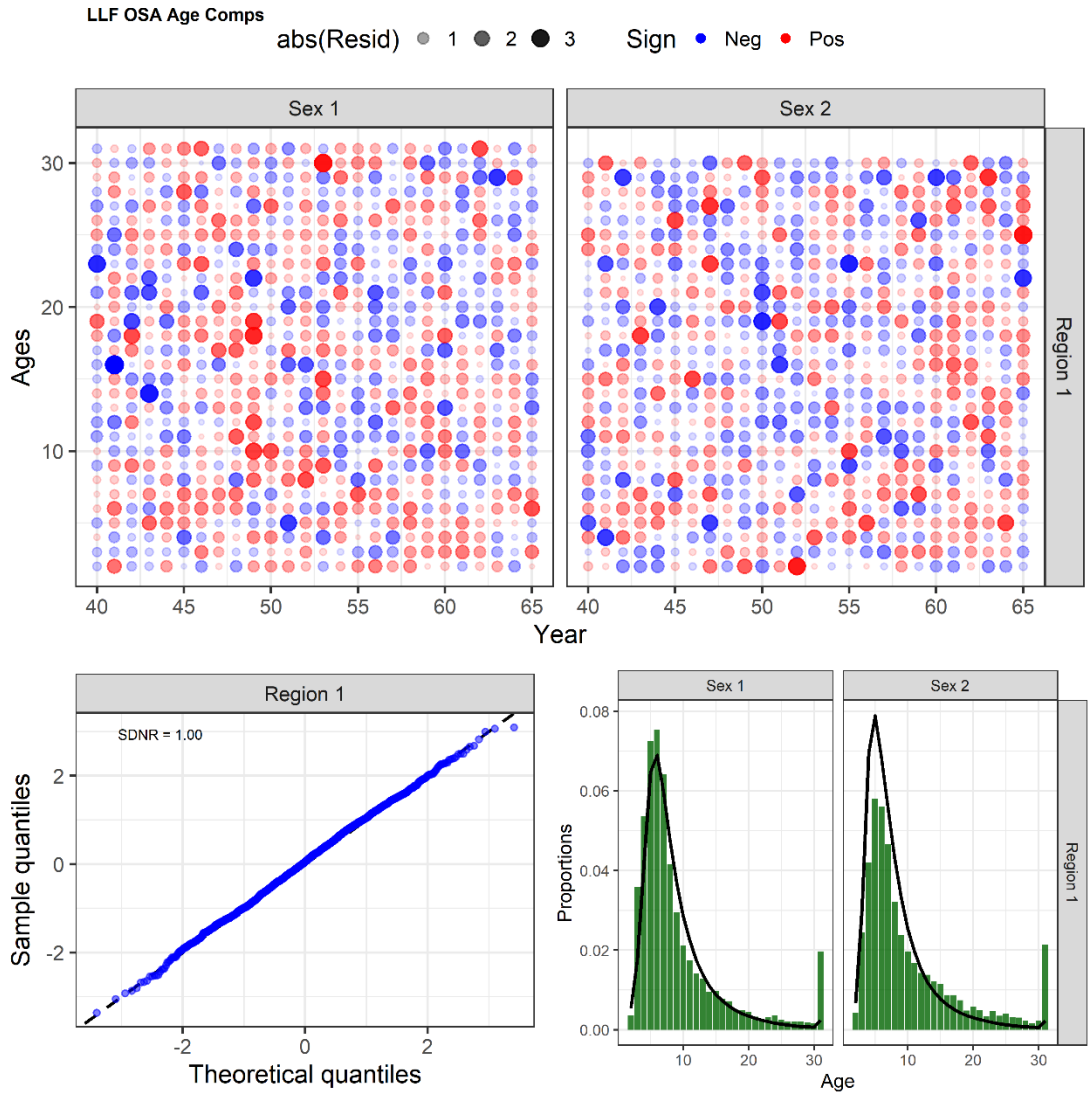

Figure SM3C.49. Fits to fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

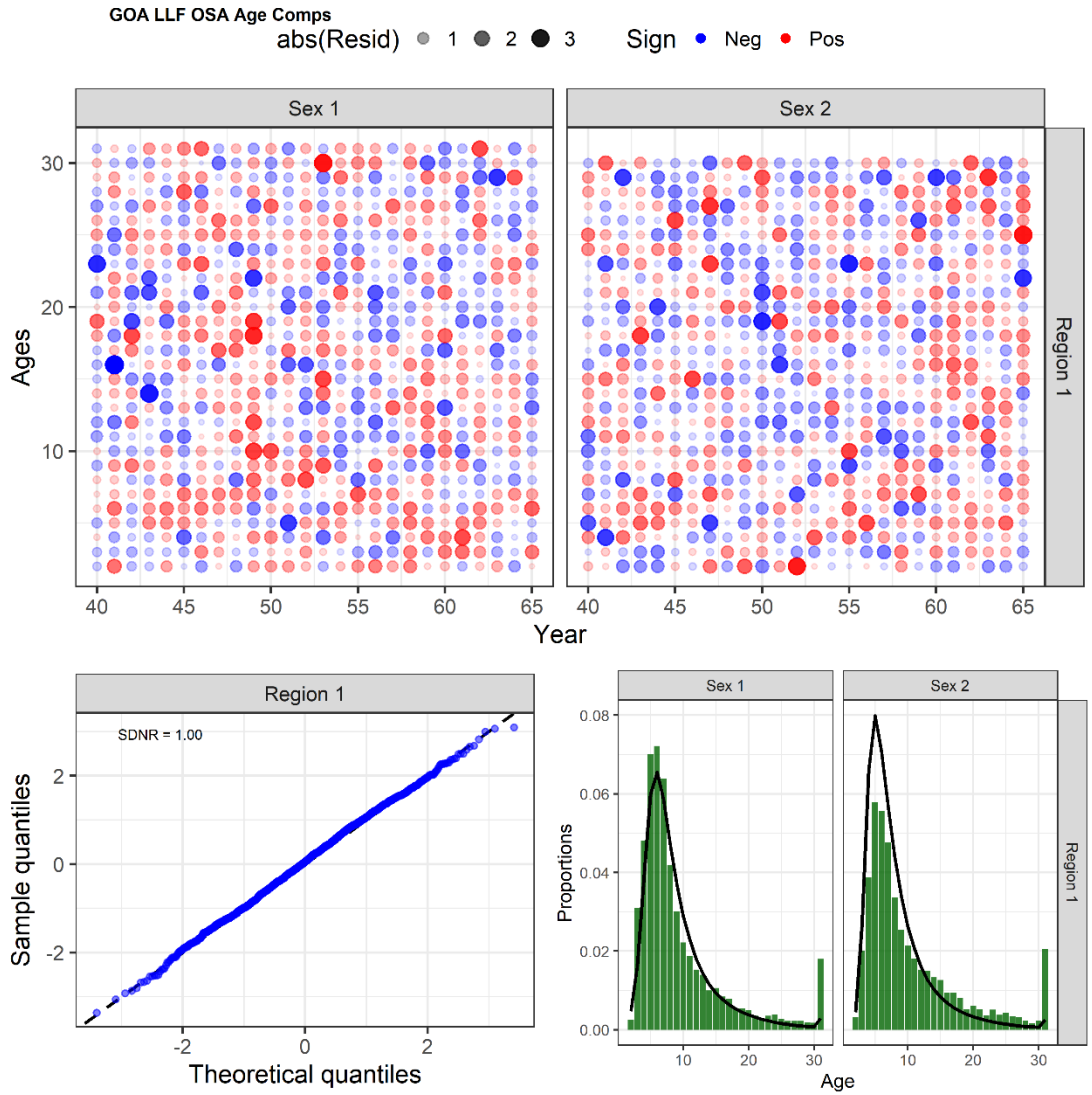

Figure SM3C.50. Fits to GOA fixed gear fishery age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

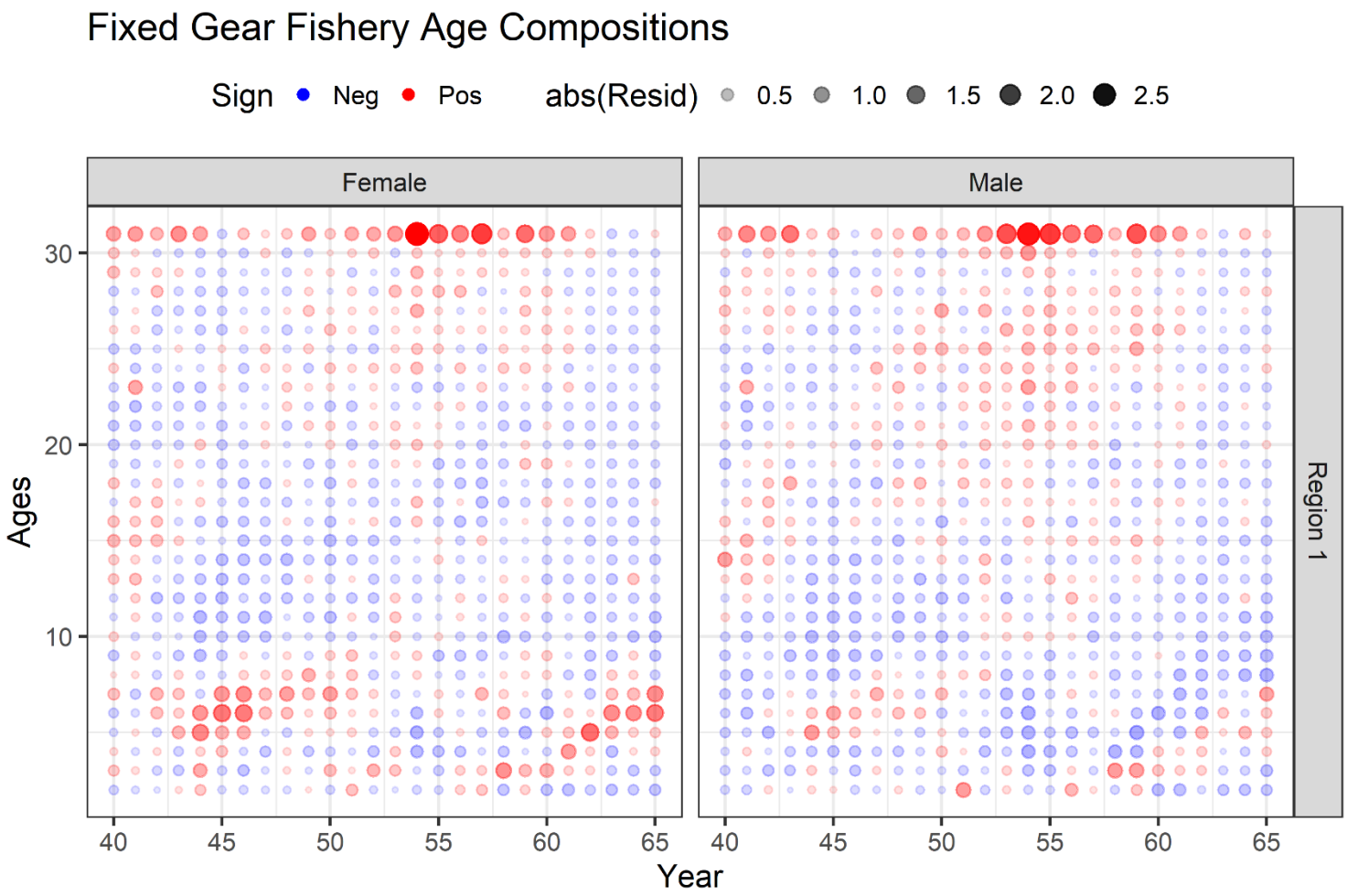

Figure SM3C.51. Fits to the fixed gear fishery age composition data based on Pearson residuals for model *26.1_1Reg*.

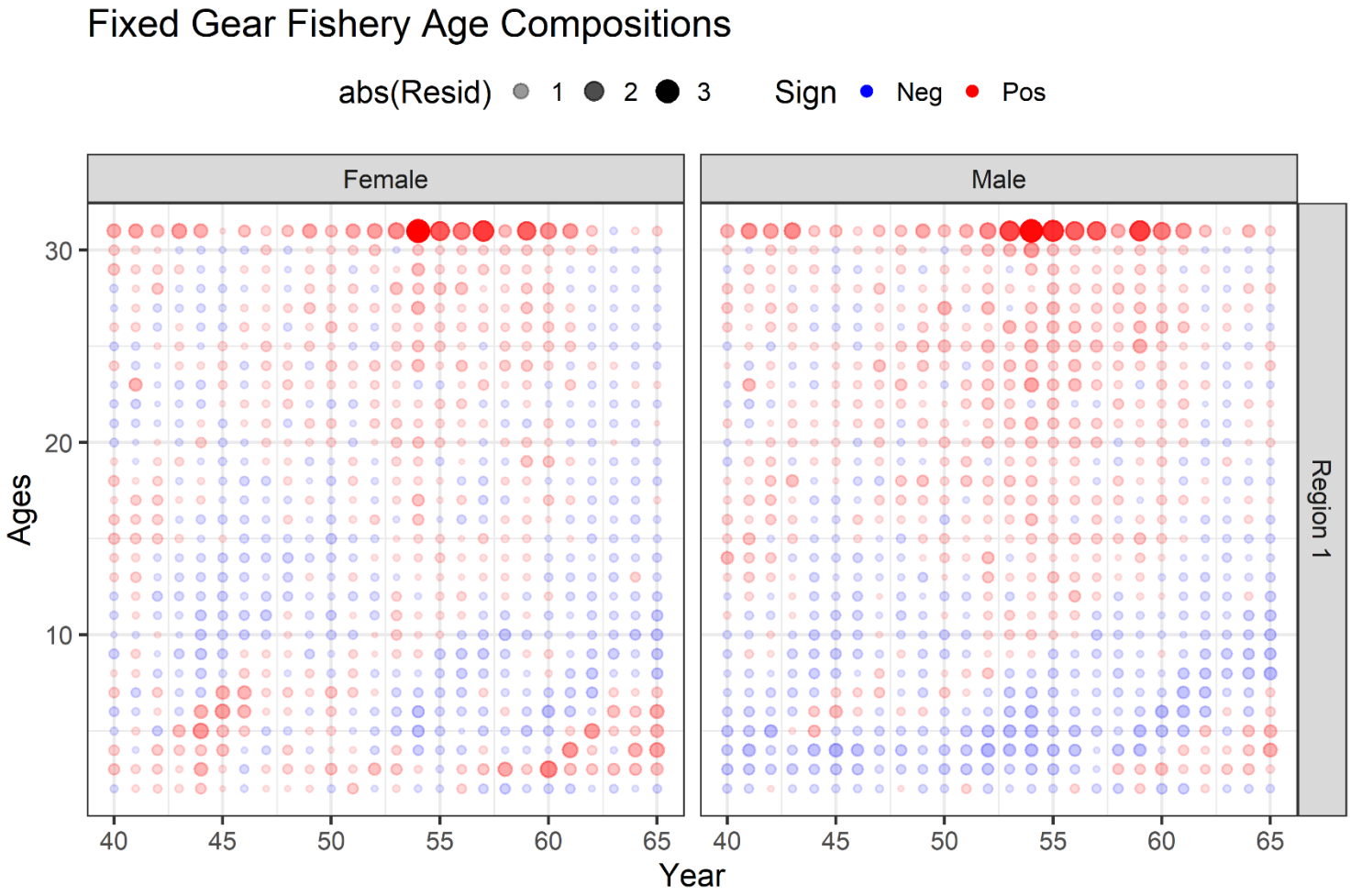

Figure SM3C.52. Fits to the fixed gear fishery age composition data based on Pearson residuals for model *25. 2_FAA*.

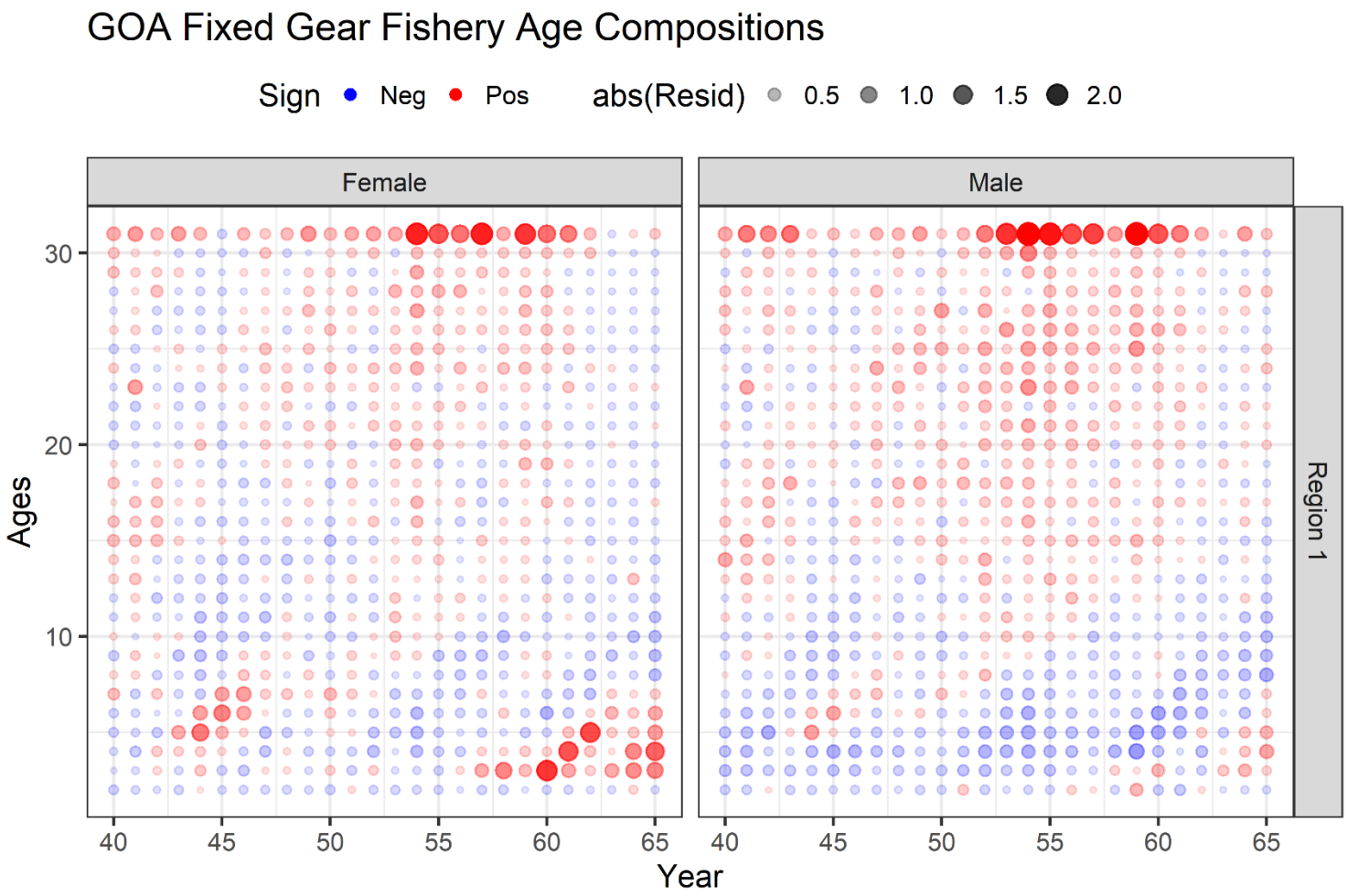

Figure SM3C.53. Fits to the GOA fixed gear fishery age composition data based on Pearson residuals for model *25. 2_FAA_Fsh+Srvy*.

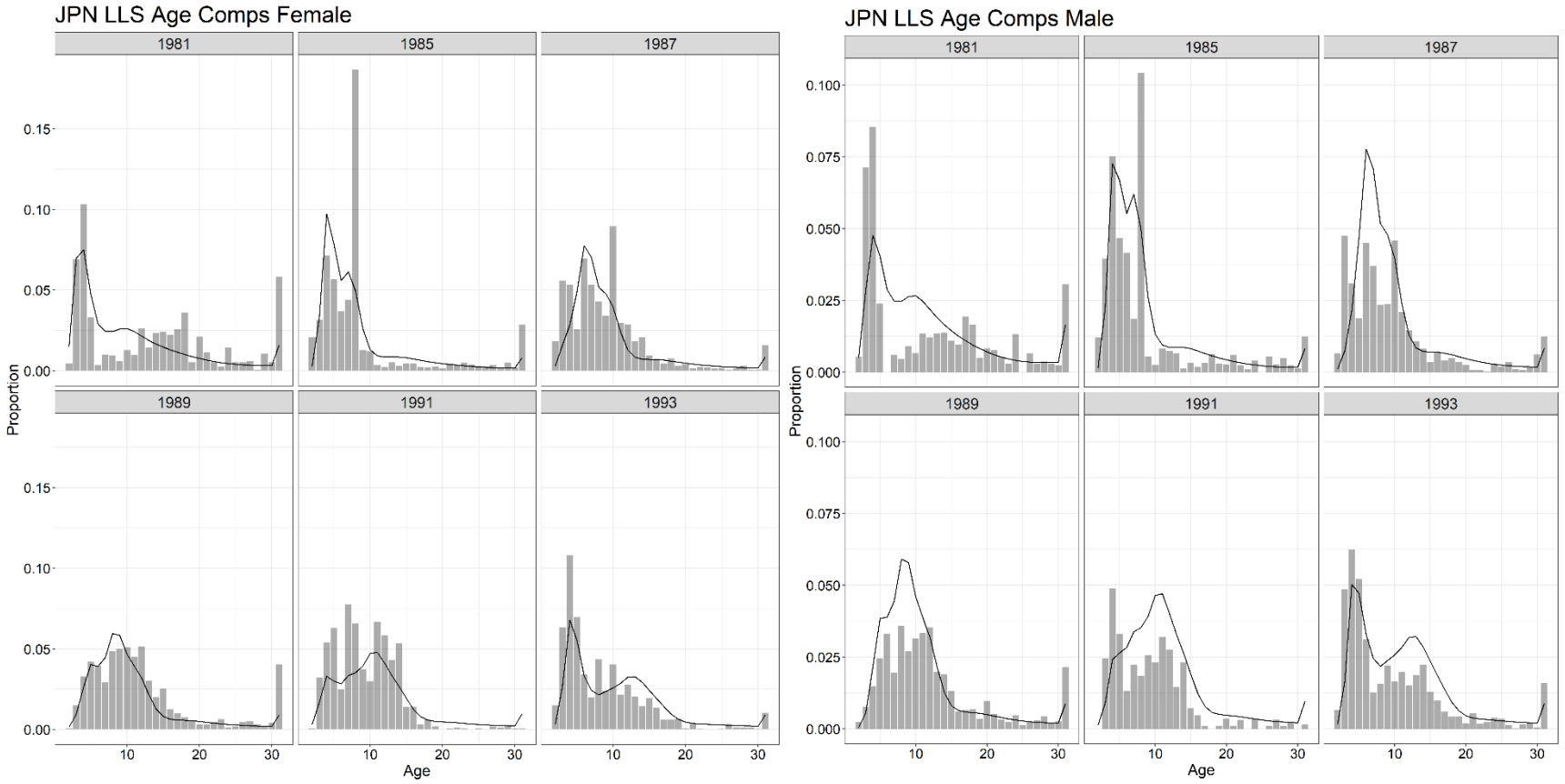

Figure SM3C.54. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *26.1_1Reg*.

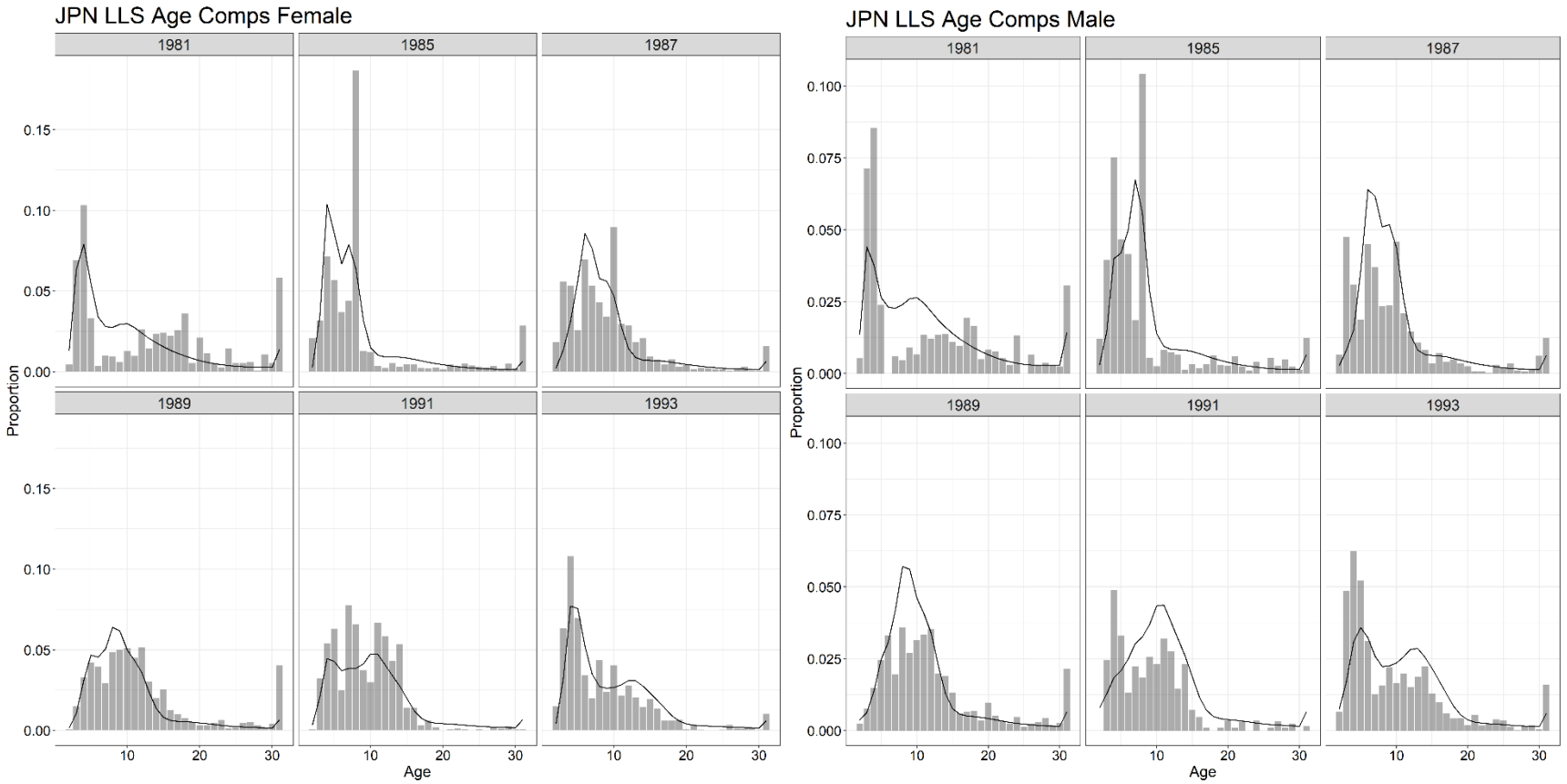

Figure SM3C.55. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *26.2_FAA*.

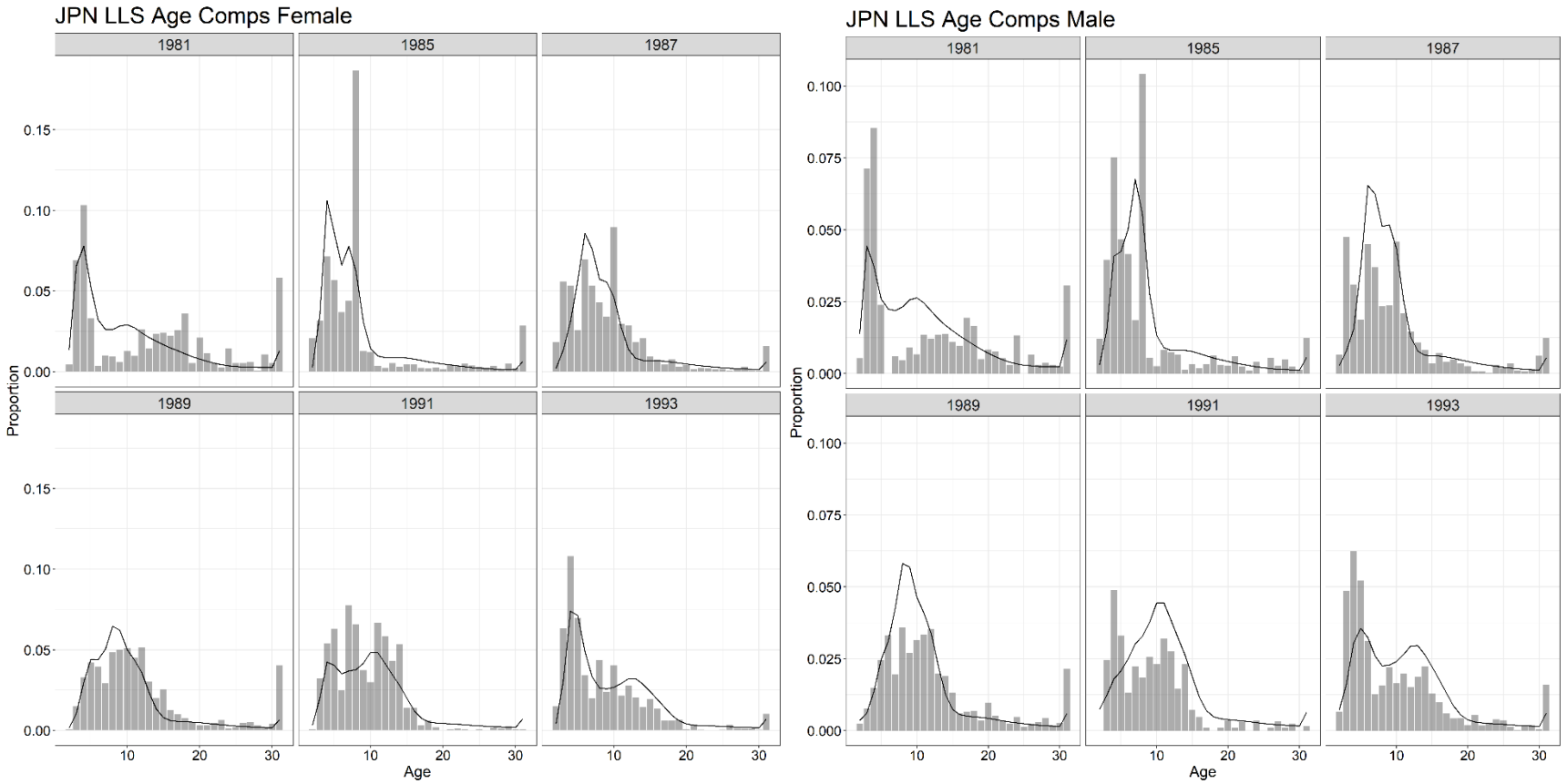

Figure SM3C.56. Fits to Japanese Longline Survey age composition data by year and sex (females left and males right) for model *26.2a_FAA_Fsh+Srvy*.

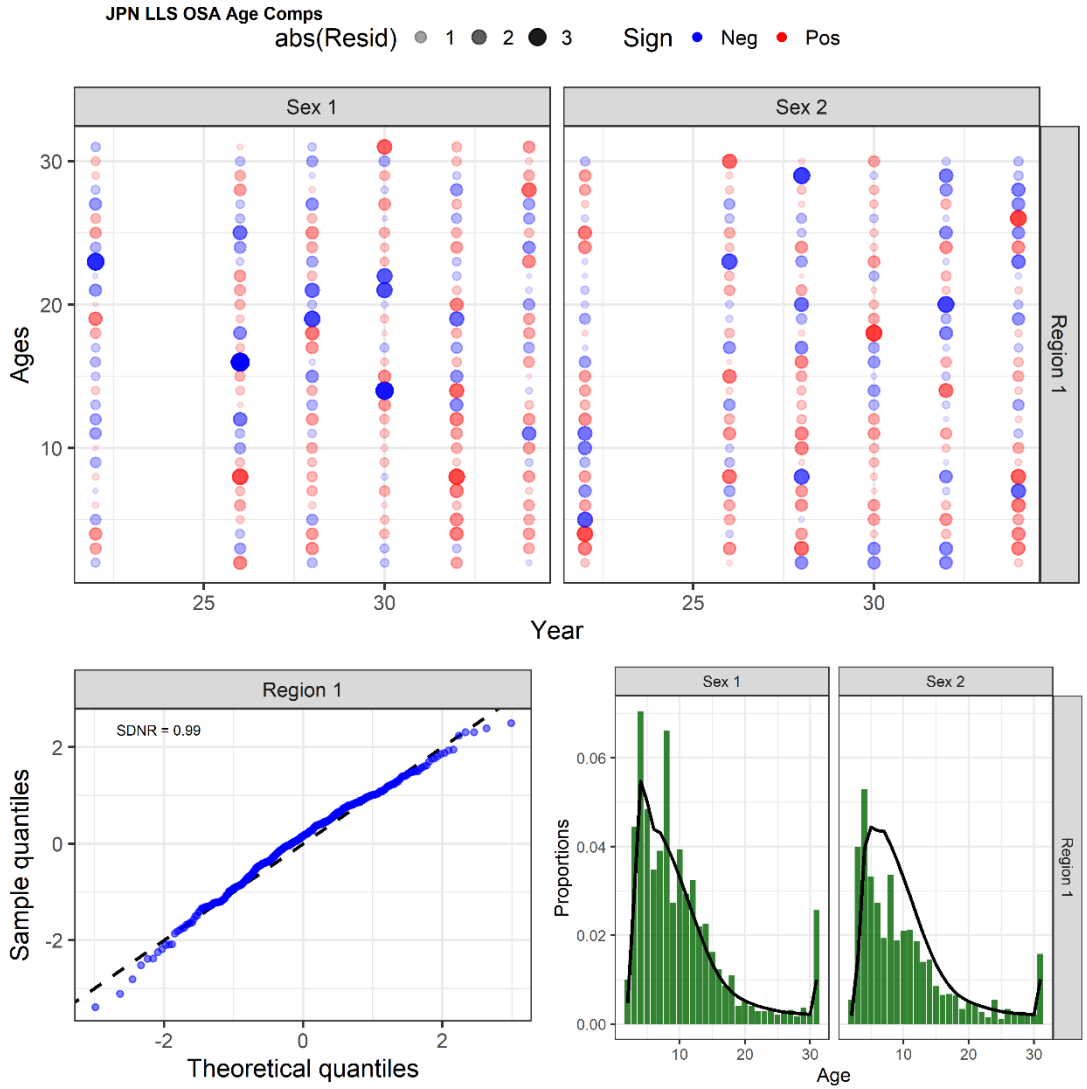

Figure SM3C.57. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

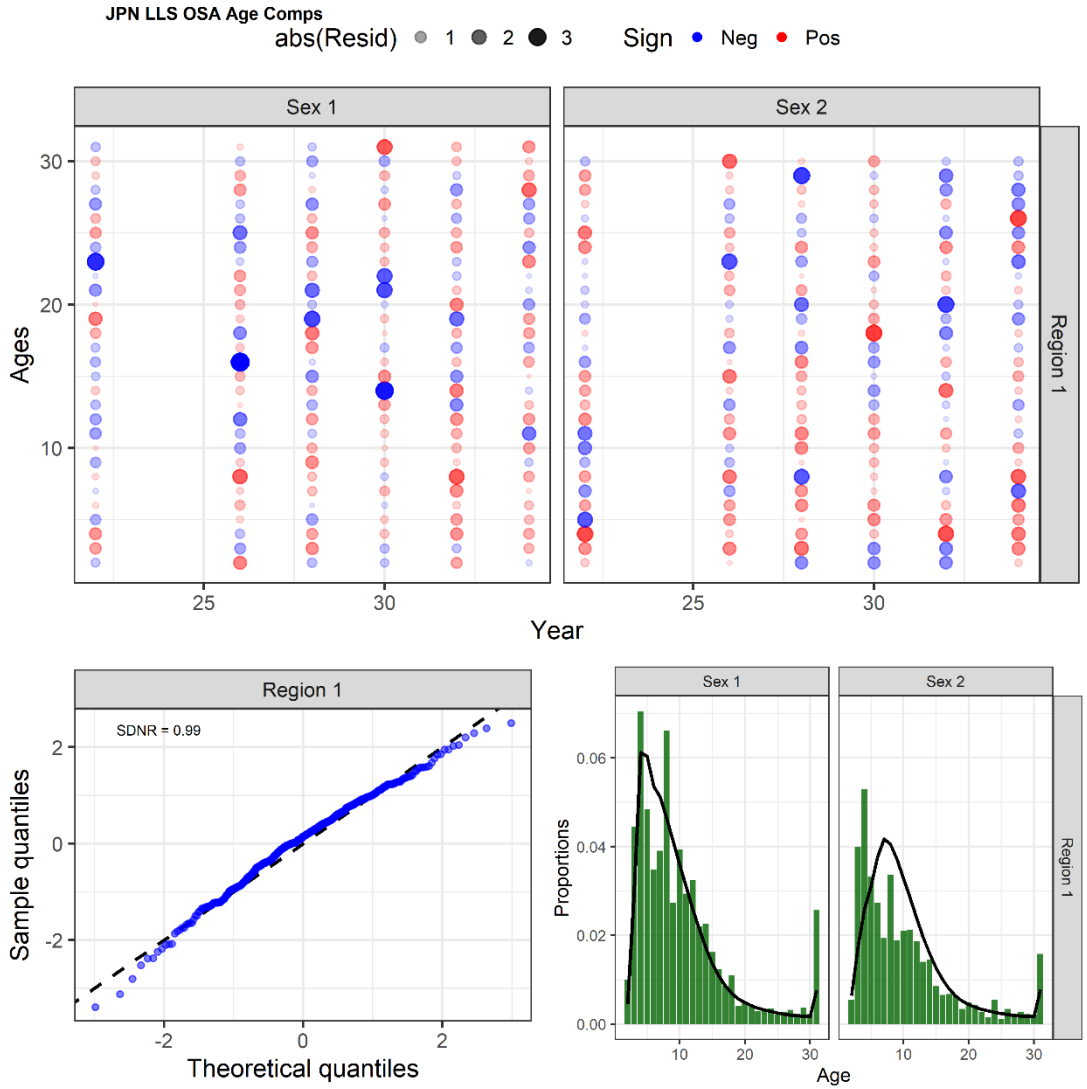

Figure SM3C.58. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

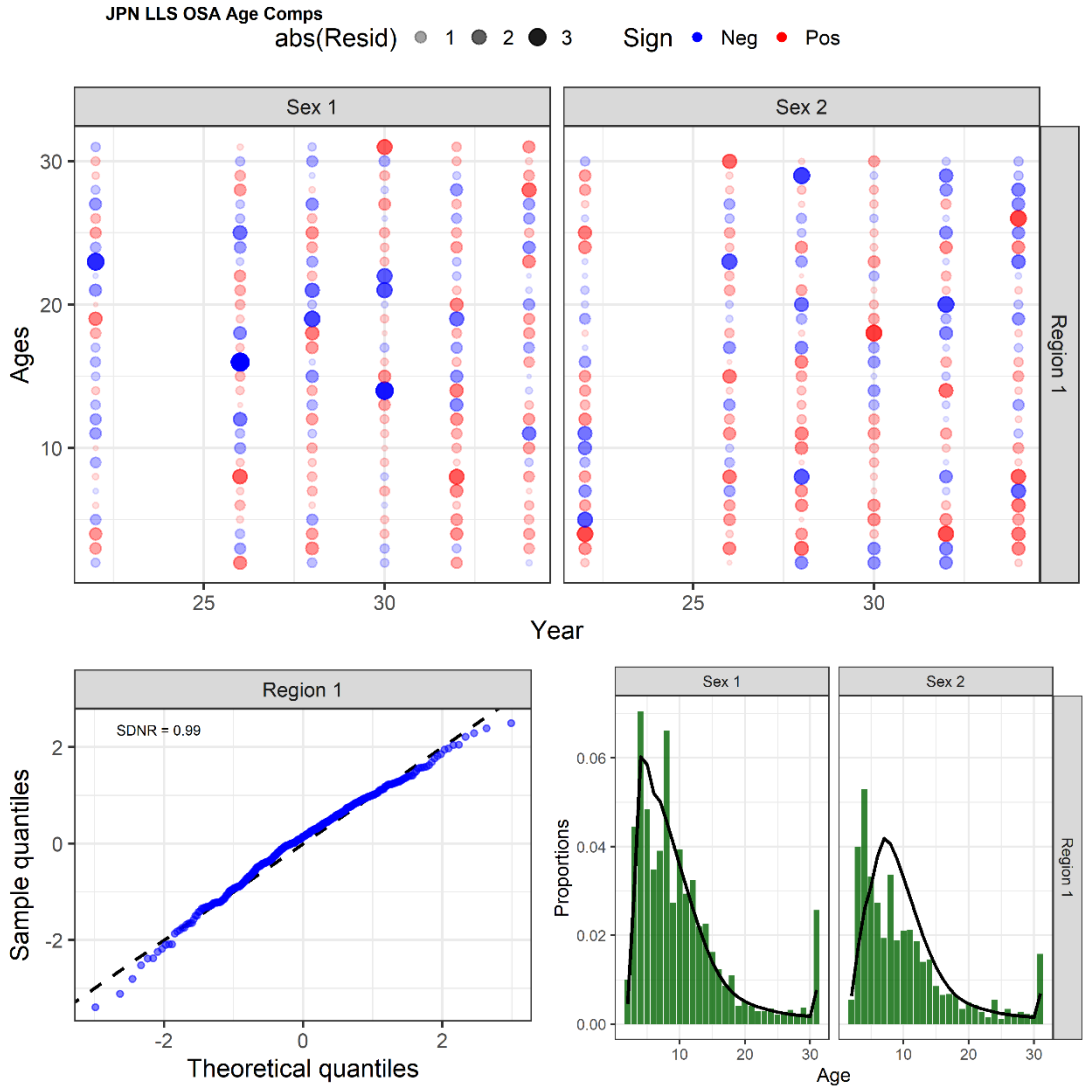

Figure SM3C.59. Fits to Japanese longline survey age composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

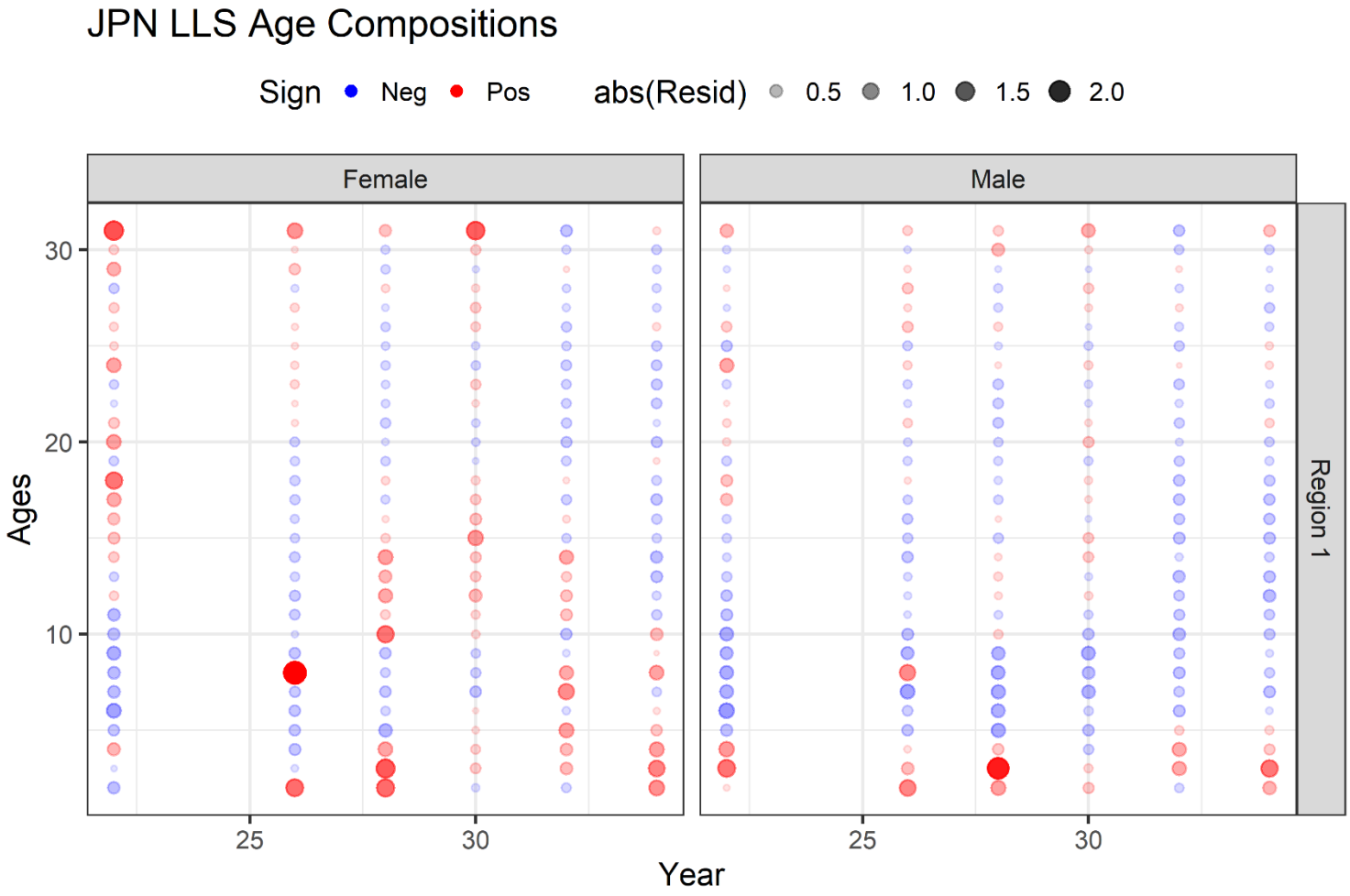

Figure SM3B.60. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for model *26.1_1Reg*.

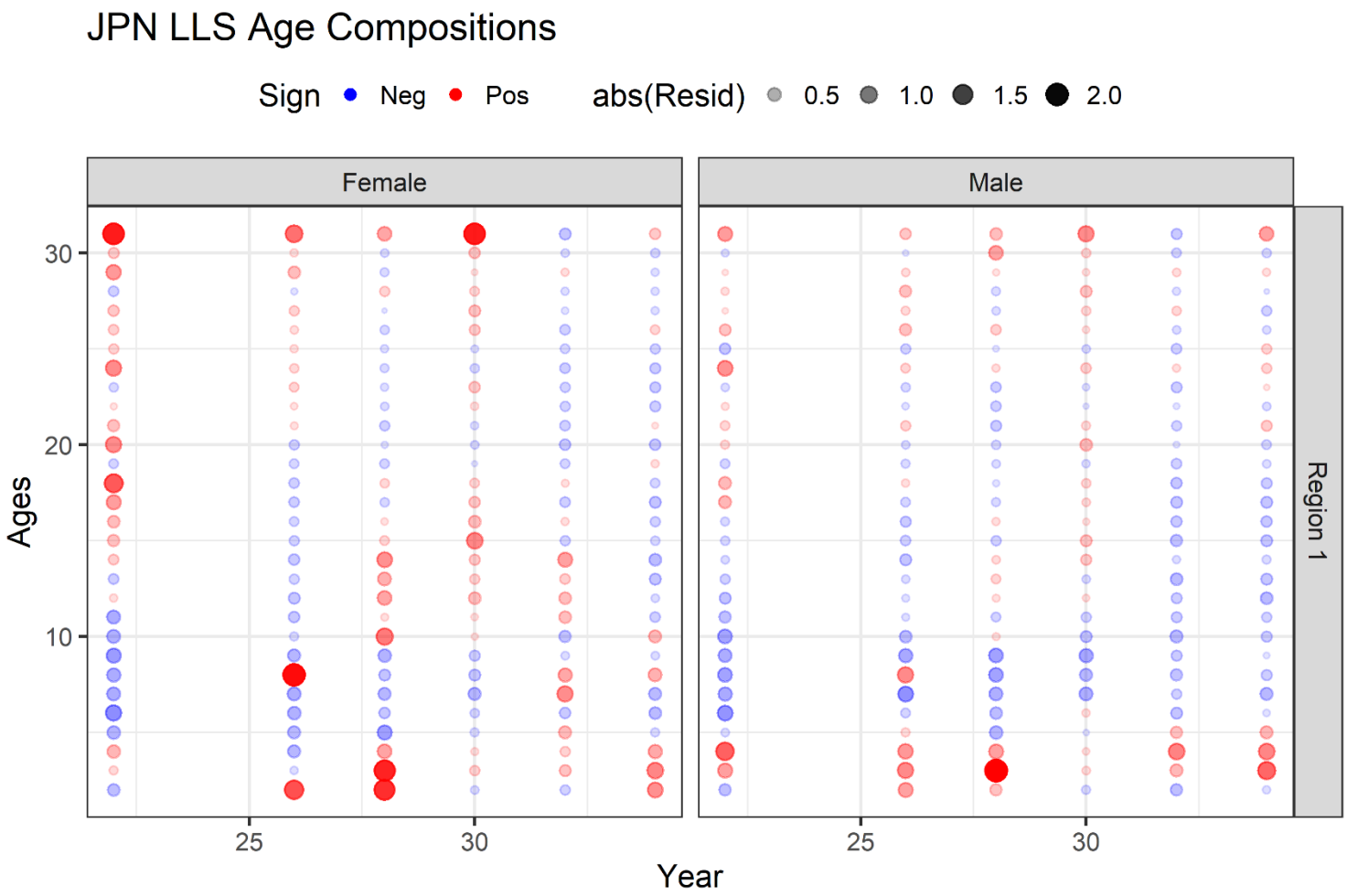

Figure SM3B.61. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for model *26.2_FAA*.

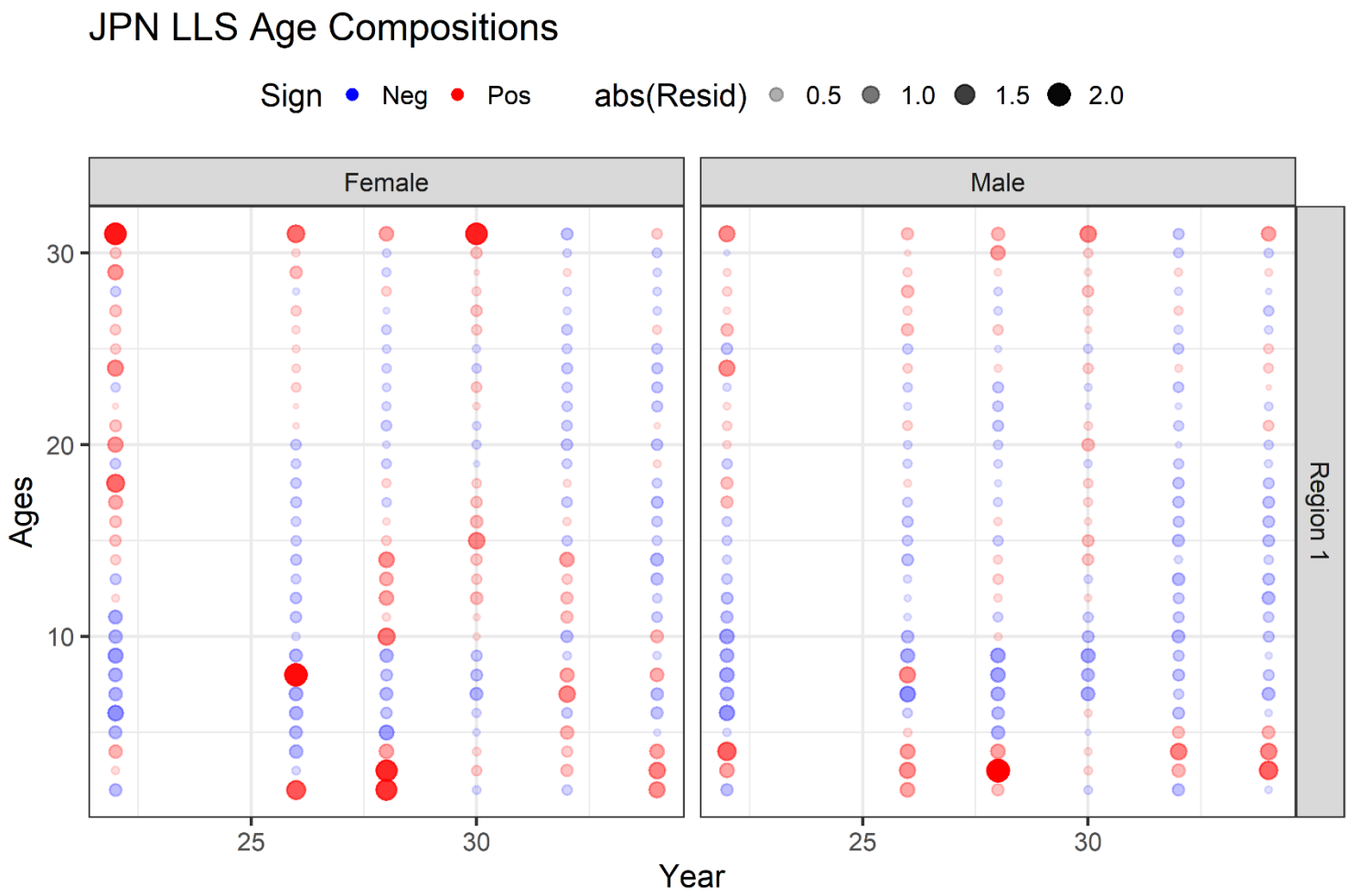

Figure SM3B.62. Fits to the Japanese Longline Survey age composition data based on Pearson residuals for model *26.2a_FAA_Fsh+Srvy*.

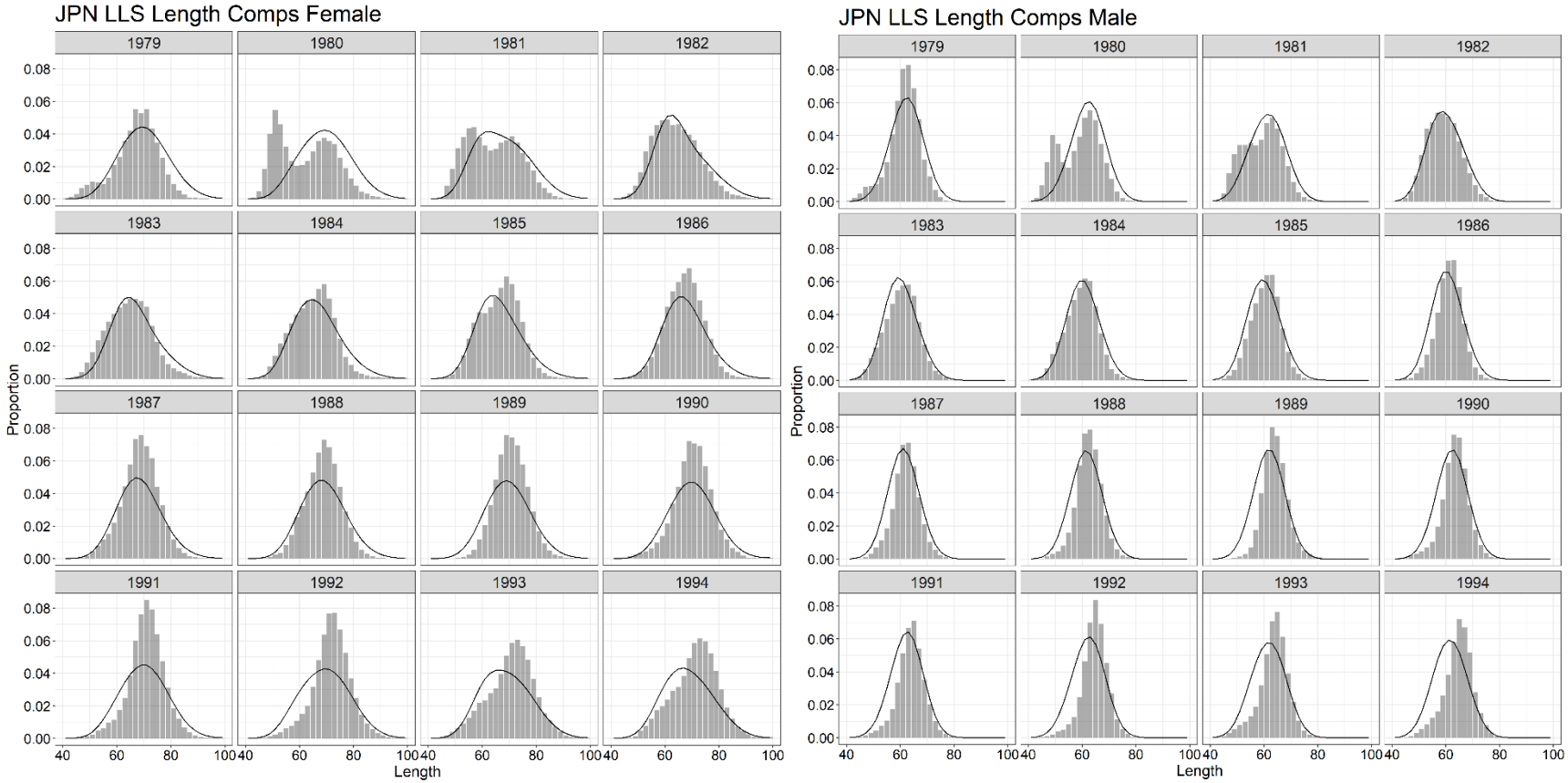

Figure SM3C.63. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

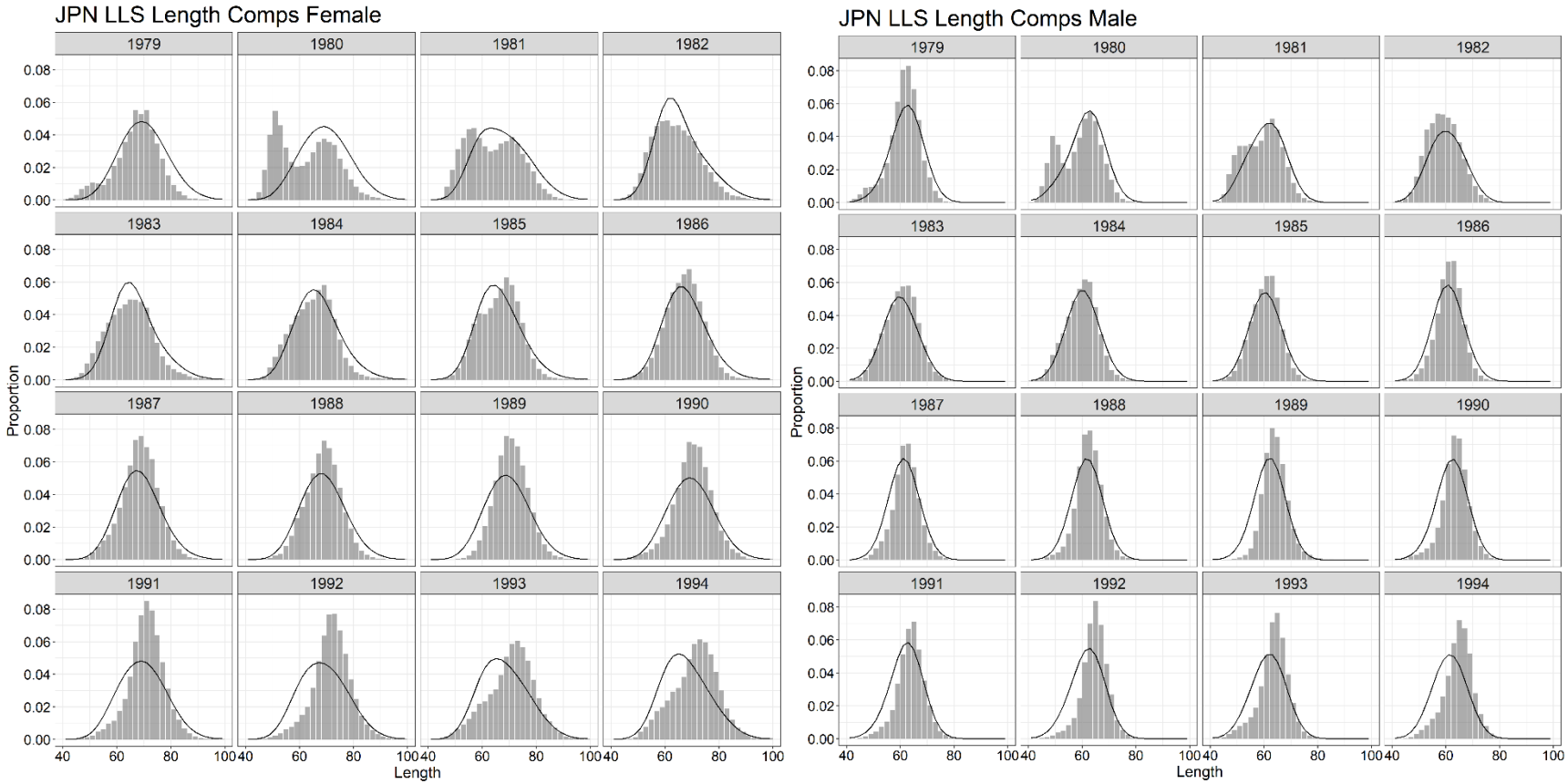

Figure SM3C.64. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.2_FAA*.

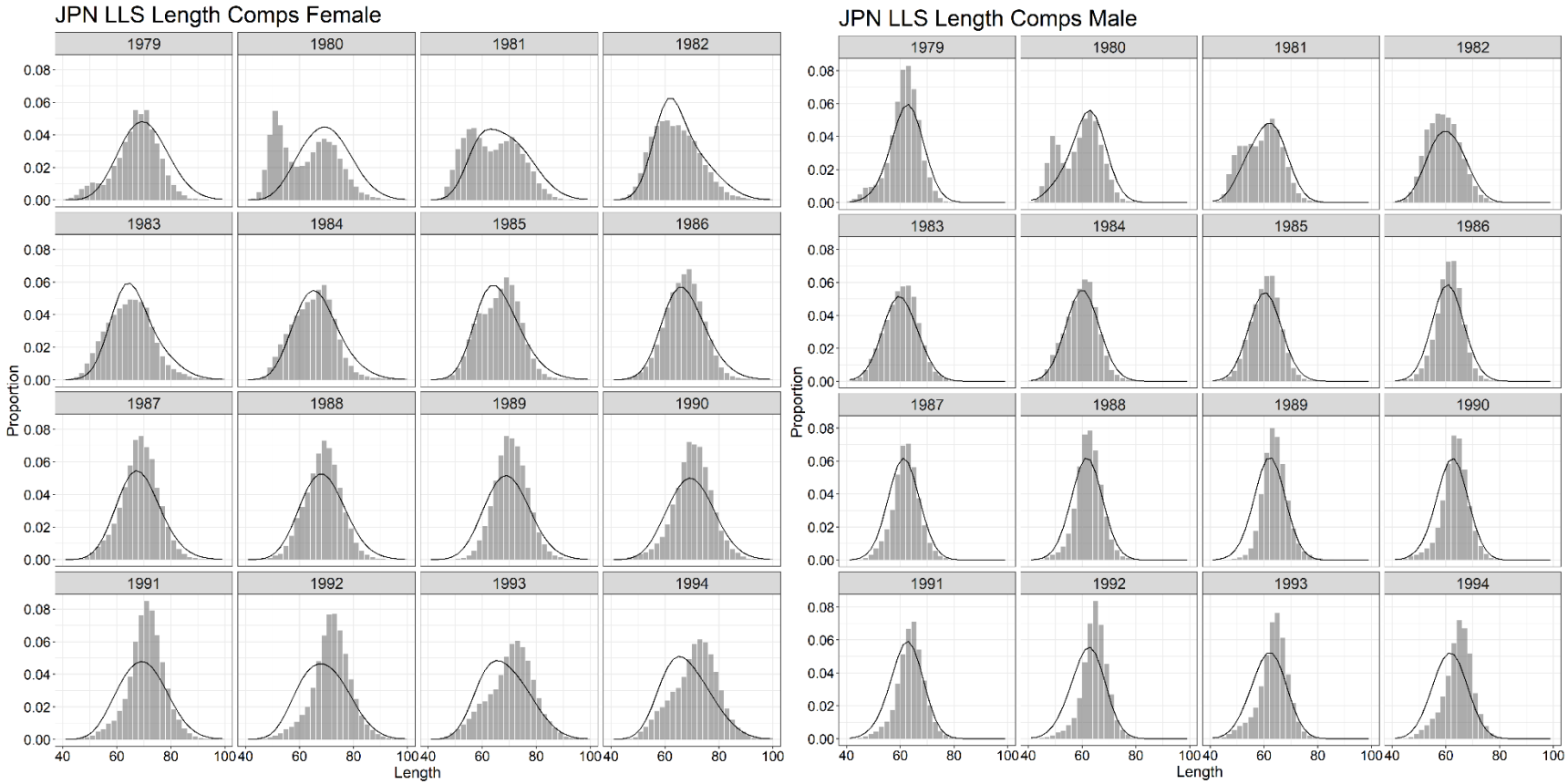

Figure SM3C.65. Fits to Japanese Longline Survey length composition data by year and sex (females left and males right) for Model *26.2a_FAA_Fsh+Srvy*.

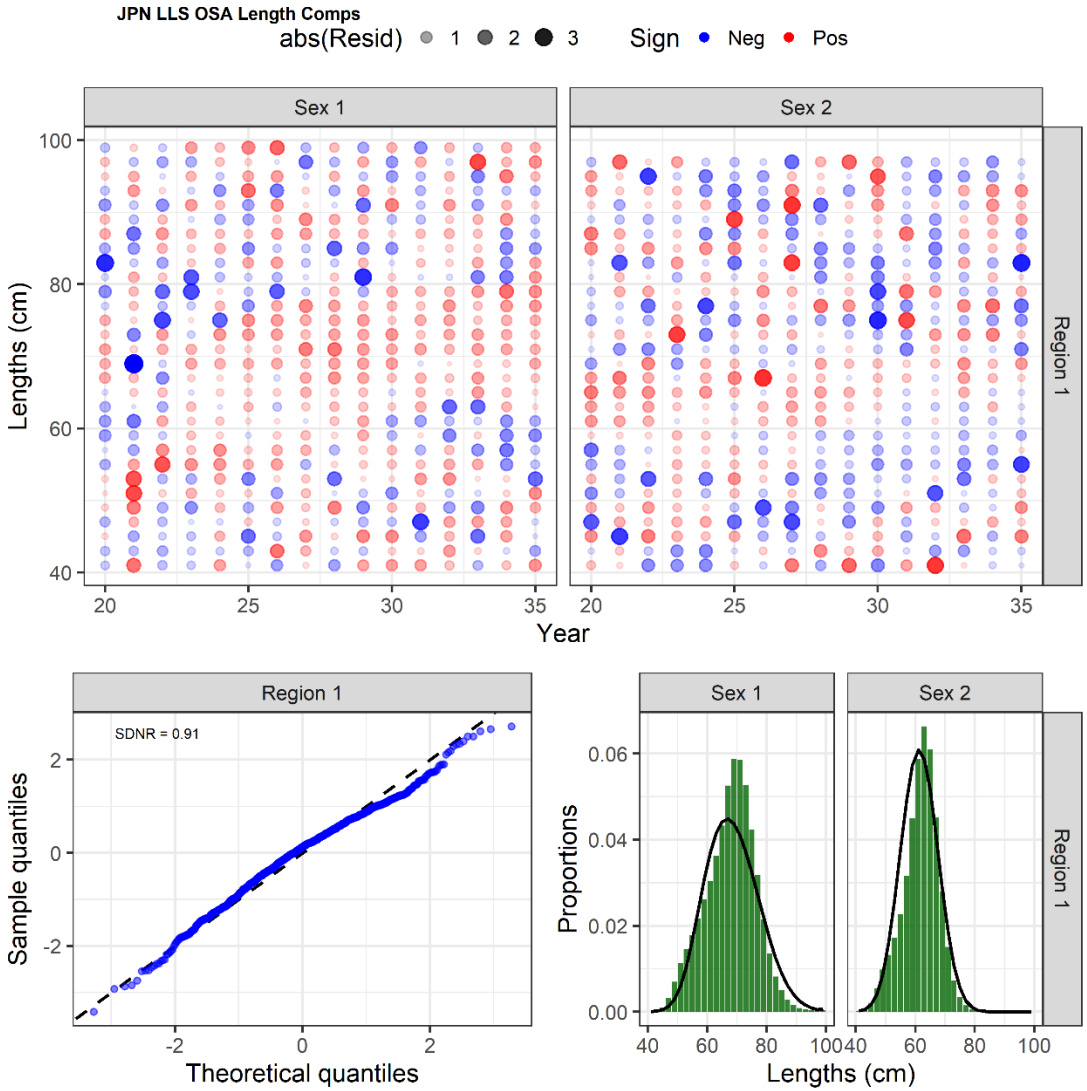

Figure SM3C.66. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

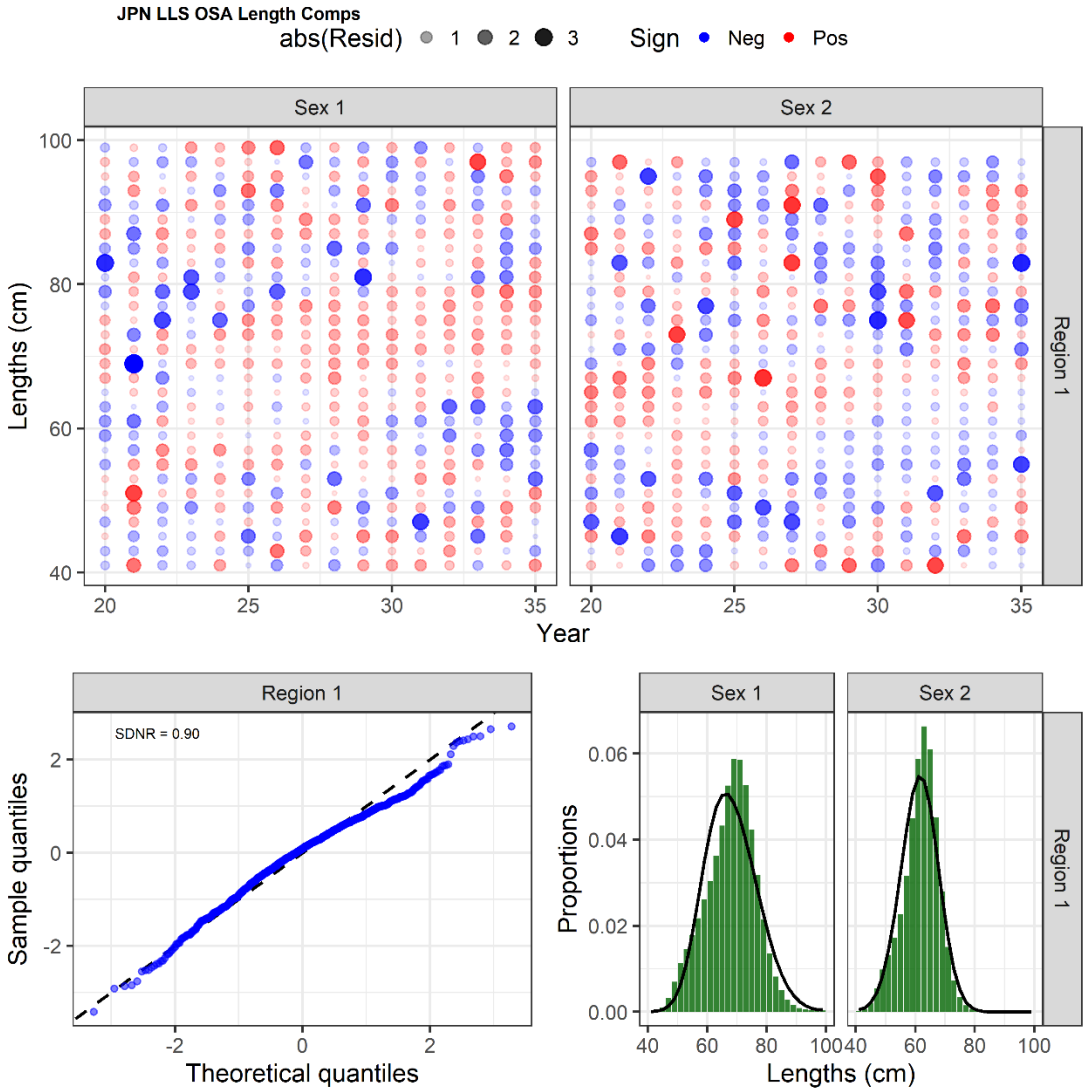

Figure SM3C.67. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

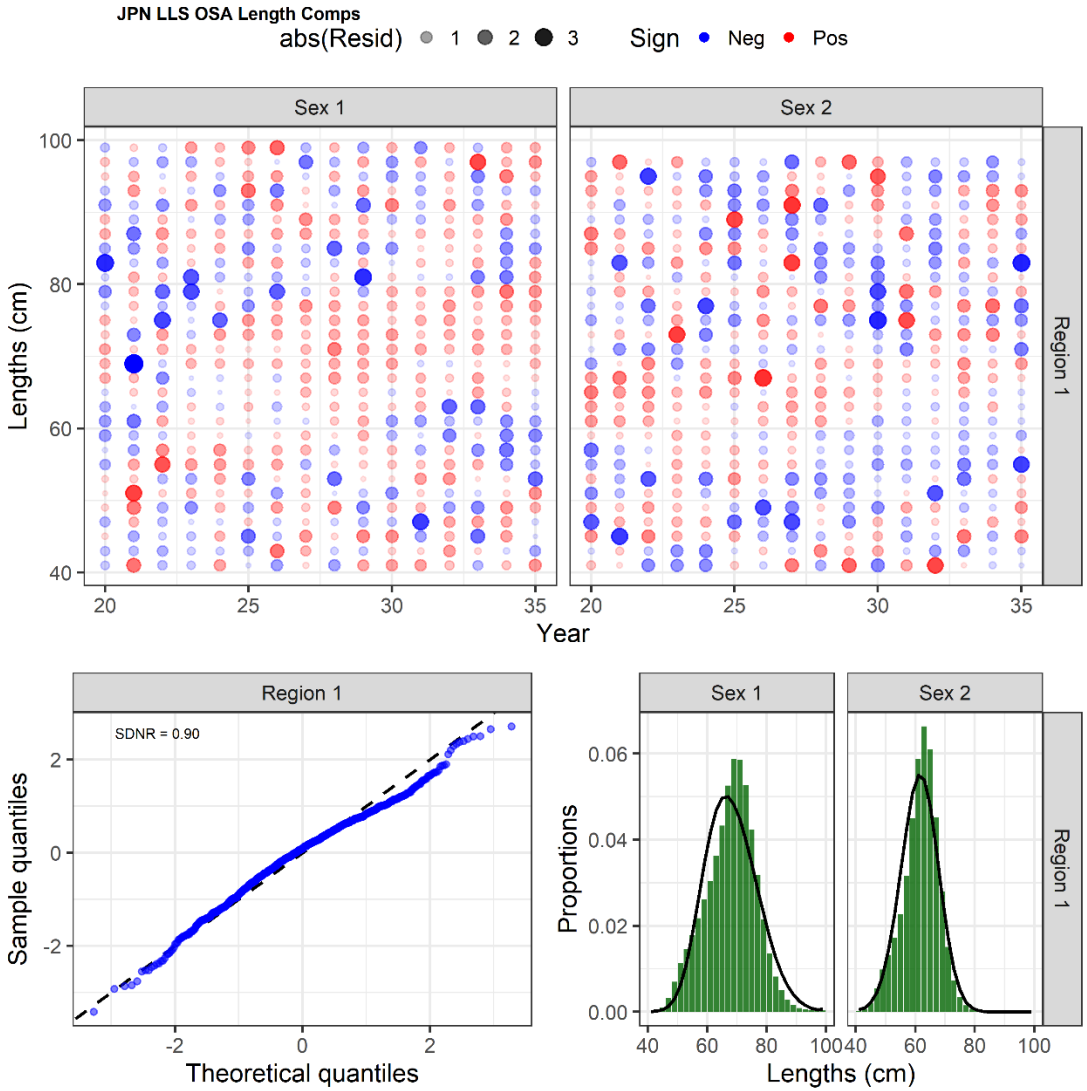

Figure SM3C.68. Fits to Japanese longline survey length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

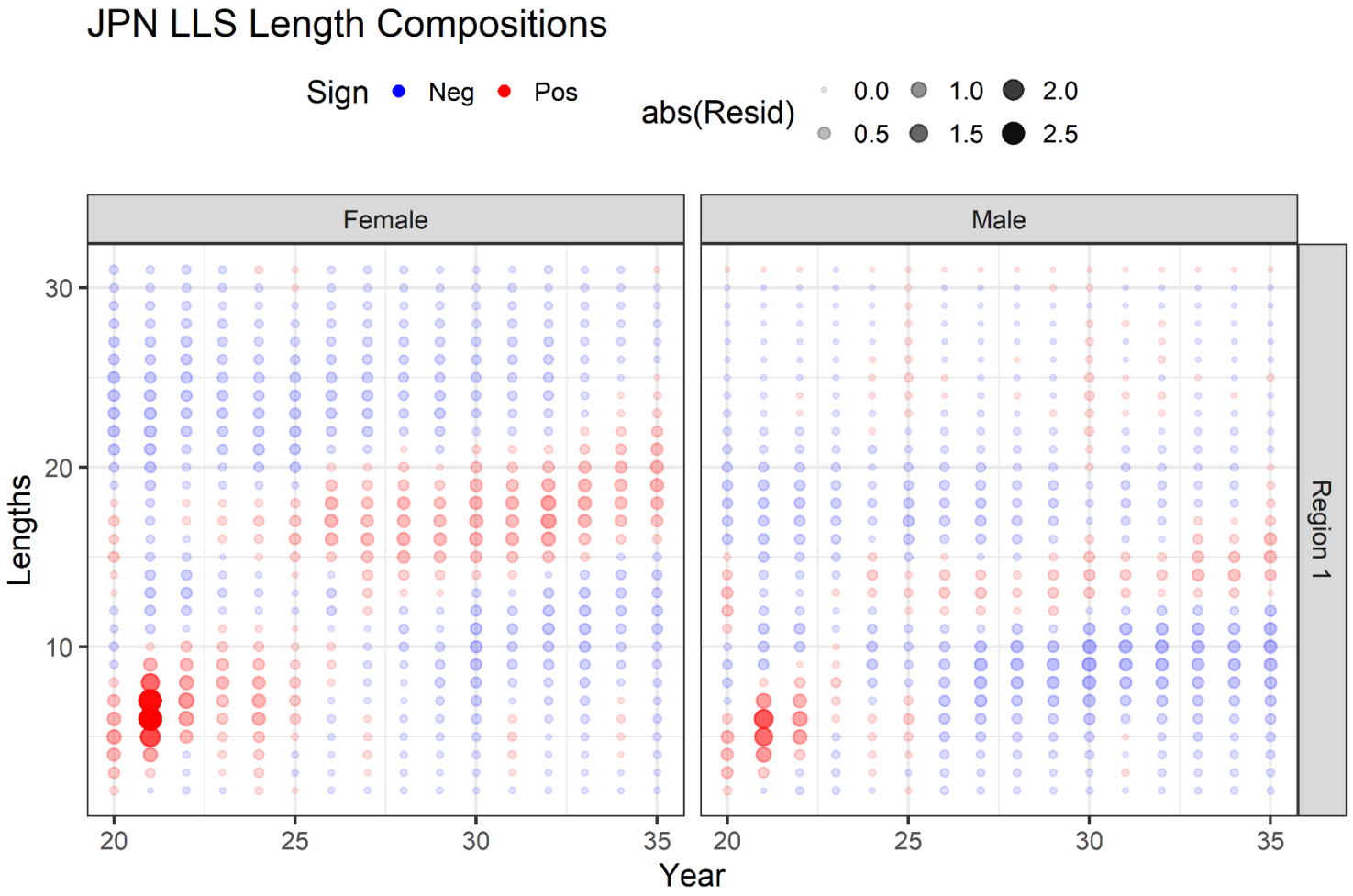

Figure SM3C.69. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.1_1Reg*.

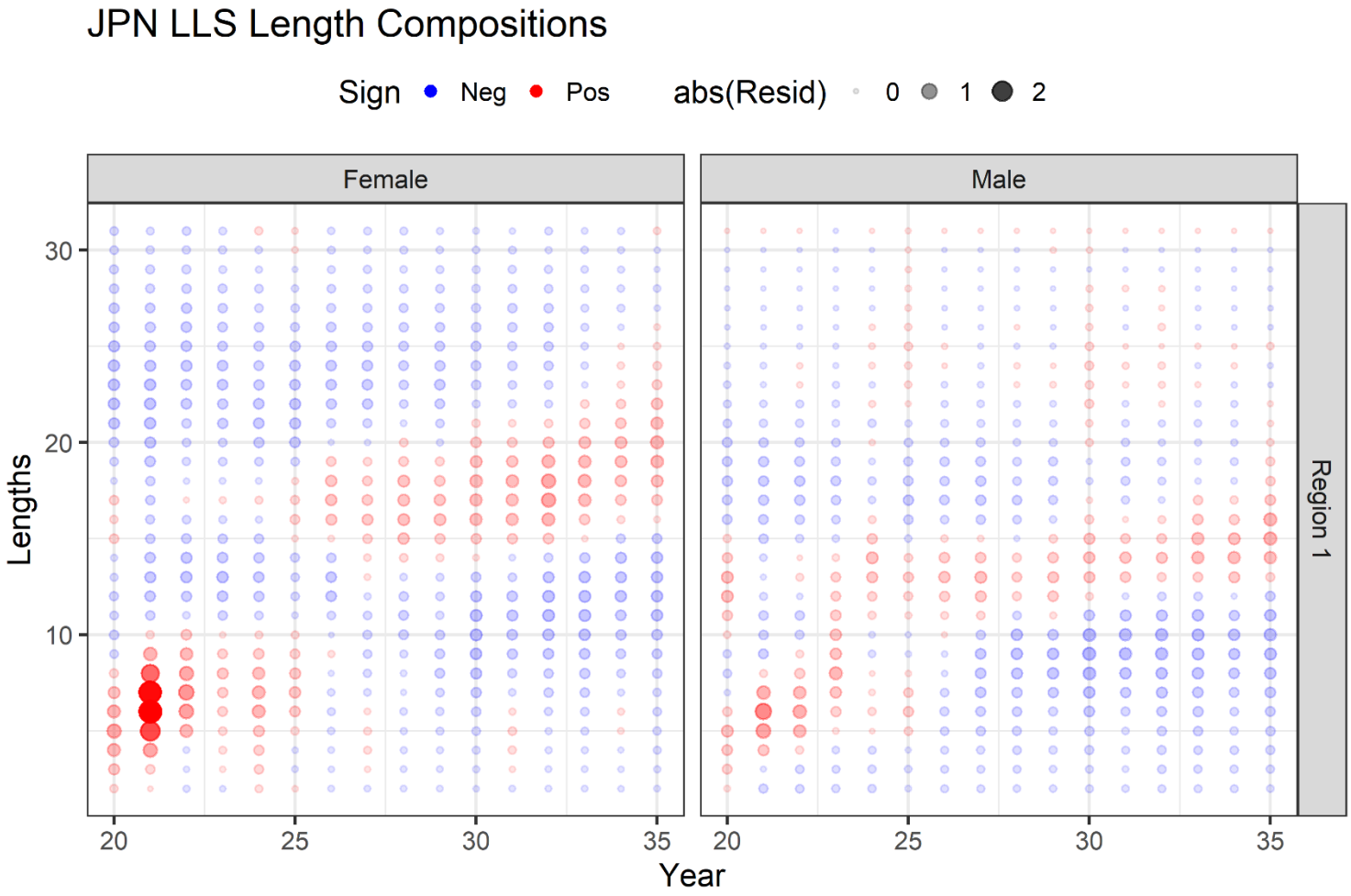

Figure SM3C.70. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.2_FAA*.

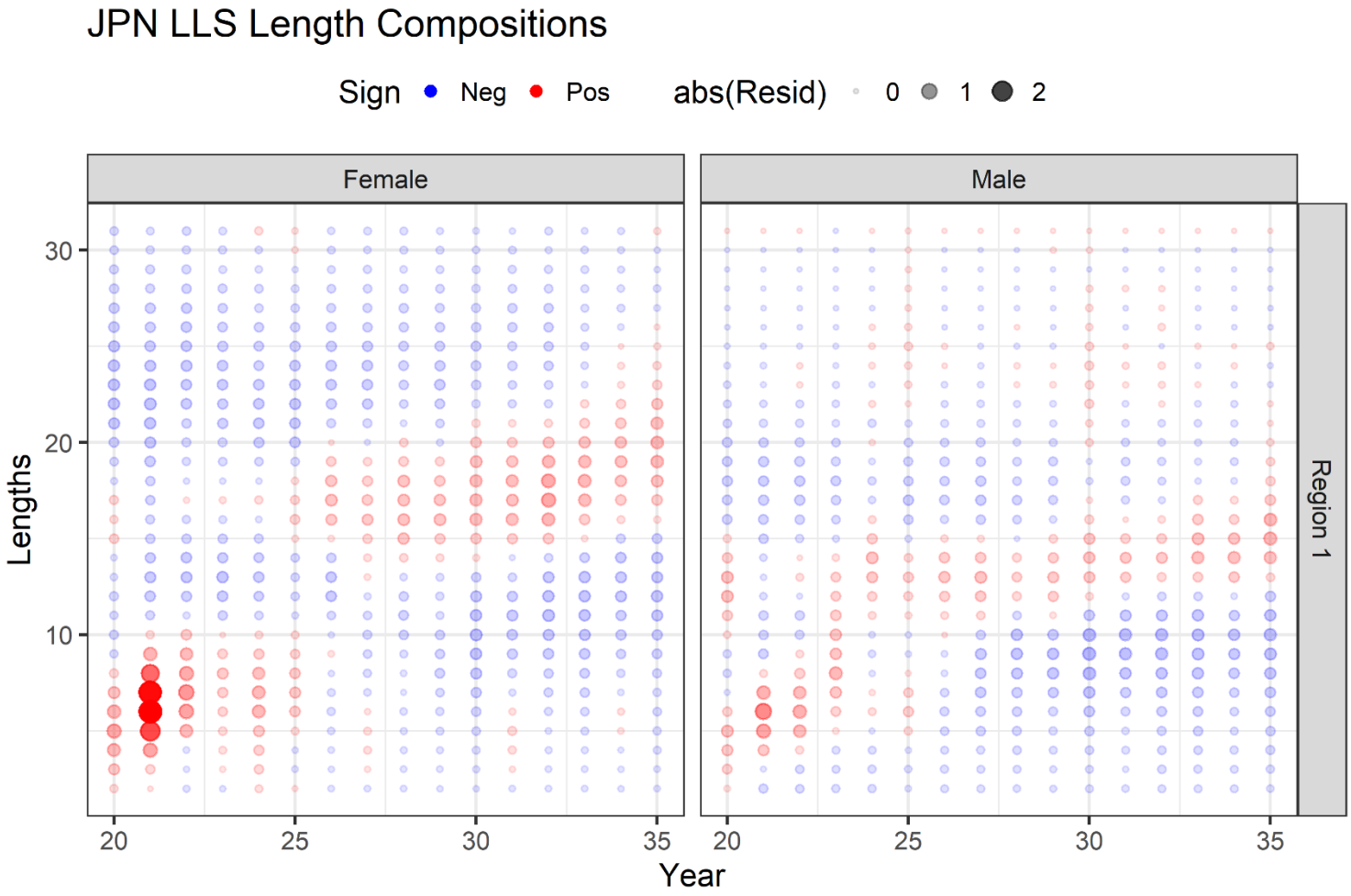

Figure SM3C.71. Fits to the Japanese Longline Survey length composition data based on Pearson residuals for Model *26.2a_FAA_Fsh+Srvy*.

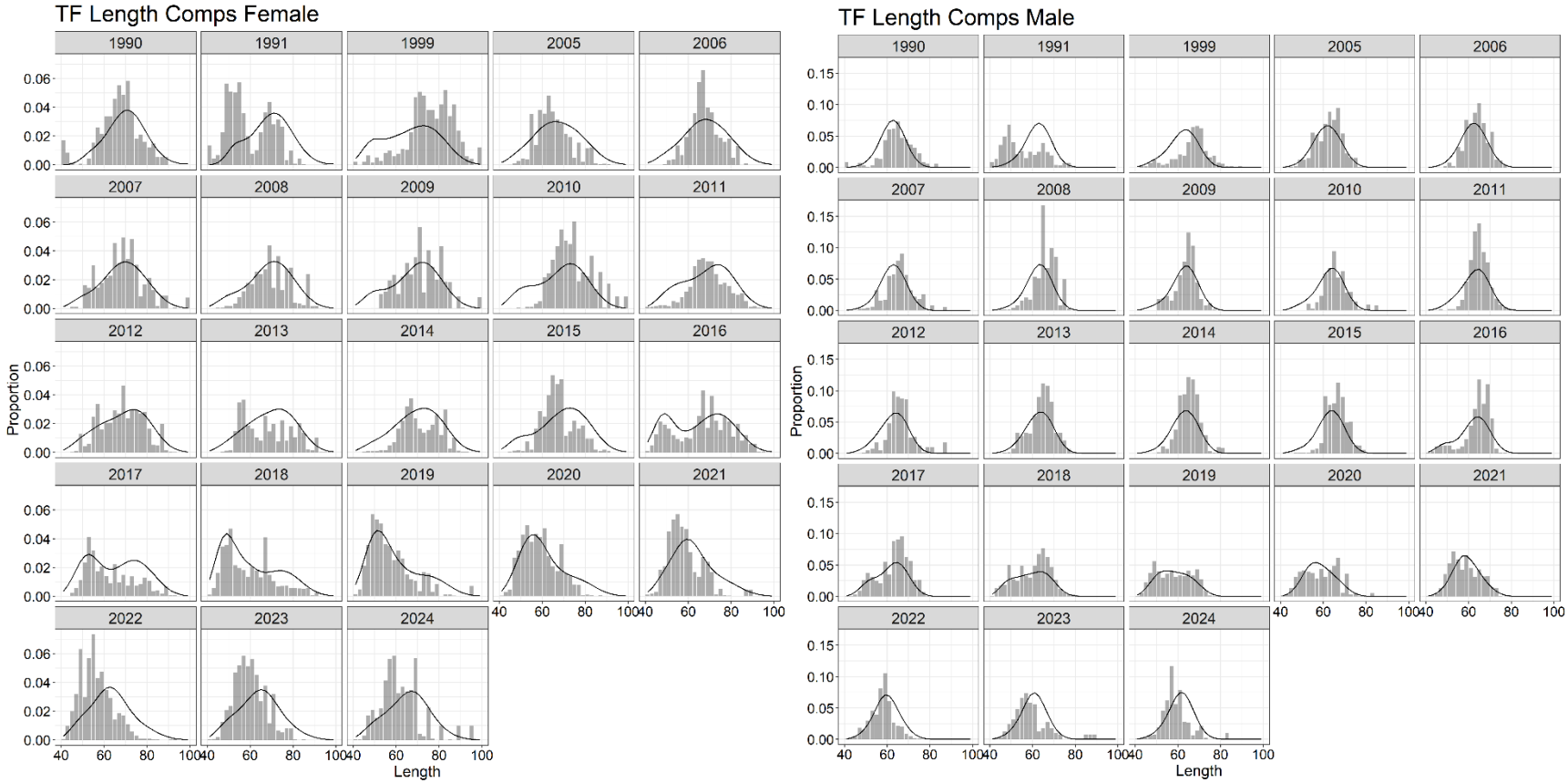

Figure SM3C.72. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.1_1Reg*.

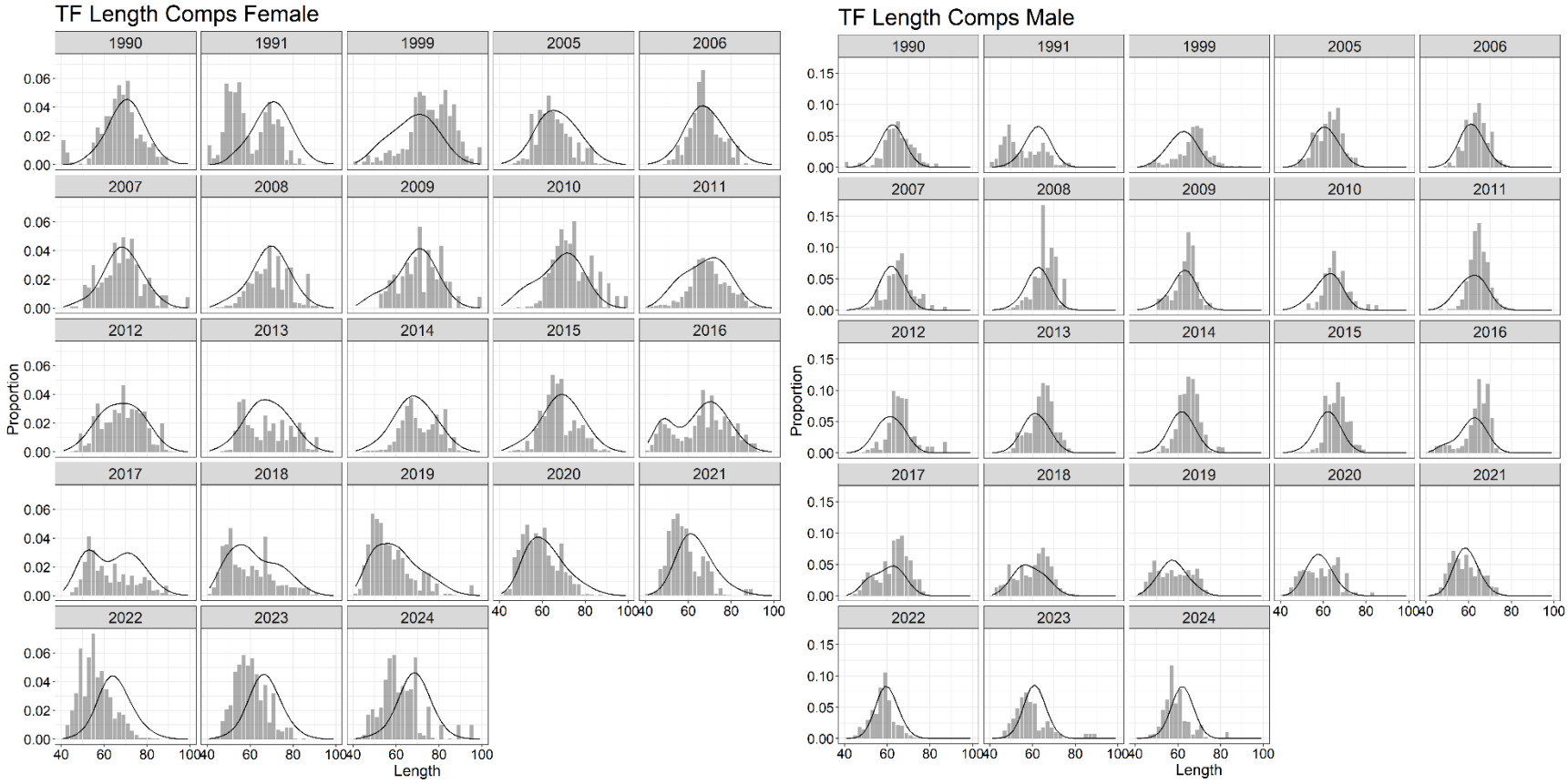

Figure SM3C.73. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.2_FAA*.

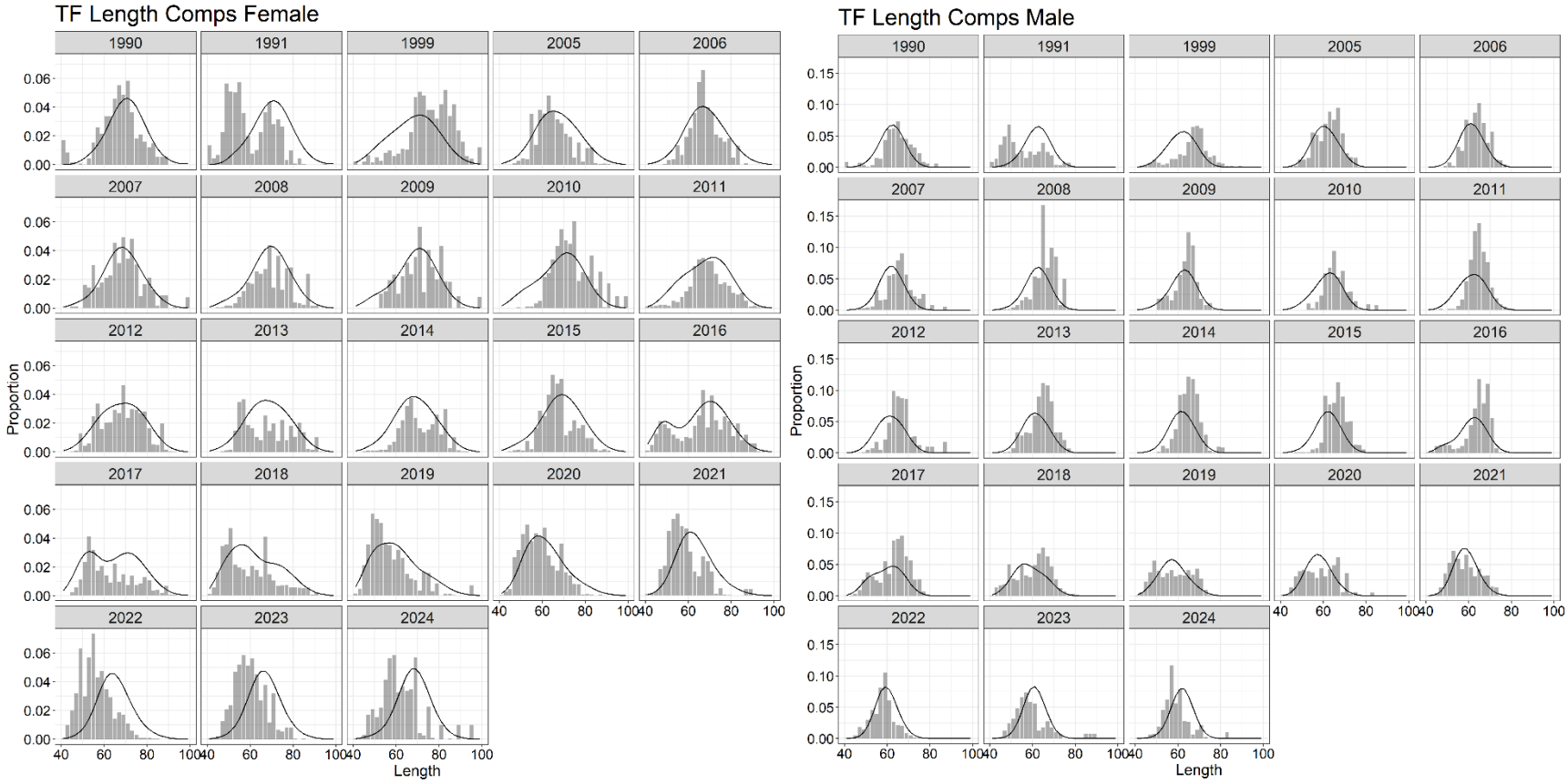

Figure SM3C.74. Fits to the trawl fishery length composition data by year and sex (females left and males right) for Model *26.2a_FAA_Fsh+Srvy*.

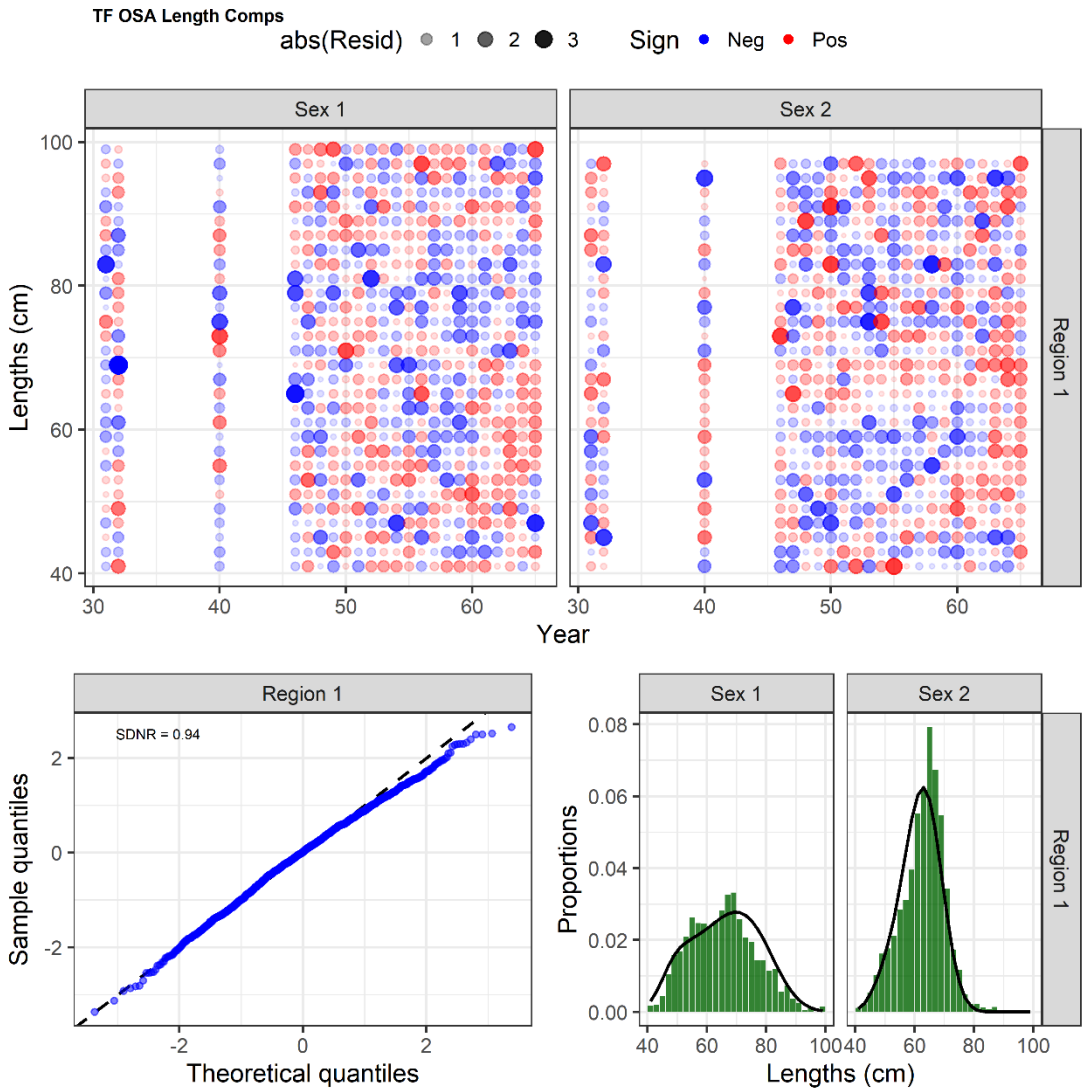

Figure SM3C.75. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.1_1Reg*.

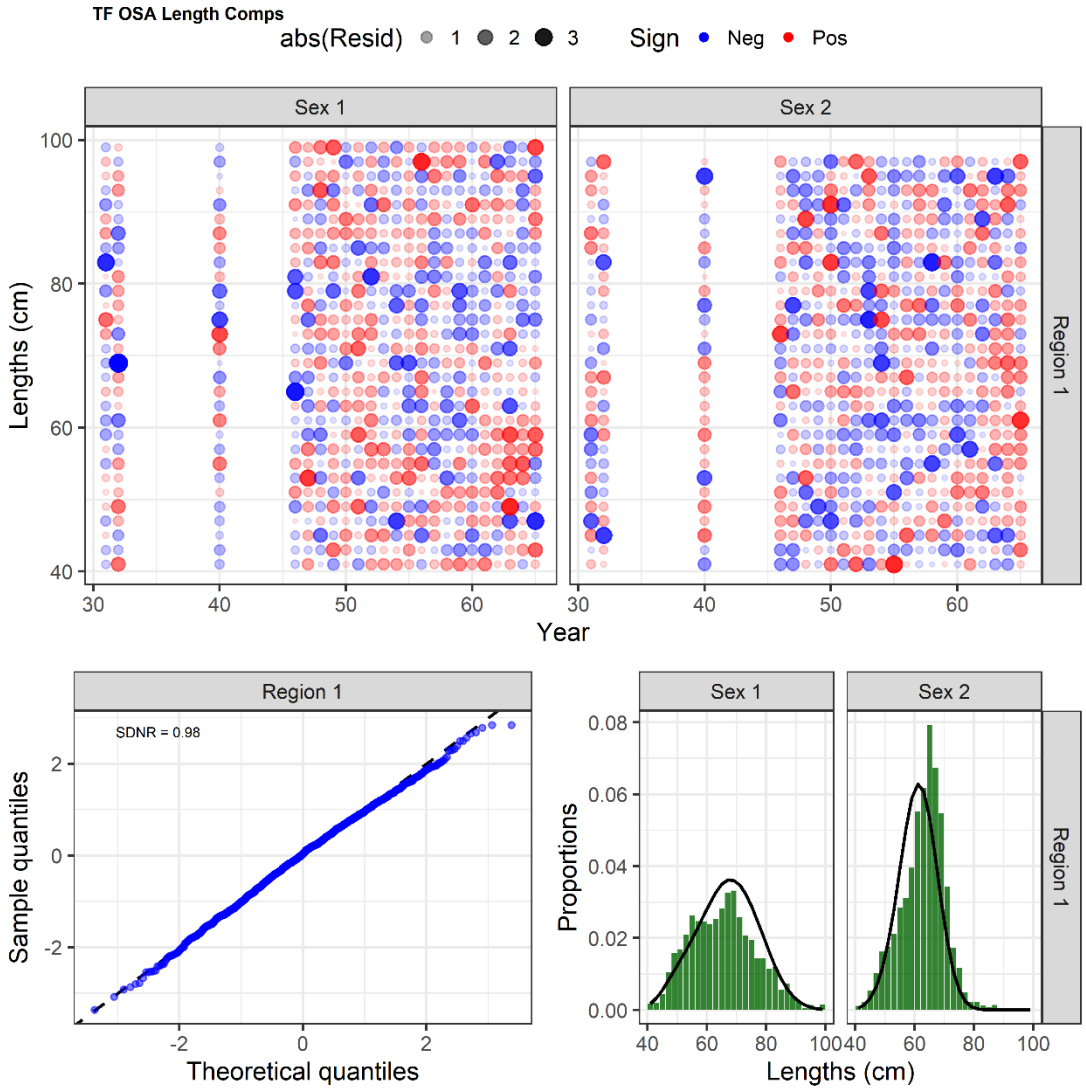

Figure SM3C.76. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2_FAA*.

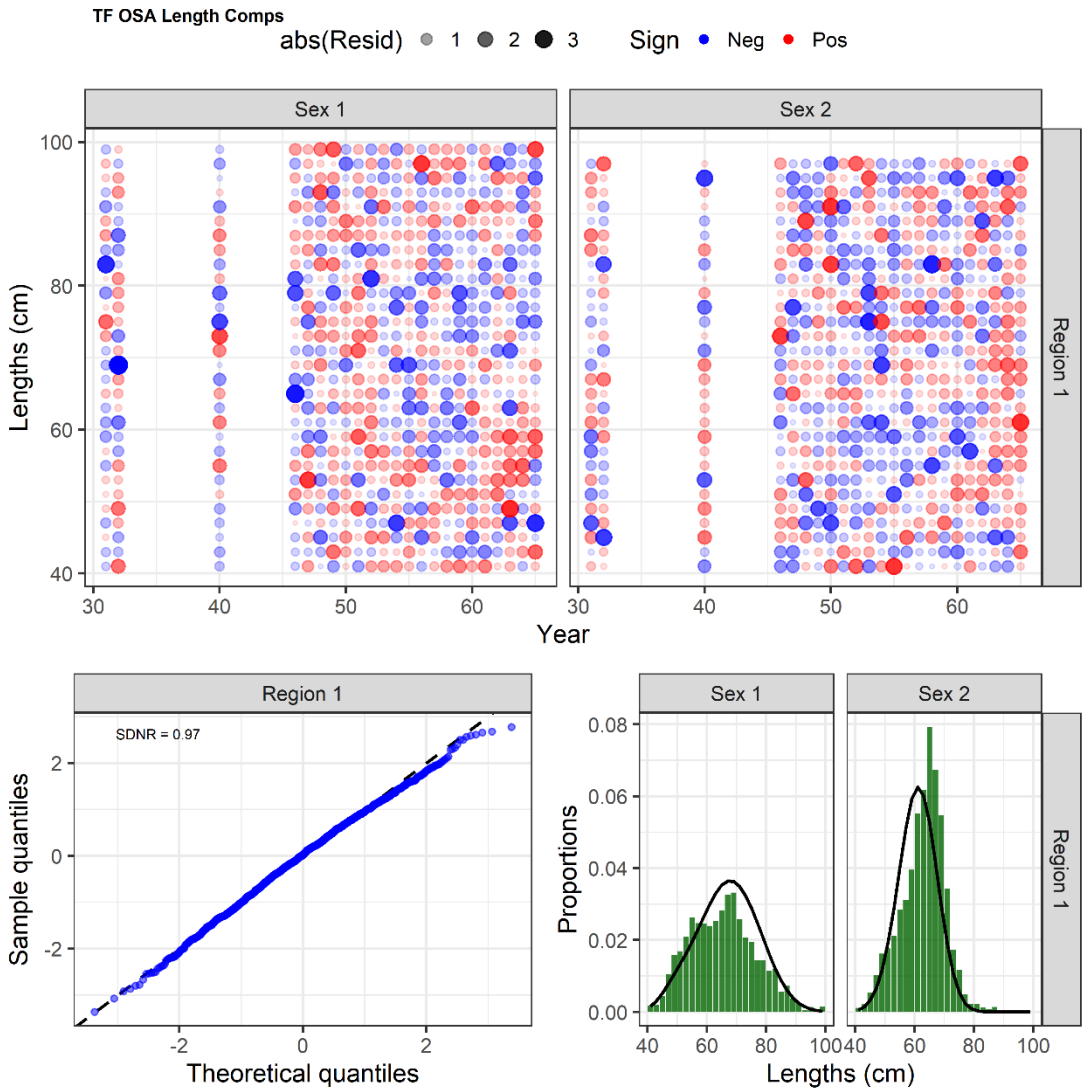

Figure SM3C.77. Fits to trawl fishery length composition data, including one step ahead (OSA) residuals (top panel), quantile residuals (bottom left panel), and aggregate (across all years) fit to the age compositions (bottom right panel; green bars are observed proportions and black lines are expected model estimated proportions), for Model *26.2a_FAA_Fsh+Srvy*.

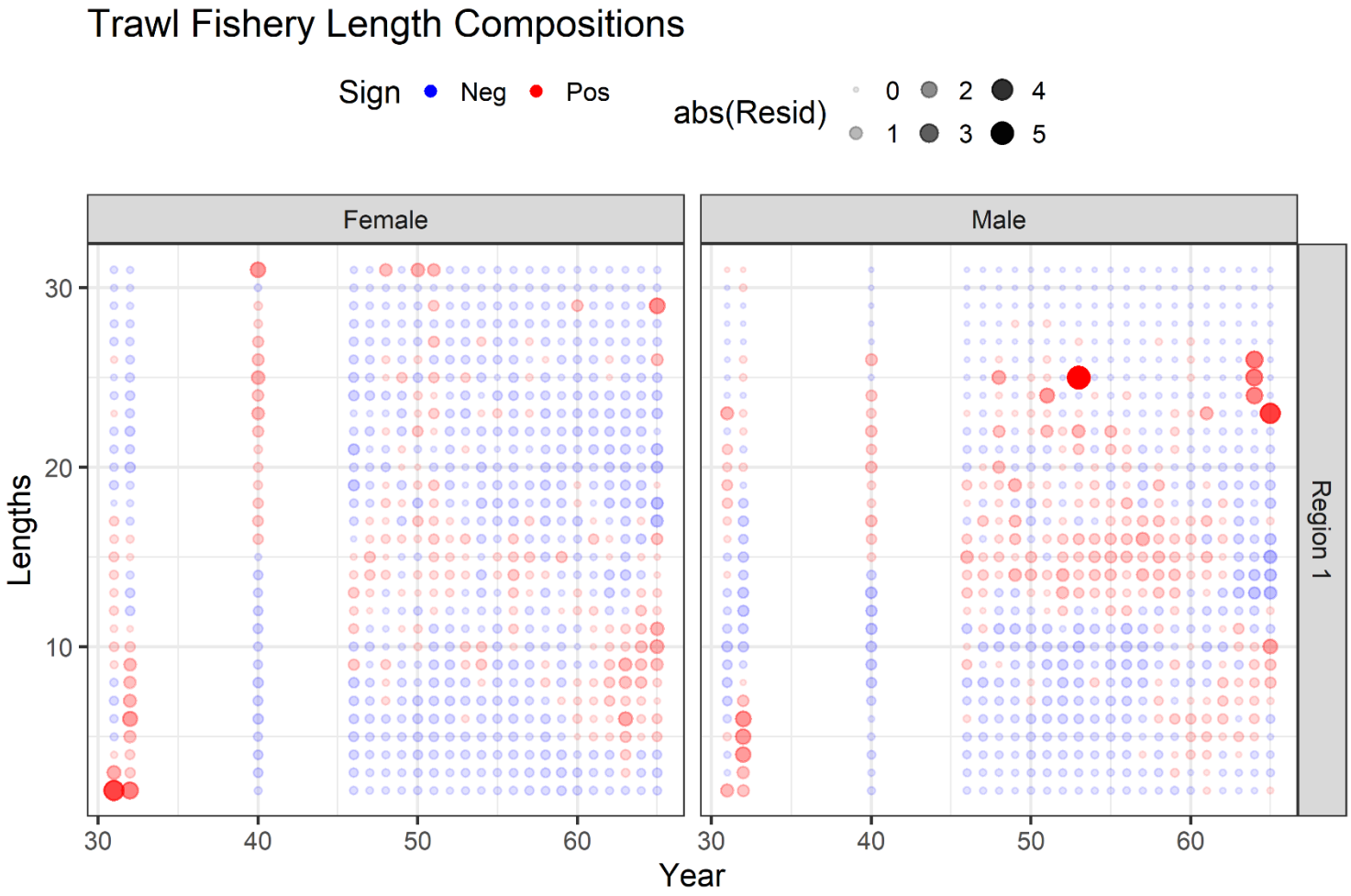

Figure SM3C.78. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.1_1Reg*.

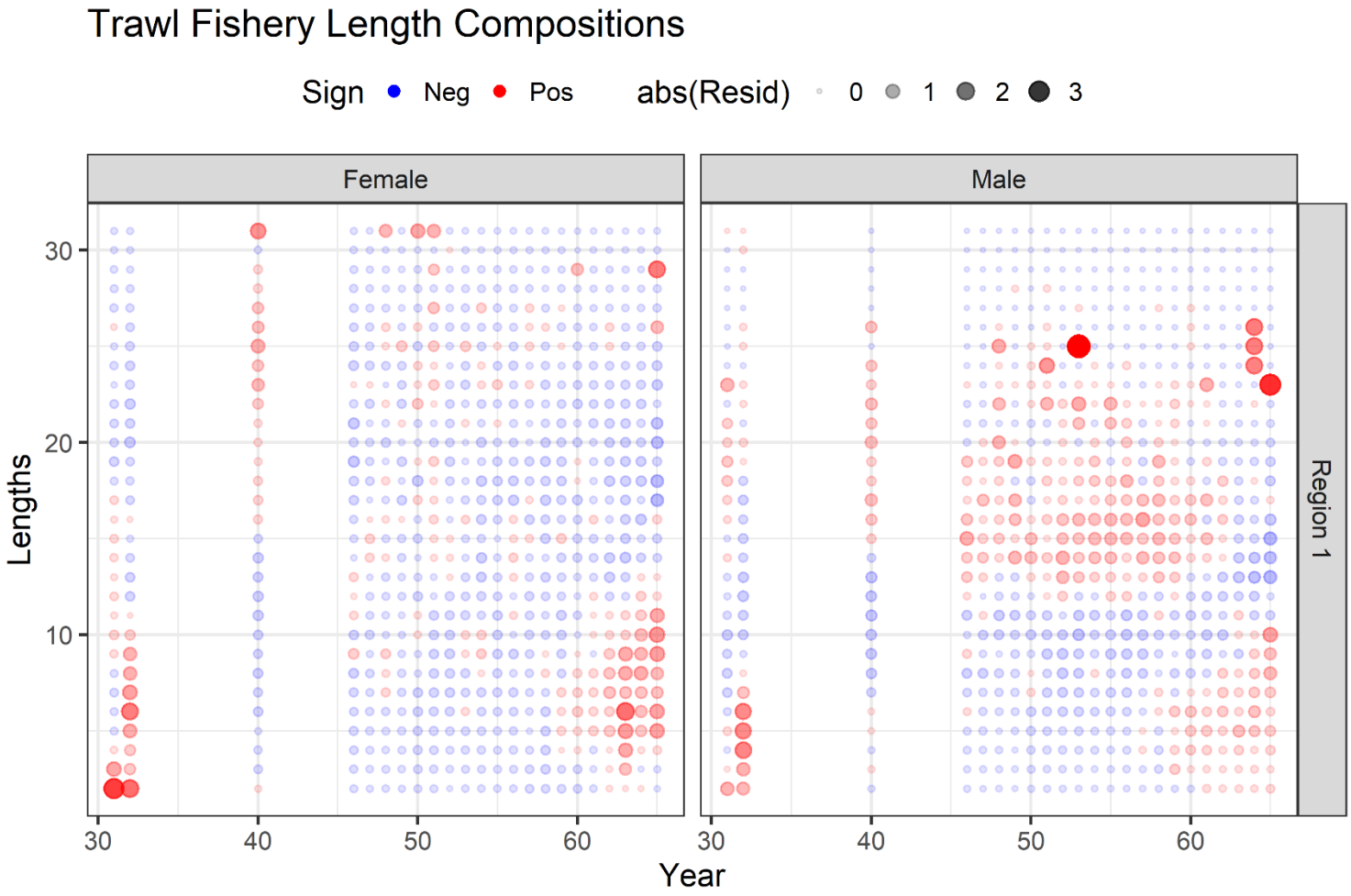

Figure SM3C.79. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.2_FAA*.

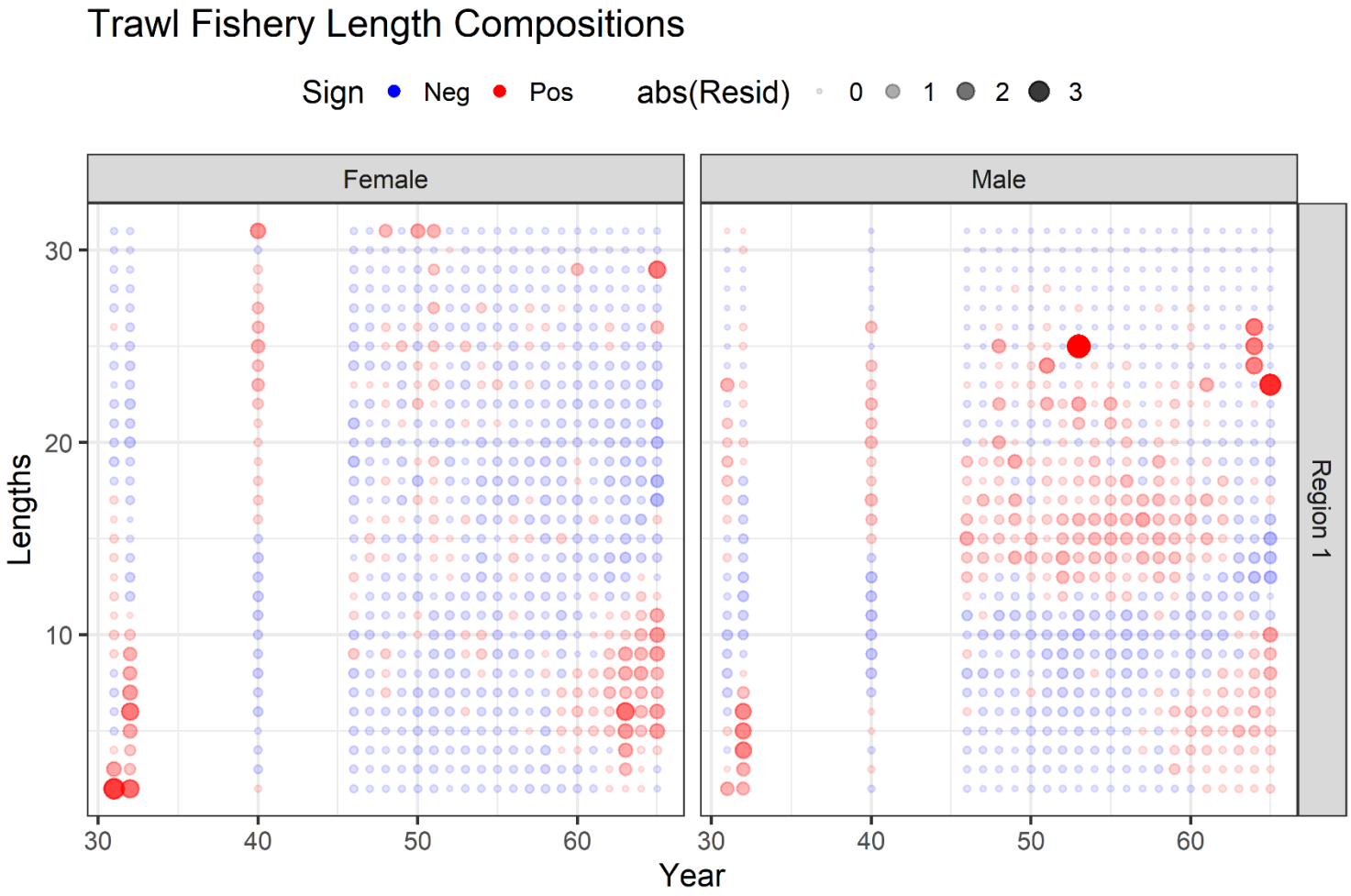

Figure SM3C.80. Fits to the trawl fishery length composition data based on Pearson residuals for Model *26.2a_FAA_Fsh+Srvy*.

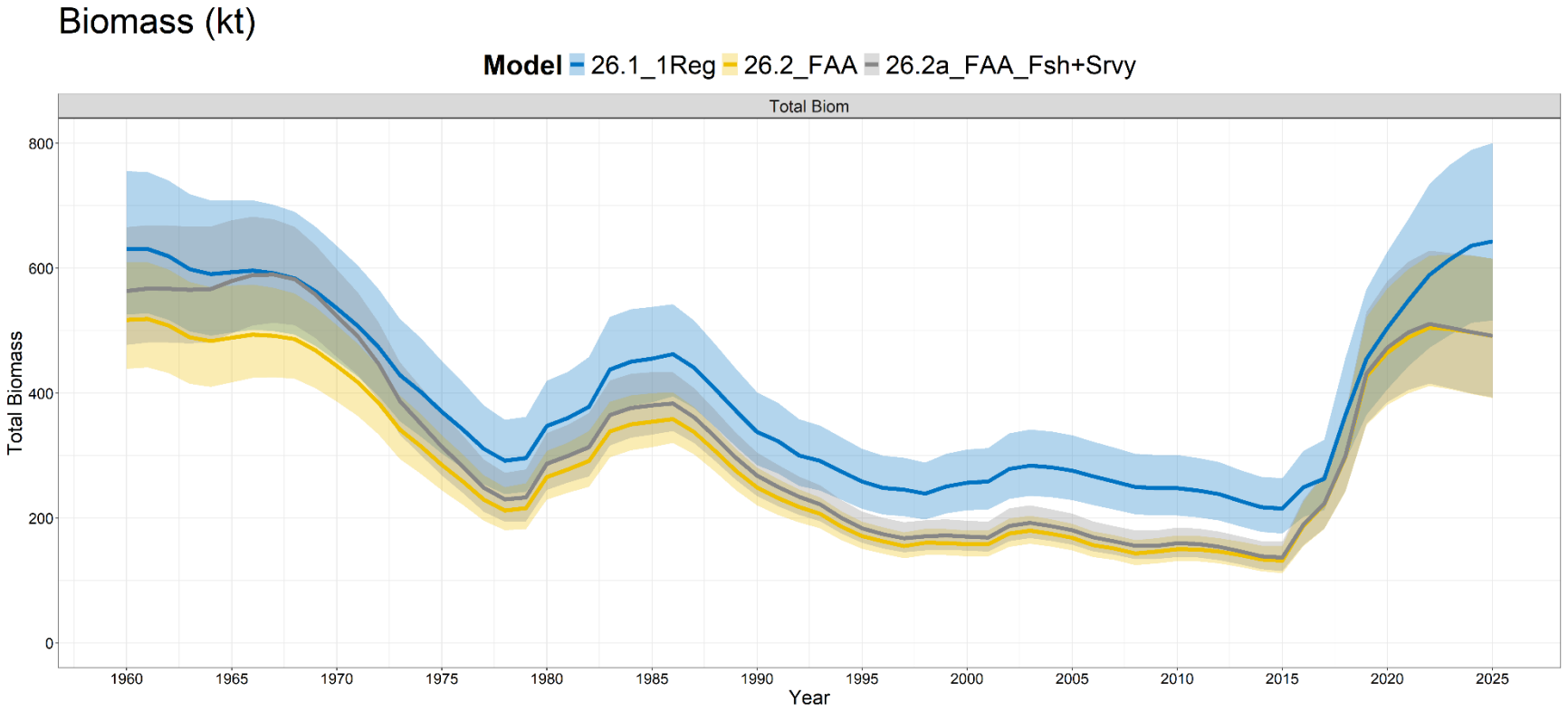

Figure SM3C.81. Time series estimates of total biomass (kt) for each model.

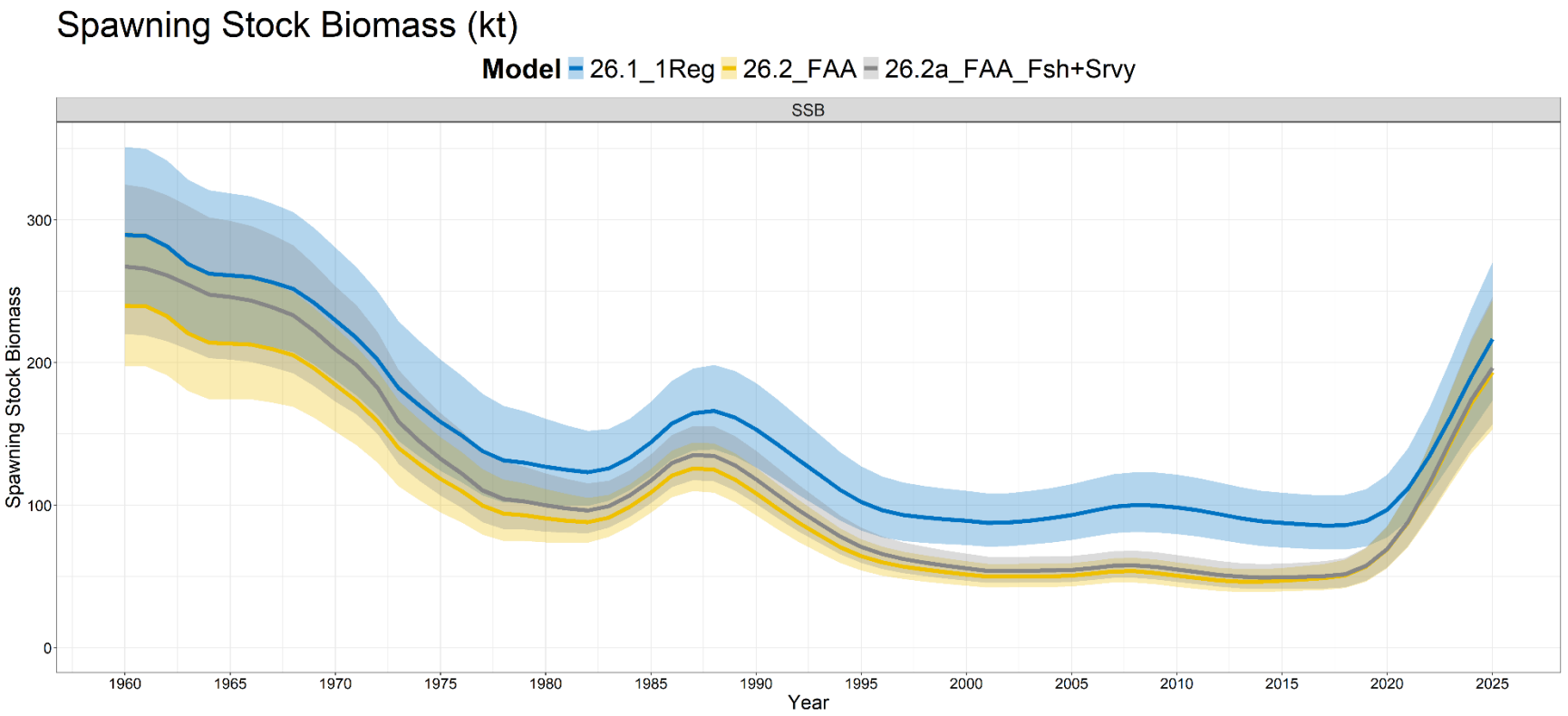

Figure SM3C.82. Time series estimates of spawning stock biomass (kt) for each model.

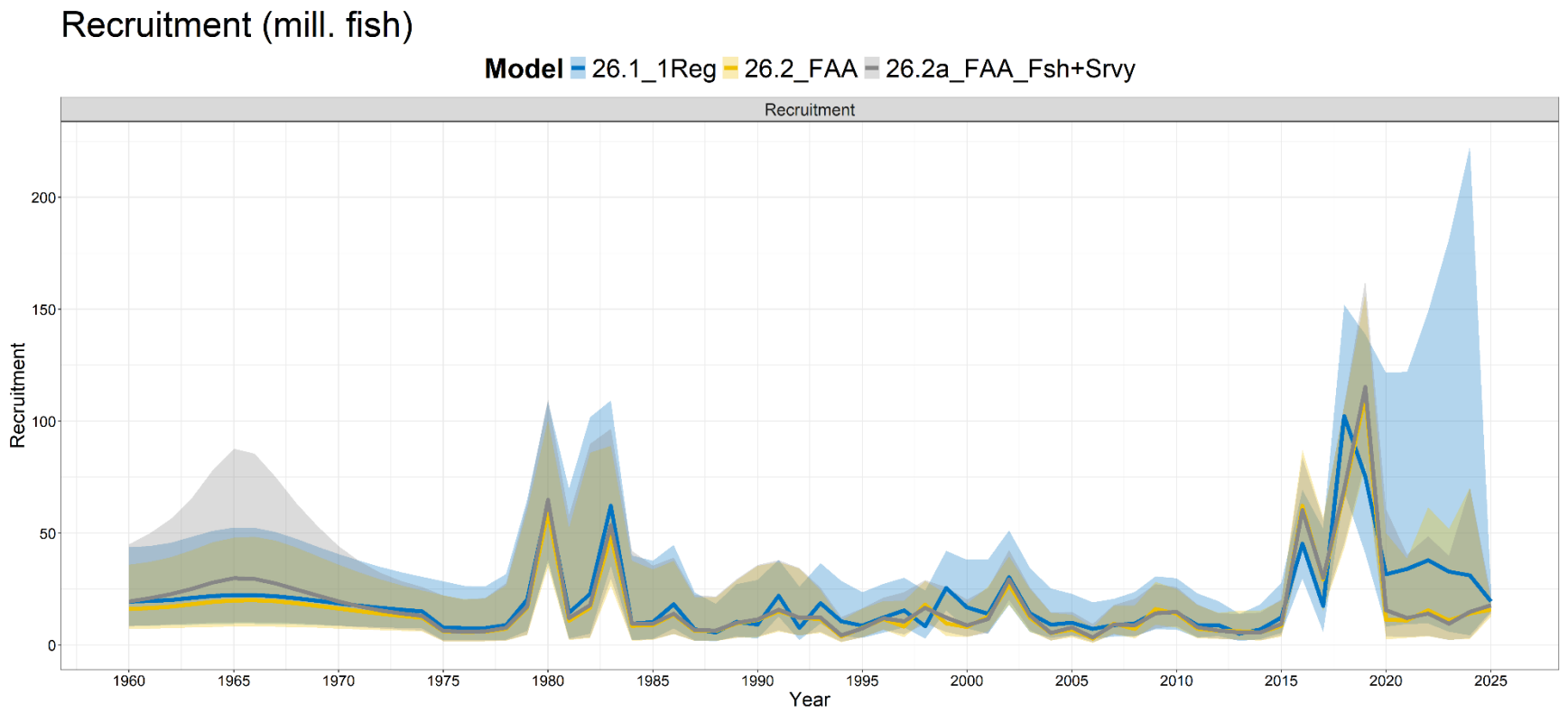

Figure SM3C.83. Time series estimates of recruitment (millions of fish) for each model.

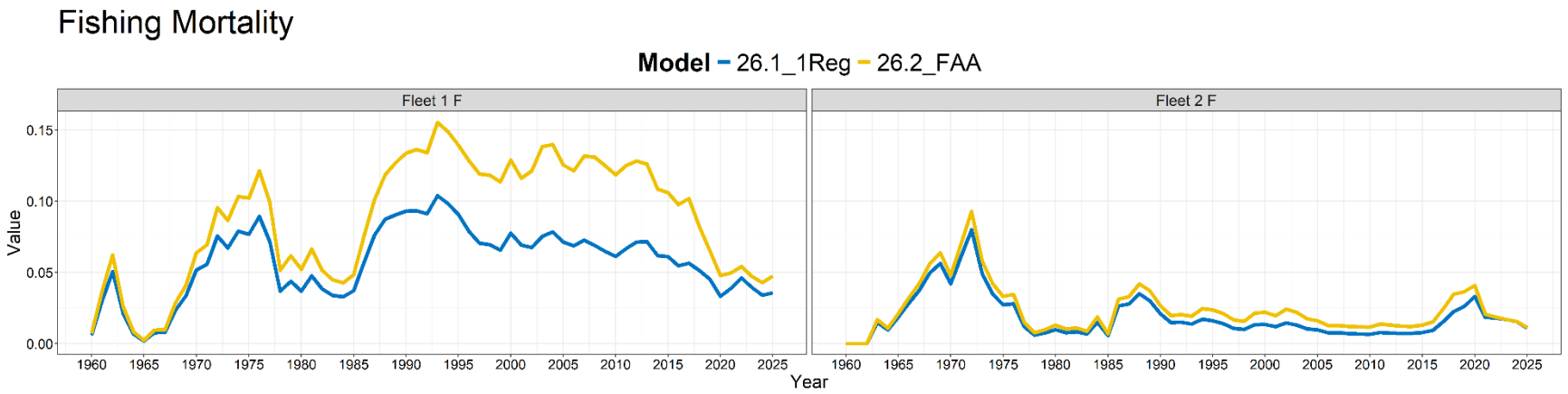

Figure SM3C.84. Time series estimates of fishing mortality (yr^{-1}) for Model *26.1_1Reg* and Model *26.2_FAA*. Fleet 1 is the fixed gear fishery and fleet 2 is the trawl fishery.

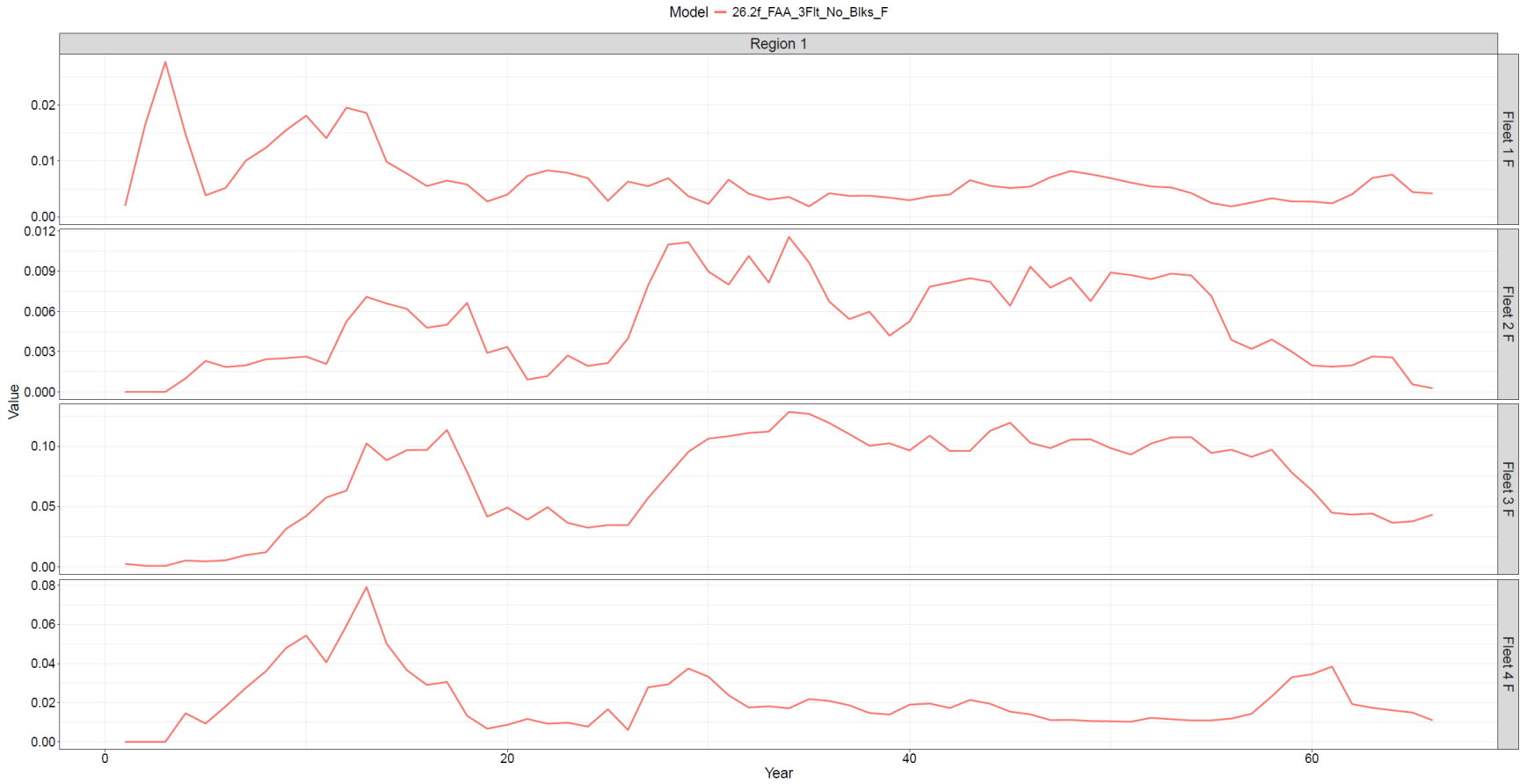

Figure SM3C.85. Time series estimates of fishing mortality (yr^{-1}) for Model *26.2a_FAA_Fsh+Srvy*. Fishery Fleet 1 is the BS fixed gear fleet, Fishery Fleet 2 is the AI fixed gear fleet, Fishery Fleet 3 is the GOA fixed gear fleet, and Fishery Fleet 4 is the trawl fleet. Note the difference in y-axis scales. Year 1 is 1960.

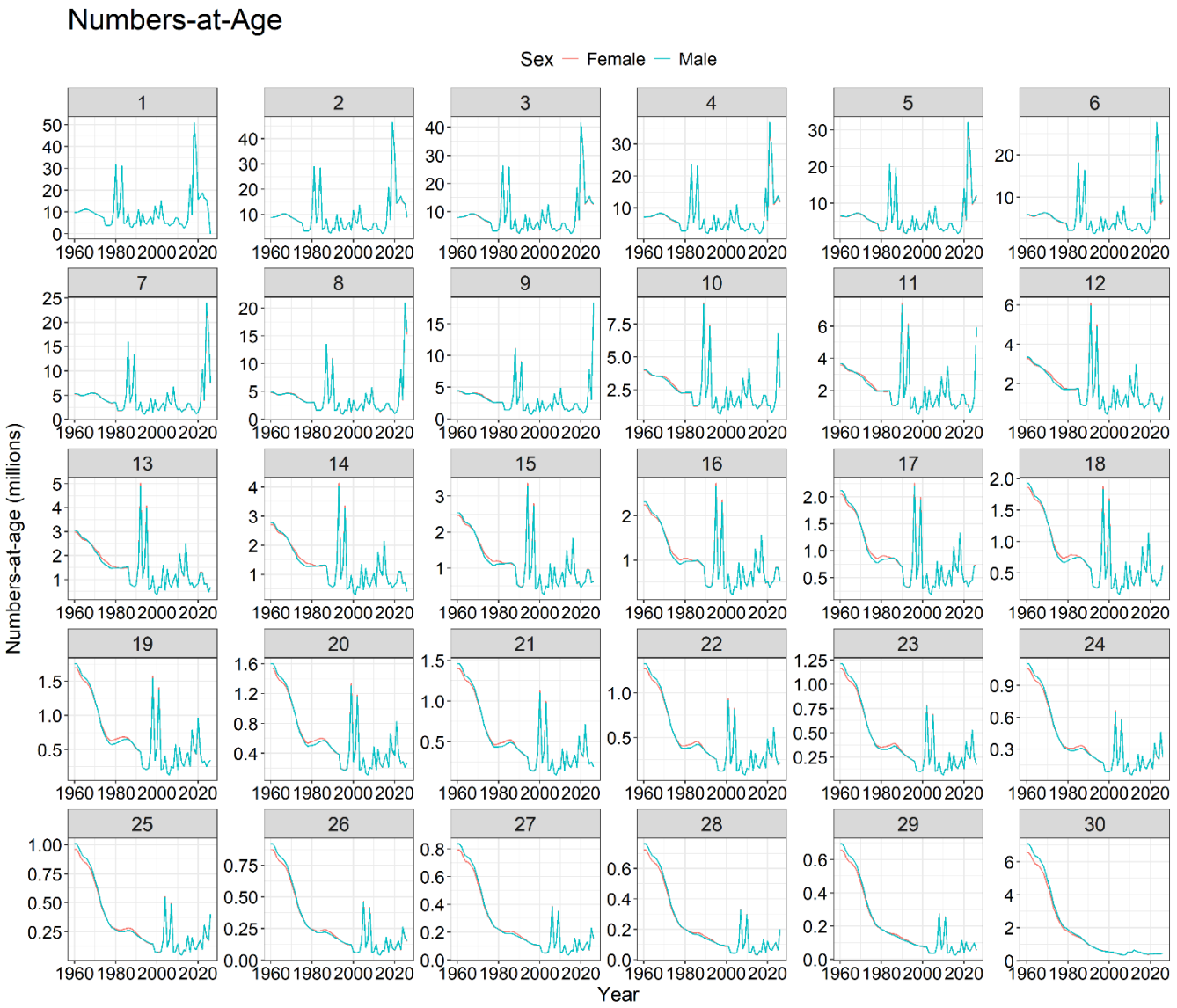

Figure SM3C.86. Time series estimates of numbers-at-age (millions of fish) by sex for model *26.1_1Reg*.

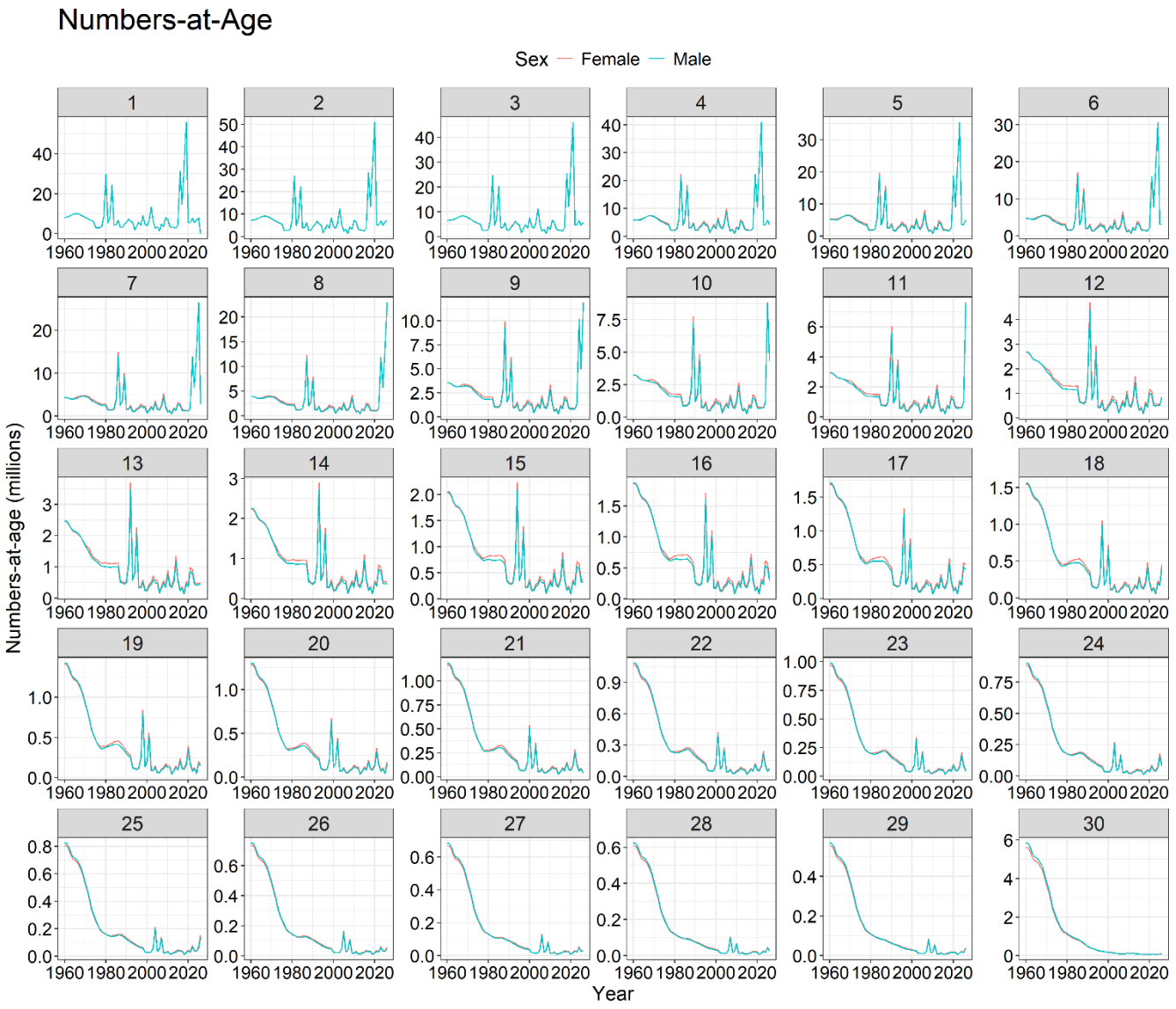

Figure SM3C.87. Time series estimates of numbers-at-age (millions of fish) by sex for model *26.2_FAA*.

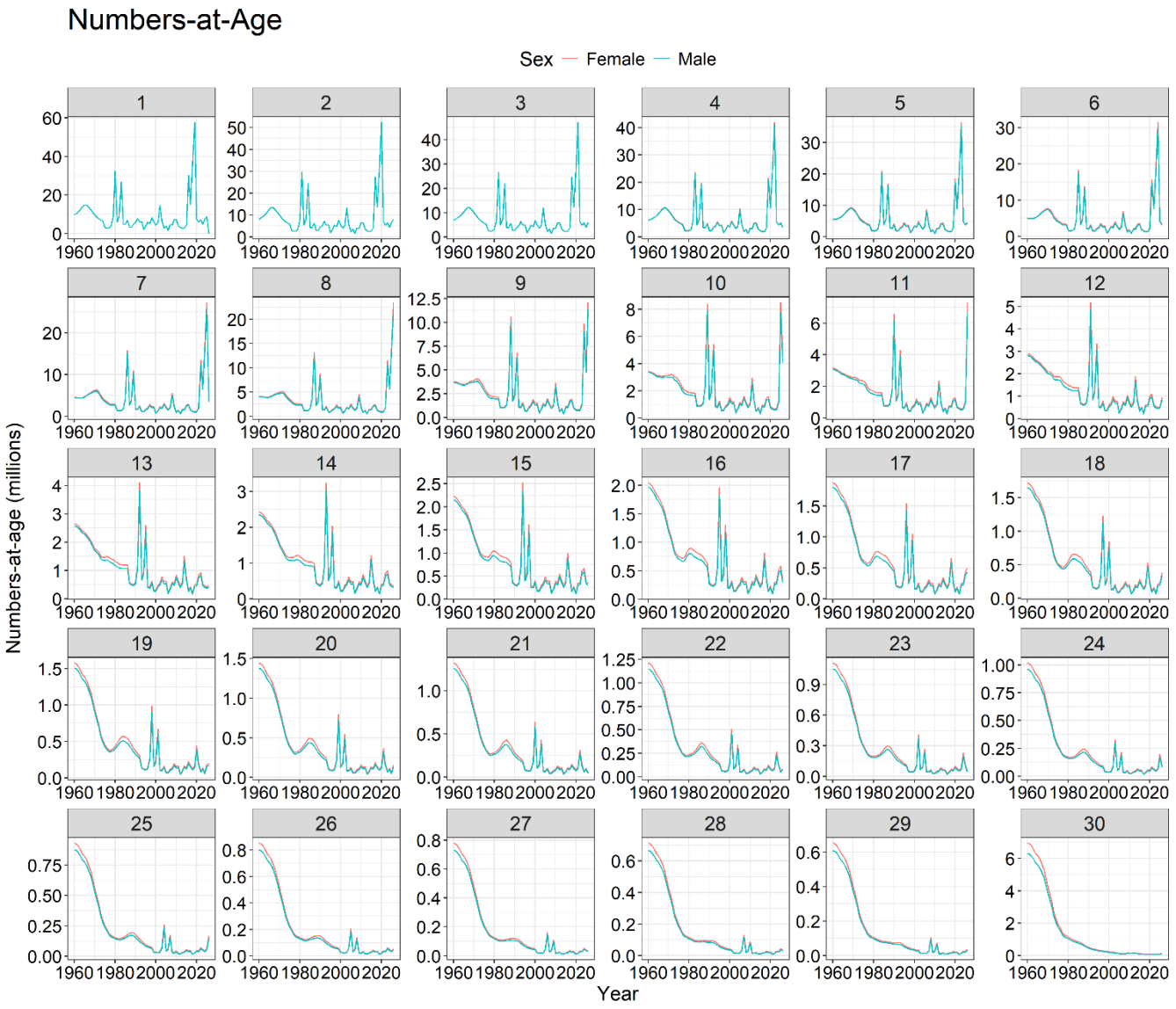

Figure SM3C.88. Time series estimates of numbers-at-age (millions of fish) by sex for model *26.2a_FAA_Fsh+Srvy*.

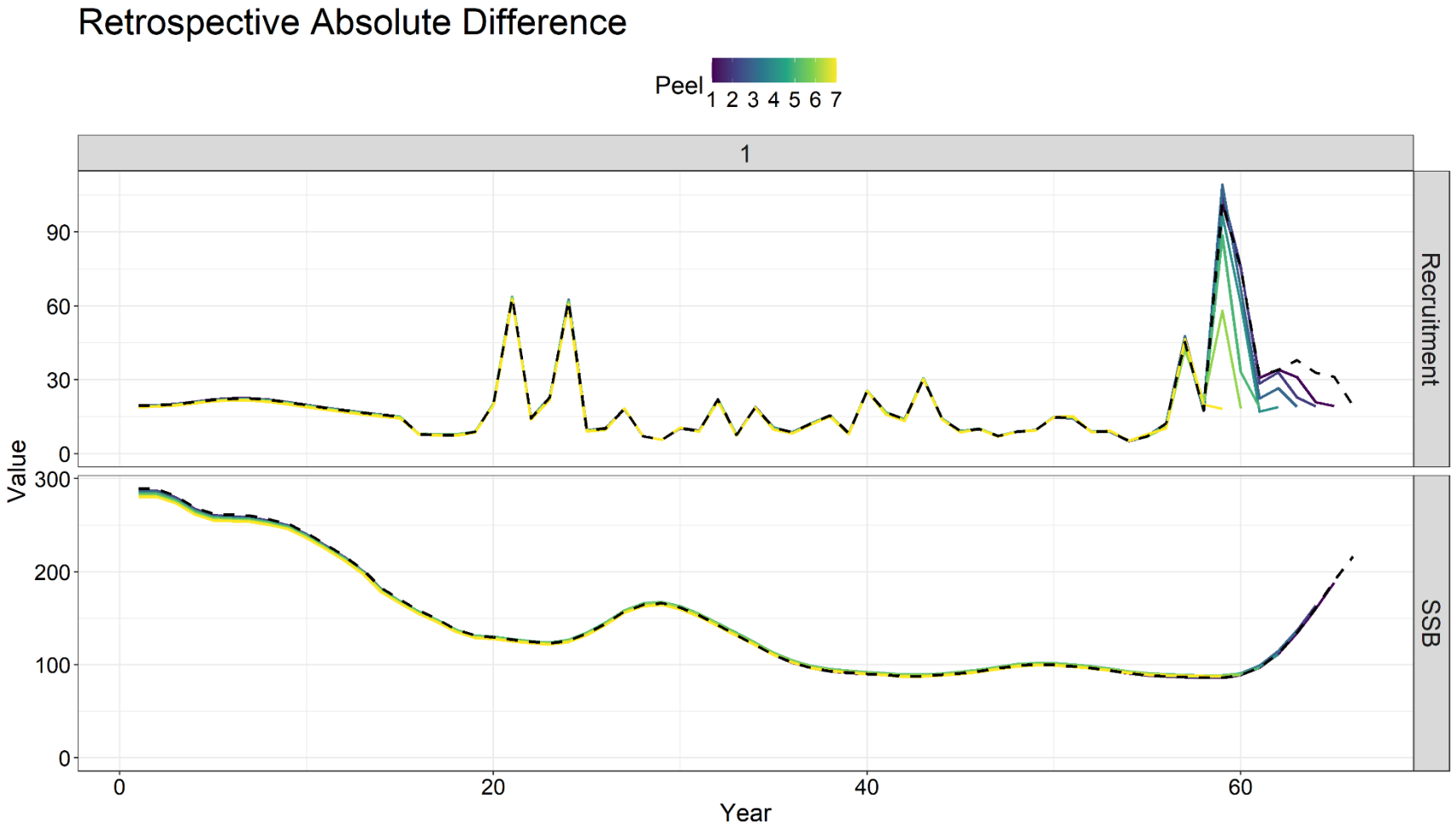

Figure SM3C.89. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven year retrospective analysis for Model *26.1_1Reg*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.0054.

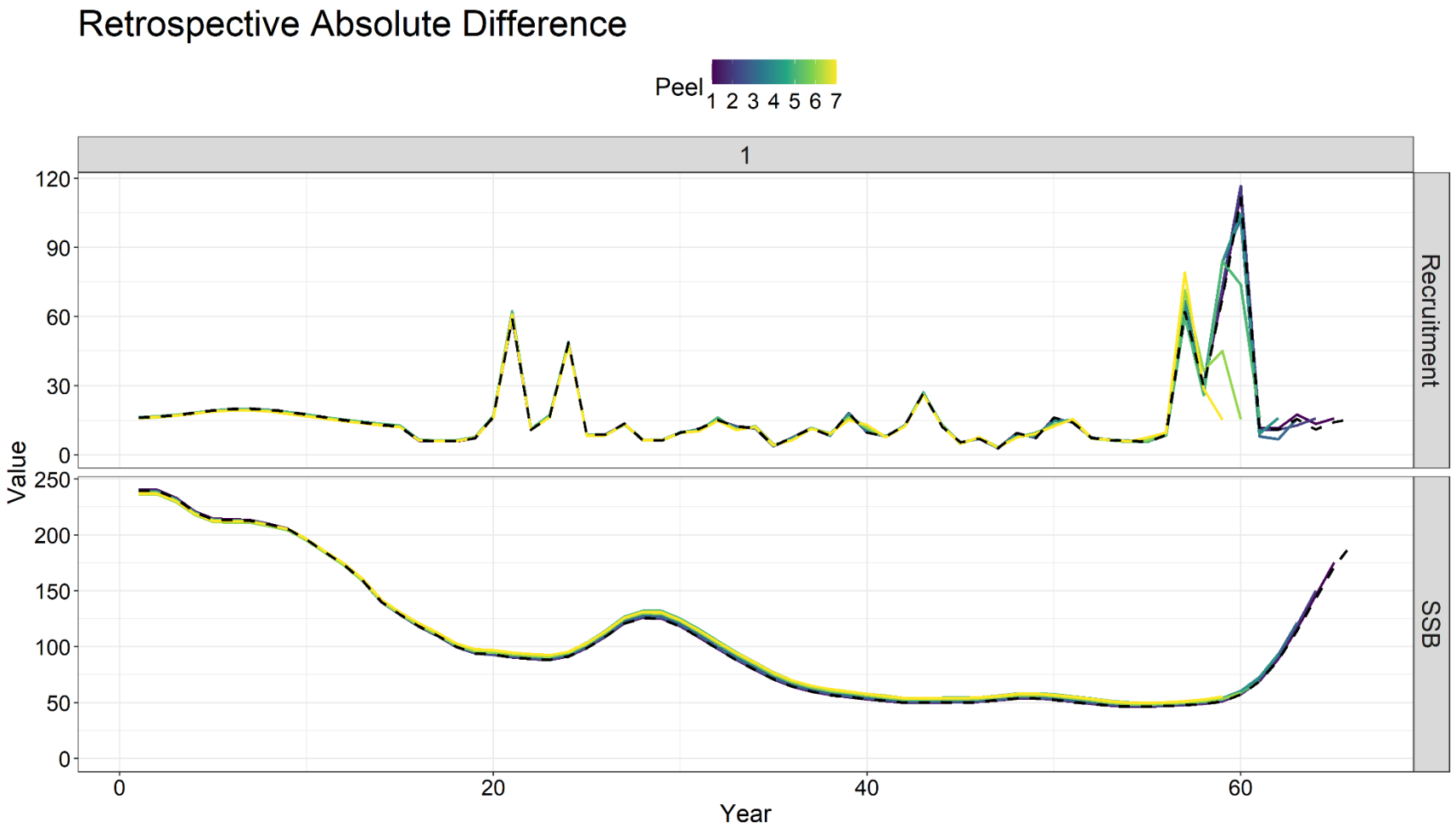

Figure SM3C.90. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2_FAA*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.046.

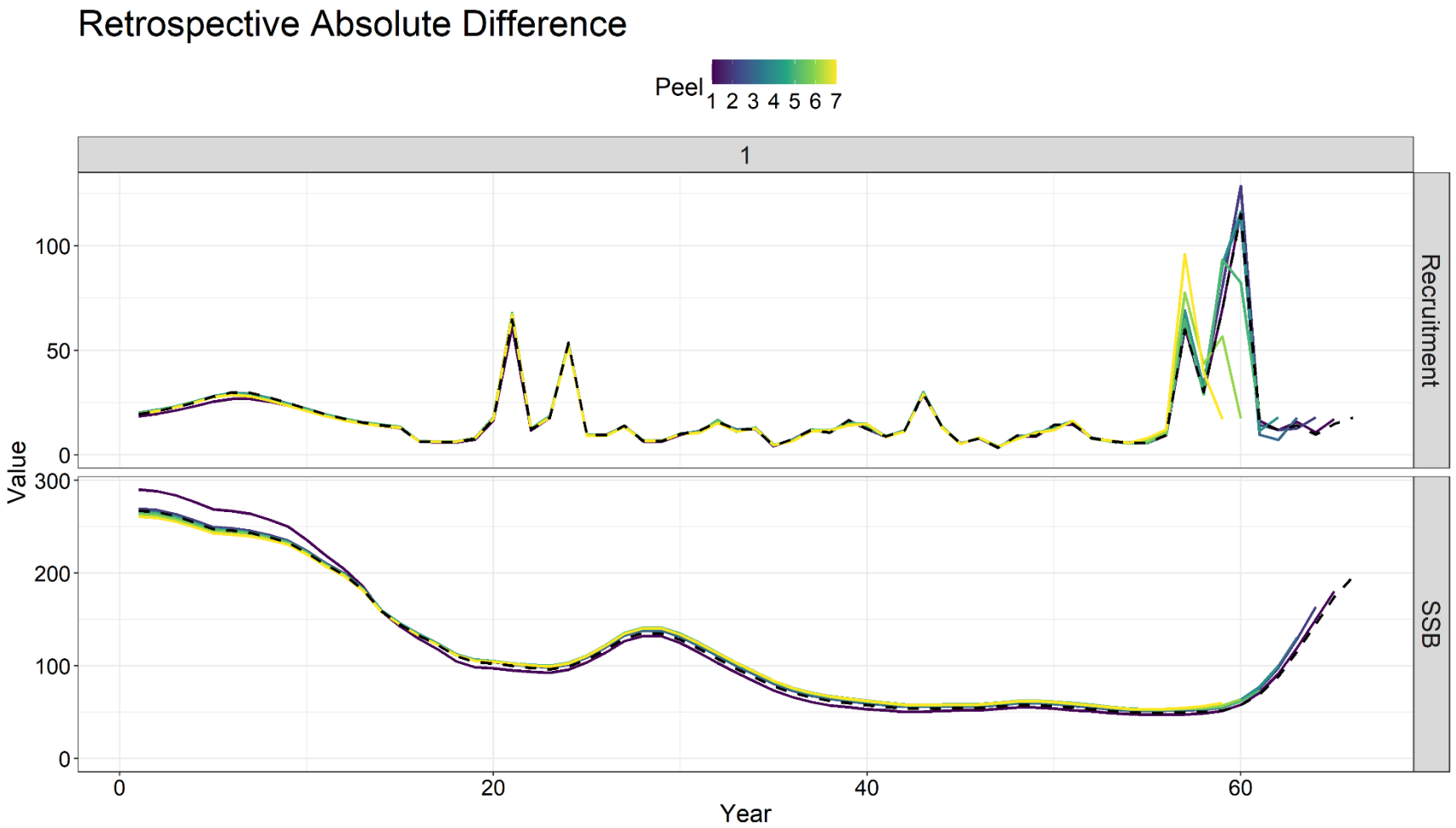

Figure SM3C.91. Comparison of recruitment (millions of fish; top panel) and spawning stock biomass (SSB; bottom panel) time series estimates from a seven-year retrospective analysis for Model *26.2a_FAA_Fsh+Srvy*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Mohn's rho for SSB was 0.11.

Figure SM3C.92. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven-year retrospective analysis for model *26.1_1Reg*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Cohort 56 represents the 2015 year class (2017 recruitment event).

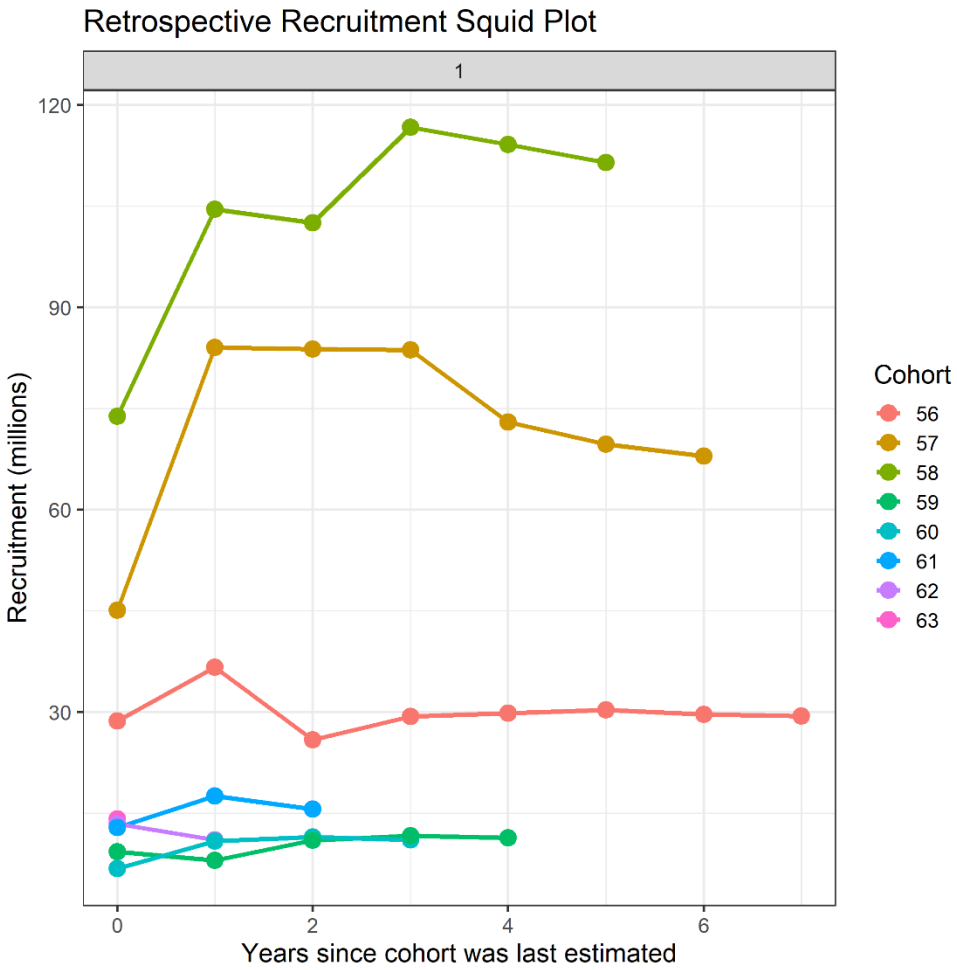

Figure SM3C.93. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven-year retrospective analysis for model *26.2_FAA*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Cohort 56 represents the 2015 year class (2017 recruitment event).

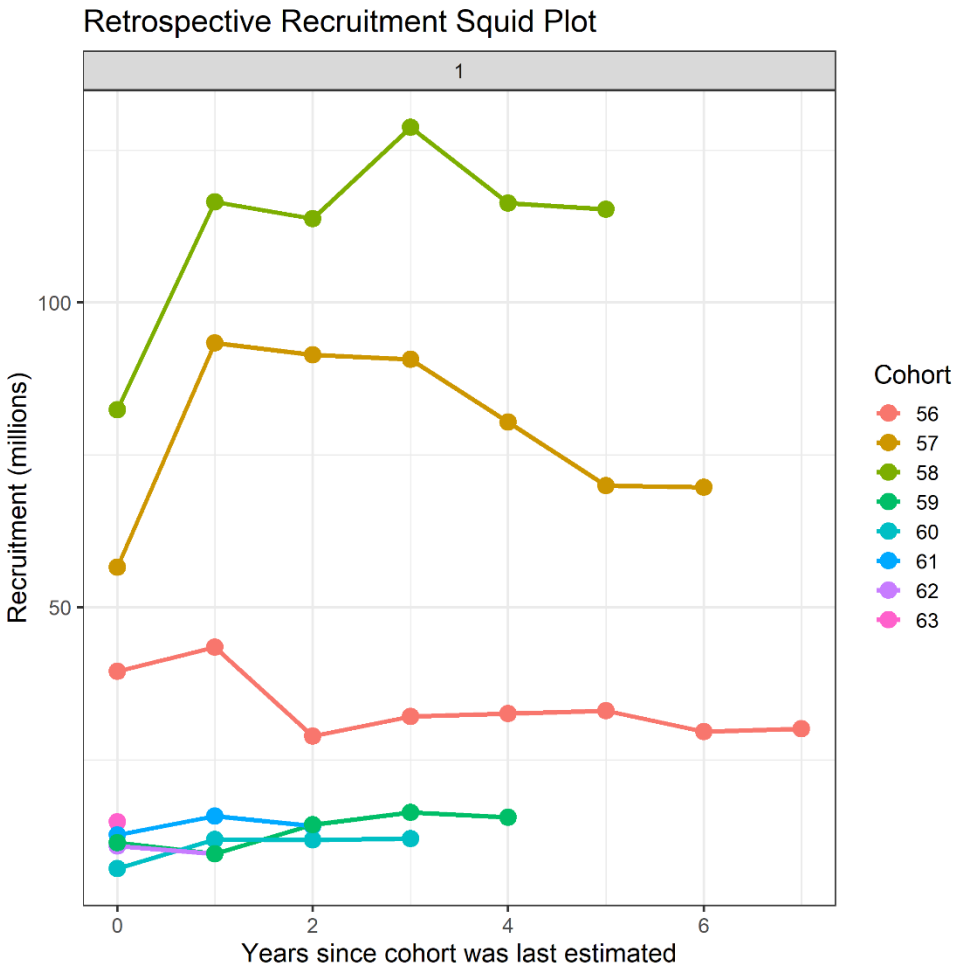

Figure SM3C.94. Comparison of a recruitment retrospective 'squid' plot representing how estimates of a given year class change as data are added in a seven-year retrospective analysis for model *26.2a_FAA_Fsh+Srvy*. Francis reweighting was undertaken for each peel to ensure consistency and to match how an assessment would be performed in a real-world situation. Cohort 56 represents the 2015 year class (2017 recruitment event).

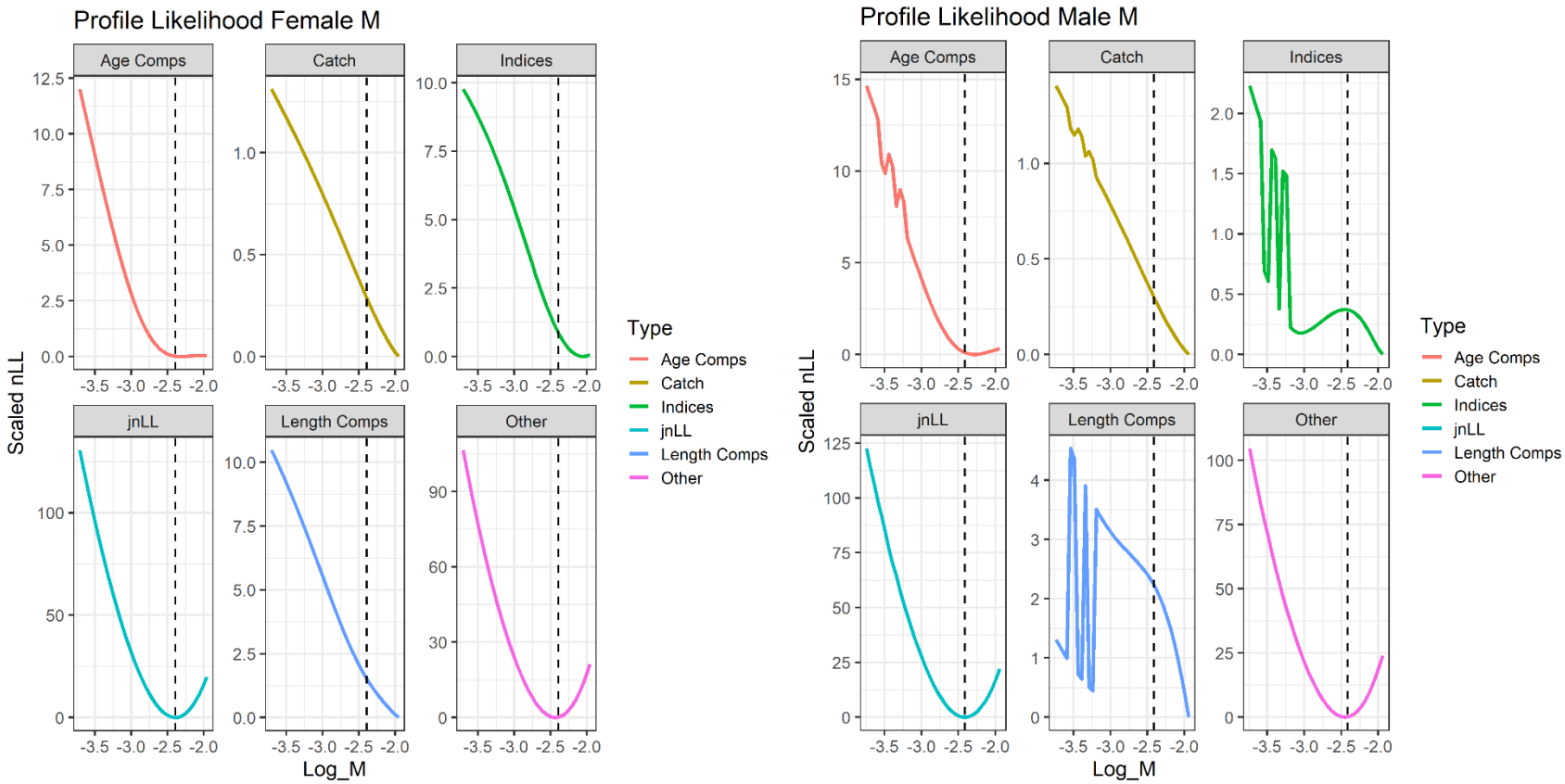

Figure SM3C.95. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for model *26.1_1Reg*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

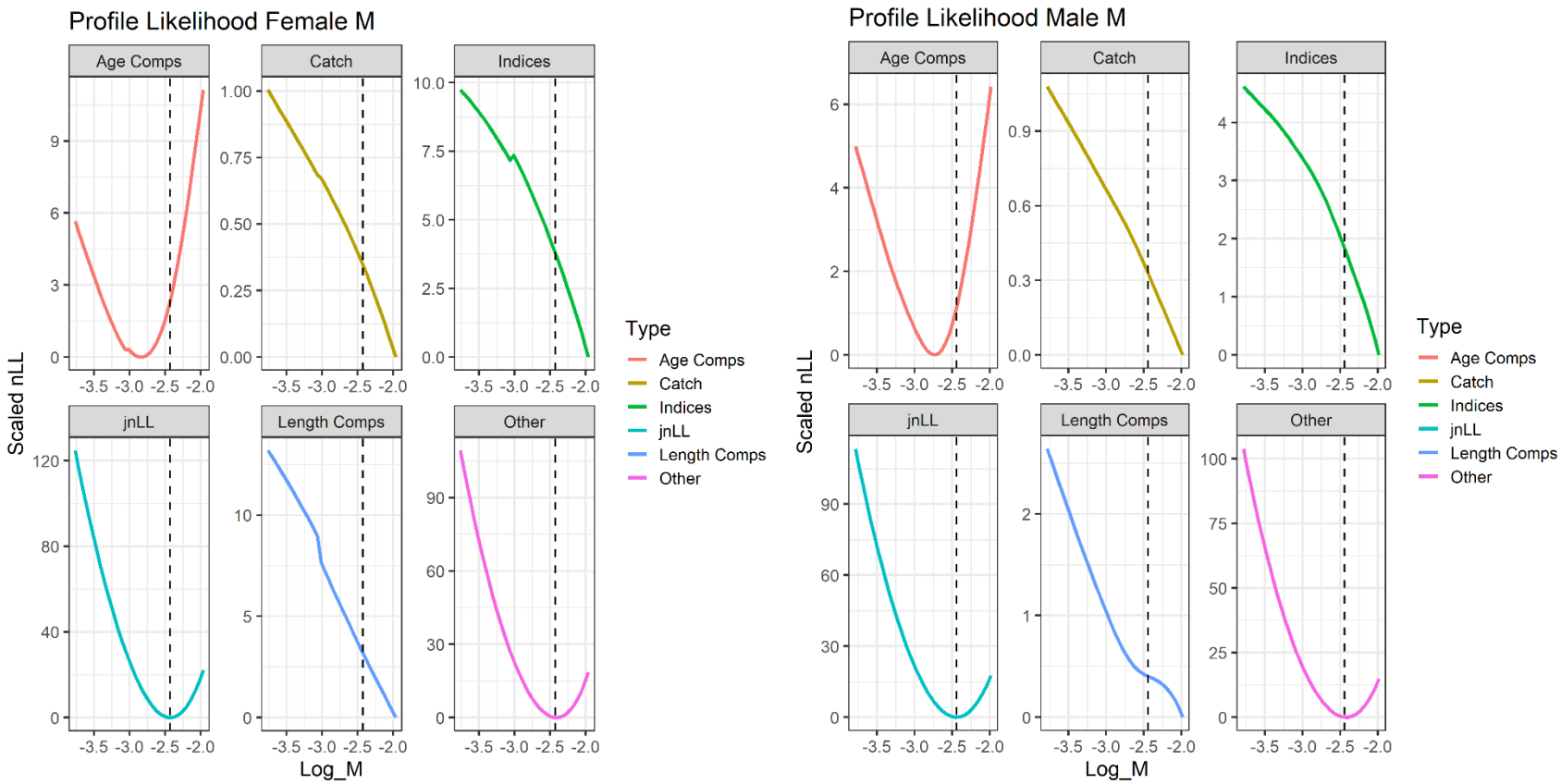

Figure SM3C.96. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for model *26.2_FAA*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

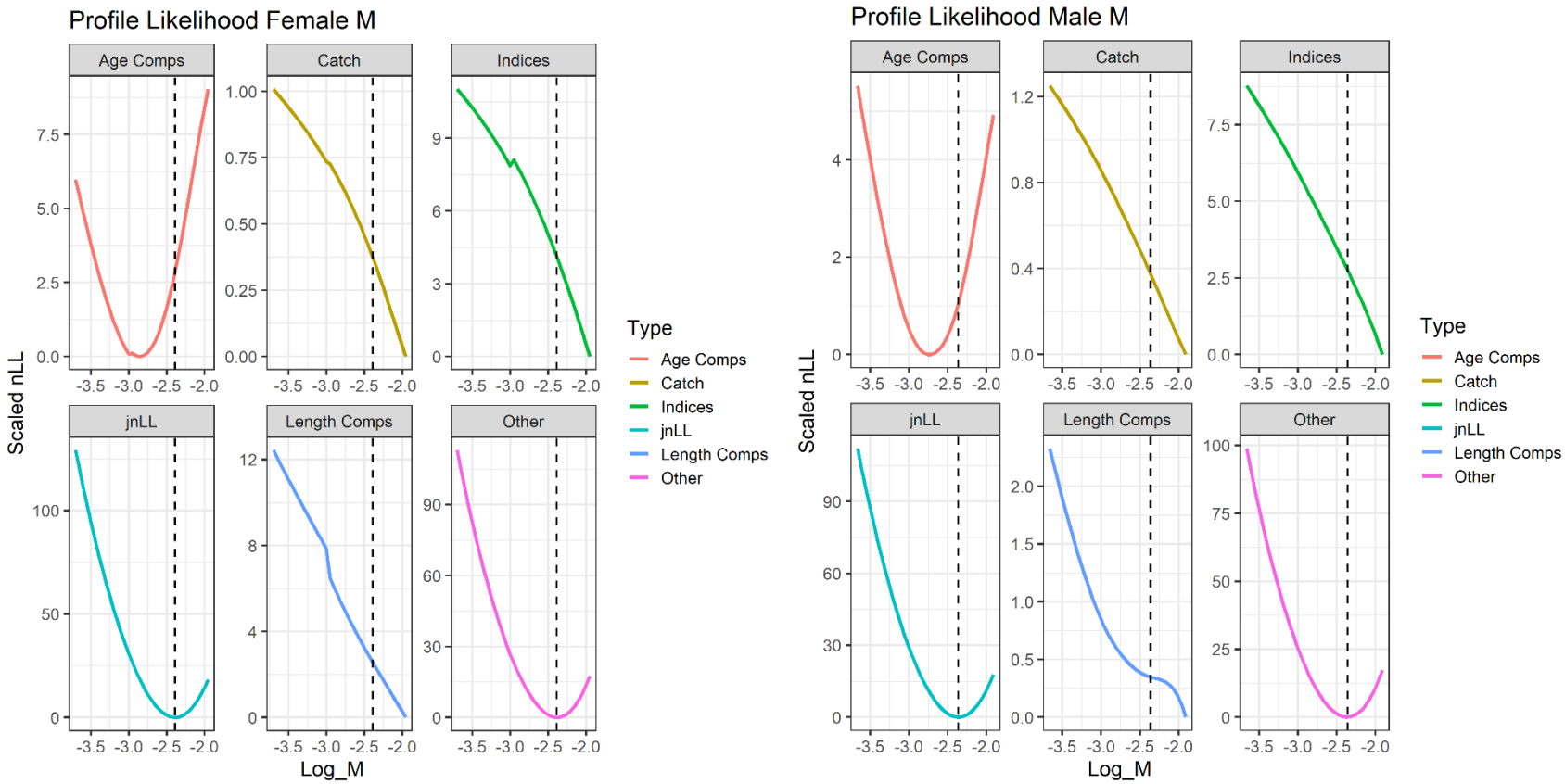

Figure SM3C.97. Likelihood profile for the natural mortality parameter (*M*) for females (left) and males (right), where panels represent the profile for each data type used in the model, for model *26.2a_FAA_Fsh+Srvy*. The 'other' category represents the priors and penalties, which is particularly dominated by the *M* prior.

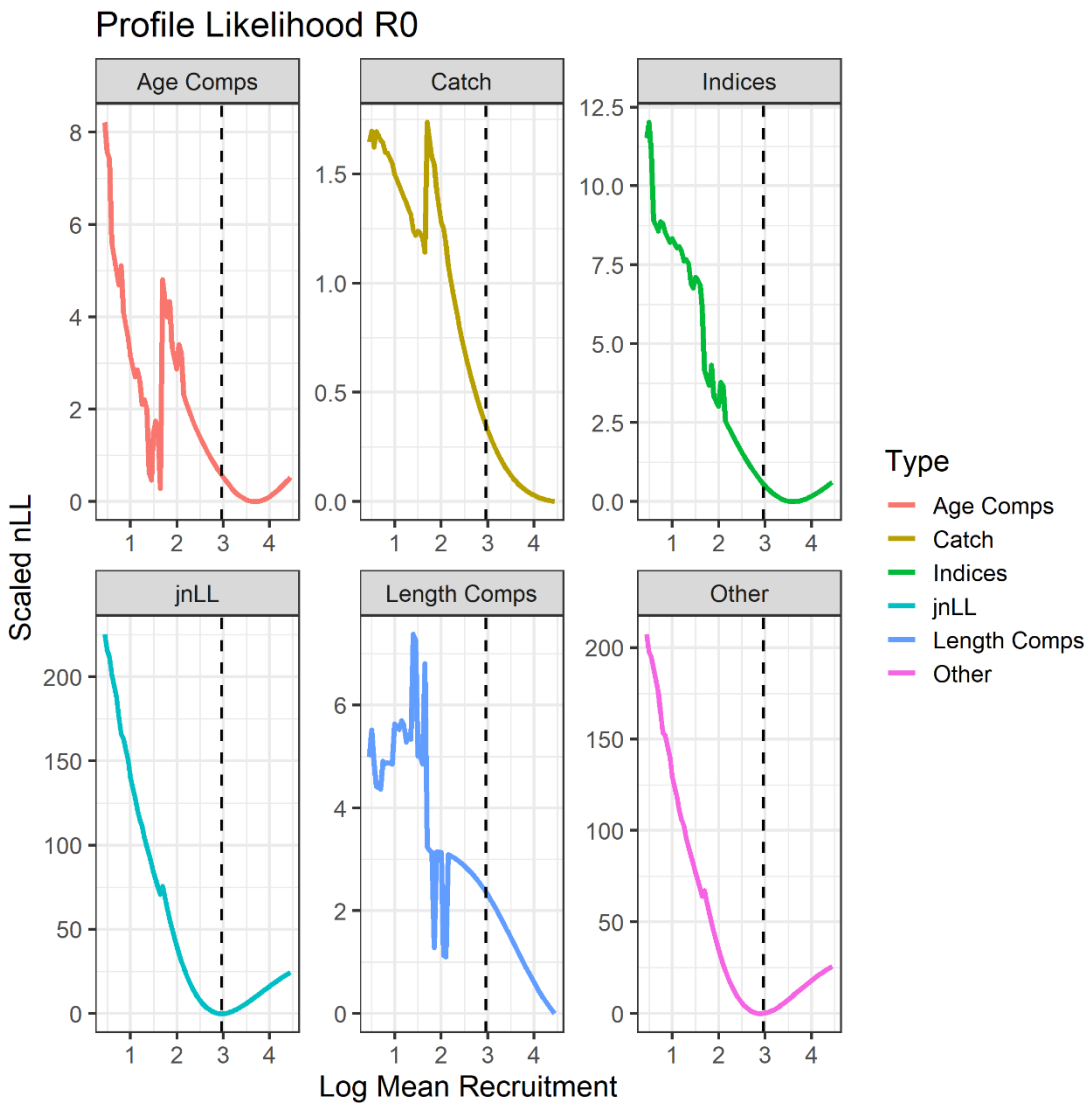

Figure SM3C.98. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for model *26.1_1Reg*.

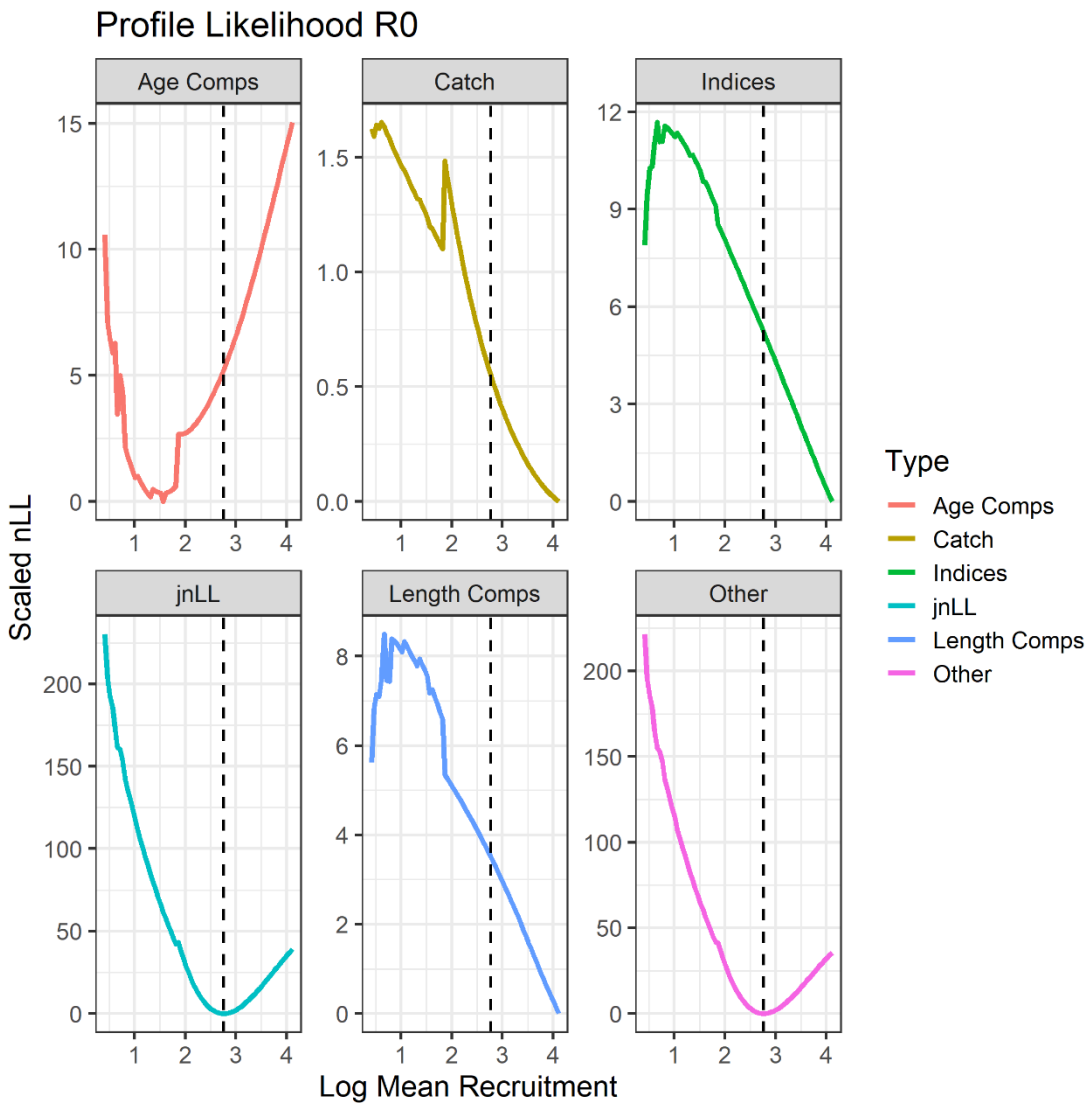

Figure SM3C.98. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for model *26.2_FAA*.

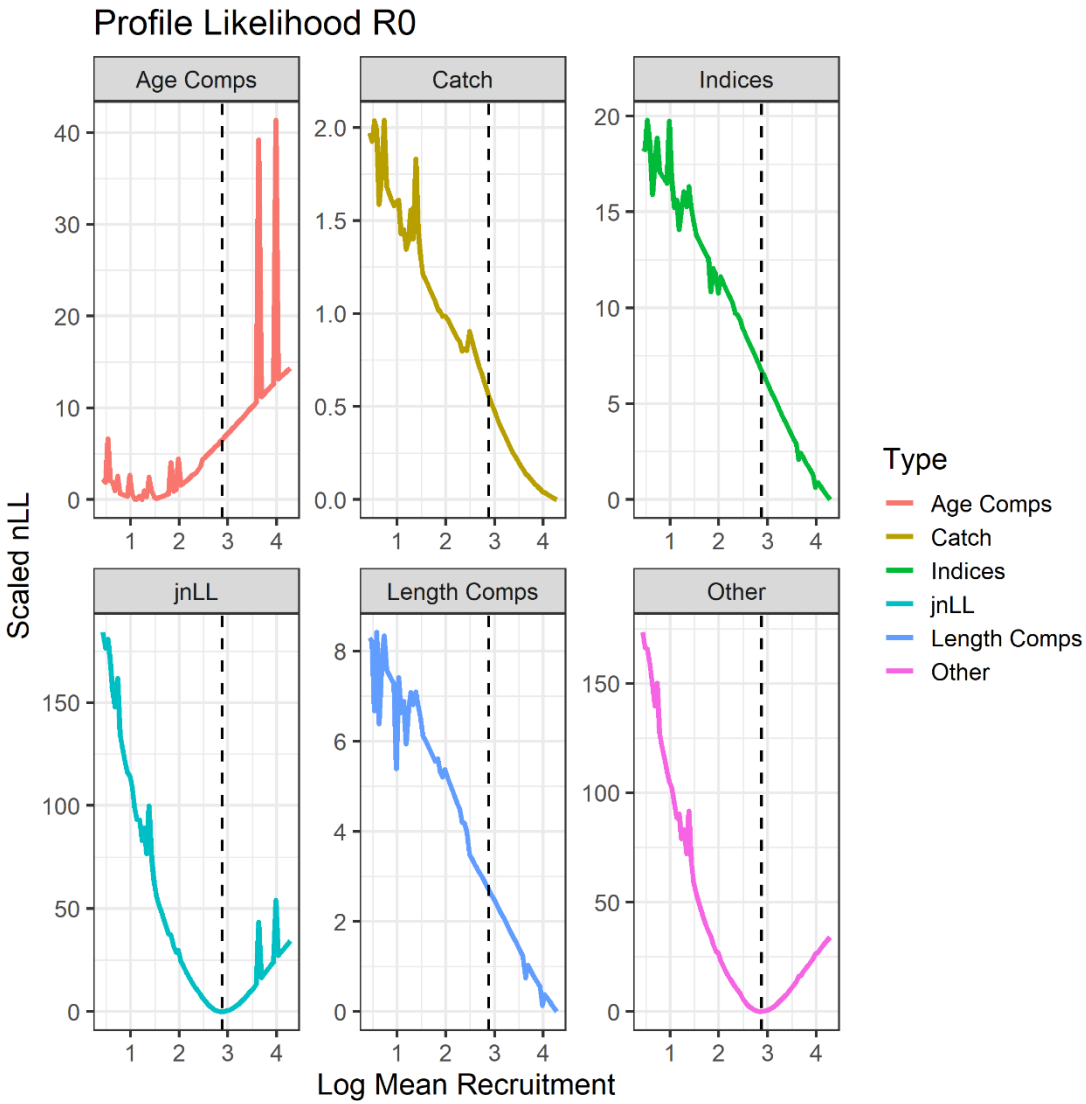

Figure SM3C.99. Likelihood profile for the mean recruitment (R_0), where panels represent the profile for each data type used in the model, for model *26.2a_FAA_Fsh+Srvy*.

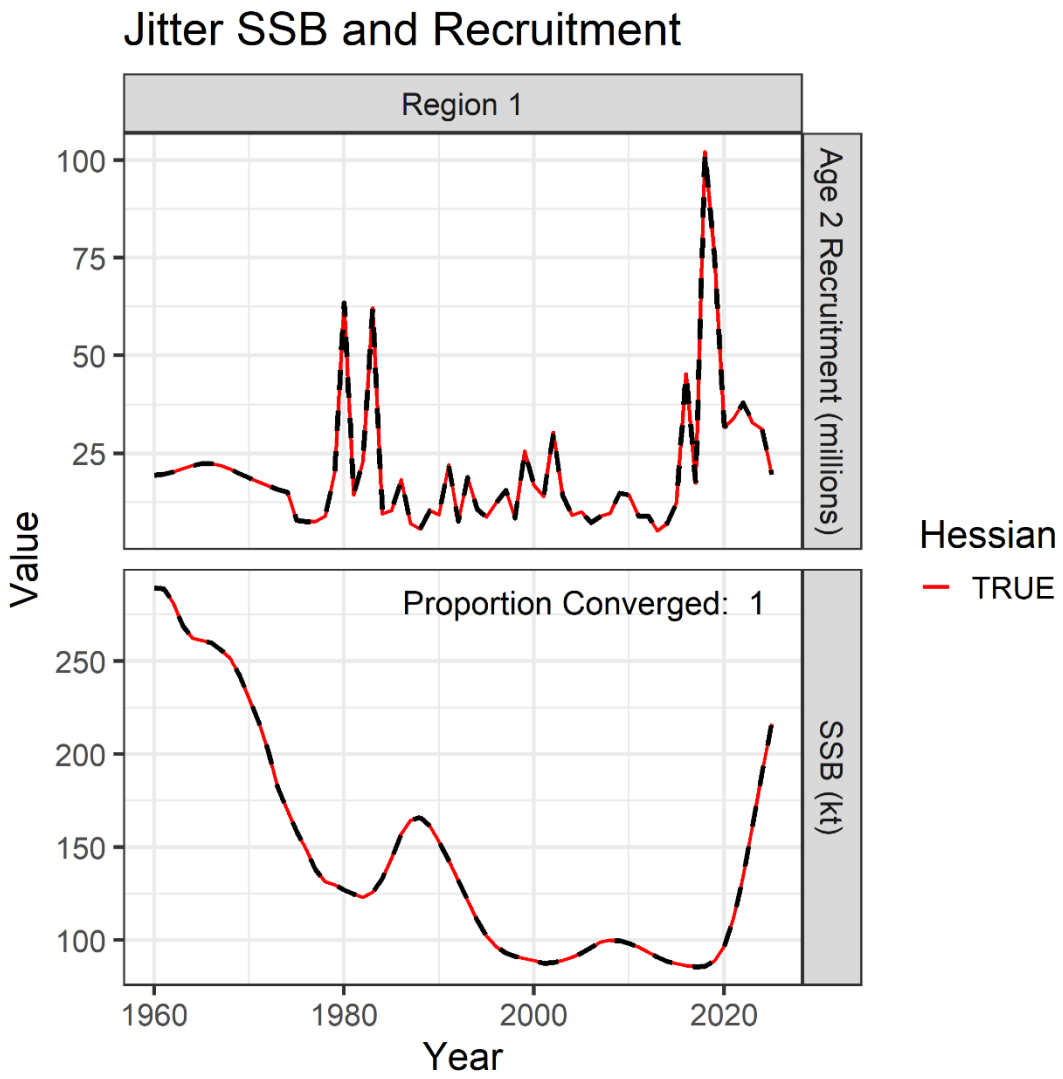

Figure SM3C.100. Results of a jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

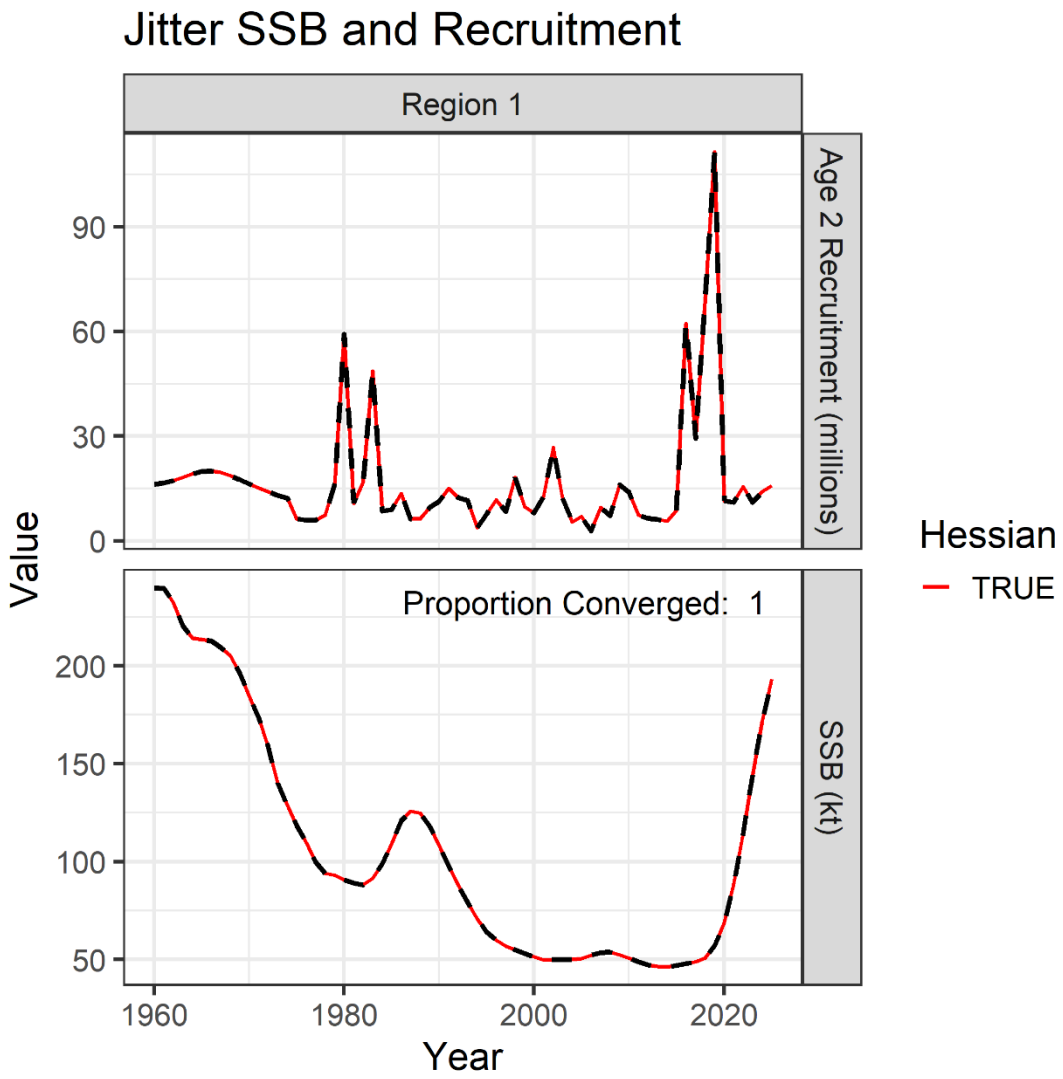

Figure SM3C.101. Results of a jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.2_FAA*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

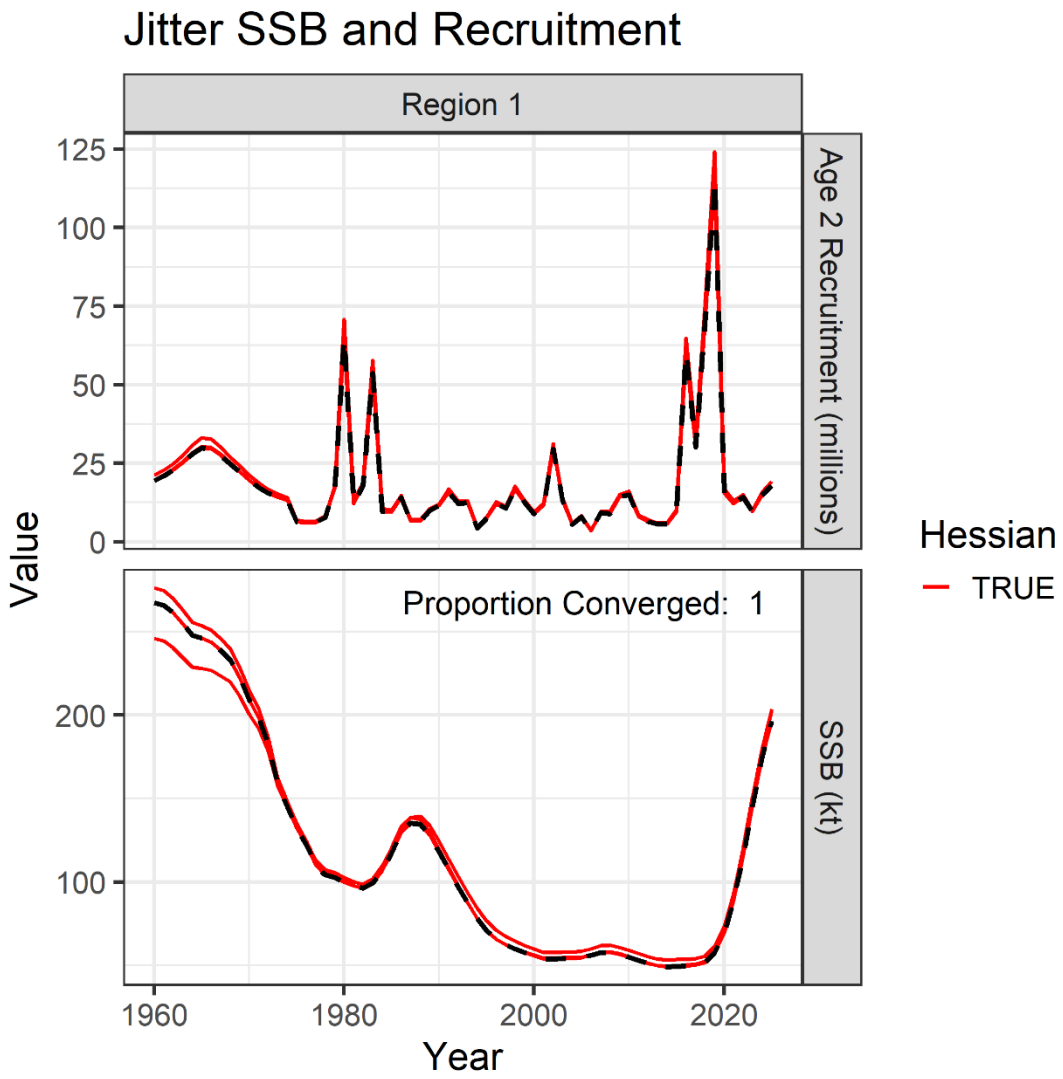

Figure SM3C.102. Results of a jitter analysis showing recruitment (top panel) and SSB (bottom panel) for each jitter iteration for Model *26.2a_FAA_Fsh+Srvy*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

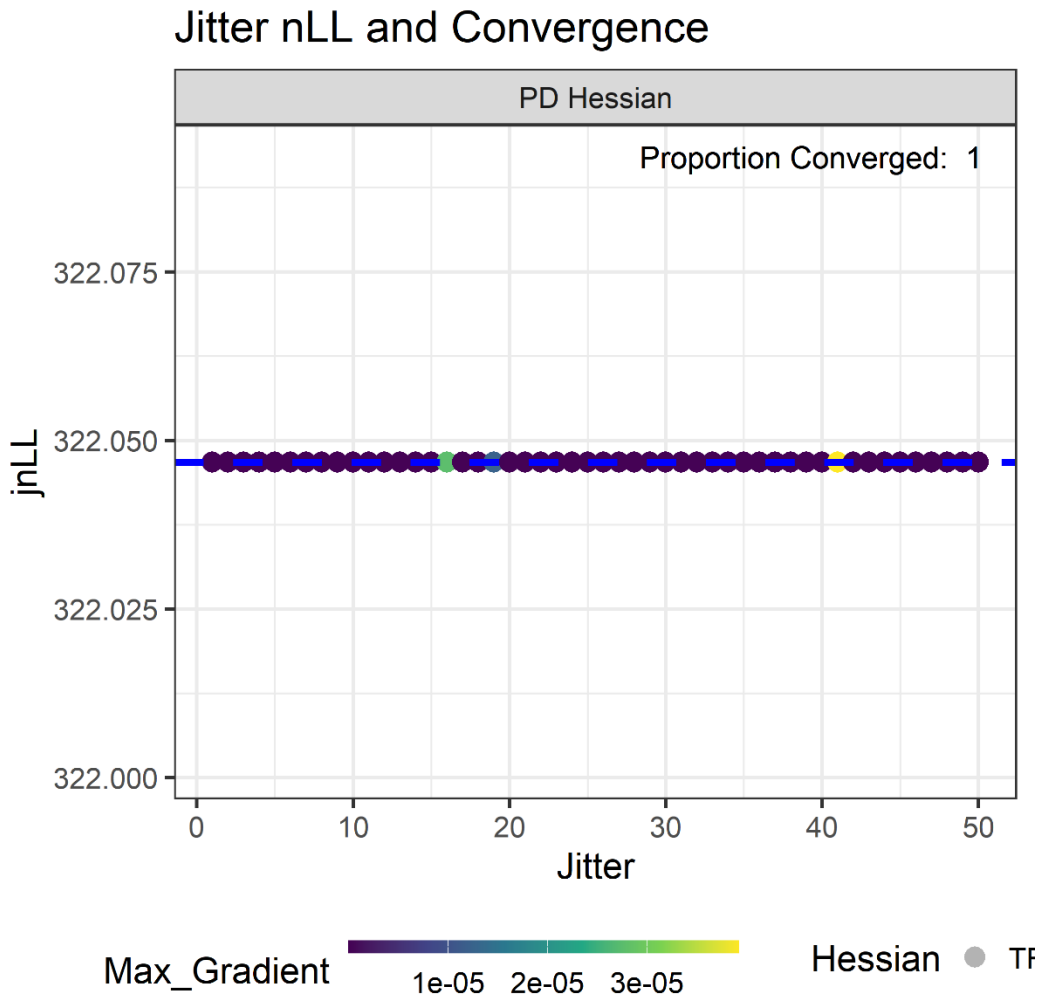

Figure SM3C.103. Results of a jitter analysis showing the joint negative log-likelihood values for converged (Positive Definite, PD, Hessian) and non-converged runs (non-Positive Definite Hessian) for model *26.1_1Reg*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

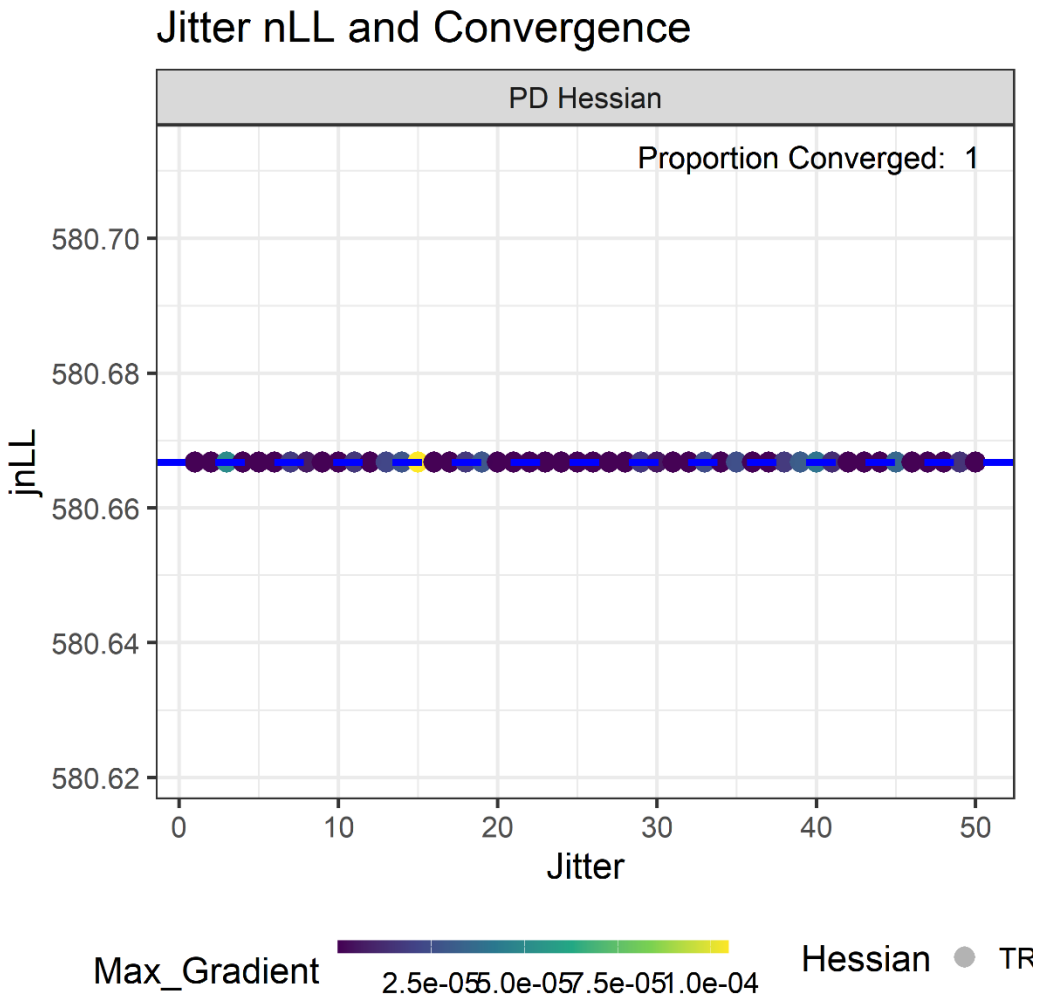

Figure SM3C.104. Results of a jitter analysis showing the joint negative log-likelihood values for converged (Positive Definite, PD, Hessian) and non-converged runs (non-Positive Definite Hessian) for model *26.2_FAA*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.

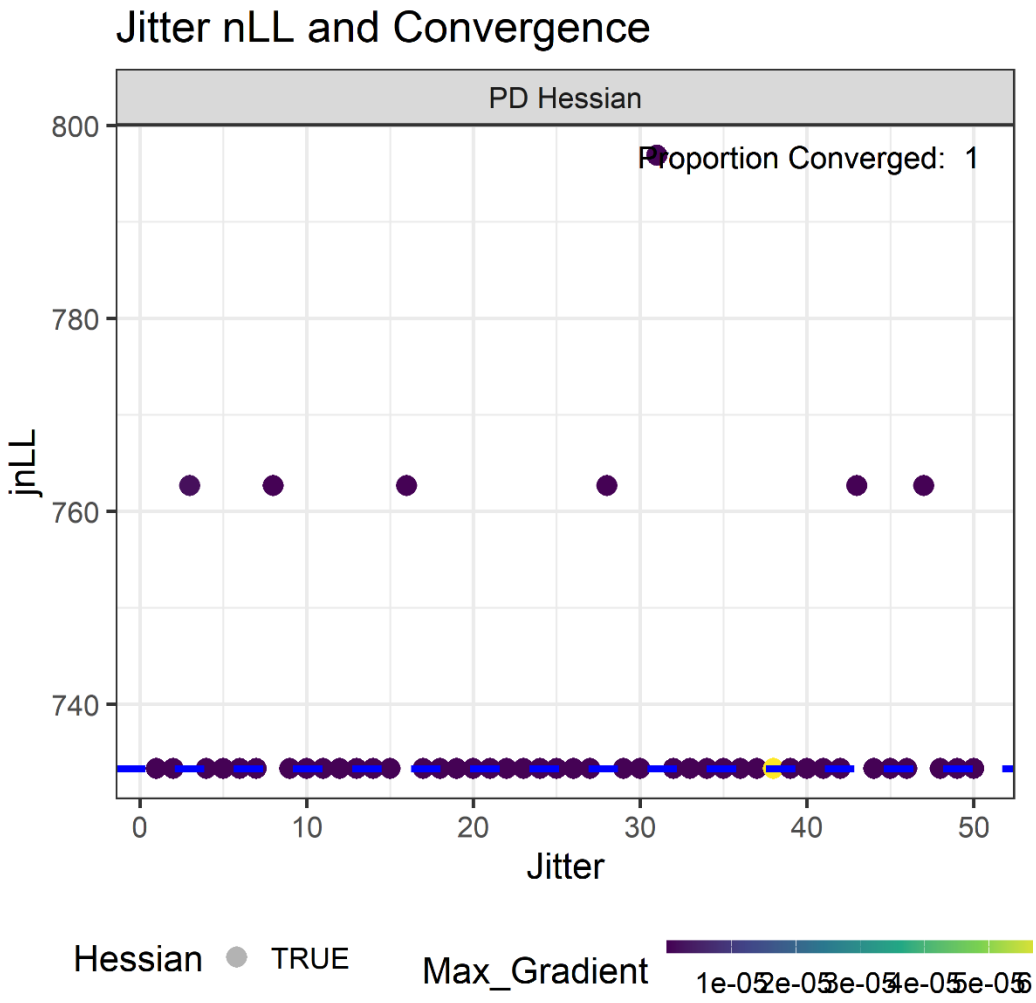

Figure SM3C.105. Results of a jitter analysis showing the joint negative log-likelihood values for converged (Positive Definite, PD, Hessian) and non-converged runs (non-Positive Definite Hessian) for model *26.2a_FAA_Fsh+Srvy*. The starting parameter values were randomly jittered assuming ~$N(0,0.1^2)$.